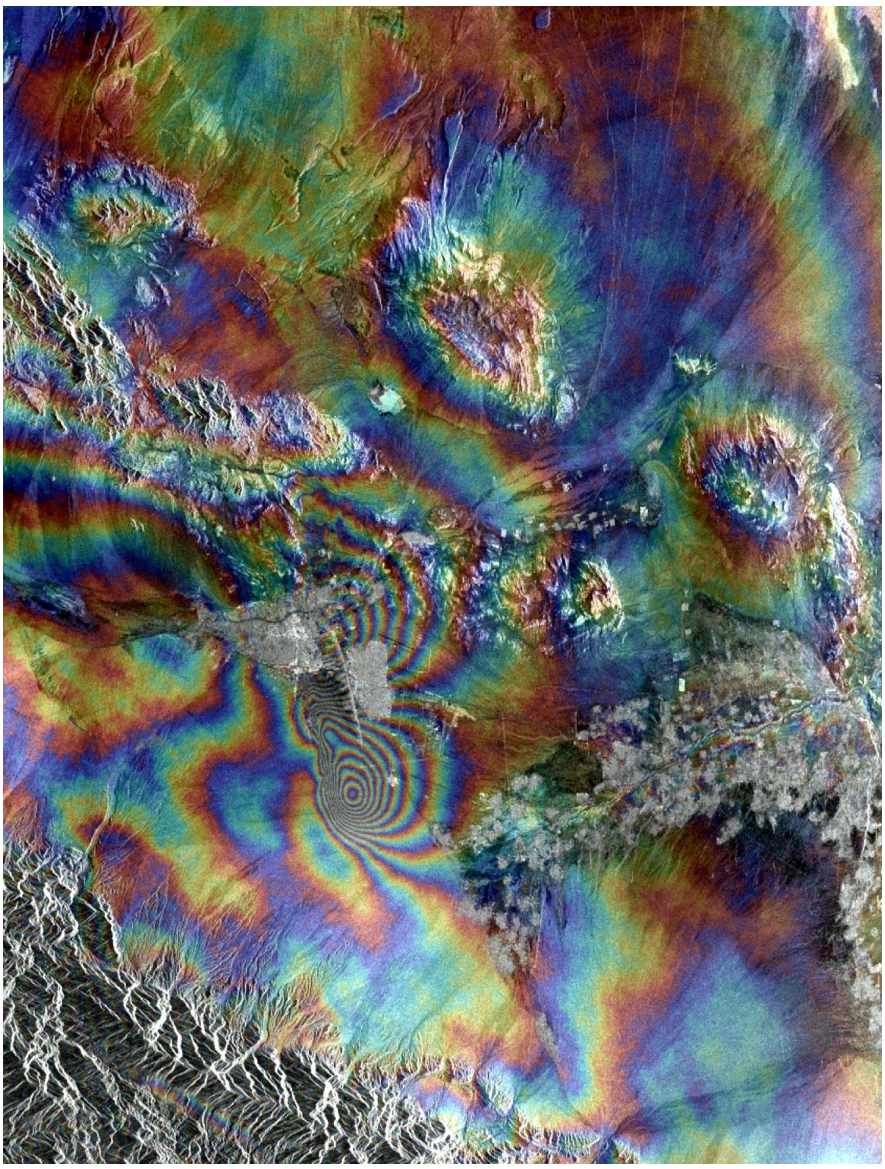
Electromagnetic Imaging

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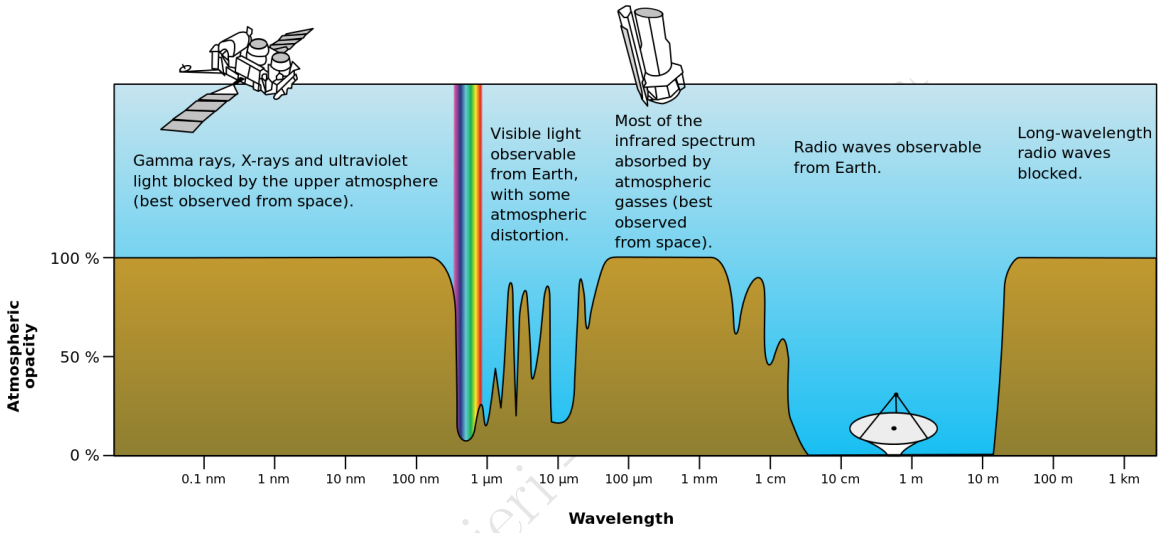
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Preface

The current textbook is a support for students attending the "Radar Imaging" course, that is part of the "Geophysical and Radar Imaging," dedicated to the acquistion systems, calibration, processing and imaging generation and interpretation in the whole EM spectrum, from the microwaves up to visible. The focus is both remote sensing and close, and the goal is to guide the engineer in the use of such systems for their professional activity and may be the design of new customized solution.

The atmospheric attenutation, here schematized, leave some open window for trasmitting significant level of energy: the microwaves are addressed by RADAR, in the second part of this book, while the InfraRed (IR) to visible are addressed in the first part.



The author wish to thank prof. Claudio Prati, whos text-book, "Note di fondamenti di telerilevamento" is at the base of the first part of the course.

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Milano, Oct 2015
Andrea Monti Guarnieri

Part I

Elements of radiometry

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Chapter 1

Wave equation and plane waves

The EM wavefield should be a solution of the Maxwell equations, in particular In a region with no charges and no currents, like in vacuum, they reduce to the EM wave equations:

$$\frac{\partial^2 e(x,y,z,t)}{\partial x^2}+\frac{\partial^2 e(x,y,z,t)}{\partial y^2}+\frac{\partial^2 e(x,y,z,t)}{\partial z^2}=\left(\frac{1}{c}\right)^2\frac{\partial^2 e(x,y,z,t)}{\partial t^2} \tag{1.1}$$

$$\nabla^2\mathbf{E}=\left(\frac{1}{c}\right)^2\frac{\partial^2}{\partial t^2}\mathbf{E}$$

where the second line is the compact vector formulation and c is the velocity of light in vacuum.

The most useful solution of wave equation is a plane wave:

$$e(x,y,z,t)=e^{j2\pi f_{x0}x}e^{j2\pi f_{y0}y}e^{j2\pi f_{z0}z}e^{\pm j2\pi f_0 t} \tag{1.2}$$

$$=e^{j\mathbf{k}_0'\times\mathbf{r}}e^{\pm j2\pi f_0 t} \tag{1.3}$$

while the linearity ensures that all other solutions could be superposition of plane waves. Expression (1.2) has been written as scalar product between the wavelength vector and the space vector:

$$\mathbf{k}_0=2\pi\begin{bmatrix}f_{x0}\\f_{y0}\\f_{z0}\end{bmatrix};\ \mathbf{r}=\begin{bmatrix}x\\y\\z\end{bmatrix}\Rightarrow\mathbf{k}'\times\mathbf{r}=2\pi f_{x0}x+2\pi f_{y0}y+2\pi f_{z0}z$$

The scalar product is zero in the plane orthognal to $\mathbf{r}$, leading to constant phase: that is plane waves. The weight of the linear combination are provided by Fourier Transform o the field, in fact the FT of (1.2) is a Dirac in the 4D frequency domain. To get a simple idea, if the EM field is constant over two direction, say y and z, the solution of the wave equation is, in the (x,t) space, and taking for example the real part

$$\frac{\partial^2 e(x,y,z,t)}{\partial x^2}=\left(\frac{1}{c}\right)^2\frac{\partial^2 e(x,y,z,t)}{\partial t^2}$$

$$e(x,t)=\cos\left(2\pi\left(f_{x0}x\pm f_0 t\right)\right) \tag{1.4}$$

the is a cosinusoid propagating rightward in direction of increasing x if the sign is negative, and opposite if the sign is positive.

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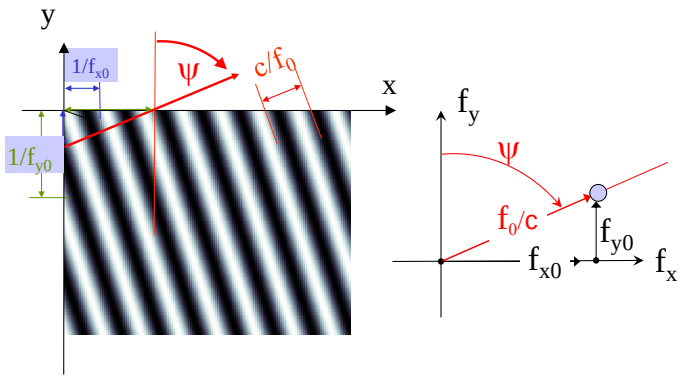


Figure 1.1: Representation of a plane wave in the space (left) and the Fourier domain (right)

The plane wave shown in Fig.1.1 has wavelengths $\lambda_{x0} = 1/f_{x0}$ along x-axis and $\lambda_{y0} = 1/f_{y0}$, and the wavelength that results in the direction orthogonal to the constant phase fronts is $\lambda = \sqrt{\lambda_{x0}^2 + \lambda_{y0}^2}$, and to be a solution of wave-equation, this value should be the same as c/f_0.In the frequency (transformed domain), the figure shows that the same relation holds with frequencies. In the general 3D case we have:

$$\sqrt{f_{x0}^2 + f_{y0}^2 + f_{z0}^2} = \frac{|f_0|}{c} \tag{1.5}$$

The direction of the plane orthogonal to the constant phase defines the polarization, that can be horizontal, vertical, as in the case shown in Fig 1.1 and Fig. 1.2 left, or a combination of the two, as shown in Fig. 1.2, right.

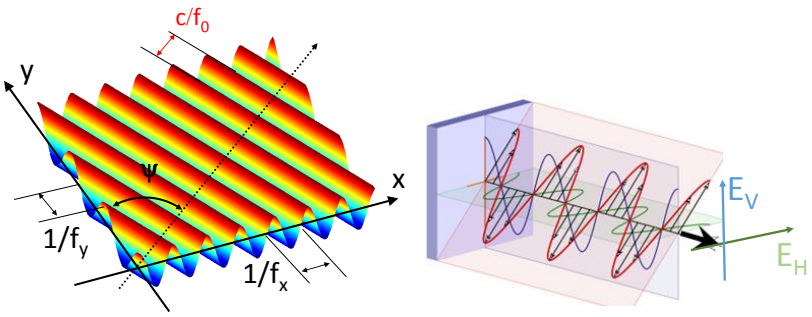


Figure 1.2: Left: 3D of the plane wave in the previous figure, the polarization is vertical. Right: decomposition of a plane wave into its vertical (E_V), and horizontal polarized components (E_H)

Chapter 2

Black body radiation: Planck's law

A Black Body in thermal equilibrium with the environment, is a body that absorbs all the incident energy, without reflecting it (like a mirror, or a white surface would do), nor transmitting (like a transparent body would). In such conditions, the BB irradiates, in the average, the same amount of energy in form of electromagnetic waves. Each instant, it emits an Electro Magnetic field that is the result of random motion of a huge number of electric charges. Such field is emitted as a random superposition of modes, that corresponds to stationary waveform that are null at the extreme of the body, like the wave generated in a pool of water between two rocks.

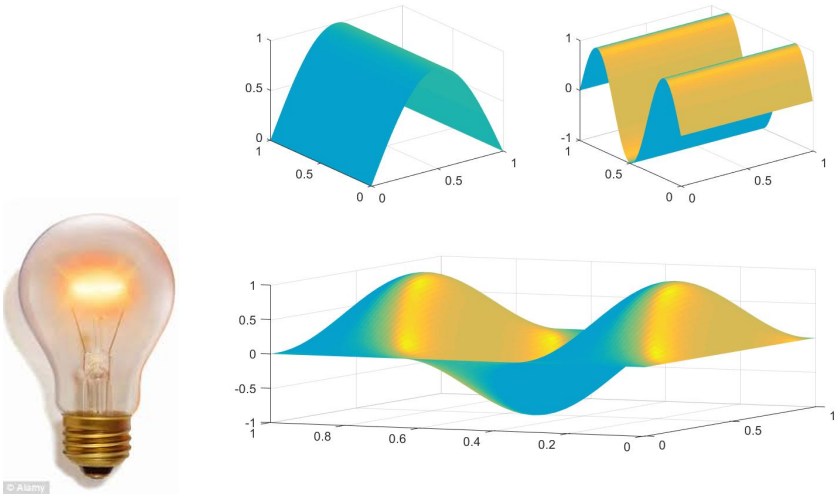


Figure 2.1: Left: a bulb as approximation of a black body. Right: example of a stationary waveform resonant at the boundaries, corresponding in the 2D to the mode: $n_x = 1, n_y = 3$

The EM filed is random, but its mean energy can be derived by combining the number of freedom degrees, that is the different, independent, modes in which an electromagnetic field can be generated, with the mean energy associated with each mode.

For one axis, say x, the stationary wavelengths are:

$$\lambda_x = \frac{2L}{n_x}$$

where $n_x$is a positive integer. This should hold by generating a proper combination of plane

waves in (1.4). As an example, in the 2D case the following signal

$$f(x,y;t) = \sin(2\pi f_x x)\sin(2\pi f_x x)\exp(j\omega t)$$

has the property to be null at the edge of the boxes, say for x=0, x=1, y=0, y=1 and to be a linear combination of plane waves, in fact:

$$\sin(a)\sin(b) = \frac{\exp(ja)}{2j}\left\{\frac{\exp(jb)}{2j} - \frac{\exp(-jb)}{2j}\right\} - \frac{\exp(-ja)}{2j}\left\{\frac{\exp(jb)}{2j} - \frac{\exp(-jb)}{2j}\right\}$$
$$2j\sin(a)\sin(b) = \exp(j(a+b)) - \exp(j(a-b)) - \exp(-j(a-b)) + \exp(-j(a+b))$$

As an example, in Fig 2.1 on the left, the mode n_x=1,n_y=3 has been generated with the following Matlab code:

r=0:0.01:1; [X,Y] = meshgrid(r, r);
*fx=2*pi*0.5*X; fy=2*pi*0.5*3*Y;*
subplot(221); surfl(r,r,sin(fx)); shading('flat')
subplot(222); surfl(r,r,sin(fy)); shading('flat')
*subplot(212);surfl(r,r,sin(fx).*sin(fy)); shading('flat')*

The transmitted frequency is bounded by (1.5) to be:

$$\frac{1}{2L}\sqrt{(n_x^2+n_y^2+n_z^2)} = \frac{f}{c} \tag{2.1}$$

and this means that the number of modes in the spectrum up to $f \leq f_0$ increases as the volume of a cube (actually n_x, n_y, n_z are all the integer positions internal to a cube of radius $\sqrt{2L}\cdot f_0/c$. As this volute is proportional to the cube of the radius, the total number of modes $N_T \propto f_0^3$and then the increment with frequency is expected to be

$$\Delta N_T \propto \frac{\partial f_0^3}{\partial f_0}\Delta f_0 \Rightarrow \frac{\Delta N}{\Delta f} \propto \left(\frac{f}{c}\right)^2 \tag{2.2}$$

If we assume that for each mode is associated an average energy

$$E_0 = k_B T, \tag{2.3}$$

where

$$k_B = 1.38\times 10^{-23}[\mathrm{J/K}]$$

is Boltzmann constant, that comes form Maxwell-Boltzmann probability density function energy of each particle (the same applies in thermodynamic for the energy of each freedom degrees in a gas), than we wold get from (2.1) that the energy would diverge at highest frequencies. The result would be a shift of the energy power spectrum beyond the ultraviolet, for the very simple reason that the number of available modes diverges, causing the so called "ultraviolet catastrophe", that is clearly incompatible with observations.

It was just to justify the observations, that Planck postulated the fact that energy cannot get arbitrary values, but it is quantized:

$$E_n = n\cdot hf \tag{2.4}$$

where n is integer, and h is Plank constant:

$$h = 6.626\times 10^{-34}\,[J\times s]$$

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therefore, when the frequency diverges the emission from the BB cannot happen since it has not enough energy to build a quantum, say n=1. If we account for the discretization of energy, the expected average energy for each freedom degrees turns to be:

$$E_0 = \frac{hf}{e^{\frac{hf}{k_B T}} - 1} \tag{2.5}$$

Note that n is expected to be a big number at low frequency, in fact when

$$hf \ll k_B T \Rightarrow f \ll \frac{k_B T}{h} \tag{2.6}$$

that is $\sim$ 6 THz at 300 °K (indeed a low temperature), the approximation $\exp(x) \sim 1 + x$ reduces (2.5) to (2.4). That is, in the full range of microwave, the classical approximation $E \propto f^2$ is well justified.

If we multiply the mean energy for the number of freedom degrees, and then we compute the power that is isotropically irradiated per unit surface by a BB in thermal equilibrium (that is the total emitted and received powers are equal), we get the following expression, **Planck Law**:

$$\frac{P(f)}{A \cdot df} = \frac{2\pi f^2}{c^2} \frac{hf}{e^{\frac{hf}{k_B T}} - 1} \left[W/m^2/Hz\right] \tag{2.7}$$

The power spectrum can be integrated over the whole spectrum, giving the total power emitted, per unit surface of the BB, that is **Stefan-Boltzmann law**:

$$\frac{P}{A} = \sigma T^4 \left[W/m^2\right] \tag{2.8}$$
$$where \sigma = 5.67 \times 10^{-8} \left[W/m^2/K^4\right]$$

that is named Stefan–Boltzmann constant or Stefan's constant. The expression can be used also to compute the energy that is transferred from a BB to the environment where there is not a thermal equilibrium. If the BB is at $T_{BB}$and the environment T_{ENV} the flow of power per unit area is:

$$\frac{P}{A} = \sigma \left(T_{BB}^4 - T_{ENV}^4\right) = \sigma T_{BB}^4 \left(1 - \left(\frac{T_{ENV}}{T_{BB}}\right)^4\right)$$

where we notice that the second term can be ignored when environment temperature is, say half of that of the BB.

The Plank Law is often expressed in terms of wavelengths, in place of frequencies:

$$\frac{P(\lambda)}{A \cdot d\lambda} = \frac{P(c/f)}{A \cdot df} \cdot \frac{df}{d\lambda} = \frac{2\pi}{\lambda^3} \frac{hc}{e^{\frac{hc}{\lambda k_B T}} - 1} \frac{d}{d\lambda}\left(\frac{c}{\lambda}\right)$$
$$= \frac{2\pi}{\lambda^5} \frac{hc^2}{e^{\frac{hc}{\lambda k_B T}} - 1} \left[W/m^2/m\right] \tag{2.9}$$

2.1 Example: sun and Earth

The sun sun in is a good approximation of a BB, even though not exactly in thermal equilibrium (it is colding), is represented, together with its relevant parameters, temperature and size, in Fig. 2.2.

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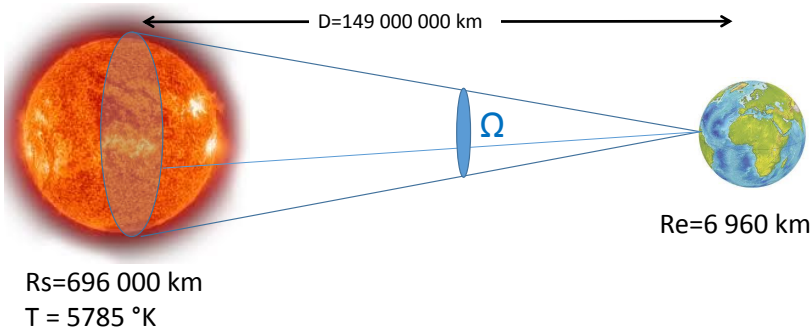


Figure 2.2: Sun and Earth (not in scale)

We can use Stefan-Boltzmann law to compute the total power irradiated in thermal equilibrium:

$$P_s = \sigma T^4 \cdot 4\pi R_s^2 = 3.8 \times 10^{26} W \tag{2.10}$$

that is indeed an extraordinary huge power. However the fraction that gets to the Earth surface is minimal:

$$\frac{P_e}{A} = \frac{P_s}{4\pi D^2} = 1380 W/m^2 \tag{2.11}$$

but still significant to operate solar cells. Actually there is an 8% variation due to the change of distance along the year.

We can then compute the total power that gets to the Earth and the mean Earth temperature:

$$\begin{aligned} P_e &= \frac{P_s}{4\pi D^2}\pi R_e^2 = 2.1 \times 10^{17} W \\ &= \sigma T_e^4 A_e \Rightarrow T_e = \sqrt[4]{\frac{1}{\sigma}\frac{P_e}{A_e}} = \sqrt[4]{\frac{1}{\sigma}\frac{P_s \pi R_e^2}{4\pi D^2 4\pi R_e^2}} \\ T_e &= \sqrt[4]{\frac{1}{\sigma}\frac{1380}{4}} = 279^\circ K = 6^\circ \mathrm{C} \end{aligned} \tag{2.12}$$

This is the power we would get under ideal conditions. However, the true balance is different for many reasons.

First, we have to account for the **reflectivity** of the Earth surface, that makes of Earth a gray body rather than a BB. We assume that the **albedo** that is the percentage of reflected energy is 0.3. This would further reduce the temperature to

$$T = 279 \cdot \sqrt[4]{0.7} = 255^0 K = -18^\circ C$$

As second instance, we have to account for the effect of atmosphere. We observe, from the radiation power spectra (expressed in "radiance", see next section) of sun and Earth, that for the first the atmosphere is almost transparent at the optical frequencies, where most of the energy is concentrated, whereas for the Earth, the atmosphere is only partially transparent - only in the MIR (medium infrared), as shown in Fig. 2.3 This means that the energy irradiated by earth, differently form the one reflected, is kept in the lower parts of the atmosphere, contributing, with this green-house effect, to the increase of average temperature to 288°K∼15°C. This value would still neglect the energy radiated by the atmosphere itself

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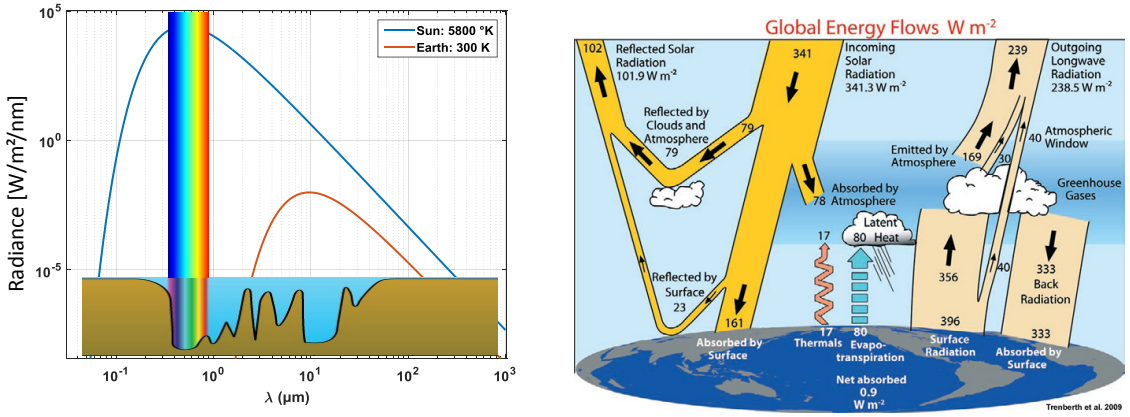


Figure 2.3: Left: radiance power spectrum from sun and Earth (assuming T=300°K), superposed to the atmosphere attenuation. Right: global annual mean energy balance for March 2000 to May 2004, from "Introduction to Tropical Meteorology", http://www.goes-r.gov/users/comet/tropical.

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(that is also a gray body, named "path radiance") in the direction of the Earth, the major fluxes of energy being summarized in the figure.

Indeed Earth ecosystem is highly dependent on slight variation of temperature, that in turns is dependent on the atmosphere attenuation at IR, and the variation of gas in the atmosphere play a fundamental role, not yet fully understood. Notice how the atmosphere is effective to screen the highly energetic photon, those at UV and further. This does not happen in the same way at sea level or in the mountain, where solar irradiation is much more dangerous.

2.1.1 Solar cells

Solar cells are rather common, nowadays. Panels of 35 ×20 cm are commercially available for 20€ delivering 10 W. The power per unit area in (2.11) of $1380 W/m^2$ is too optimistic, as the atmosphere absorbs a part of that energy (even in absence of clouds). We may assume a value of 1000 W/m^2, at sea level, clear day, that would lead to a total :

$$P = 1000\,(0.35 \times 0.2) = 70W$$

that reduces further if the surface is not orthogonal to the irradiation: as an example we assume to leave it flat and a sun elevation of 20° (that occurs at our latitudes in December at noon), that leads to an offset angle of 70° and reduction of the surface of cos(70°)=0.34.

Furthermore, solar cell does not capture the whole spectrum, and this reduces the efficiency, say for cheapest and most used in polycrystalline silicon, to 15%. This makes $70 \times 0.34 \times 0.15 = 3.5$W, whereas the value declared, 10 W, applies only for perfectly oriented cells. Nonetheless, that would still be enough to charge in the order of a couple of hours, a ˜2000 mA/h, 4 V battery of a smartphone.

2.1.1.1 Motorbike and greenhouses

The saddle of the motorbike, left in sun light, comes to thermal equilibrium getting a power that depends on the season. In the previous exercise we computed a factor 0.34 in winter at

noon, whereas in summer (21-june) sun elevation is 67° (at our latitudes, 45°), leading to a factor 0.92. The in the two seasons::

$$\frac{P_W}{A} = 1000 \cdot 0.34 = \sigma T_W^4 \Rightarrow T_W = \left(\frac{340}{\sigma}\right)^{0.25} = 278.2°\text{K}=5°\text{C}$$
$$\frac{P_S}{A} = 1000 \cdot 0.92 = \sigma T_S^4 \Rightarrow T_S = \left(\frac{920}{\sigma}\right)^{0.25} = 357°\text{K}=83°\text{C}$$

where we ignore the transportation of heat by convention and conduction. Notice that in winter we can enhance the temperature by using windows, that block the emission at medium infrared, thus elevating the temperature in the greenhouse.

2.2 Radiance

So far, (2.8) and (2.9) express the total power and the power spectrum per unitary area of the emitting BB. However, it is more convenient to relate this two magnitude to the position of the sensor measuring, or using, that power. Following (2.10) and (2.11) the total power per **receiving surface** at a distance D is:

$$\frac{P_e}{A} = \sigma T^4 \cdot \frac{4\pi R_s^2}{4\pi D^2} = \sigma T^4 \cdot \frac{\Omega}{\pi}$$

where Ωis the solid angle under which the source is observed, as shown in Fig. 2.2. From this expression we introduce the definition of **Radianc**e as the power per area (of the sensor) per unit solid angle:

$$L = \frac{P_e}{A\Omega} = \sigma T^4 = \frac{\sigma T^4}{\pi} \left[W/m^2/sr\right] \tag{2.13}$$

In the same way, the Radiance power spectrum follows from (2.9):

$$L(\lambda) = \frac{2hc^2}{\lambda^5} \frac{1}{e^{\frac{hc}{\lambda k_B T}} - 1} \left[W/m^2/m/sr\right] \tag{2.14}$$

that we can as well express in frequencies, like in (2.7):

$$L(f) = \frac{2f^2}{c^2} \frac{hf}{e^{\frac{hf}{k_B T}} - 1} \left[W/m^2/Hz/sr\right] \tag{2.15}$$

An example of radiance from two BB at temperature comparable with sun and earth are in Fig. 2.3.

2.3 Color of the radiation and Wien law

Wien law represent a quick and very rough way to approximate the temperature with a color. In fact it computes the peak of the power spectrum, as function of the temperature:

$$\frac{\partial L(\lambda)}{\partial \lambda} = 0 \Rightarrow \lambda_M \simeq \frac{2898}{T} [\mu m]$$
$$\frac{\partial L(f)}{\partial f} = 0 \Rightarrow f_M \simeq 5.87 \cdot 10^{10} T[Hz]$$

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As an example, for the sun, T=5785°K, we get a peak of the radiance power per frequency at $f_M = 3.4 \cdot 10^{14} (\lambda \sim 0.88 \mu m)$, whereas the peak of the radiance power per wavelength is at $\lambda_M \sim 0.5 \mu m$. It is not a surprise that the two peaks are different: this is due to the fact that the transformation between frequencies and wavelength is not linear, and the power spectrum is smooth around the peak.

A much better way to associate the temperature of a BB with the color perceived by or eyes, is to weight the power spectrum by the color matching functions, like those shown in Fig. 2.4 on the left. The result, shown in the same figure on the right, also known as Plankian locus, provides a more likely conversion of temperature to color. Notice that only subset of the whole colors belongs to the one of a BB.

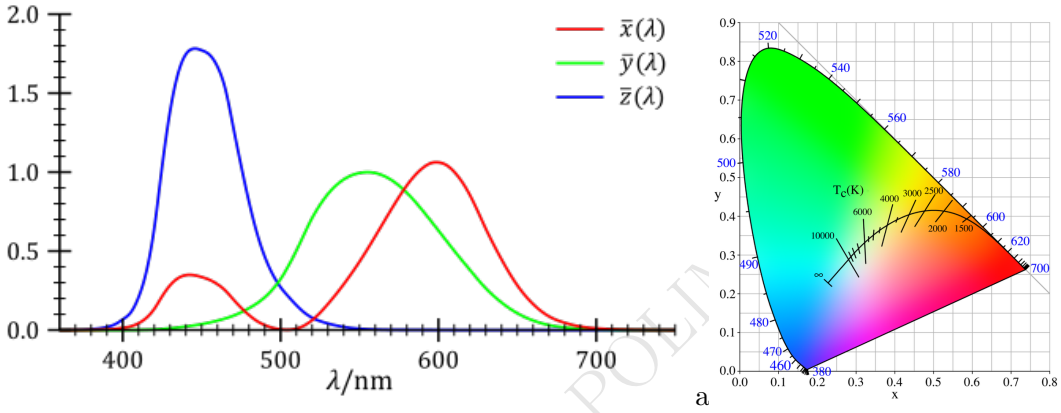


Figure 2.4: Left: color matching function for CIE space and, right, the Plankian locus.

However, do not expect a simple relationship between color and temperature, in some cases, like for combustion, chemical elements with particular absorption index can change the perceived color. In most cases, Wien law provides a way to find temperature of stars given the color.

2.4 Kirchhoff law: the gray body

BB is an ideal case of a body emitting 100% of the received energy. Real bodies, "Gray" Bodies (GB) reflect part of the incident power, so that the one emitted as EM field is lower than unitary:

$$\frac{P_e}{A} = \epsilon \sigma T^4 \; 0 \leq \epsilon \leq 1$$

ϵ is the emissivity (re-irradiated). The reflected power is then a portion ρ so that, in thermal equilibrium the outgoing energy, sum of emitted and reflected, equals the ingoing:

$$\frac{P_t + P_e}{A} = (\epsilon + \rho)\, \sigma T^4 = \sigma T^4 \Rightarrow \rho = 1 - \epsilon \tag{2.16}$$

That is, reflectivity, also called "albedo" is complementary to emissivity. A total reflecting GB, is white ("albus" in Latin). Typical values are reported in 2.1, notice that in general emissivity change with frequency and the polarization, however balances like (2.16) concern energies and are then intended integrated over the whole spectrum (and polarization). An example is the polyethylene carbon bag shown in Fig. 2.5 is transparent to the IR, but not the visible.



Figure 2.5: Polyethylene carbon bag observed by optics and IR, from physics.stackexchange.com.

Material	Emissivity
Aluminum foil	0.03
Aluminum, anodized	0.9
Asphalt	0.88
Brick	0.9
Concrete, rough	0.91
Copper, polished	0.04
Copper, oxidized	0.87
Glass, smooth (uncoated)	0.95
Ice	0.97
Limestone	0.92
Marble (polished)	0.89 to 0.92
Paint (including white)	0.9
Paper, roofing or white	0.88 to 0.86
Plaster, rough	0.89
Silver, polished	0.02
Silver, oxidized	0.04
Snow	0.8 to 0.9
Water, pure	0.96

Surface	Typical albedo
Fresh asphalt	0.04
Worn asphalt	0.12
Conifer forest	0.08 to 0.15
Deciduous trees	0.15 to 0.18
Bare soil	0.17
Green grass	0.25
Desert sand	0.4
New concrete	0.55
Ocean ice	0.5–0.7
Fresh snow	0.80–0.90

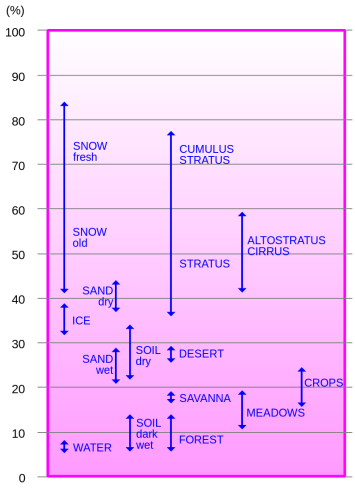


Table 2.1: Typical emissivity (left), albedos (middle), and sunlight reflections (right) from Wikipedia

Notice that fresh snow has very high albedo, however melting changes its water content and grain size to much lower value. This in turn accelerate the melting as more heath is kept. A similar mechanism is causing the fast melting of polar ice.

This result is Kirchhoff law of thermal radiation, that predicts that when a GB has high emissivity, $\epsilon \to 1$, the reflection is minimized, $\rho \to 0$, i.e. the BB is the best transmitter. Vice-versa, a white body does not absorb energy, nor it transmits.

So far, we have ignored the case when a GB in partly transparent, that is, transmits a portion $\alpha \leq 1$of the incoming energy. In this general case, the outgoing energy is the sum of the conducted, the emitted and the reflected, and this balance the ingoing one:

$$P_I = \alpha P_I + \epsilon P_I + \rho P_I \Rightarrow \alpha + \epsilon + \rho = 1$$

The three ideal cases are shown in Tab.

In many remote sensing applications, addressing properties of the Earth and atmosphere, measures are taken from the signal emitted by sun and reflected from the earth, that is named "reflectance" in similarity with radiance. The advantage is to exploit sum light, that has high energy in all spectral components from mid infra-red to visible, with exception to

Body type	absorptivity	emissivity	reflectivity
	α	ϵ	ρ
Black	0	1	0
White	0	0	1
Transparent	1	0	0

Table 2.2: Properties of ideal bodies

some frequencies as shown in Fig. 2.3, much higher energy than earth, however measures should be carried out during the day and are potentially effected by cloud cover. An example of a radiance and a reflectance images is given in Fig. 2.6.

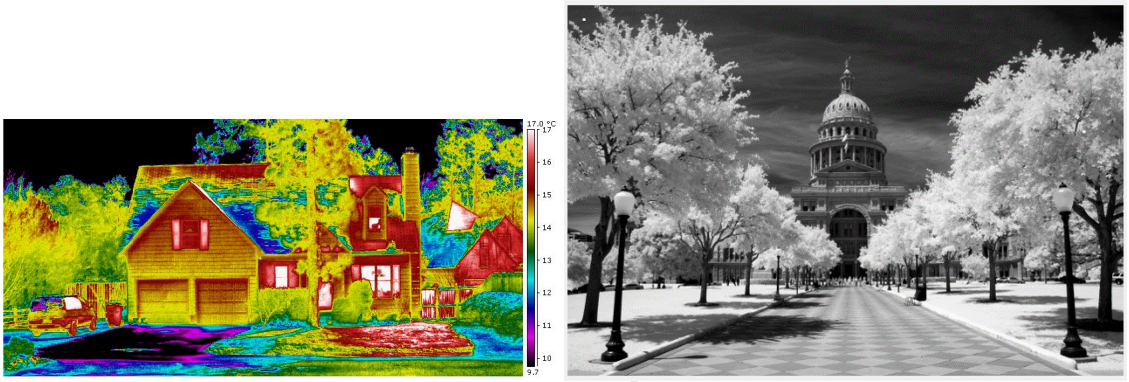


Figure 2.6: Radiance (left) and reflectance (right) images taken in the Infra-Red - source: web.

In general, we observe two different type of reflections, all combined in Fig. 2.7 on the left: **specular** and **diffuse**. The first, that are originated by coherent mechanism that will be discussed in next section, reflectance will be nearly zero at all angles except at the appropriate reflected angle computed by optics. In the second case, the light is diffused everywhere. A particular case of this is when reflectance is uniform in all angles, then the intensity will depend upon the cosine of the angle between the direction of the incident light and the surface normal, also Lambert cosine law. In the figure on the center, Lambert shading has been generated in Matlab:

[X,Y,Z] = sphere(200);
surfl(X,Y,Z);
shading flat;
axis('off');axis('equal')

The property of that Lambertian surface is that light intensity depends upon the direction of illumination, but not the direction where the observer is placed. Notice that the moon, on the right, light is almost uniform all over the surface facing the sun, and this does not make a Lambertian reflector.

2.4.1 Example: Lambertian reflectance

Let us compute the total sun energy reflected by the moon, when the sun it is illuminating the entire half (full moon), by assuming a total albedo $\rho = 0.12$.We should provide an angular profile of reflectivity of the surface: here we assume the uniform, that is a Lambertian surface

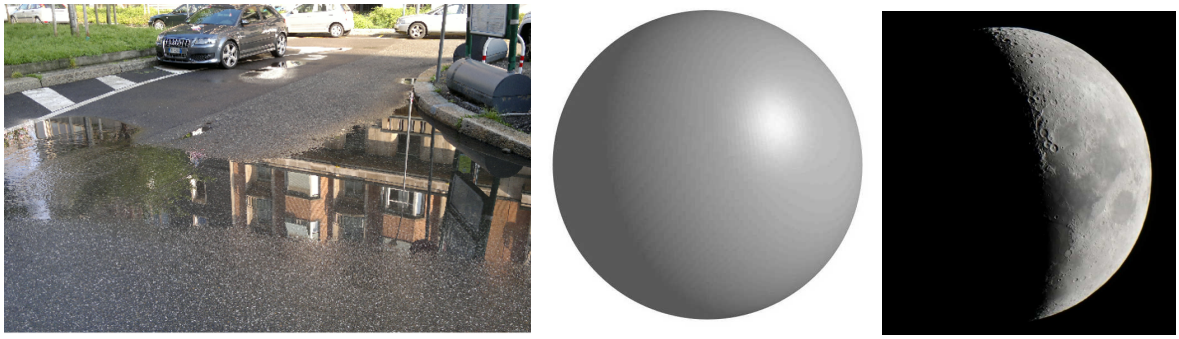


Figure 2.7: Left: specular and diffused reflections, center: Lambertian shading, right: moon.

even if we have shown that it is a rough approximation.

The energy per surface at the distance of the moon is assumed as that for Earth, in (2.11) 1380 W/m^2, then we account for the fraction reflected, and that is proportional to the cos of the elevation angle, $0 \leq \theta \leq \pi/2$ (for the hemisphere) and the elementary surface of the moon sphere:

$$ds_M = R_M \cdot d\theta \cdot R_M \sin\theta \cdot d\phi$$

we combine the two expression and the cos law:

$$\frac{P_r}{A_M} = 1380 \cdot \rho \frac{\int_0^{\pi/2} \int_{-\pi}^{+\pi} R_M^2 \cos\theta \sin\theta d\phi d\vartheta}{\pi R_M^2} = 165.6$$

That eventually we convert into reflectance per solid angle as for (2.13):

$$L = \frac{P_e}{A_E d\Omega} = \frac{P_s}{\pi A_M} = 52 W/m^2/sr \tag{2.17}$$

that we can compare with the Sun radiance

$$L_S = \frac{\sigma T_S^4}{\pi} = 2{\times}10^7 \mathrm{W}/m^2/\mathrm{sr}$$

it is quite stronger than moon.

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2.4.2 Reflection and diffraction

The principles of reflection and diffraction are shown in Fig. 2.8. The reflection and diffraction can be thought as the contribution of spherical wavefronts emitted by a continuous distribution of isotropical reflectors, like P in the figure. The different radii of the circles are related to the different time, and then phase (for monochromatic wavefronts) that hits the target. The reflected waveform travels in the specular direction since this corresponds to iso-delay, iso-phase: notice, for example, that the distance OA, is equivalent to the travel path BP+PC. This means that the rays recombine in phase on the waveform AC, as results also from the envelop of the green circles. In the gray medium, the propagation velocity is smaller, and this accounts for red circles that have progressively smaller radii. The result is that the refracted wavefront (red) has a smaller angle. One can reconstruct the second Snell law:

$$\frac{\sin\theta_r}{\sin\theta_i} = \frac{n_1}{n_2}$$

where

$$n_r = \frac{c}{c_r} = \frac{\lambda}{\lambda_r} = \frac{c}{f\lambda_r} \tag{2.18}$$

is the refractive index in the medium "r".

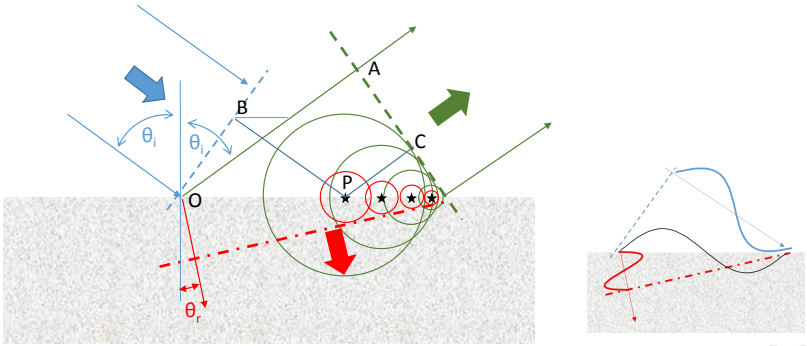


Figure 2.8: Principle of reflection and diffraction (left) and correspondence between incident and diffracted wavelengths.

The refractive index of homogeneous transparent bodies is frequency dependent, causing different color to refract at different angles, as it appears from a prism, or a rainbow: these are regarded as **dispersive** media, as shown in Fig. 2.9.

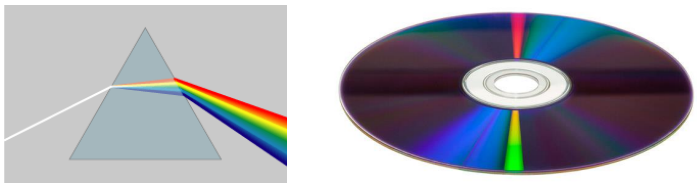


Figure 2.9: Dispersive media. Left: in prism, diffraction is frequency dependent. Right: in a DVD, the refraction grating is disperse the light according to incidence angle.

Refraction can also be generated by gratings, that are grids of regularly placed reflectors working with the same principle of the elements marked with stars in Fig. 2.8. As an example, a plane wave incident to the grating with angle θ_i , and assuming the medium is non dispersive (refractive index=1) will generate a wavelength:

$$\lambda_x = \frac{\lambda}{\sin\theta_i} = \frac{d_x}{M} \Rightarrow \sin\theta_i = \frac{M\lambda}{d_x}$$

if $\lambda_x = d_x/M$with M integer. This equation compares with (2.18), with the difference that $\lambda_x$can assume only a discrete number of values. The diffracted ray will have maxima in direction that

$$\lambda_x = -\frac{\lambda}{\sin\theta_M} \Rightarrow \sin\theta_n = -\frac{N\lambda}{d_x}$$

That can be combined together:

$$d_x\left(\sin\theta_i + \sin\theta_n\right) = (M+N)\,\lambda$$

giving all the possible angles (with integer K)

$$\sin\theta_n = \frac{K.\lambda}{d_x} - \sin\theta_i$$

2.4.3 Diffuse reflections: scattering from rough surfaces

The specular reflection occurring in Fig. 2.8 requires that all the targets are perfectly aligned, to get in-phase constructive superposition in the direction conjugate to incidence. If this condition does not holds, as when the reflecting elements are dispersed in a volume, the reflected energy is **diffused** in every direction.

This is also the case of rough surfaces. A quantitative assessment of "roughness" can be derived by considering the single step in Fig. 2.10. We assume that the wavefront has incidence angle θ_i and the step height is ΔH. In absence of the step, ad assuming the surface large with respect to the wavelength, we assume a pure forward scattering in the direction specular to the incidence.

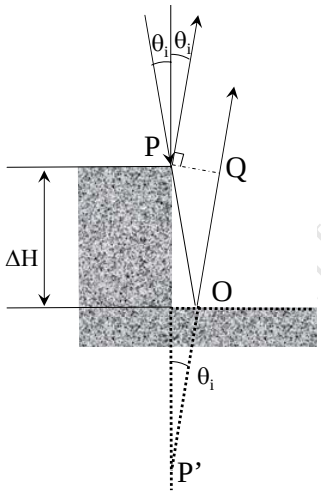


Figure 2.10: A step in a smooth surface is responsible of a widespread scattering.

However, due to the step, the wavefront, orthogonal to the assumed forward propagation, has contributions coming from different paths. By referring to the figure, this path length difference is:

$$\overline{PO} + \overline{OQ} = \overline{P'Q} = 2\Delta H \cos\theta_i$$

The corresponding phase difference is then:

$$\Delta\phi = \frac{4\pi}{\lambda}\Delta H \cos\theta_i$$

Due to this difference, the amplitude of the forward scattering is reduced, the higher the step the stronger the attenuation. For very rough surfaces, the energy is scattered everywhere almost isotropically, as shown in Fig. 2.11 shows.

Rayleigh criterion define the limit condition under which we may still retain the **surface smooth**, by imposing that the phase difference is bounded into[1] $\pm\pi/2$:

$$|\Delta\phi| = \frac{4\pi}{\lambda}\Delta H \cos\theta_i < \frac{\pi}{2}$$

We get the limit step size:

$$\Delta H = \frac{\lambda}{8\cos\theta} \tag{2.19}$$

[1]This value is the same considered in the evaluation of an antenna directivity, that leads to the definition of Rayleigh criterion for resolving to targets.

In the case of rough surface, we replace ΔH by the rms roughness. The figure 2.7 is a clear example of this.



Figure 2.11: Scattering from surfaces at different roughness.

2.4.4 Polarimetric properties

The amplitudes of the reflected field and the incident are ruled by Fresnel equations, that are derived from Snell laws and imposing continuities of the fields at the interface. With reference to the **horizontal polarized field** in Fig, 2.12, the relation is the following

$$\frac{E_r^H}{E_i^H} = \frac{n_1 \cos\theta_i - n_2 \cos\theta_t}{n_1 \cos\theta_i + n_2 \cos\theta_t}$$

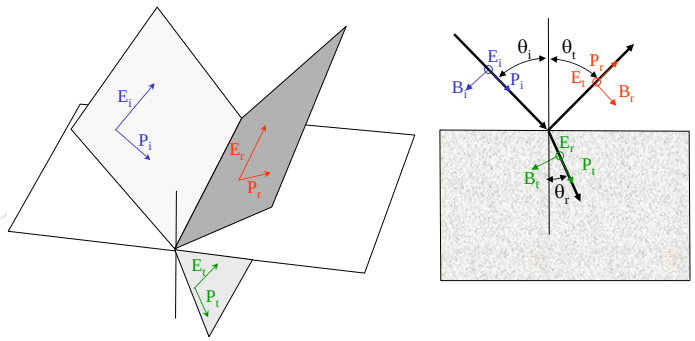


Figure 2.12: Electric and magnetic fields in the case of reflection with horizontal polarization.

For a perfect conductor ($v, n_2 \to 0$) the ratio is 1, there is total reflection as there cannot be a field traveling in the medium.

In the case of **vertical polarization,** shown in Fig. 2.13.a, the Fresnel equation is the following:

$$\frac{E_r^V}{E_i^V} = \frac{n_1 \cos\theta_t - n_2 \cos\theta_i}{n_1 \cos\theta_t + n_2 \cos\theta_i} \tag{2.20}$$

A particular case occurr when the verical component of the reflected field is null: $\theta_i = \arctan(n_2/n_1)$. For a perfect conductor the above ratio is 1, that is total reflection.

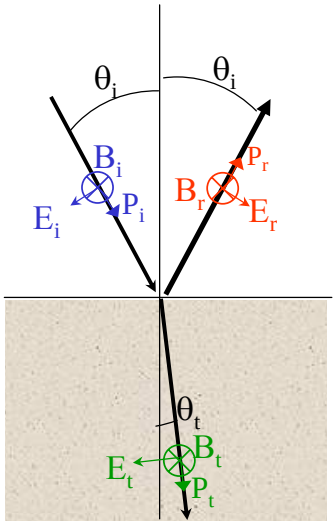


Figure 2.13: The electric field, **E**, and the magnetic field, **B**, for a surface reflection of an horizontal polarized wave

2.4.5 Backscatter in Lidar, Radar and Sonar

The properties of rough surfaces to diffuse energy is fully exploited by Lidar, Radar and other active devices. Such sensors, in the monostatic version, observe the signal in the same direction they transmitted, then they rely in the **backscatter**. This occurs in non-flat surfaces, where roughness is related to wavelength. For the laser image in Fig. 2.4.5, nearly all the surfaces are rough. For a Radar, operating wavelength, roughness starts from millimeters.

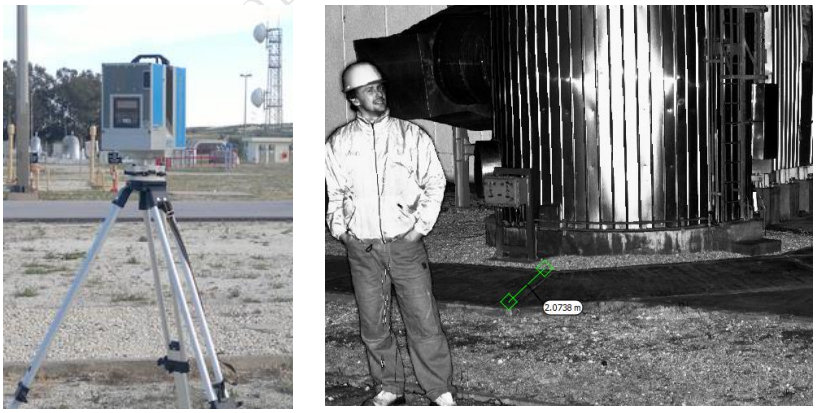


Figure 2.14: Left: Lidar sensor, capable of scanning the space in spots as small as 1×1 mm. Right: laser reflectivity, rendering for 3D bubble data: notice the capabilities to measure distances (in green), as well as placing each pixel in 3D.

2.4.5.1 Example: backscatter and oil spill detection by radar imaging

The properties of rough surfaces to diffuse energy is fully exploited by Radar for many applications. A microwave RADAR, like the C band sensors like the European Space Agency's

SENTINEL, $\lambda = 5.6$ cm, $\theta = 23^o$ would be sensitive to a surface roughness up $\Delta H_{RMS} = 0.7$ cm, thus the slightest wave in a sea produces a backscatter, but not so if sea surface is covered by an oil spill:, as shown in Fig. 2.4.5.1.

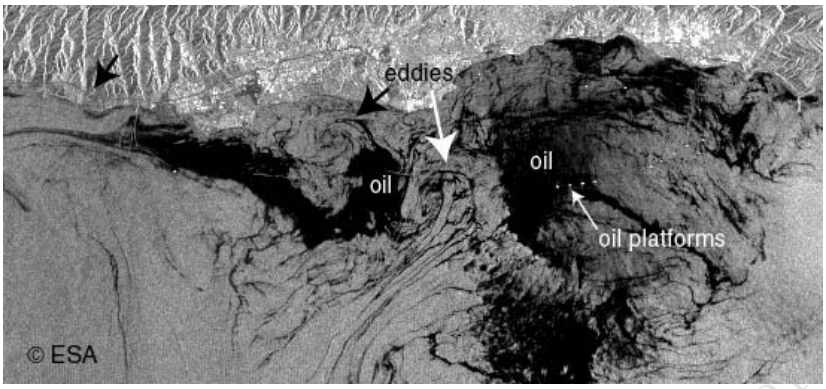


Figure 2.15: This SAR image shows a natural oil slick in the Santa Barbara Channel; it also demonstrates that small-scale eddies, due to ocean circulation can impact the distribution of this oil.

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Chapter 3

Antennas as coherent sources

The electron vibration within a BB excites the many modes with random, independent distributed pulses generating a wide bandwidth noise signal. This is not the most efficient way to transmit (and then to receive) signal, given an effective constant energy. To understand the principle, let us assume that N transmitters produce their own signal, acting incoherently. The energy of the sum of N random independent is, in the average:

$$\begin{aligned} e_I &= E\left[\left(\sum_{i=1}^{N} x_i\right)^2\right] = \sum_{i=1}^{N}\sum_{j=1}^{N} E\left[x_i x_j\right] = NE\left[x_i^2\right] + \left(N^2 - N\right) E\left[x_i\right] E\left[x_j\right] \\ &= NE\left[x^2\right] = N\sigma_x^2 \end{aligned}$$

Whereas, if the act coherently, all the signals add together leading to an energy:

$$e_C = E\left[\left(\sum_{i=1}^{N} x\right)^2\right] = E\left[(Nx)^2\right] = N^2\sigma_x^2$$

The ratio between the coherent and incoherent generated waveforms is so large as to diverge when the number of elements gets bigger and bigger, and the difference between a laser and a common BB - say a bulb or a LED is the clear evidence. Coherence should apply in both space and time, as we will see later, and that is at the basis of antenna for transmitting and receiving electrical fields.

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3.1 The antenna array

Let us first assume that a Uniformly-spaced, Linear Array of small elements, all of them emitting an EM field consisting of a monochromatic waveform (that is all the energy is concentrated in one frequency), that propagates in the space as spherical EM field:

$$e\left(\mathbf{r}, t\right) = A\frac{\exp\left(j\left(kr - \omega t\right)\right)}{r} \tag{3.1}$$

where A is the amplitude, k is the spatial wavenumber, the number of radians per meters:

$$k = \frac{\partial\phi}{\partial r} = \frac{2\pi}{\lambda}$$

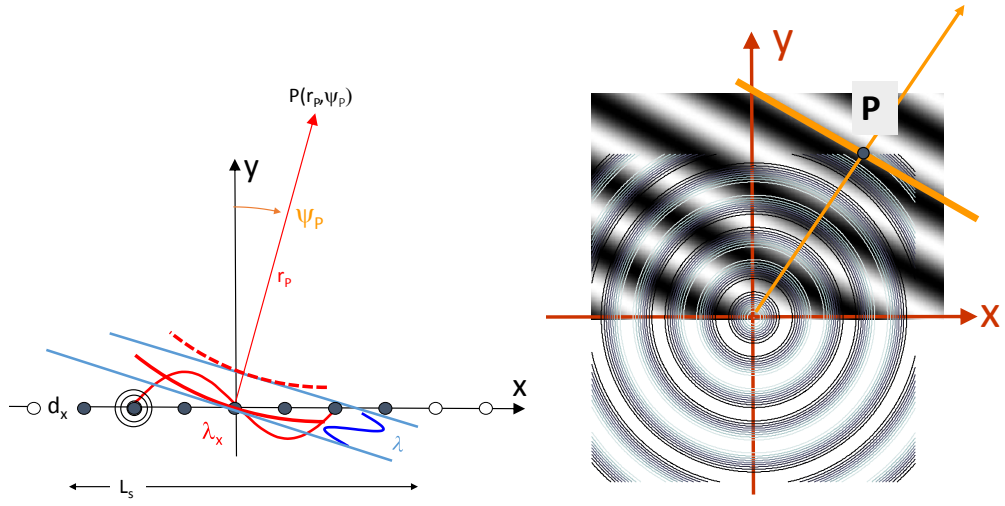


Figure 3.1: Left: geometry for computing the antenna directivity pattern. Right: the circular wavefront is approximated by plane waves whose frequency depends on the direction of arrival.

We compute the field at a point $P(r_p, \psi_p)$, where the angle ψ_P is measured with respect to the vertical, the space by the linear superposition of the field emitted by the 2N+1 elementary antennas:

$$y_P(t) = \sum_{n=-N}^{N} \frac{A}{r_n(P)} \cdot \exp\left(j\left(k \cdot r_n(P) - \omega t\right)\right) \tag{3.2}$$

where the distance between the point P and the location of the $n-th$ antenna can be approximated, in the far field, as:

$$r_n(P) \approx r_P + x_n \sin\psi_P = r_P + n \cdot d_x \sin\psi_P \tag{3.3}$$

that corresponds to ignore the curvature of the wavefront, approximation that holds if the length of the array is small compared to the distance of P (**far field**):

$$L_x = (2N+1)\, d_x \ll R_0$$

In addition, we ignore the variation of the amplitude, the factor $1/R_n(p) \simeq 1/r_p$.The field $y_P(t)$ is then:

$$\begin{aligned} y_P(t) &= \frac{A}{r_P} \sum_n \exp\left(j\left(k\left(r_P + n \cdot d_x \sin\psi_P\right) - \omega t\right)\right) \\ &= \frac{A}{r_P} \exp\left(j2\pi\frac{r_p}{\lambda}\right) \cdot \sum_{n=-N}^{N} \exp\left(j2\pi\frac{\sin\psi_P}{\lambda} n d_x\right) \exp\left(-j\omega t\right) \\ &= \exp\left(j\phi_P\right) \cdot \exp\left(-j\omega t\right) \cdot \sum_{n=-N}^{N} \exp\left(j2\pi\frac{\sin\psi_P}{\lambda} n d_x\right) \\ &= e(r_p, t) \cdot G\left(\psi_P\right) \end{aligned}$$

where the phase term, $\phi_P = 2\pi r_p/\lambda$, that is the number of cycles from the origin to P. Then, we get two components:

1. the waveform emitted by an elementary antenna in the center of the array, attenuated and changed in phase as seen by a distance r_p

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2. a factor that depends on the angle, ψ_P, $G(\psi_P)$,that accounts for the coherent superposition of the N contributions

This last term accounts for the directivity of the antenna, and, in this case, can be computed in closed form by noticing that it is the discrete Fourier transform of a constant:

$$G(\psi) = \sum_{n=-N}^{+N} \exp\left(-j2\pi f_x \cdot n d_x\right)$$
$$f_x = -\frac{\sin\psi}{\lambda}$$

where $d_x$is the sampling in the space and

$$f_x = -\frac{\sin\psi}{\lambda}$$

is the spatial frequency. In practice the approximation of the curved wavefront with a linear one, corresponds to the approximation of the spherical wave with a plane one, whose wavenumber is

$$k_x = \mathbf{k} \times \mathbf{u}_x$$

then, by using the Fourier Transform formulas (or by summing the geometrical tuncated series):

$$G(\psi) = \sum_{n=-N}^{+N} \exp\left(-j2\pi f_x \cdot n d_x\right) = \frac{\sin\left(\pi f_x N_s d_x\right)}{\sin\left(\pi f_x d_x\right)}$$
$$= \frac{\sin\left(\pi \frac{\sin\psi}{\lambda} N_s d_x\right)}{\sin\left(\pi \frac{\sin\psi}{\lambda} d_x\right)} \text{where} N_s = 2(N+1)$$

3.1.1 Main lobe, sidelobes and grating lobes

For small aperture, the directivity can be approximated by a *sinc* function:

$$G(\psi) \underset{\psi \ll \lambda/d_x}{\simeq} \frac{\sin\left(\pi \frac{\sin\psi}{\lambda} N_s d_x\right)}{\pi \frac{\sin\psi}{\lambda} d_x} = N_s \,\mathrm{sinc}\left(\frac{\sin\psi}{\lambda} N_s d_x\right)$$
$$= N_s \,\mathrm{sinc}\left(\frac{\sin\psi}{\lambda} L_s\right)$$

whose **main lobe width** is approximately (for small angles):

$$\Delta\psi = \frac{\lambda}{L_s}$$

and there are *sidelobes* at angles multiple of λ/L_s. The ratio between the main lobe and the sidelobes is not that large: 20*log10(abs(sinc(3/2)))=13 dB. If a very directive antenna is to be designed, to confine the field in a very small aperture the width (that is the number of N of elements) should large, but this does not reduces the sidelobes. To that goal, a different weight, w(n), should be applied to each elementary antenna, so that the resulting directivity is:

$$G(\psi) = \sum_{n=-N}^{+N} w(n) \cdot \exp\left(-j2\pi f_x \cdot n d_x\right) = W(f_x)$$

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also known as the Fraunhofer approximation for the far field: the directivity is always the Fourier Transform of the antenna weight, or "illumination". An interesting case is that when the main lobe is to be rotated, say by an angle ψ_d, meaning that the array is **pointed in a different direction**. This corresponds to a shift in the frequency domain, and then adding a linear phase to the array:

$$w(n) \cdot \exp\left(j2\pi f x_0 n d_x\right) \leftrightarrow W((f - f_{x0})$$

Moreover the directivity is periodical, as being the Fourier Transform of a sequence, of multiple of $f_{sx} = 1/dx$, and this generates grating lobes at the angles:

$$f_x = \frac{\sin\psi}{\lambda} = \frac{K}{d_x} \Rightarrow \psi = K \arcsin\left(\frac{\lambda}{d_x}\right) \tag{3.4}$$

These lobes are much more relevant than sidelobes, since the amplitude is the same as the main lobe. Their mitigation is achieved only by densifying the number of elementary antenna, then increasing the number. To avoid them, one solution is to dimension $d_x$so that the sampling frequency is lager than twice the maximum frequency:

$$f_{sx} = \frac{1}{d_x} > 2f_{xmax} = 2\frac{\sin\psi_{max}}{\lambda} \geq \frac{2}{\lambda}d_x$$
$$\Rightarrow d_x < \frac{\lambda}{2\sin\psi_{max}} < \frac{\lambda}{2}$$

This could lead to a large number of antenna, a different solution is in the next exercise.

3.1.2 Example

Let as assume to generate an antenna with a main lobe of 5°, centered around -5° and accept a grating lobe only for $|\psi - \psi_0| > 30°$,assuming that λ is 3 cm (X band).

The total antenna length should be dimensioned according to the main lobe:

$$L_x = (2N+1)\, d_x = \frac{\lambda}{\Delta\psi} = \frac{0.03}{\frac{5^\circ}{180^\circ}\pi} = 0.34$$

The sensors spacing is found by placing the gating lobe position at 30° from the array pointing:

$$d_x \leq \frac{\lambda}{|\sin 35^0|} = 0.052$$
$$d_x \leq \frac{\lambda}{|\sin(-25^0)|} = 0.07$$

that accounts for 4.8 and 6.5. Let us assume 5 elements, that is not conservative, but it would reduce the number of elements.

The weighting vector should steer of ψ_0 =-5°:

$$w(n) = \exp\left(j2\pi\frac{\sin\psi_0}{\lambda}nd_x\right)$$

The pattern is shown in Fig. 3.2, compared with the closed form (Fraunhofer approximation for the far field):

$$G(\psi) = \frac{\sin\left(\pi\frac{\sin\psi - \sin\psi_0}{\lambda}N_s d_x\right)}{\sin\left(\pi\frac{\sin\psi - \sin\psi_0}{\lambda}d_x\right)} \tag{3.5}$$

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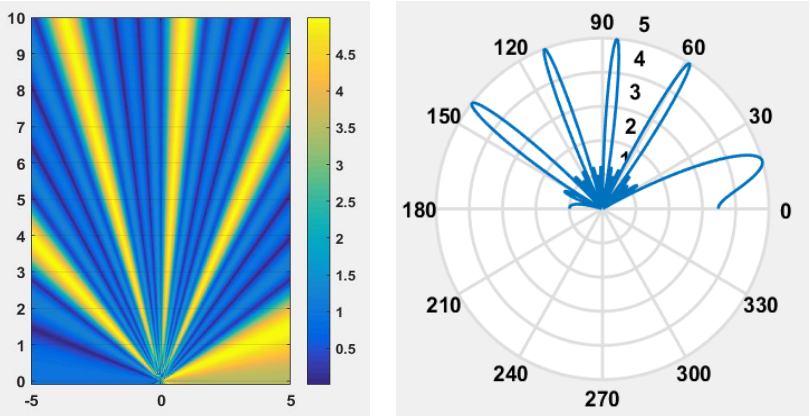


Figure 3.2: Left: evaluation of the wavefield as linear superposition of spherical wavefronts, generated at each array position. Right: pattern computed by Fourier Transform (Fraunhofer approximation for the far field).

3.2 Real antenna

The case of a real antenna, say an aperture of size L_x is quite similar to the array, at the limit when $N \to \infty$. The summation (3.2) becomes the integral:

$$\begin{aligned} y_P(t) &= \int_{-L/2}^{+L/2} \frac{A}{R(P,x)} \cdot \exp\left(j\left(kR(P,x) - \omega t\right)\right) \\ &\simeq \frac{A}{r_P} \exp\left(j2\pi \frac{r_p}{\lambda}\right) \cdot \int \exp\left(+j2\pi \frac{\sin\psi}{\lambda} x\right) \exp\left(-j\omega t\right) dx \\ &\simeq e(r_p, t) \cdot G\left(\psi\right) \end{aligned}$$

where the directivity gain is now the Fourier Transform of the rectangular illumination:

$$\begin{aligned} G(\psi) &= \int_{-L/2}^{+L/2} \exp\left(-j2\pi f_x \cdot x\right) dx = \operatorname{sinc}\left(L \cdot f_x\right) \\ &= \operatorname{sinc}\left(\frac{L}{\lambda} \sin\psi\right) \end{aligned}$$

We have a very similar gain with the exception of the grating lobes. Notice that the sinc(x) function has its 3 dB lobe for x=±0.44, therefore the 3 dB lobe of a uniformly illuminated antenna of size L is

$$\Delta\psi_{3_{dB}} = 0.88 \cdot \arcsin\left(\frac{\lambda}{L}\right) \simeq 0.88 \cdot \frac{\lambda}{L}$$

assuming directive antenna, thus $\lambda \ll L$. Another measure used for the beam-width is the angle at which directivity halves, that is roughly: $0.6\ \lambda/L$.

However, it is widely adopted the following definition of "antenna main lobe width", also *Rayleigh aperture*:

$$W = \frac{\lambda}{L} \tag{3.6}$$

that actually expresses the equivalent energy beam-width

$$\int_{-\infty}^{+\infty} G^2(\psi) d\psi = G^2(0) \cdot W$$

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3.2.1 Bidimensional directivity pattern

In the general case when the antenna is a surface, the properties so far derived extends to the 2D. The directivity pattern is the 2D Fourier Transform of the aperture, that in the case of uniform illumination with length D_x, D_y,extends to:

$$G(\psi,\theta) = FT\left\{\text{rect}\left(\frac{x}{D_x}\right)\cdot\text{rect}\left(\frac{y}{D_y}\right)\right\} f_x = \frac{\sin\psi}{\lambda}; f_y = \frac{\sin\theta}{\lambda}$$
$$= \text{sinc}\left(\frac{D_x\sin\psi}{\lambda}\right)\text{sinc}\left(\frac{D_y\sin\theta}{\lambda}\right)$$

that we can approximate like a rectangular windowing in the angle domain, withing the solid angle of size

$$\delta\Omega = \frac{\lambda}{D_x}\frac{\lambda}{D_y} = \frac{\lambda^2}{A}$$

"A" being the area of the antenna.

Another case of interest is the circular aperture for which the FT is symmetric, then:

$$G(\psi,\psi) = FT\left\{\begin{array}{cc} 1 & for \quad \sqrt{x^2+y^2} < D \\ 0 & elsewhere \end{array}\right\}_{f_x=f_y=\frac{\sin\psi}{\lambda}}$$
$$= \frac{\text{J}_1\left(\pi\frac{D\sin\psi}{\lambda}\right)}{\pi\frac{D\sin\psi}{\lambda}}$$

where $J_1$is the first order Bessel function. The first zero of the aperture is at

$$\sin\psi_{null} = 1.2\frac{\lambda}{L} \tag{3.7}$$

and this is frequently used as definition of resolution for a circular aperture, like in optics. The -3 dB is instead quite close to Raleigh.

3.2.2 Example

Let us compute the radiance out of a Laser with 1 mW power, $\lambda = 1\mu m$ and emitting area A=1 mm^2.

We use the relation

$$L = \frac{P_R}{A_R\Delta\Omega_T}$$

where the ration between the power received at a distance R, and the receiving surface can be computed by assuming that all the power goes to the surface whose size is $\Omega_L R^2$:

$$\frac{P_T}{A_R} = \frac{P_T}{\Omega_L R^2}$$

where

$$\Omega_L = \frac{\lambda^2}{A_T} = \frac{10^{-12}}{10^{-6}} = 10^{-6}sr$$

is the angle of the laser beam.

The solid angle under which the source observe the laser is:

$$\Delta\Omega_T = \frac{A_T}{R^2}$$

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The radiance is then

$$L = \frac{P_T}{A_R \Delta\Omega_T} = \frac{P_T}{\Omega_L R^2}\frac{R^2}{A_T} = \frac{P_t}{\Omega_L A_T} = \frac{10^{-3}}{10^{-12}} = 10^9\, W/m^2/sr$$

The temperature is

$$L = \frac{\sigma T^4}{\pi} = 10^9 \Rightarrow T = \left(\frac{10^9\pi}{\sigma}\right)^{0.25} = 15000°\text{K}$$

Notice that it is much more than the one of the Sun: it is not a real temperature, but an equivalent one, since the Laser is a coherent source that is emitting all the power within a single frequency. Nonetheless, the energy per surface and angle is several times larger. This is the case when a coherent emission, in both space and time, corresponds to a huge power, when compared with an incoherent BB.

3.3 The far field

The approximation introduced in omitting the term responsible of the *curvature of the wave-front*:

$$R_{\max} - r_P = \sqrt{y_P^2 + \left(\frac{L}{2}\right)^2} \ll \lambda$$

If we assume a maximum limit as $\lambda/2$, we get:

$$\sqrt{y_P^2 + \left(\frac{L}{2}\right)^2} - y_P < \frac{\lambda}{2}$$

$$y_P\left(1 + \left(\frac{L}{2y_P}\right)^2 + \dots\right) - y_P < \frac{\lambda}{2}$$

$$\frac{L^2}{2r_P} < \lambda$$

As an example, if $r_P = 1$ km and $\lambda = 3.5$ cm, we get $L < \sqrt{0.035 \cdot 1000} = 6$ m. This limit (Fresnel zone) imposes that larger antennas cannot be straight, for example parabolic.

re $G(f_x)$ is the Fourier transform of the antenna illumination function. , and just express the field received by the antenna as the field emitted by P, multiplied times the FT of the antenna illumination, times P_0 that accounts for the loss of energy (spherical divergency), the term r^{-1}, and for the phase change due to the distance.

3.4 Multi-beam scanning

Whatever the choice, there is a strong advantage in favor of such antenna arrays: the capability of making an **electronic steering**, as shown in (3.5). The major advantage, with respect to the physical rotation of the real antenna, is that here the re-phasing of the array can be done simultaneously for all the frequencies involved, means that the steering in all the angles is done at once, with **no need of any rotation**. Furthermore, the operator needed to steer the array can be as simple as a FT.

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Note that, for ULA, the directivity worsens with the angle:

$$\psi = -\arcsin(\lambda f_x)$$
$$\Delta\psi = -\frac{\partial \arcsin(\lambda f)}{\partial f}\Delta f = \lambda\frac{\Delta f}{\cos\psi}$$

A constant frequency resolution Δf causes a severe loss of angular resolution say for $\psi > 60°$. In that case, a circular array is recommended.

3.4.1 Example

Dimension the array for a microwave RADAR placed on a ship should, to scan a field of view that is 60° horizontal × 20° vertical with an horizontal resolution of 10 m at 1 km, the frequency is 9 GHz.

The total array horizontal length is driven by the resolution

$$\rho_H = \Delta\psi_h \cdot R = \frac{\lambda}{L_H}\cdot R = 10 \Rightarrow L_H = \frac{\lambda}{\rho_H}\cdot R = \frac{0.03}{10}1000 = 3\,m \tag{3.8}$$

The single elementary antenna is then dimensioned to limit the horizontal beam-width:

$$\Delta\psi_H = \frac{\lambda}{l_H} = \frac{60°}{180°}\pi \rightarrow l_H = 0.03\frac{3}{\pi} \sim 2.86\,cm$$

and the vertical beam-width

$$\Delta\psi_V = \frac{\lambda}{l_V} = \frac{20°}{180°}\pi \rightarrow l_V = 0.03\frac{9}{\pi} \sim 8.5\,cm$$

The total array in then 3 m × 8.5 cm, and we notice that the number of antenna, 3 m / 2.86 cm = 104, is equal to *the number of independent pixels* we would resolve:

$$N = \frac{60°/180°\cdot\pi\cdot 1000}{10} = 104$$

3.5 Antenna temperature

Let us now assume that the antenna observes a BB with spectral radiance$L(\psi,\theta,f)$, with a directivity $G(\psi,\theta)$ that is a fraction of the total solid angle. The power received by an antenna with directivity area A_d is the integral over the band of interest and of the product of the antenna directivity aperture and the radiance:

$$P = \int df \int d\theta \int d\psi A_d(\psi,\theta,f)L(\psi,\theta,f)df$$

where

$$A_d = \frac{\lambda^2}{4\pi}G(\psi,\theta)$$

Eventually, for frequencies much lower than 6 THz, like in the microwave, wee take advantage of (2.6) to approximate the radiance (2.15):

$$L(\psi,\theta,f) \approx \frac{2f^2}{c^2}k_B T$$

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then, for a bandwidth B,

$$P = \int df \int d\theta \int d\psi \frac{1}{2\pi} G(\psi,\theta) k_B T df = \frac{k_B T \cdot B}{2\pi} \cdot \int \int d\theta d\psi G(\psi,\theta)$$

The region of integration is the region where the antenna see the BB. As an example, for a spherical antenna in a cavity:

$$P = \frac{k_B T \cdot B}{2\pi} \cdot 4\pi = 2 k_B T \cdot B \tag{3.9}$$

This under the assumption that the antenna receives the full EM field emitted by the BB in the bandwidth B, for both polarization. Is the antenna is single polarized, the power would halve:

$$P = k_B T \cdot B$$

Furthermore, if the antenna is not totally surrounded by a BB, and its directivity can be assumed constant over the solid angle Ω_A, larger than the solid angleΩ_{BB} under which the BB is observed:

$$P = 2 \cdot k_B T \cdot \frac{\Omega_{BB}}{\Omega_A} B$$

The fact that the noise power is independent on the atenna area is a conbination of source energy and receiver gain. If the recevier antenna gets bigger, its directivity increase with surface, but then gathers noise from a portion of the BB that is reduced by the same amount.

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3.5.1 Example

With reference to the array in section 3.1.2, dimension each single antenna to avoid the grating lobe located close -30°, and assuming an elevation aperture of 10°. In the design we assume L=0.34 m, N=5 elements and $\lambda = 0.3$.The closest grating lobes are located at:

$$\frac{\sin\psi - \sin\psi_0}{\lambda} d_x = 1 \rightarrow \sin\psi = \sin\psi_0 - N\frac{\lambda}{L} = 0.0872 \pm 5\frac{0.03}{0.34} = \begin{matrix} 0.52 \\ -0.35 \end{matrix}$$

The text refers to the negative one:

$$L_a = \frac{\lambda}{|\sin\psi|} = 0.057$$

that is shown in Fig. 3.3. Notice that the width of each elementary antenna, should be less than the antenna spacing. The height of each antenna is the same as the whole array length, that is 0.17 m.

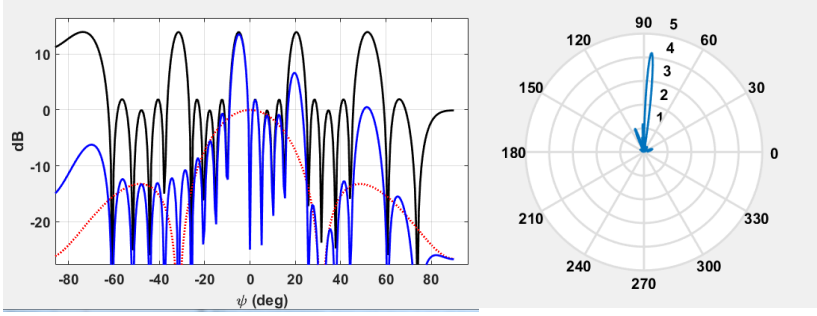


Figure 3.3: Left: directivity of the overall array as product of the array (black) and the single element pattern (red). Right: polar plot.

Chapter 4

Radiometers: devices and applications

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4.1 Radiometers and spectroradiometers

As an example, the next generation of hyperspectral sensors, like Italian PRISMA or German ENMAP will have a swath width of 300 km, with a pixel (intended as resolution) size of 30 $\times$ 30 m.

Radiometers and spectrometers are devices to measure radiance and reflectance and produces maps of these. The elements of a radiometer are:

1) a directive antenna with a very small beamwidth, to get the desired resolution

2) a scanning mechanism to explore the entire Field of View

3) a sensor, or multiple sensors that are detect the EM radiation in the bandwidth of interest

4) an analog or digital signal processing to measure either the energy in each elementary cell (a pixel) or the energy in each frequency bin (spectroradiometers).

The measured quantities, that are images, are then interpreted as physical parameters, and used for very different applications. Also our eyes can be though in this framework, where the antenna is the pupil, the array of sensors are the photoreceptor on the retina, that perform different spectral weighting to detect the primary colors.

In some cases the interest is not the scene imaged, but the medium. Microwave radiometers measures water vapor in the atmosphere by tuning the frequencies of water vapor absorption. Water by vapor, that in turn is one of the most critical elements influencing the atmosphere, entering in the thermal balance of Earth, ruling precipitations and heath transfer, than agriculture, storm and weather forecasts, climate changes.

Multispectral and hyperspectral instruments do measures these quantities within different bandwidths, producing one image for each bandwidth. According to the frequency, they have different sensitivities to elements of primary importance, besides water vapor: burning a fires, ocean color, terrain cover (vegetation, snow, etc.), pollution (carbon dioxide, methane) and atmosphere chemistry.

4.2 Scanning the space: the Field Of View

EM imaging systems produces images, each of them being a set of pixels observed within the Field Of View (FOV) of the instruments. The number of pixels is in principle larger than the

ratio between the FOV and the system resolution, see example 3.4.1. In the case of airborne and spaceborne system, a continuous image is formed along-track, by putting together slices observed at each time. An example of different platform for remote sensing is in Fig. 4.1. Platform at low altitude are quite suited for local sensing: the swath, as limited by incidence angles, and then shadowing is in fact in the order of the elevation itself. Systems like drones are becoming quite popular as being cheap. On the other hand, when global observation is needed, sensors at high altitude provide much bigger swath, and *world-wide coverage and access*.

A satellite in Low Earth Orbit (160 - 2000 km) have typical velocity of 7-8 km/s and turn round the Earth in few hours. For near circular orbits, the period, that can be computed by balancing the gravity and the centripetal acceleration:

$$T = 2\pi\sqrt{\frac{(h + 6.36 \cdot 10^6)^3}{4 \cdot 10^{14}}} \tag{4.1}$$

that ranges from 90 minutes to two hours. The very low orbits are unfavored by atmosphere and gravity drag, whereas for the very high orbit the image performances degrades due to the large distance: most of remote sensing instruments are placed around 600-700 km. Nonetheless, the high orbits, particularly the Medium Earth Orbit up to the Geosynchronous one are becoming quite attractive for global observations: typical examples are the GNSS like GPS, GLONASS and GALILEO in Medium Earth Orbit MEO, ˜20000 km, and 12 hours orbit, and telecommunication system, in GEO-synchronous orbit, 36000 km and 24 hours, that becomes GEO-stationary if both inclination and ellipticity are null, so circular orbit in the equatorial plane.

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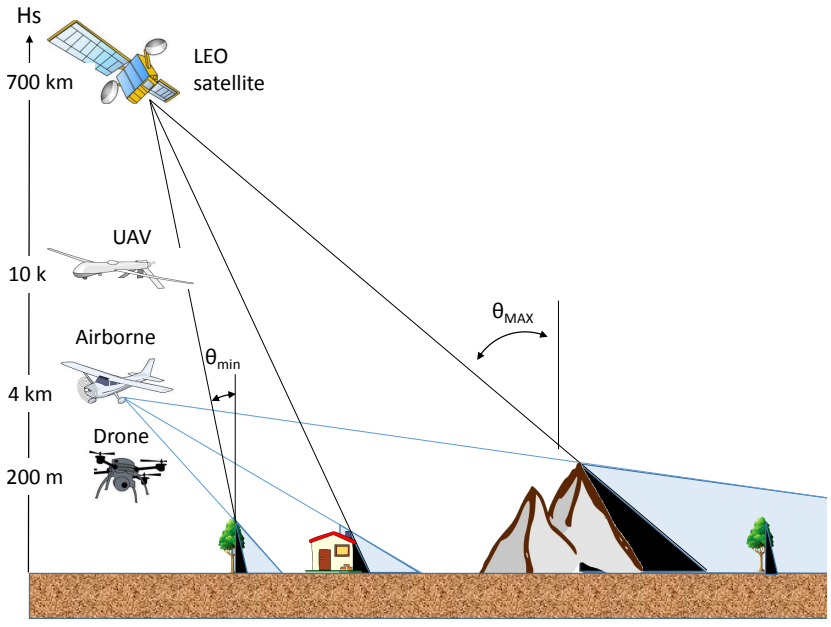


Figure 4.1: Example of remote sensing platforms.

4.2.1 Scanning systems

In the microwaves the long wavelength makes difficult to achieve a reasonable resolution. An antenna of 10 m, placed on a LEO satellite at some 800 km distance from the observed scene

would end in an along-track resolution:

$$\rho_{AT} = \frac{\lambda}{L}R = \frac{0.03}{10}800000 = 2.4km \tag{4.2}$$

in the most favorable case of X band, that is one of the lowest wavelength, $\lambda = 3$ cm. Such resolution would be proper for a Radiometer, to measure water vapor cells, or a Scatterometer, that measures wind direction and speed, but not really for imaging system. In that case, a very long antenna should be synthesized with the same principle of the array, but implementing it a single antenna moving along track and transmitting, receiving and recording the EM field in each position. The synthesis of such long antenna - up to kilometers, is achieved by a proper processing of the recorded echoes. The result in the image shown in the up-right panel in Fig. 4.2.

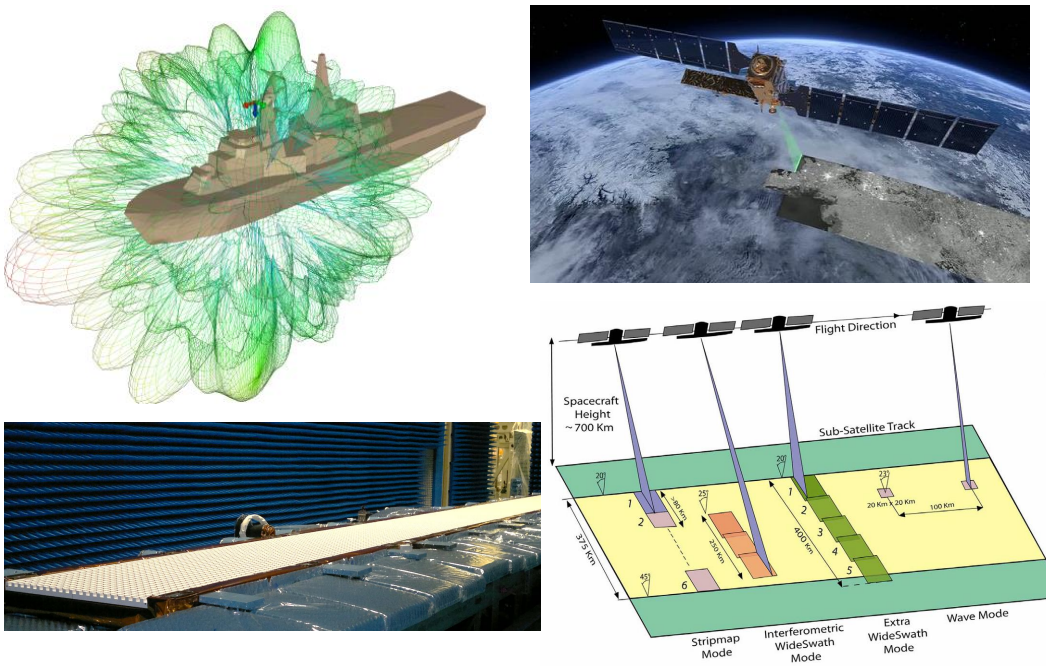


Figure 4.2: Scanning by antenna arrays used in microwave RADARs. Left: beams of a synthetic array (from http://www.antennamagus.com). Right: ESA Sentinel-1 spaceborne Synthetic Aperture Radar (up: artist representation) and different image modes (from European Space Agency).

Nonetheless, antenna array are still quite useful, either for close sensing, like in the Radar mounted on vehicles or ships, see Fig. 4.2 on the left, or to move the field of view in different "sub-swaths", like in the Sentinel 1 modes shown in the same figure on bottom right. In this case, the Radar antenna, shown in anechoic chamber in the bottom left picture, is made by a phased array arranged as a matrix of $20 \times 14 = 280$ Tx/Rx modules, for a total of size 12.3 $\times$ 0.8 m and a weight of 880 Kg. Such antenna is quite expensive and leads to a heavy and expensive satellite (and launch).

4.2.2 Trading resolution for coverage

The typical mean velocityv_s of a LEO system, ranging from 7.8 km/s to 6.9 km/s, impose constraint in the scanning, since the time available for observing an across-track line is limited

by the along-track resolution

$$\Delta T \leq \frac{\rho_{AT}}{v_s} \tag{4.3}$$

We assume here the *whisk broom* scanning, shown in Fig. 4.3 on the left, where an entire column of N_{XT} resolution cells should be observed before moving further along track:

$$N_{XT} = \frac{L_{XT}}{\rho_{XT}}$$

L_{XT} being the across-track FOV. If we assume that a single resolution cell should be observed for some time T_o:

$$\Delta T = N_{XT} \cdot T_o = \frac{L_{XT}}{\rho_{XT}} \cdot T_o \tag{4.4}$$

Eventually we combine (4.3) and (4.4) and we assume the same resolution along and across track, $\rho_{AT} = \rho_{XT} = \rho$:

$$\begin{aligned} \frac{L_{XT}}{\rho_{XT}} \cdot T_o &\leq \frac{\rho_{AT}}{v_s} \\ \Rightarrow \frac{L_{XT}}{\rho^2} &\leq \frac{1}{v_s T_o} \end{aligned} \tag{4.5}$$

This is a fundamental inequality bounding the FOV and the inverse of the resolution cell area. In principle we wish the first to be huge, and the second to be tiny, but in practice we can only improve one at the expenses of the other. The overall limits is the inverse of the distance covered in the observation time, that is reduced by the fast velocity of LEO orbit.

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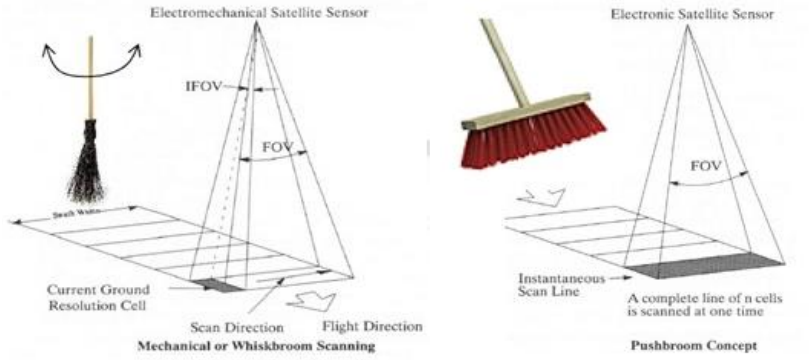


Figure 4.3: Scanning schemes: whisk broom, left, and pushbroom, right. Image courtesy of Florian Hillen.

In the case of on-ground systems, the mechanical rotation of antenna, or of a reflector illuminated by a phased array is quite common, as shown in Fig. 4.4.

If the constraint imposed by (4.5) becomes too much demanding, the sole solution is the use of a multi-sensor system, such that an entire across-track line can be imaged simultaneously, then scanning along track according to the push-broom method shown in Fig. 4.3 on the right. Multi-sensors systems are quite expensive in the microwave, but not in IR and visible, where the smaller wavelength (three order of magnitude), makes then more compact, as shown in Fig. 4.5.



Figure 4.4: Scanning methods for Radars. Left: Radar mounted on Australian frigate FFG01. Notice the rotating reflector illuminated by an array. Right: the spherical radome protect the rotating antenna in Trentino Meteo weather Radar.

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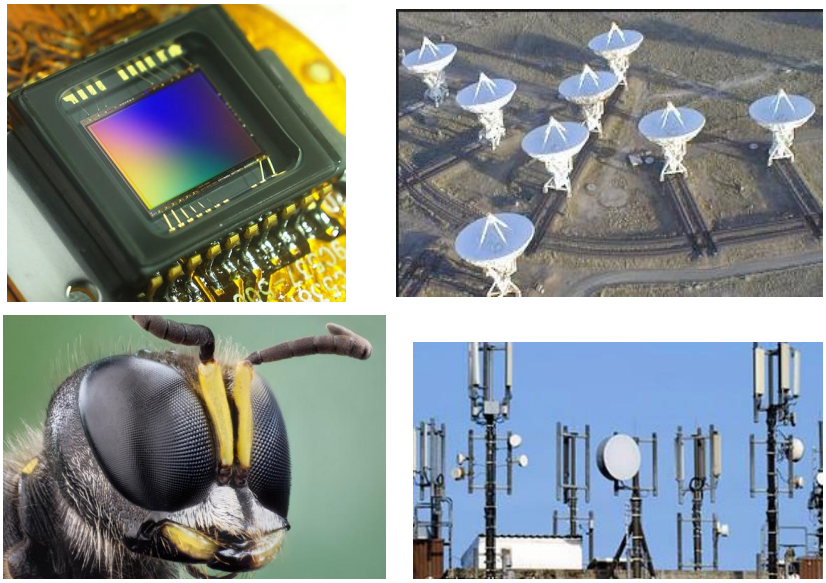


Figure 4.5: Examples of arrays of sensors in optics (left) and microwaves (right).

4.3 Optical systems

Modern CCD are capable of packing millions of elementary sensors within few centimeters, like the 20 Mpixel one shown in 4.5 that is mounted on some smartphones. The very simple principle is the pinhole camera shown in Fig. 4.6 on the left. The camera is a box in which a small hole is drilled: the light coming from the hole project to the opposite face of the box - the image plane- a reduced image of the scene. The target, whose height is H at distance R appears in the image plane reversed and of size:

$$h = H\frac{L}{R}$$

where L is the distance between the image plane and the hole, and it is directly proportional to the magnification. The camera has a couple of disadvantages, due to the fact that the hole should be narrow to have focused image, then both the energy is low - making difficult to image dark scenes, and the resolution is limited by diffraction - we use the formula for circular apertures (3.7):

$$\rho = 1.22\frac{\lambda}{D}R \tag{4.6}$$

where D is the diameter of the hole.

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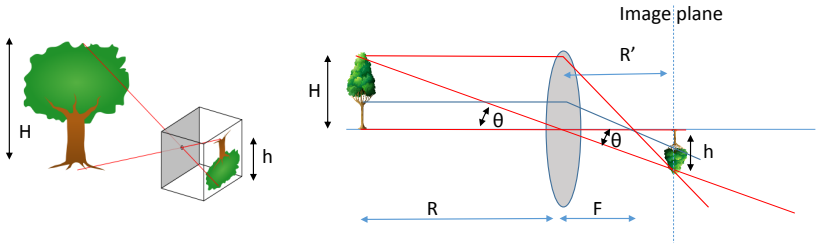


Figure 4.6: Left: the pinhole camera. Right principle of the lens.

The lens, shown in the same figure on the right, implement the same principle but allowing for large pupils. The representation of the scene on the right can be constructed by assuming that the horizontal rays (orthogonal to the lens plane) converge all in the focus, at a distance F from the optical center. Furthermore, the rays directed trough the optical center are not refracted. For a lens, the relationship between the object distance, R, the image plane distance, R', and the focal, F is:

$$\frac{1}{R} + \frac{1}{R'} = \frac{1}{F} \Rightarrow \frac{1}{R'} = \frac{1}{F}$$

where we assume that R is huge, compared to R'. Then, the image plane is very close to the focal and the magnification is such that:

$$h = H\frac{R'}{R} \approx H\frac{f}{R} \tag{4.7}$$

4.3.1 Example: iphone

Apple products, iphone and PC, introduced the concept of "retina display". The principle is that, upon tests with many observers, human eye is capable to resolve not better than 300

ppi (pixel-per-inch) at d=30 cm distance. The camera sensor of Iphone 5 has a CCD 4.5 × 3.4 mm, with 3264 × 2448 pixels, ˜8 Mp, sensitive to the three colors. The focal length is 4.1 mm, as this is critically related the the width of the smartphone. The total angular aperture, say the FOV is

$$\Delta\theta = 2\arctan\frac{h}{F} = 2\arctan\frac{(4.5/2)}{4.1} \sim 60^0$$

and at a distance of 30 cm, the pixel spacing is

$$d = \frac{4.5}{3264}\cdot\frac{300}{4.1} = 0.1mm$$

that is 10 pixel-per-mm or 254 ppi, as for retina display. We can check if this matches the resolution: the lens has aperture f/2.2, that is 1.8 mm, then resolution at 10 m is

$$\rho = 1.2\frac{0.55\cdot 10^{-6}}{2.1\cdot 10^{-3}}10 \simeq 3\,mm$$

The FOV is ˜10 m, that leads to the sensor pixels. The human pupil, when closed, is about 2-4 mm, then very similar resolution.

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4.3.2 Example: Skybox

In 2014 Google bought Skybox Imaging, for 500 millions $. The company plans to delivers 24 satellites in LEO orbit, allowing for taking three images a day of each target on the Earth. The imagery specifications and an example of one image are shown in Fig. 4.7. The system scans in pushbroom mode, but video can be made rotating the platform in Spot mode to continue pointing the same site while moving along track.



Specifications

Image Bands	Pan	450 - 900 nm
	Blue	450 - 515 nm
	Green	515 - 595 nm
	Red	605 - 695 nm
	Near-IR	740 - 900 nm
Panchromatic GSD	90 cm at nadir	
Multispectral GSD	2 m at nadir	
Swath Width	8 km at nadir	
File Format	16-bit GeoTIFF	

Figure 4.7: Skybox imagery specifications and example of image, from skybox data sheet (http://www.skyboximaging.com).

The swath width is 8 km at nadir, and we assume a nominal resolution of 70 cm, that demands for 8000/0.7 = 11500 resolution cells. We assume the CCD to be an array of 16000 (swath) × 128 (track) elements, that is 2 Mpixels. The 128 track elements allows for an increase time for swath imaging: $\Delta T = 128\cdot 0.7/6900 = 12ms$, where 6900 is the speed of the swath on the ground. Notice that the 6900 m/s corresponds to 9900 resolution cells per second, along track, × 16000 pixels across gives a rate of 160 Mpixels/s.

The resolution of 70 cm leads to an aperture D=52 cm, computed by 4.6 and assuming green, $\lambda = 500$nm and a nominal altitude of 600 km. The focal length is computed by (4.7) and assuming that the CCD pixel size is 3.5 μm, leading to an height h=$3.5 \cdot 10^{-6} \cdot 16000 =$ 56mm, then F=$0.056 \cdot 600/8 = 4.2$m . This is achieved by a telescope, like a Cassegrain reflection telescope, that exploits mirrors to make the physical size smaller than the focal.

4.4 Instrument: block scheme, signal and noise

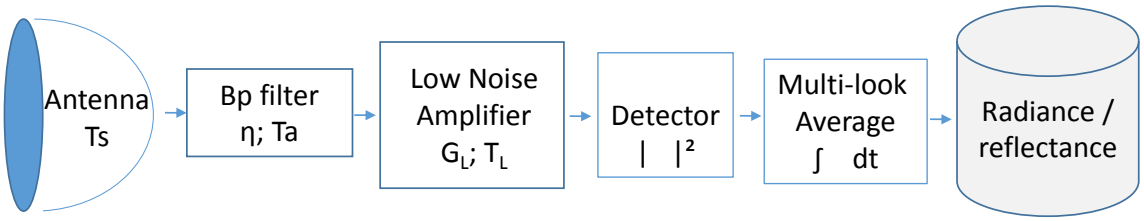


Figure 4.8: Block scheme of a radiometer.

Both radiometers and spectroradiometers measures the average power collected by the directive antenna, that scans the space with some scheme, just discussed, and interpret the energy as radiance or reflectance. The principle is shown in Fig. 4.8: the signal collected by the antenna is fist filtered, to suppress RF interferences and to limit the thermal noise contributions, then detected and integrated over some time to provide a reliable energy evaluation. The Low Noise Amplifier amplifier, that is as close to antenna as possible (like for TVSAT), amplify the signals fro the following stages with introduction of minimal noise.

4.4.1 Sources of thermal noise

All the blocks in the scheme introduce noise, in form of thermal, AWGN noise due to electron fluctuations. This noise is relevant only in the microwaves say << 6 THz, where it can be assumed white and with power spectrum density:

$$P_N = k_B T$$

that is the same power emitted by a BB at that temperature in single polarization.

However in the block scheme of Fig. 4.8 active devices introduces noises different from thermal like shot, flicker etc, that can be well modeled as AWGN, though temperature is intended to be equivalent (not physical). For such **active devices**, manufacturers usually specify the *noise factor, F* in place of the equivalent temperature. This factor accounts for the total noise in the case when the device is placed at a reference temperature $T_0 = 290$ °k. From this, we can derive the equivalent noise temperature:

$$P_W = k_B T_0 B + k_B T_E B = F k_B T_0 B \Rightarrow T_E = (F-1)\, T_0$$

In the case of a **passive device**, placed at a temperature T_A, a noise term is introduced that can be evaluated by looking at thermal balances between input and output interfaces, as shown in 4.9.

The input noise power, P_I,appears a the output interface scaled by the gain η, that is less than unitary for a passive. The output interface is in energy balance with the noise

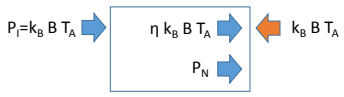


Figure 4.9: Evaluation of noise introduced by a passive device.

power from the environment, and such balance implies an additional noise source (the one introduced by the passive device):

$$P_N + \eta k_B T_A B = k_B T_A B \Rightarrow P_N = (1-\eta)\, k_B T_A B$$

Eventually we account for such noise as an equivalent source at the input of the device:

$$T_E = \frac{1-\eta}{\eta} T_A$$

The way a noisy device is handled as a noiseless one plus an additional noise source is shown in Fig. 4.10.

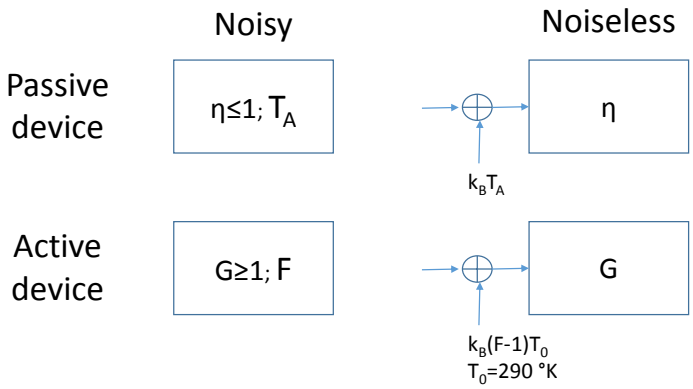


Figure 4.10: Relations between noisy and noiseless devices for passive and active systems.

This allows us to simplify the radiometer in Fig. 4.8 by the equivalent diagram in Fig. 4.11.

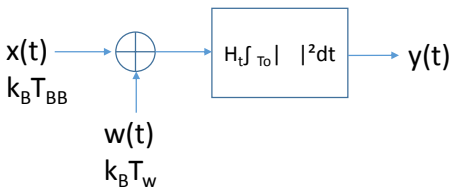


Figure 4.11: Equivalent scheme of a radiometer.

In the figure, the emission of the BB, x(t) is added with the additional total noise contribution, w(t), and then the energy is measured by integrating over the observation time, T_0,that is limited by the necessity of scanning multiple beams, to provide geometric resolution. In the average, different spectra like the one shown on the left, should converge to the

measure of energy thanks to the ergodicity of the process:

$$E\left[|y|^2\right] = lim_{T\to\infty}\frac{1}{T}\int_T |y(t)|^2\,dt = H_t\left(P_x + P_w\right)$$

this last expression accounting for the independence between radiance and thermal noise. Notice that the proper estimation of T_{BB}requires the exact knowledge of the overall system gain, H_t, and the thermal noise temperature:

$$T_{BB} = \frac{\frac{E\left[|y|^2\right]}{H_t} - P_w}{k_B \cdot B}$$

This procedure is known as **calibration**. Calibration is made much more complex is, in place of the temperature, a physical quantity is to be retrieved, like the water-vapor.

4.4.1.1 Example

Let us evaluate the equivalent input temperature, T_E, as from Fig. 4.8, by assuming that the filter has attenuation 1 dB, at environment temperature 300 °K, and the amplifier has a noise factor F=2 dB.

The contribution of the amplifier, moved back at the input of the filter (passive device) is computed by scaling by the inverse of the filter power gain:

$$T_{Ea} = \frac{(F-1)\,T_0}{\eta} = \frac{\left(10^{0.2}-1\right)\cdot 290}{10^{-0.1}} = 213.53°K$$

The contribution of the passive filter is

$$T_{Ep} = \frac{1-\eta}{\eta}T_A = \frac{1-10^{-0.1}}{10^{-0.1}}300 = 77.6°\text{K}$$

the total noise is then $T_E = 213.53 + 77.6 = 291°$K.

4.5 Radiometric resolution and looks

Both signal, emitted by the BB and thermal noise can be assimilated to a bandpass process, like the one plotted in Fig. 4.12. In the real case, the fractional bandwidth:

$$B_f = \frac{B}{f_0} \tag{4.8}$$

is much smaller than the one in the figure, and this means that the envelope is extremely slow, its correlation time being ˜1/B, when compared to the fluctuations of the sinusoid, with period $T_0 = 1/f_0$. Therefore, the simplest measure of energy can be implemented by rectifying the signal - leading to the red one in the figure, and then averaging in time much larger than $T_0$to remove the sinusoidal fluctuations leaving the envelope (green).

4.5.1 The IQ demodulator

The detection and average instrument shown in Fig. 4.11 is the simplest - purely analogue - and most used in the case that average energy is the sole signal of interest. CCD in cameras and photo-receptors in eyes work with the same principle.

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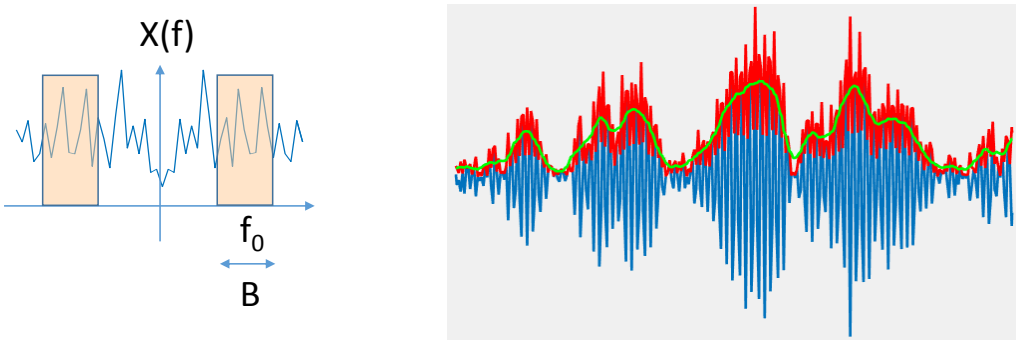


Figure 4.12: Left: representation of the spectrum of a noise realization. Right: plot of bandpass noise before (blue) and after detection (red) and averaging (green).

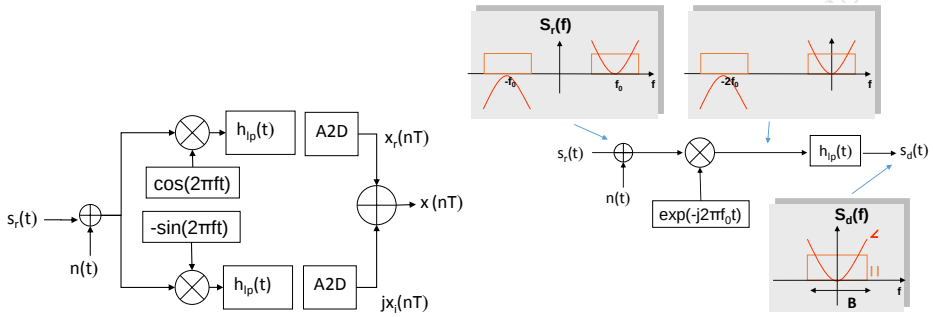


Figure 4.13: Left: scheme of an I+Q coherent demodulator. Right: signal model.

However, is not only the power, but also the signal is of interest, like for Radars, Sonars etc, a coherent demodulator and digitizer is used. The scheme of the demodulator and the equivalent signal model are shown in Fig. 4.13.

The demodulated signal is obtained by feeding the input into a couple of quadrature demodulators, cosinus and sinus (that means one is phase shifted by $\pi/2$ with respect to the other). The outputs are bandpass filtered, digitized and combined into a complex signal, whose real part is the one from the cosinus, that is the **I**n phase channel, and the imaginary part is the output from the sinus, the **Q**uadrature channel. The model for the complex output is of the block diagram in Fig.4.13 on the left is:

$$\begin{aligned} x\left(t\right) &= \left(s(t)+n(t)\right)\cos\left(\omega_0 t\right) * h_{lp}(t) - j\left(s(t)+n(t)\right)\sin\left(\omega_0 t\right) * h_{lp}(t) \\ &= \left(s(t)+n(t)\right)\cdot\left(\cos\left(\omega_0 t\right) - j\sin\left(\omega_0 t\right)\right) * h_{lp}(t) \\ &= \left(s(t)+n(t)\right)\cdot\exp\left(-j\omega_0 t\right) * h_{lp}(t) \end{aligned}$$

The frequency domain interpretation, shown on the right side of the figure, is a simple shift into baseband and filter to remove the component at frequency $-2f_0$:

$$X(f) = \begin{cases} S(f+f_0)+N(f+f_0) & for \quad -B/2 \le f \le B/f \\ 0 & elsewhere \end{cases}$$

In the end, the received signal can be modeled as a complex signal and a complex noise:

$$x(t) = s_d(t) + n_c(t) \tag{4.9}$$

4.5.2 The passband noise

In Radar and radiometric applications we have them to face with bandpass signal and noise, that are baseband demodulated as in (4.9). The complex noise n_c (and the same for the signal, in case of a Radiometer), is obtained after the demodulator:

$$\begin{aligned} n_c(t) =& \{n(t) \cdot \cos(2\pi f_0 t) - j \cdot n(t) \cdot \sin(2\pi f_0 t)\} * h_{lp}(t) \\ =& n_i(t) + j n_q(t) \end{aligned} \tag{4.10}$$

Notice that both components within the braces, then before low-pass filtering, are non stationary, but cyclostationary, as appears from the realizations plotted in Fig. 4.12 on the right. We can show that the two components are mutually uncorrelated.. We can compute the averaged power of the I channel (real part) noise $n_i(t)$

$$\begin{aligned} \overline{\sigma^2_{ni}(t)} =& \frac{1}{T}\int_T E\left[n^2(t) \cdot \cos^2(2\pi f_0 t)\right] dt \\ =& \frac{\sigma_n^2}{2} = \frac{N_0}{4} 2B = \frac{N_0}{2} B \end{aligned} \tag{4.11}$$

The complex noise n_c in (4.10) is however the superposition of two of such processes, mutually uncorrelated, then its power - time averaged - is

$$\begin{aligned} E\left[n_c(t) n_c^*(t)\right] =& \frac{N_0}{2} B + \frac{N_0}{2} B = N_0 B \\ =& k_B T \cdot B \end{aligned}$$

Therefore, when we account for a complex noise, we have to consider a power spectrum density of $k_B T$ and a bandwidth B, that is totally equivalent to a real noise with power spectrum density$N/2 = k_{0B} T/2$ and bandwidth 2B.

4.5.3 Power estimation: the speckle noise

The estimate of temperature can then be assimilated to estimate of average power from a noisy process, we can think of a circular Gaussian distributed random process :

$$x = x_r + j x_i \tag{4.12}$$

$$x_r, x_i \sim N\left(0, \sigma^0/2\right)$$

$$E\left[x_r x_i^*\right] = 0$$

$$\Rightarrow \sigma_x^2 = \sigma_{xr}^2 + \sigma_{xi}^2 = \sigma^0 \tag{4.13}$$

In the microwave radiometers, $\sigma^0$is proportional to the temperature, $k_B T_{BB} B$ to be measured. However, the instantaneous signal, squared is not the power, but just fluctuating randomly. This **random fluctuation is a noise**, if we are interested in power, that is completely different form the thermal noise, the contribution n_c in (4.9). We call it "speckle" noise due to its peculiar "speckled" structure shown in Fig. 4.15.

4.5.3.1 Speckle statistics

As we are interested in power, we measure the intensity $I(x) = |x|^2$. This intensity is not constant at all - as we wish - since it is due to the random contributions of huge number of charges. The central limit applies, leading to x as Gaussian, zero mean. Then I is distributed

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as *Chi-Squared* (χ^2) with one ore two freedom degrees, depending if x is real or complex (i.e. the output for the IQ demodulator just discussed).

In general a *Chi-Squared* distribution with N_F freedom degrees is the distribution of the summation of N_F Independent Identically Distributed (iid) rand variables:

$$I = \sum_{n=1}^{N_F} x_n^2$$

where $x_n$are Gaussian distributed $x_n \sim N\left(0, \sigma^2\right)$. The distribution has mean and standard deviation:

$$E\left[I\right] = N_F\sigma^2;\ VAR\left[I\right] = 2N_F\sigma^4 \Rightarrow \frac{E\left[I\right]^2}{VAR[I]} = \frac{N_F}{2} \tag{4.14}$$

(this last is usual result for the summation of iid variables).

Notice that we are interested in estimating the mean intensity, $E\left[I\right]$,however we get it with a dispersion - an error - proportional to the standard deviation. In the following, we will assume the complex processes, that leads to an Intensity that is $\chi^2$with $N_F = 2$ freedom degrees:

$$I = x_r^2 + x_i^2 \tag{4.15}$$

$$f_I(I) = \frac{1}{\sigma^0}\exp\left(-\frac{I}{\sigma^0}\right) \tag{4.16}$$

$$\mu_I = \sigma^0;\ \sigma_I^2 = \left(\sigma^0\right)^2 \Rightarrow \frac{\mu_I^2}{\sigma_I^2} = 1 = \frac{E\left[I\right]}{STD\left[I\right]}$$

This last expression shows that the dispersion of the measure is equal to the value we want to measure, leading to potential 100% error, the measure is not significant.

4.5.3.2 Multi-look average: Equivalent Number of Looks and radiometric resolution

The need is for improving the estimate of temperature: ideally we wish the dispersion of the measure to be zero, so that we can appreciate the tiniest fluctuation, like the variation of gray in a greyscale image. The improvement can be achieved by averaging M independent measures of look:

$$\widehat{I} = \frac{1}{M}\sum_{n=1}^{M} I(n) = \frac{1}{M}\sum_{n=1}^{M}\left\{x_r^2(n) + x_i^2(n)\right\} \tag{4.17}$$

we then need M complex samples, that usually are derived from neighboring pixels - say a boxcar filtering, assuming that the temperature is homogeneous within the averaging window. That's to say, we trade geometric for radiometric resolution. The improvement on radiometric resolution appears evident in the plots in Fig. 4.14, that shows χ^2PDF with $N_F = 2M$ freedom degrees (also known as Erlang distributions with form factor M). The performances of this estimates increases with the number of looks, with M=100 the standard deviation would be 10% of the mean value, a well known results in power estimation problems.

In general, we are not sure that the number of pixels averaged are independent, therefore we define the Equivalent Number of Looks (ENL) as:

$$L_{ENL} = \frac{\left[\sigma^0\right]^2}{VAR\left[\widehat{\sigma^0}\right]} \Rightarrow \frac{STD\left[\widehat{\sigma^0}\right]}{\sigma^0} = \frac{1}{\sqrt{L_{ENL}}}$$

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that in practice is the ratio between the power and its random fluctuation of its estimate, when this is made basing on L_{ENL}indpendent samples. In the ideal case of infinite samples, $L_{ENL} \to \infty$, and the estimate of the power would be without errors.

This definition is quite general, and it is applied to any "power" estimator, where power can be related to temperatures, in this case, but also to other physical quantities, like backscatter in a RADAR.

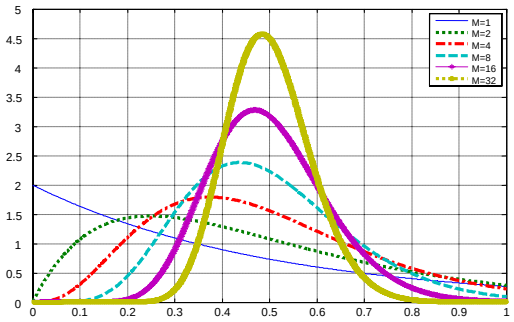


Figure 4.14: PDF of the M-looks backscatter estimator, that is Erlang with form factor M. The graphs plotted here for E[I]=0.5.

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However, the figure of quality usually adopted is not the equivalent number of looks, but the *radiometric resolution*:

$$\rho_{RAD} = 10\log_{10}\left(1 + \frac{STD\left[\widehat{\sigma^0}\right]}{\sigma^0}\right) = 10\log_{10}\left(1 + \frac{1}{\sqrt{L_{ENL}}}\right)$$

The **radiometric resolution** is then 3 dB in the worst case of a single look image, and gets close to 0_{dB} as the number of looks goes to infinity. Quite good images are obtained by exploiting $\gg 10$ looks, say 40, that leads to $\rho_{RAD} \sim 0.1$ dB.

4.5.3.3 Geometric and radiometric resolution: the speckle filter

An example of the improvement of the radiometric resolution with different number of looks in given in Fig. 4.15. In that case, looks have been generated as independent realization of the process, therefore the improvement of radiometric resolution has not been achieved at the cost of a loss in geometric resolution - but, say, with an increase in the time of measure - that is not possible in fast moving platforms, like LEO system

4.5.3.4 The multiplicative model

The previous example has shown that the signal we are interested, say temperature, is not constant, but varying from space to space. The problem can be formulated like estimating the instantaneous power of $x(P)$, P being any point in the scene: the examples of Fig. 2.6, Fig. 4.7, and Fig. 4.15 correspond to the case. The reason why in the first two images we do not see noise is due to the fact that number of looks averaged is pretty large (we will show an example).

The problem can be formulated as the estimate of the space-varing signal, $\sigma^0(P)$ form:

$$x\left(P\right) = \sqrt{\sigma^0(P)}w(P) \Rightarrow I(P) = \left|x(P)\right|^2 = \sigma^0(P) \cdot w_a(P) \tag{4.18}$$

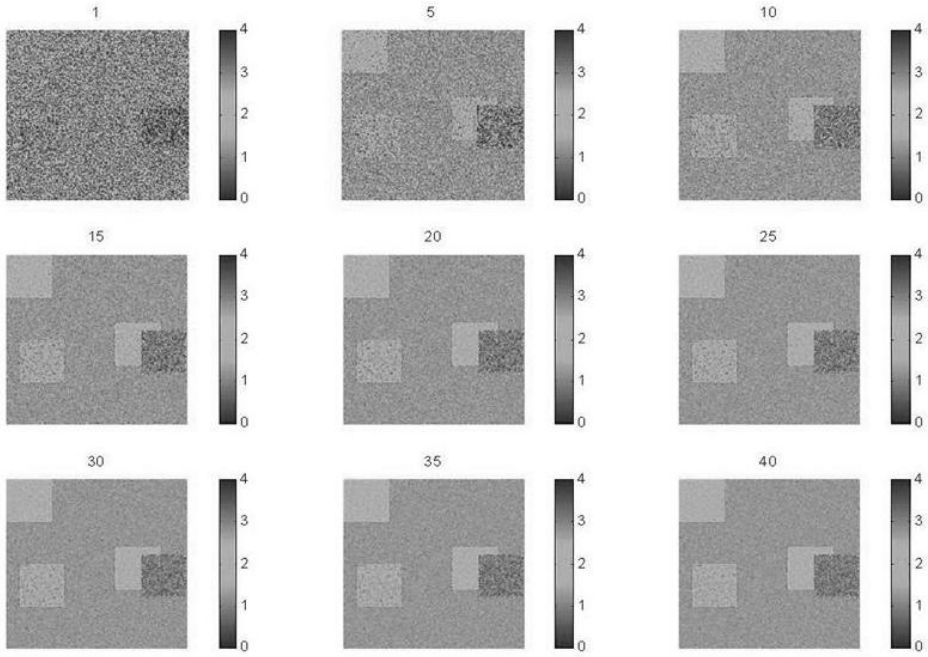


Figure 4.15: Simulation of multi-look averaged images for different Equivalent Number of Looks.

where $w_a(t)$ is a stationary, random noise, that is χ^2distributed. We are now facing the case of a *multiplicative noise*, in place of the additive more frequently found, that is if the signal increase, the noise (intended as fluctuation) increase in the same order. It can be shown that, if the signal is constant, $\sigma^0(P) = \sigma^0$ the moving average implied by (4.17), is indeed the Maximum Likelihood estimator.

However, in the general case, different methods may be used, like the Lee filter, that often defined as "speckle noise" filters.

4.5.3.5 The noisy case

So far we have assumed that the signal x is free from thermal additive noise, only affected by its own fluctuations. However we observed that the radiometer signal is affected by noise, like the case of thermal noise on the receiver discussed in the section 4.4. Let us look model the complex observables as a signal, $s(n)$ and a noise $w(n)$ contributions :

$$y(n) = s(n) + w(n)$$

where both are circular, zero mean, random, complex, with variances σ_s^2 and σ_w^2. We are interested in measuring $\sigma^0 = \sigma_s^2$, however this measure is affected by the following errors:

1. a bias, due to the additional contribution of noise:

$$VAR\left[y^2\right] = VAR\left[s^2\right] + VAR\left[w^2\right] \Rightarrow BIAS = \sigma_s^2 - VAR\left[y^2\right] = \sigma_w^2$$

   In the case of a temperature, the one we estimate is slightly larger than the true one, the additional term being the equivalent thermal noise temperature in the receiver, like $T_E$in section 4.4.1.1;

2. a statistical fluctuation that is the result of both signal and noise, for the $N_F = 2$ freedom degrees χ^2PDF:

$$VAR\left[\sigma_y^2\right] = E\left[\sigma_y^2\right]^2 = \left(\sigma^0 + \sigma_w^2\right)^2$$

This fluctuation corresponds to the "speckle" noise defined in previous section, but for the fact that it comes also from the noise contribution $w(n)$ and this is much more relevant when that noise becomes comparable with the signal, that is when

$$SNR = \frac{\sigma^0}{\sigma_w^2} = \frac{\sigma_s^2}{\sigma_w^2}$$

is close to unitary, or less. The equivalent number of looks, say out of a multi-look average, is then:

$$\begin{aligned} L_{ENL} &= \frac{\left[\sigma^0\right]^2}{VAR\left[\widehat{\sigma^0}\right]} = \frac{\left[\sigma^0\right]^2}{\frac{(\sigma^0+\sigma_w^2)^2}{M}} \\ &= M\left(\frac{1}{1+\frac{\sigma_w^2}{\sigma^0}}\right)^2 \\ &= M\frac{1}{(1+SNR^{-1})^2} \end{aligned} \tag{4.19}$$

As an example, at SNR=0 dB the ENL halves with respect to the noiseless case.

The radiometric resolution is then

$$\rho_{RAD} = 10\log_{10}\left(1 + \frac{1+SNR^{-1}}{\sqrt{M}}\right) \tag{4.20}$$

For example, let us assume that we have a patch of 10 x 10 (independent) pixels with resolution 5 × 5 m., with SNR=3 dB. We assume to estimate the temperature as a bare averaging of all the 10×10 pixels, that is 100 looks, for a radiometric resolution $\rho_{RAD} = 10\log_{10}(1.1) \sim 0.4$dB if we ignore noise. The geometric resolution thus drops to 50 × 50 m. However, if we do account for noise the equivalent number of looks is, form (4.19):

$$ENL = M\frac{1}{(1+SNR^{-1})^2} = \frac{100}{\left(1+\frac{10^{-2.3}}{10^{-2}}\right)^2} \sim 44$$

giving a radiometric resolution of $\rho_{RAD} = 10\log_{10}(1+1/\sqrt{44}) \sim 0.6$dB

4.6 Accuracy for LEO microwave radiometers

In the LEO case, the radiometric accuracy is affected by the problem of a limited observation time, due to the fast acquisition. If the time each resolution cell is observed is T_o and the desired bandwidth is B, then the equivalent number of looks is like the number of independent samples we have, that is the ratio between the duration and the sampling frequency: $M = B \times T_o$ (time-bandwidth product). Then we can upgrade (4.5) as follows:

$$\frac{L_{XT}M}{\rho^2 B} \leq \frac{1}{v_s} \tag{4.21}$$

The expression now relates the basic requirements (or quality factor) of a radiometer:

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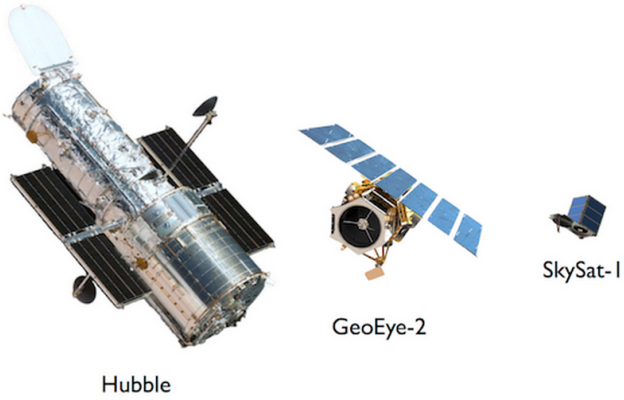


Figure 4.16: A size comparison of imaging satellites currently in orbit. The Hubble is pointed toward the heavens, not Earth, while GeoEye satellites are what Google uses today for imaging data. (Source: Skybox)

1. the FOV, or **image size** across track, L_{XT}, that we desire as large as possible;
2. the **radiometric resolution**, ρ_{RAD}, that demands the equivalent number of looks M in (4.19), to be as large as possible;
3. the **geometric resolution**, ρ, that we need as fine (small) as possible
4. the **chromatic resolution,** related to B, that we need as small as possible

The obvious-result is that we cannot improve one of those without loosing the other, although, the total value can be increased by using a more complex instrument, say a pushbroom scanning or the multi-track system discussed in the example in section 4.3.2.

4.6.1 Radiometric accuracy in Skybox

Let now compute the ENL in case of an optical system, like Sky-box satellite, see section 4.3.2 .

As the system is operating in the visible, we express the luminance observed in the visible as

$$L = \frac{P}{A \cdot \Omega \cdot \Delta\lambda} = 0.06\,[W/nm/sr]$$

the numeric value on the right has been computed as the energy spectrum that comes to the earth form sun in the visible, $1W/m^2/nm$, then we assumes that the gray object has an albedo 0.2, and we account for $1/\pi$ to convert in luminance, by assuming a Lambertian reflection (like the example of the moon).

The sensor has a geometric resolution of 70 cm, and the LEO orbit height is 600 km, so that the solid angle under which a single resolution surface on the Earth is observed is

$$\delta\Omega = \frac{0.7^2}{600000} = 1.3 \cdot 10^{-12} sr$$

The lens has been already dimensioned to the resolution to 56 cm diameter, section 4.3.2 , corresponding to an area of 0.24 m^2. We compute the total power by integrating over the bandwidth, that for a single color is 60-90 nm:

$$P = L \times A \times \Omega \times \Delta\lambda = 0.06 \times 0.24 \times 1.3 \cdot 10^{-12} \times 70nm = 1.3 \cdot 10^{-12} W \tag{4.22}$$

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that results, in the integration time $\Delta T_{AT} = 12ms$, into an energy

$$E = 1.3 \cdot 10^{-12} \times 0.012 = 1.5 \cdot 10^{-14} Joule$$

that leads to a total number of photons:

$$N = \frac{E}{hf} = \frac{1.5 \cdot 10^{-14} \times 0.55 \cdot 10^{-6}}{6.62 \cdot 10^{-34} \times 300 \cdot 10^{6}} = 43500$$

Notice that we get this value thanks to the supposed 128 pixels along track of the CCD. Older devices w<ere using a single line CCD. Finally, is we assume an efficiency of the detector equal to 10%, we get the effective number of looks 4300. However, this number may change according to the surface reflectivity: a white body will get 100%, then 5 times more looks, whereas non reflective surfaces leads to so few photons that the quantization effect is clearly visible in granule of colors.

However, the white surface is not the brightest one, that would be the case of a mirror in specular direction with the sun light. We assume a mirror of size D=1 cm, say with the same albedo 0.2, so that the reflected power is:

$$P_R = 0.06 \cdot \frac{\pi D^2}{4} = 4.7 \cdot 10^{-6} W$$

This power is spread over a surface, at the distance of the satellite:

$$A = \Delta\Omega \cdot R^2 = \frac{\lambda^2}{\frac{\pi D^2}{4}} R^2 = 1400 m^2$$

and then the fraction 0.24/1400 of the reflected power corresponds to $4.7 \cdot 10^{-6} \cdot 0.24/1400 = 8 \cdot 10^{-10}$that is 800 larger than the one coming from the resolution cell, 70×70 cm with the same albedo.

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Chapter 5

Applications

5.1 Radar, Lidar and Sonar: when speckle is signal

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Radar, Sonar and Lidar are coherent imaging systems exploiting quasi monochromatic observations respectively in the microwave and IR to visible. In laser space coherence is ensured by a resonant chamber, and by exploiting photons with same energy, temporal coherence - the monochromatic behavior is achieved by reducing the number of excited modes. In Radars, an ultra stable oscillator is exploited for "long" term temporal coherence.

It has pointed out that these devices exploits the diffusion of surfaces: the back-scatter, to receive echoes from the emitted signal. We show here that a dispersion of reflectors contributing in the same resolution cell introduce a random fluctuation of the amplitude that appears exactly like the speckle noise we have discussed in radiometers, with the difference that this is temporally stable. Here speckle is the fingerprint of the surface, that would always appears identical if observed by the same geometry and source (frequency, polarization).

An example of speckle is given in Fig. 5.1, that represents (on the right) an high resolution Radar image taken when the satellite was passing over the receiving station of Kiruna. The cross in the middle of the image corresponds to the impulse response of the Radar system, that is bandpass (then a 2D sinc in space when converted into baseband), resulting from the specular reflection is the big dish of Kiruna station. The grainy surface around is the effect of speckle "signal"

In order to understand speckle, let us assume a rough surface, or a random distribution of scatterers (reflectors) in the system resolution cell. This distribution should be such that no one prevails over the other (like for the parabolic dish), that may be the case of homogeneous media. The resulting signal gathered by the Lidar or the Radar, can be modeled by a stochastic random process, independent from pixel to pixel. The random fluctuations of the backscattered energy give rise to the so called "speckle". More precisely, the case when no scatterer prevails is defined as "fully developed speckle".

Let us assume to have N independent targets in the resolution cell, each target with its own, reflectance, say:

$$r(n) = \sqrt{\sigma_n}\exp(j\psi_n) \qquad n = 0 \dots N-1$$

as pictorially represented by the trees in Fig. 5.2. Here we have assumed the general case where each scatterer has a different amplitude and a phase - that may be to its dielectric property. However, we would get speckle if amplitude is constant and phase null.

The major assumption we make here is the *linear* behavior of the composite target, that holds as far as the backscattered field can be computed according to the *first Born*

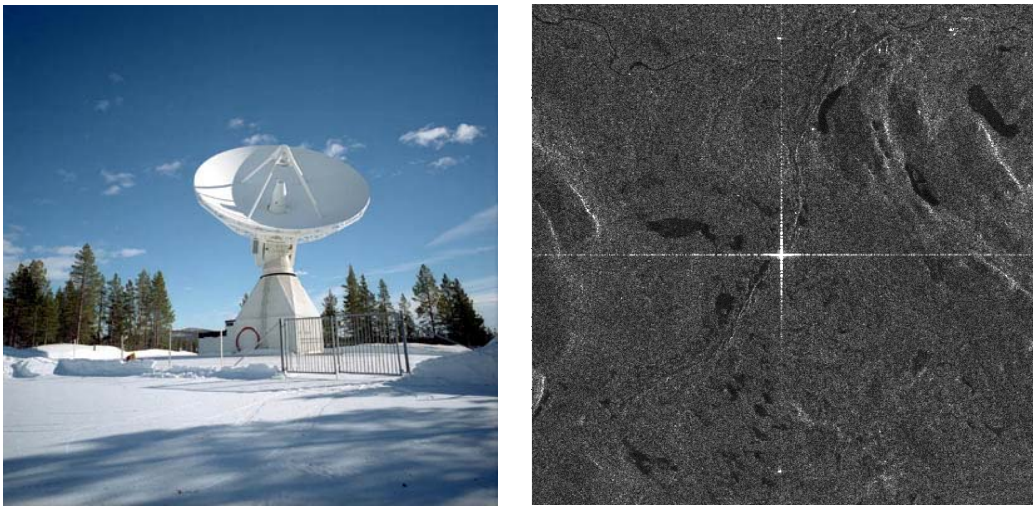


Figure 5.1: The ESA receiving station at Kiruna, on the left, behaves like a point target in the SAR image on right. The image size is 12x12 km and the sensor is ESA satellite ERS-2.

approximation. In practice we assume that contributions from double and multiple bounces are so small that can be ignored.

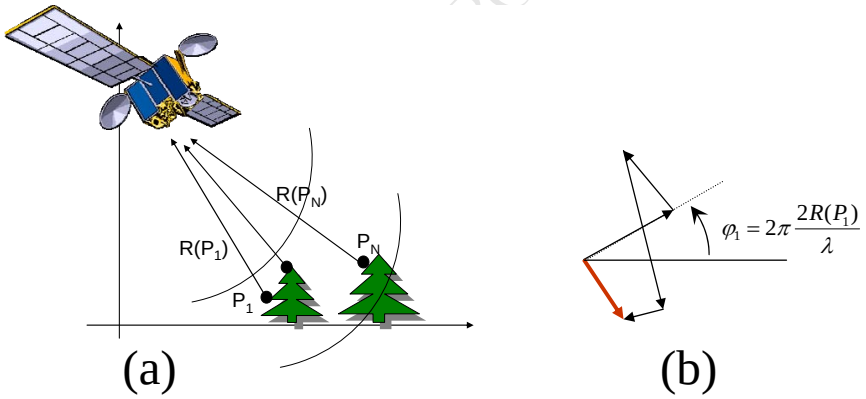


Figure 5.2: (a) A distributed target is represented by several independent scatterers in the resolution cell. (b) The backscattered signal is the coherent superposition of all the returns, with phases that changes according to the precise location of each target..

The backscattered filed in then the superposition of the N returns from the N targets, each delayed by the different target-sensor travel path, $R(P_n)$ in the figure. We can then model all the returns from the same resolution cell - say the pixel as:

$$s\left(t = 2R_0/c\right) = \sum_{n=1}^{N} \sigma_n \exp\left(j\psi_n\right) \exp\left(j2\pi f_0 \left(t - \frac{2R(P_n)}{c}\right)\right)$$

R_0 being the mean distance of the resolution cell from the device and$\sigma_n$the amplitude single complex return. Notice that we omitted the attenuation in the two way, that does not change so much for very close targets. The signal is bandpass, yet after baseband conversion, see

Fig. 4.13:

$$
\begin{aligned}
y(t) &= s(t) \cdot \exp(-j2\pi f_0 t) \qquad (5.1)\\
y(t = 2R_0/c) &= \sum_{n=1}^{N} \sigma_n \exp(j\psi_n) \exp\left(-j\frac{4\pi}{\lambda} r_n\right)\\
&= \sum_{n=1}^{N} \sigma_n \exp(j\phi_n)\\
&\text{where } \phi_n = \psi_n - \frac{4\pi}{\lambda} r_n
\end{aligned}
$$

The point is that $\phi_i$is a random variable that tends to be uniformly distributed in $-\pi \ldots +\pi$ is the dispersion of the distances of the scatterers in the resolution cell becomes bigger than half the wavelength. Therefore, *if the resolution cell is bigger than the wavelength*, or, equivalently, the *fractional bandwidth is small,* the reflectivity -as sum of independent contributions (scattered sparse int he resolution cell), becomes a random number whatever the contributions. This occurs as shown in Fig. 4.13, it is known in statistics as "random walk", that is the summation of contributions with independent directions (phases). As for the central theorem, the distribution of the resultant tends to be Gaussian, and, from homogeneous scatterers, we have a Brownian motion (a particular case of random walk that is stationary in increments). We say that we are in presence of *fully developed speckle*. The mean and covariances of the complex process are:

$$
\begin{aligned}
E[y] &= \sum_{n=1}^{N} \sigma_n E[\exp(j\phi_n)] = 0\\
E[yy^*] &= \sum_{i=1}^{N}\sum_{k=1}^{N} E[\sigma_i \sigma_k \exp(j(\phi_i - \phi_k))]\\
&= \sum_{n=1}^{N} E\left[\sigma_n^2\right] = N \cdot E\left[\sigma_n^2\right] = N \cdot \sigma^0 \qquad (5.2)
\end{aligned}
$$

The process is then zero mean, circular Normal distributed with variances of real and imaginary part $\sigma_r^2 = \sigma_i^2 = \sigma^0/2$:

$$
\begin{aligned}
f_{y_r,y_i}(y_c) &= \frac{1}{\sqrt{2\pi}\sigma_r^2} \exp\left(-\frac{y_r^2}{2\sigma_r^2}\right) \times \frac{1}{\sqrt{2\pi}\sigma_i^2} \exp\left(-\frac{y_i^2}{2\sigma_i^2}\right) \qquad (5.3)\\
&= \frac{1}{\pi\sigma^0} \exp\left(-\frac{|y|^2}{\sigma^0}\right)\\
&= N_c\left(0, \sigma^0\right)
\end{aligned}
$$

the variance being the product between the variance of each scatterer - say their reflectivity - and the number of scatterers in the resolution cell. We define the former as *backscatter coefficient* (or simply backscatter):

$$
\sigma^0 = E\left[\sigma_n^2\right] = \frac{E[yy^*]}{N} = \frac{\sigma_y^2}{N} \qquad (5.4)
$$

A remarkable example of this is given in Fig. 5.3, that is a SAR image of crops. Each field, being made of the same type of scatterers, as for height, water content, moisture, represents

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an homogeneous scatterer, than wit a specific σ^0that actually changes with the position. We remark that, the noisy appearance should not confuse the reader: is the very same scene is observed at the same frequency from the same point of view, the "noise" structure will be replicated exactly.

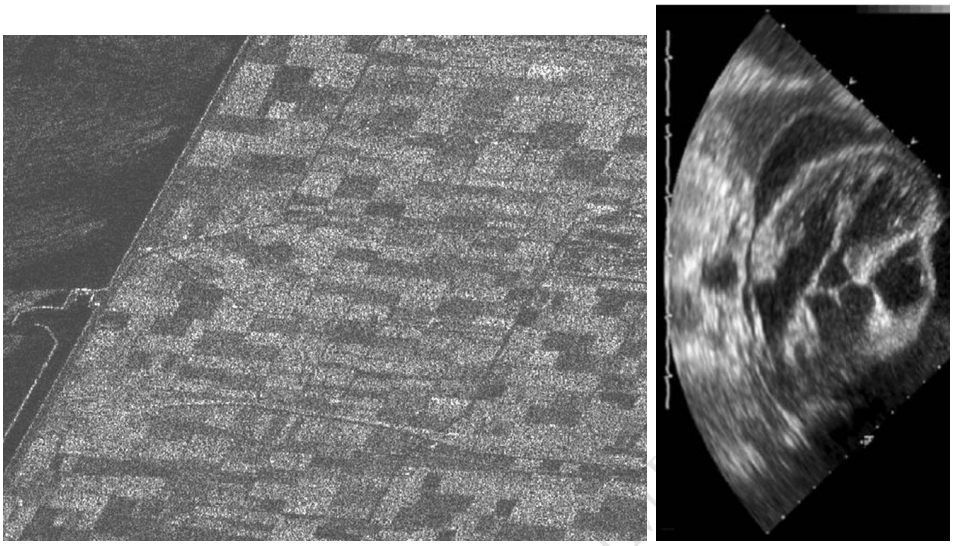


Figure 5.3: Speckle. Left: Synthetic Aperture Radar intensities images gathered near Flevoland, Holland. The detected amplitude is here represented, resolution is 20 x 20 m per pixel. Right: ultrasound image of an heart. Notice the degradation of vertical resolution from near ranger (rigth in the image) to far range (left).

5.1.1 Change detection: coherence

The speckled nature is used for very different applications, like classifications, feature extraction, motion tracking, change detection. An example, is the coherent change detection shown in Fig. 5.4. Two SAR images have been taken for an airborne system 10 minutes apart. The amplitude of the images, one of which is shown on the left, does not provide sensible differences. However the change in speckle structure, evidenced by coherence, on the left, show the passage of ships and a truck.

5.1.2 Definition, evaluation and use of coherence

The detection of such slight changes in the texture, and the measure of quality in coherent imaging systems, like Sonar, Radar and Lidar, is provided by coherence, the geophysics definition for the normalized correlation index:

$$\gamma_s(r) = \frac{E\left[s_M(P)s_S^*(P)\right]}{\sqrt{E\left[|s_M(P)|^2\right]E\left[|s_S(P)|^2\right]}} \tag{5.5}$$

The definition applies to two complex processes: here $s_m(P)$ and $s_s(P)$ represents the two complex observations of the same target, located in P, say in slight different time (see the case above), or maybe different view points, as we will see for interferometry. We know from statistics, that $|\gamma| = 1$ if and only if the two images are linearly related, that is they differ

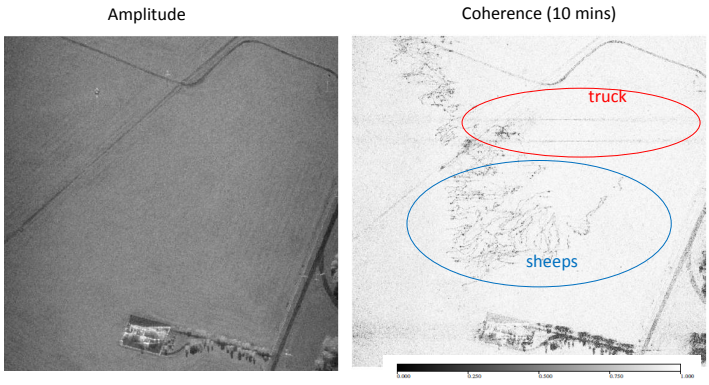


Figure 5.4: Coherent change detection, based on two images taken at 10 minutes. Left: amplitude of one of the two, right: coherence. Courtesy of: "Coherent Change Detection: Theoretical Description and Experimental Results" Mark Preiss and Nicholas J. S. Stacy Intelligence, Surveillance and Reconnaissance Division Defence Science and Technology Organization.

for just a complex scale factor, whereas $|\gamma| = 0$ for incorrelated processes, that is when the images do not carry any mutual information.

5.1.2.1 Decorrelation due to additive noise

Let us assume that the two complex images are identical, but for a pure additive noise, like AWGN thermal noise:

$$\begin{aligned} y_1 &= s + w_1 \\ y_2 &= s \cdot \exp(j\phi) + w_2 \end{aligned} \tag{5.6}$$

Coherence results from the definition:

$$\begin{aligned} \gamma_{th} &= \frac{E\left[y_1 y_2^*\right]}{\sqrt{E\left[y_1 y_1^*\right] E\left[y_2 y_2^*\right]}} = \frac{\sigma_s^2 \exp(j\phi)}{\sqrt{\left(\sigma_s^2 + \sigma_{w_1}^2\right)\left(\sigma_s^2 + \sigma_{w_2}^2\right)}} = \frac{\exp(j\phi)}{\sqrt{\left(1 + \frac{\sigma_{w_1}^2}{\sigma_s^2}\right)\left(1 + \frac{\sigma_{w_2}^2}{\sigma_s^2}\right)}} \\ &= \frac{\exp(j\phi)}{1 + SNR^{-1}} \end{aligned} \tag{5.7}$$

where $\sigma_s^2, \sigma_{w_1}^2$ and $\sigma_{w_2}^2$ are respectively the signal power and the powers of the two noises superposed, that we assumed these to be identical. As an example, SNR=0,1,∞ corresponds to $\gamma = 0$ (pure noise), 0.5, 1 (noiseless). It's interesting to notice that the phase of the coherence is equal to the different phase between the two signals, in absence on noise. This eventually leads to the statistically optimal Maximum Likelihood estimate of $\phi$from (5.6). We notice that noise can be measured from coherence by reversing (5.7):

$$\begin{aligned} SNR^{-1} &= 1 - \gamma_{th} \\ SNR &= \frac{1}{1 - \gamma_{th}} \end{aligned}$$

5.1.2.2 Decorrelation due to image changes

Coherence do not account only for additive noise, but also changes in the two images. Let us assume that one is the other multiplied by a random process, like in the case shown in

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Fig. 5.4, where just a few pixels were changed from one image to the other. We can model this as each of the elementary contribution in one resolution cell are multiplied by a random variable$k$

$$y_1 = k \cdot s$$
$$y_2 = s$$

Coherence is then:

$$\gamma_k = \frac{E\left[k \cdot s \cdot s^*\right]}{\sqrt{E\left[k \cdot s \cdot k^* \cdot s^*\right] E\left[s \cdot s^*\right]}} = \frac{E\left[k\right]}{\sqrt{E\left[|k|^2\right]}} = \frac{E\left[k\right]}{\sqrt{VAR\left[k\right] + E\left[k\right]^2}}$$
$$= \frac{1}{\sqrt{1 + \frac{VAR[k]}{E[k]^2}}} \tag{5.8}$$

Then, is the change is just a fixed scaling, $VAR\left[k\right] = 0$ and coherence is 1. However, we may take the case where a fraction p of the scene is kept constant, and a fraction (1-p) is canceled. Then the pdf is, like for tossing one coin,

$$pdf(k) = (1-p) \cdot \delta(k) + p \cdot \delta(k)$$
$$E[k] = p$$
$$E\left[k^2\right] = p \Rightarrow \gamma = \frac{p}{\sqrt{p}} = \sqrt{p}$$

then coherence is the square root of the fraction of unchanged scatterers in the resolution cell.

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5.1.2.3 Example: small decorrelation

As an example, we can evaluate which decorrelation due to changes corresponds to that of an additional noise with SNR=20 dB. From (5.7):

$$|\gamma_{th}| = \frac{1}{1 + SNR^{-1}} \underset{SNR \gg 1}{\simeq} \left(1 - \frac{1}{SNR}\right) \Rightarrow 1 - |\gamma_{th}| = \frac{1}{SNR}$$

that is a decorrelation of 0.01, or $|\gamma_{th}| = 0.99$. We would get the same result with a fraction

$$p = \gamma^2 = (1 - 0.01)^2 \sim 1 - 0.02$$

that is a change occurring to 2% of the targets.

5.1.2.3.1 Temporal decorrelation We may now assume that changes occur due to random motion of scatterers is the resolution cell, like leaves subject to wind, or those represented in Fig. 5.2. We assume Brownian motion, that is increments are Gaussian stationary, and we assume n steps of duration T. The displacement after a total time $\Delta t = nT$ is:

$$\mathbf{d}(t = nT) = \mathbf{r}(nT) - \mathbf{r}(0) = \sum_{m=0}^{n-1} \mathbf{v}(m) \cdot T$$

where $\mathbf{v}(m)$ is the mean velocity vector in the m-th step. If we assume a 2D motion, and $\mathbf{v}$ a circular Normal distributed variable $N_c\left(0, \sigma_v^2\right)$, then the resulting displacement is distributed as for Brownian random walk, see section 4.5.3.1:

$$\begin{aligned} E[\mathbf{d}(nT)] &= 0 \\ \sigma_d^2 = E[|\mathbf{d}(nT)|^2] &= nT^2 \cdot \sigma_v^2 = \Delta t \cdot T \cdot \sigma_v^2 \\ \mathbf{d}(\Delta t = nT) &\sim N_c\left(0, \sigma_d^2\right) \end{aligned} \tag{5.9}$$

We model the two Radar observations taken at different time as:

$$\begin{aligned} s_1 &= \rho \exp\left(j\phi\right) \\ s_2 &= \rho \exp\left(j\left(\phi + \Delta\phi\right)\right) \end{aligned}$$

where, for Brownian motion,

$$\begin{aligned} \Delta\phi = \frac{4\pi}{\lambda}\Delta r_{LOS} = \frac{4\pi}{\lambda}\mathbf{d}\cdot\mathbf{u}_{LOS} &\sim N\left(0, \sigma_{\Delta\phi}^2\right) \\ \sigma_{\Delta\phi}^2 = \left(\frac{4\pi}{\lambda}\right)^2 \frac{\sigma_d^2}{2} & \end{aligned} \tag{5.10}$$

Coherence can be computed by noticing that, for normal distribution of the phase $\Delta\phi$:

$$\begin{aligned} \gamma_s &= \frac{E\left[s_1 s_2^*\right]}{\sqrt{E\left[\left|s_1\right|^2\left|s_2\right|^2\right]}} = E\left[\exp\left(-j\Delta\phi\right)\right] \\ &= \exp\left(-\frac{\sigma_{\Delta\phi}^2}{2}\right) \\ &= \exp\left(-\frac{\Delta t}{\tau}\right) \end{aligned} \tag{5.11}$$

where the time constant τ can be derived by combining (5.9) and (5.10):

$$\tau = \frac{4}{T\cdot\sigma_v^2}\left(\frac{\lambda}{4\pi}\right)^2 \tag{5.12}$$

The final result predicts an exponential decorrelation, where, for short term, the decorrelation is proportional to the time elapsed, and a time constant:

$$1 - \gamma \underset{\gamma\to 1}{\simeq} \frac{\Delta t}{\tau}$$

It has been shown in [1]that for the C band, urban area, τ is 40 days.. An example of temporal decorrelation is provided by the coherence maps in Fig. 5.5, each of them computed by two images taken one day apart. Notice the decorrelation due to snow melting in spring and winter, and to a probable rain in August.

5.1.2.4 Cascade of decorrelation sources

The usefulness of coherence is than due to the fact that it accounts for all changes occurred, including but not limiting to additional noise. When more than one decorrelation mechanism is present, it can be shown that the total coherence is the product of all those mechanism:

$$|\gamma| = |\gamma_{th}| \times |\gamma_k| \times |\gamma_s| \ldots$$

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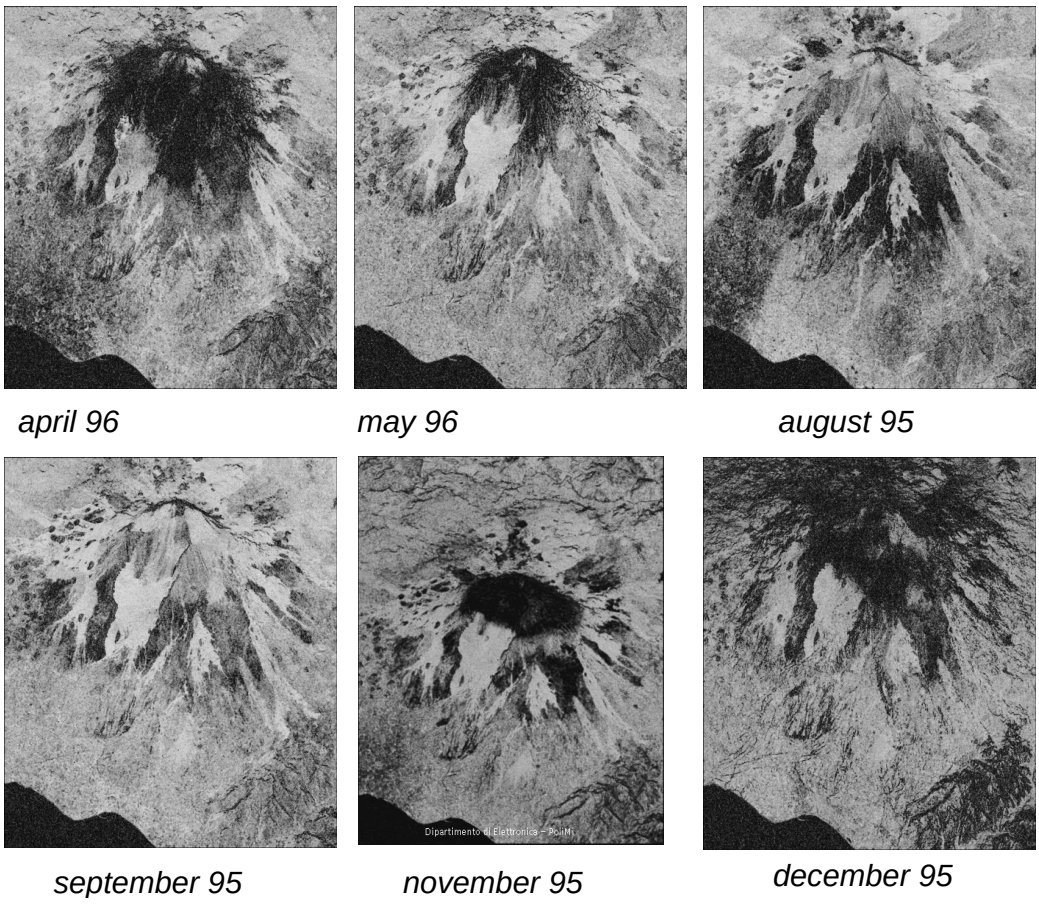


Figure 5.5: Coherence maps computed from pairs of C-band Radar images of Mt. Etna taken at time interval of one day in different month.

5.1.2.5 Coherence maps

The estimation of local coherence can be performed by exploiting the ergodicity of the process in the neighbor of a pixel, hence the "sampled covariance estimator";

$$\gamma_c(n,m) = \frac{\sum_{k,l} \mathbf{y}_1(m+k,n+l)\mathbf{y}_2^*(m+k,n+l)}{\sqrt{\sum_{k,l} |\mathbf{y}_1(m+k,n+l)|^2 \sum_{k,l} |\mathbf{y}_1(m+k,n+l)|^2}}$$

say applied on a suitable window (5×5 pixels or so).

The estimator above is the Maximum Likelihood for homogeneous processes, then, although asymptotically biased, it has a strong bias at low values of coherence when the number of looks is small. The standard deviation approached the square root of CRB for large N or high coherence:

$$\sigma_{CRB}\left(|\hat{\gamma}|\right) = \frac{1-|\gamma|^2}{2N}$$

Mean and standard deviation of the estimator are shown in Fig. 5.6.

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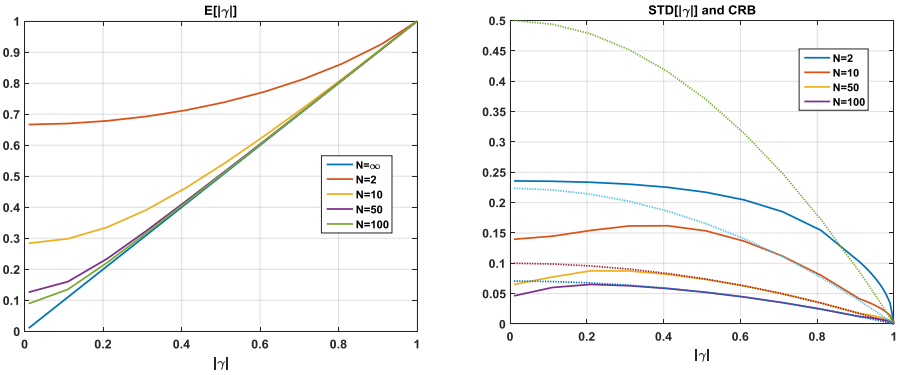


Figure 5.6: Performance of the sampled estimator of coherence magnitude for different values of the true coherence and number of looks. Left: mean value. Right: standard deviation, compared with the Cramer Rao Bound approximation (dotted line).

Examples of coherence maps from pairs of Radar images taken ad different time intervals is in Fig. 5.4 (two airborne image at 10') and Fig. 5.7 (two spaceborne image at three day). In this last case, coherence is null on water, due to fast scene decorrelation, it is high in very stable targets (urban), and gets intermediate values in vegetated areas: forests are less coherent than grass at this wavelength, due to the motion of leaves. An example of image segmentation into three classes is given in the same image.

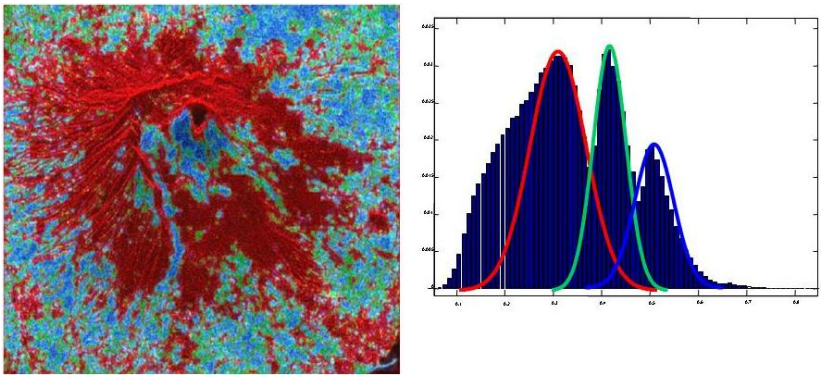


Figure 5.7: Example of image segmentation by coherence. Area of Mt. Vesuvius, ERS-data. The coherent histogram, on the right, has been interpreted as the mixture of three pdf, the highest corresponding to the targets in the recent lava flow, the lowest to highly vegetated areas (that decorrelates in C band).

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5.1.2.6 Example

With reference to the example shown in Fig. 5.4, compute the maximum revisit in order to appreciate decorrelation due to vehicle tracks.. The DSTO-Ingara SAR that took the images is an X-band system, with a 0.5 × 0.15 ground resolution. We assume that the observation window (tracks) is 1 × 2 m, then ˜25 independent looks. For that number, we get from Fig. 5.6 a bias of 0.2 and a a standard deviation of 0.1. Let then assume to appreciate a total decorrelation when scene coherence is such to leave a reasonable contrast: $\gamma > 0.5$ From SNR, we need

$$|\gamma_{th}| = \frac{1}{1+SNR^{-1}} \gg 0.5 \rightarrow SNR \gg \frac{|\gamma_{th}|}{1-|\gamma_{th}|} = 1$$

that is 0dB, not difficult to get. As for temporal coherence, we may estimate that, from Fig. 5.4 say $\gamma \sim 0.98$ in 10 minutes:

$$\frac{\Delta t}{\tau} = \frac{600}{\left(\frac{\lambda}{4\pi}\frac{\sqrt{2}}{\sigma_v}\right)^2} = \frac{600 \cdot (4\pi)^2}{2 \cdot 0.3^2}\sigma_d^2 = 0.02$$

We get $\tau \sim 8$ hours and $\sigma_d \sim 7$ cm/h, in order to get a decorrelation up to 0.5 we could wait for

$$\Delta t = -\tau \log(\gamma) = 5.5 \text{ hours}$$

after which scene decorrelation would be too much. It should be clear that with a different target than grass, say a desert, the coherence time τwould be much longer, maybe years.

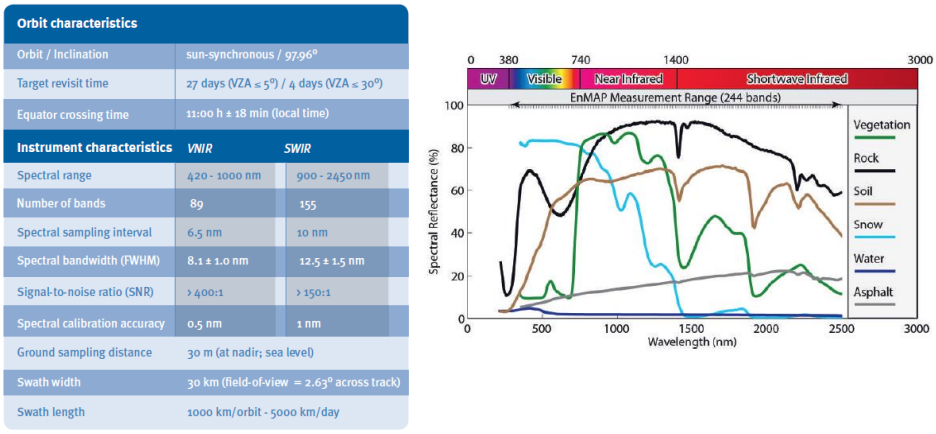
5.2 Prisma and EnMap

The Italian Prisma and German EnMAP - see Fig. 5.8 - are intended future missions for hyperspectral imaging.



Figure 5.8: Artists representations of EnMap and Prisma satellites, from DLR and SELEX-ES fliers.

Their features are quite similar and listed in Fig. 5.9 on the left (taken form EnMAP brochure). The systems record the Sun radiation, reflected from the Earth into 240-250 spectral bands, in a range comprising the visible, the near infrared and the short infrared.



Orbit characteristics		
Orbit / Inclination	sun-synchronous / 97.96°	
Target revisit time	27 days (VZA ≤ 5°) / 4 days (VZA ≤ 30°)	
Equator crossing time	11:00 h ± 18 min (local time)	
Instrument characteristics	***VNIR***	***SWIR***
Spectral range	420 - 1000 nm	900 - 2450 nm
Number of bands	89	155
Spectral sampling interval	6.5 nm	10 nm
Spectral bandwidth (FWHM)	8.1 ± 1.0 nm	12.5 ± 1.5 nm
Signal-to-noise ratio (SNR)	> 400:1	> 150:1
Spectral calibration accuracy	0.5 nm	1 nm
Ground sampling distance	30 m (at nadir; sea level)	
Swath width	30 km (field-of-view = 2.63° across track)	
Swath length	1000 km/orbit - 5000 km/day	

Figure 5.9: Main features of EnMAP missions and, right, spectral coverage, from EnMAP brochure.

The pixel size is 30 × 30 m, with a FOV of 30 km, so each image column is 1000 pixels and -say- 240 colors, the resolution of each spectral band being up to 10 nm. A panchromatic sensor is then added (Prisma) to get fine resolution images, 5 m, that can be fused with the hyperspectral ones, that are coarse but well localized in the spectrum, with a technique called pan-sharpening. Examples of classifications that results from processing spectral signatures are provided in Fig. 5.10, that relates data acquired from Airborne Visible InfraRed Imaging Spectrometer (AVIRIS) mission. The spectral signatures shown on the right, are some of the large database available at http://speclab.cr.usgs.gov/index.html. Notice the presence of narrow absorption lines due to specific components, like CO_3,H_2O the identification of some signature requires fine spectral resolution to detect those lines.

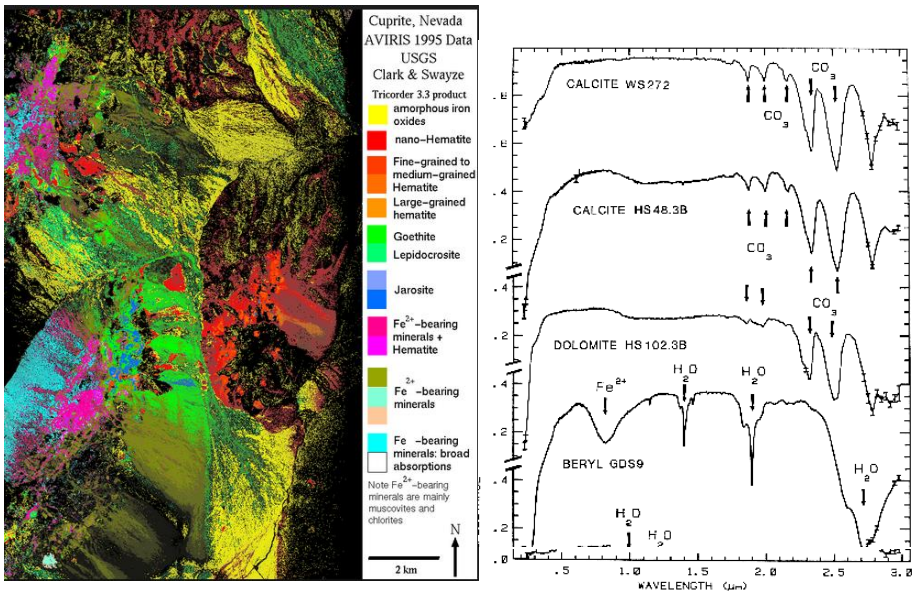


Figure 5.10: Classification of minerals from AVIRIS mission: left: false color image after flight over Nevada. Right: examples of spectral signatures http://speclab.cr.usgs.gov/index.html - notice the absorption lines due to oxide of carbon for calcite and dolomite.

A very relevant use of multi-spectral analysis is the estimation of quality and quantity of vegetation. Vegetation is in fact absorbing energy in the visible (0.4-0.7 μm), to act photosynthesis, but reflect in the NIR (0.7-1.1 μm) as those photons have not enough energy for photosynthesis but would cause heating of the plants - see Fig. 5.11 on the right. This led to the definition of the Normalized Digital Vegetation Index, NDVI as

$$NDVI = \frac{NIR - VIS}{NIR + VIS}$$

that is always int he range $-1 \ldots 1$ and provides a proxy for the amount of chlorophyll, then leaves.

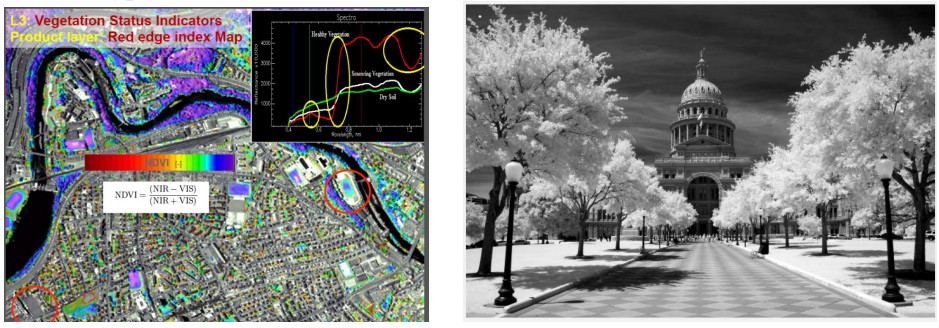


Figure 5.11: Identification of vegetation by visible and Near Infra-Red: the normalized vegetation index.

The geometry and the schematic diagram of the sensor are shown in 5.12. Notice the scanning of entire line (column) to increase the observation time, the pushbroom, that is typical of optical systems. The dispersive element, that can be grating, a prism or a combination

Quantity	Unit	Value
Sun inc. power spectrum	w/m^2/nm	1.00E+00
Albedo		2.00E-01
Reflectance	w/nm/sr	6.37E-02
Solid angle	sr	1.84E-09
Wavelength	m	7.00E-07
Lens aperture for panc	m	9.80E-02
Delta lambda	nm	1.00E+01
Total power	W	8.82E-12
Energy	J	3.78E-14
Efficiency		1.00E-01
Photons		1.34E+04
Relative radiom accuracy	dB	41.3

Table 5.1: Performance evaluation for Prisma

of both, whose role is to separate bands to be then measured by the matrix of sensors. The 2D sensor acquires an image that represents in one direction the spectrum and in the the other the across track. Therefore, by moving along-track, a full 3D image is generated where the reflectivity of each resolution cell on the ground is evaluated in each of the two hundred+ bands. The scheme is shown in Fig. 5.12.

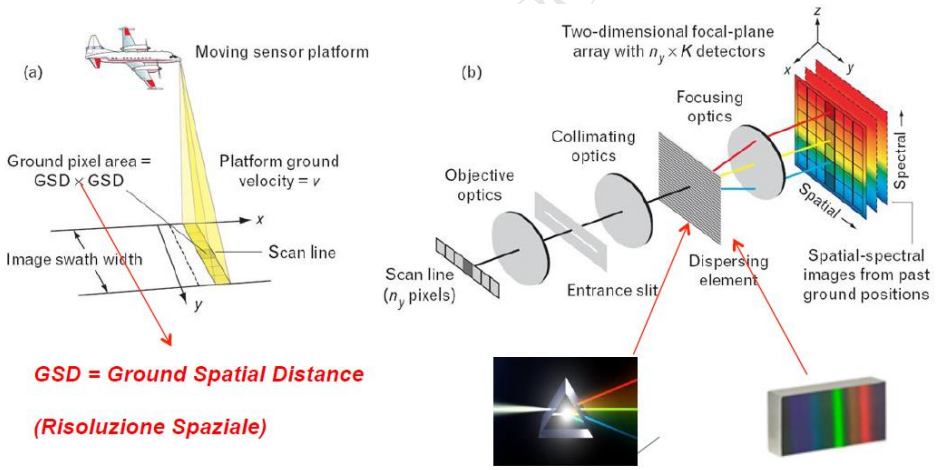


Figure 5.12: Left: geometry and, right, schematic principle of Prisma sensor form Selex-ES web site.

The evaluation of performance, carried out in Tab. 5.1, according to the example in section 4.6.1 and (4.21) provides a relative accuracy of 41 dB, intended as the standard deviation of reflectance and its actual value, that is the ENL. Notice that optical images are almost free of thermal noise, therefore the quantization of photons is the major limit.

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Part II

Radar Principles

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The aim of a **Ra**dio **D**etection **a**nd **R**anging system is to *detect* the presence of one or more *targets*, track its position with time and estimate its reflectivity, a complex parameter that is related to the target shape, size and dielectric properties. In the simplest systems, that exploit the return time of the Electro Magnetic (EM) waveform, of the acoustic waves (**SO**und **NA**vigation and **R**anging), or of photons (**Light D**etection **A**nd **R**anging) the return time is measured and from that the "Ranging", the Line-Of-Sight (LOS) distance of the target, assuming knowledge of the propagation velocity in the medium. In most cases, the tracking of the LOS position simply resolve to measure the LOS *velocity*, and reiterate these measures with time.

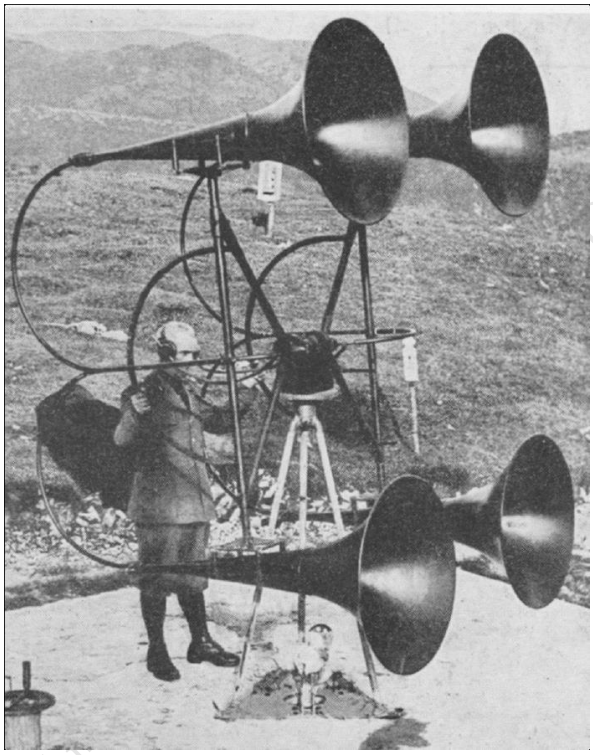


Figure 5.13: Four trumpets acoustic aerophone, used in the Italian navy up to 1940.

The principle of a RADAR system goes back to H. Hertz, in the first half of the 19^{th}century. However, the first experiment made to `detect` a target is attributed to Hulsmeyer, some fifty years later, where Marconi made a measure of target *range* in the 30th. The availability of RADAR systems were conditioned by the invention of the *Magnetron* (General Electric Labs, 1920), a tube with resonant cavities that allowed for a significant power at the microwaves (the need for large power is discussed in section 8). Several different devices follows, noticeable the *Klystron*, that quite better stability, however the Magnetron is still extensively used in modern microwave oven due it simplicity.

The basic principle of RADAR, however, is in the nature from more the fifty millions of year, as bats exploits a very light (2g), and small (3 cm) yet highly sophisticated SONAR, whose principles in target tracking, engage and capture, are close to the most modern military systems.

The selection of the wavelength is a fundamental aspect in RADAR and SONAR, that comes out from a compromise between propagation, reflection by the target and technologies. EM fields with millimetric wavelengths, with frequencies up to 10 GHz suited for observa-



Figure 5.14: The smallest microchiropter, a bumblebee bat (also hog-nose bat), *craseonycteris thonglongya*), weights less than two grams, all included (www. animalpicturesarchive. com).

Band	*Frequency, GHz*	*Wavelength, cm*
P	0.5	60
L	1	30
S	2	15
C	5	6
X	10	3
K_u	15	2
K	25	1.2
K_a	35	0.85
W	>50	<0.6

Table 5.2: Frequencies used in millimetric RADAR.

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tions of targets with similar resolution, they can propagate in the atmosphere even through clouds with moderate attenuation. Longer wavelengths are instead useful for penetration (e.g. foliage coverage, even walls or underground). The most used frequencies for millimetric RADAR are listed in Tab. II.

The attenuation of such wavelength is the atmosphere is shown in Fig. 5.15. Notice that propagation below X band are are almost unaffected, whereas above that frequency humidity strongly impair propagation. There are some local minima, like that at 35 GHz (Ka-band) exploited by geosynchronous satellites for Internet and television broadcast.

Notice that the wavelength is in many case conditioned by the constraint that should be much smaller than the object to be resolved. As an example, bats exploits ultrasounds 30-80 kHz ($\lambda = 3 - 9$ mm at 270 m/s propagation) to detect small insects, whereby tomography (and underwater systems) needs frequencies of some MHz to get the comparable resolution, due to the different velocity of the wave, typically 1500 m/s.

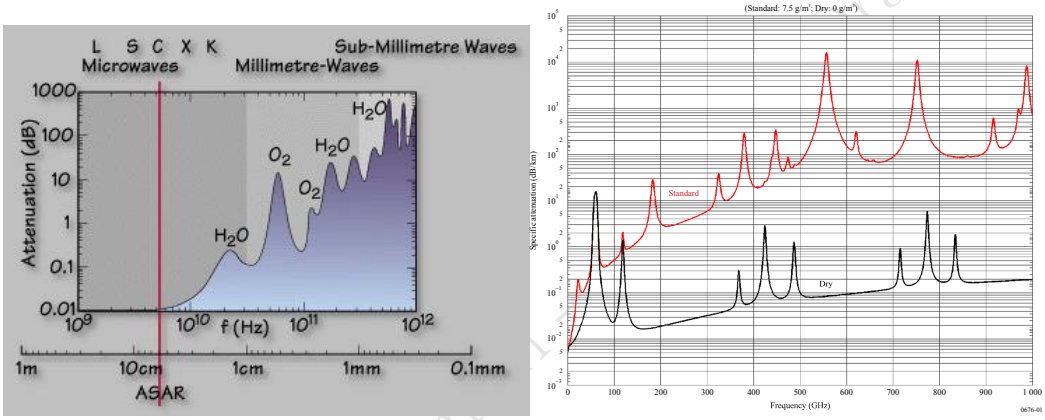


Figure 5.15: Left: attenuation of the radio frequencies in the atmosphere, due to gas and vapors, measured at sea high, T=20°. The red line corresponds to the choice of the European Space Agency (ESA) spaceborne-SARs: ERS, ENVISAT and Sentinel. © European Space Agency (ESA). Right: specific attenuation (db/Km) of the atmospheric gases as function of frequency GHz, for standard (7.5 gr/m^3water vapor) and dry air, at 15^0, from ITU-R P.6746.8.

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Chapter 6

Doppler Radar (CW)

The Radar concept we study here is a Doppler Radar, aiming to distinguish between a fixed and a moving target. This is a classical applications, leading to burglar alarms, and most of the military uses. Bats use this method as a proxy for long range, high sensitivity identification of clouds of mosquitoes.

Here we present this as a burglar alarm. The simplified geometry in Fig. 6.1, includes the transmitter, located in O, a moving intruder, in Q, and a fixed reflector (like walls, floor etc), in P. The signal transmitted can be as simple as a continuous purely sinusoidal with power P_T and frequency f_0:

$$s_T(t) = \sqrt{2P_T}\cos(2\pi f_0 t)$$

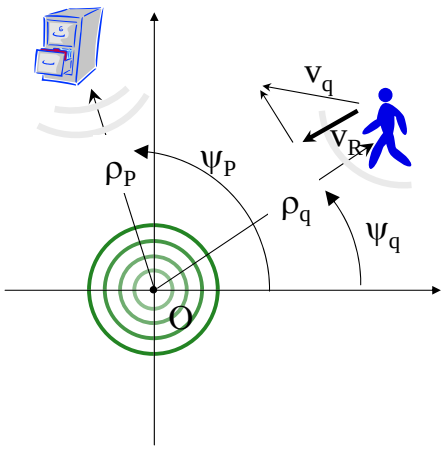


Figure 6.1: Reference geometry for the burglar alarm.

We define "signal" the return of interest, that is from moving burglar while the fixed reflector is an unwanted interference. Notice that we have assumed the 2D geometry, where the location of targets is defined the couple of polar coordinates (r, ψ) aka *range* and *azimuth.* The third dimension, that coordinate θ or elevation, is irrelevant in the example. We assume that the two targets reflect the incident wavefront as they were *point reflectors,* that reflect the Radar waveform each with its own phase and amplitude, and being in the precise location, identifies by the 3D coordinates $\mathbf{P}\,(x_p, y_p, z_p)$ and $\mathbf{Q}\,(x_q, y_q, z_q)$. We approximate the model

as linear, that is, we ignore double and multiple bounces. The received signal is then:

$$s(t) = \rho_P \cos\left(2\pi f_0\left(t - \frac{2R_P}{c}\right) + \phi_P\right) + \rho_Q \cos\left(2\pi f_0\left(t - \frac{2R_Q(t)}{c}\right) + \phi_Q\right) + n(t) \tag{6.1}$$

$$R_P = |\mathbf{P} - \mathbf{O}|$$
$$R_Q(t) = |\mathbf{Q} - \mathbf{O}|$$

where c is the propagation velocity, ρ_P and ρ_Q are coefficients that accounts for the both the propagation losses, and the amount of energy that is back propagated to the RADAR, R_P and $R_Q$are the 1-way distances of the two targets, the first changing with time and n(t) thermal noise.

6.1 Amplitude, phase and Doppler estimation

The signal is actually measured for a limited *observation time*, T. If this is small, we can approximate target's motion is linear with velocity $\mathbf{v}_Q$. Then, at time t the target range can be approximated as the initial target distance and a term proportional to the its range change with Line Of Sight velocity v_R:

$$R_Q(t) \simeq R_0 + \frac{\partial |\mathbf{R}_Q|}{\partial t} t = R_0 + v_R t \tag{6.2}$$

$$v_R = \mathbf{v}_Q \cdot \mathbf{u}_Q \tag{6.3}$$

The target contribution is then:

$$s_Q(t) = \rho_2 \cos\left(2\pi\left(f_0 - \frac{2f_0}{c} v_R\right) t + \psi_Q\right) = \rho_2 \cos\left(2\pi\left(f_0 + f_D\right) t + \psi_Q\right) \tag{6.4}$$

where ψ_Q is a constant that is both related to the target position **and** its phase, therefore cannot used for locating the target. Instead, we can notice the Doppler shift:

$$f_D = -\frac{2f_0}{c} v_R = -\frac{2v_R}{\lambda} \tag{6.5}$$

opposite to the LOS velocity (the whistle of the train increases its pitch as approaching). The object is now to *estimate the moving target.*

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6.1.1 Least Square Doppler estimation: a simple model

We will derive the estimate of the velocity, as the one that minimize the energy of the error:

$$e(t) = s_r(t) - \widehat{s}_{rt}(t; v)$$

between the received signal and its noiseless model of the return in absence of noise, $\widehat{s}_{rt}$. It can be shown that the minimization of the error energy is the optimal, Maximum Likelihood estimate when noise is Gaussian, independent, identically distributed, as in our case. Notice that the model (6.4) is too much complicated, as it would require the estimate of six parameters. We simplify it by accounting only for the moving target and noise:

$$\begin{aligned} \widehat{s}_r(t) &= \widehat{s}_{rt}(t) + n(t) \\ \widehat{s}_{rt}(t) &= \rho \cos\left(2\pi f_T t + \psi\right) \\ f_T &= f_0 + f_D \end{aligned} \tag{6.6}$$

The minimization is justified by the fact that we assume noise to be zero-mean, therefore the energy integrated in the *observation time* T_o:

$$\mathcal{E} = \int_{T_o} |s_r(t) - \widehat{s}_{rt}(t; f_T, \psi, \rho)|^2 \, \mathrm{dt}$$

this is like imposing that the model approximate the signal in L2 norm.

Then the estimate of the velocity, v_R is done from the frequency f_T, and it is implemented as follows:

$$\begin{aligned}
\hat{f}_T, \hat{\psi}, \hat{\rho} &= \arg\min \int_{T_o} |s_r(t) - \widehat{s}_{rt}(t; v(f_T))|^2 \, \mathrm{dt} \\
&= \arg\min \left\{ \int_{T_o} |s_r(t)|^2 \, dt + \int_{T_o} |\widehat{s}_{rt}(t; v(f_T))|^2 \, dt - 2 \int_{T_o} s_r(t) \cdot \widehat{s}_{rt}(t; v(f_T)) \mathrm{dt} \right\} \\
&\simeq \arg\min \left\{ \frac{\rho^2}{2} T_o - 2 \int_{T_o} s_r(t) \cdot \widehat{s}_{rt}(t; v(f_T)) \mathrm{dt} \right\} \qquad (6.7)
\end{aligned}$$

since the first term, the received signal energy, is independent on the parameters. To solve the problem, we first find f_T, ϕ to maximize the second term:

$$\hat{f}_T, \hat{\psi} = \arg\max \left\{ \int_{T_o} s_r(t) \cdot \widehat{s}_r(t; v(f_T)) \mathrm{dt} \right\}$$

that is the *maximum of the correlation between the signal and the model.* We operate as follows:

$$\begin{aligned}
\hat{f}_T &= \arg\max \int \mathrm{Re}\left[s_r(t) \cdot \rho \cdot \exp\left(j \left(2\pi f_T t + \psi \right) \right) \right] \mathrm{dt} \\
&= \arg\max \mathrm{Re} \left[\exp\left(j\psi \right) \int_{T_o} s_r(t) \exp\left(j 2\pi f_T t \right) \mathrm{dt} \right] \qquad (6.8) \\
&= \arg\max \mathrm{Re}\left[\exp\left(j\psi \right) S_r(f) \right] \qquad (6.9)
\end{aligned}$$

where we acknowledged the Fourier Transform of the data in the observation time. The maximization over ψleads to:

$$\widehat{\psi} = -\angle \left\{ S_r(f_T) \right\}$$

that we plug into (6.8) leading to the estimate of the frequency:

$$\widehat{f}_T = \max |S_r(f)| \qquad (6.10)$$

and then to the velocity

$$\widehat{v}_R = -\frac{\lambda \widehat{f}_D}{2} = -\frac{\lambda}{2} \left(\widehat{f}_T - f_0 \right)$$

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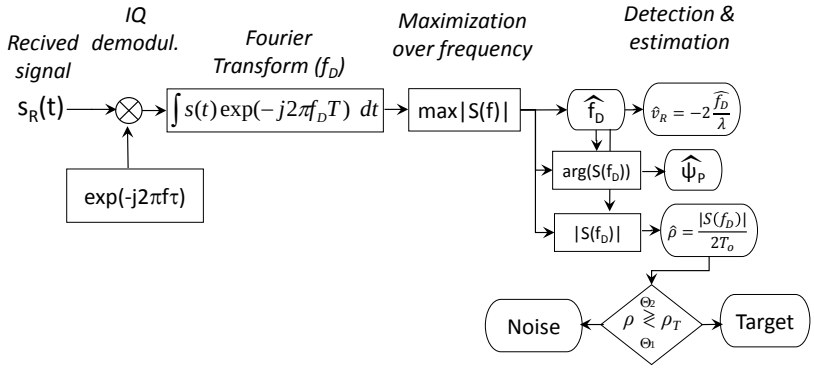


Figure 6.2: Receiver and signal processing for detection of moving target.

Finally, we can go back to (6.7) to estimate amplitude:

$$\begin{aligned}\hat{\rho} &= \arg\min\left\{\frac{\rho^2 T_o}{2} - 2\rho\int_{T_o} s_r(t)\cdot\cos\left(2\pi f_T t+\psi\right)\mathrm{dt}\right\}\\ &= \arg\min\left\{\frac{\rho^2 T_o}{2} - 2\rho\cdot\mathrm{Re}\left(S_r(f_T)\exp\left(-j\phi\right)\right)\right\}\end{aligned} \tag{6.11}$$

$$\begin{aligned}\frac{\partial}{\partial\rho} &= \rho T_o - 2\left|S_r(f_T)\right| = 0\\ \hat{\rho} &= \frac{2\left|S_r(f_T)\right|}{T_o}\end{aligned} \tag{6.12}$$

Notice that the estimate leads to the IQ demodulator discussed in section 4.5.1. The actual implementation, shown in Fig. 6.1.1, can be splitted into two steps: the first is a demodulation times the carrier, $\exp(-j2\pi f_0)$,the second step is the FT and estimation of maximum.

6.1.2 Amplitude estimate: the visibility index

To evaluate performance of the estimate, we plug the model into the estimator, say assume that the received signal is:

$$s_r(t) = \left\{\rho\cos\left(2\pi f_T t+\phi\right)+n(t)\right\}\cdot\mathrm{rect}\left(\frac{t}{T_o}\right) \tag{6.13}$$

then

$$S_r(f) = \frac{\rho}{2}\cdot T_o\left(\mathrm{sinc}\left(T_o(f-f_T)\right)\cdot\exp(j\phi)+\mathrm{sinc}\left(T_o\cdot(f+f_T)\right)\cdot\exp(-j\phi)\right)+w \tag{6.14}$$

where w is a random variable:

$$w = \int_{T_o} n(t)\cdot\exp\left(j2\pi ft\right)dt = \int_{T_o} n(t)\cdot\left\{\cos\left(j2\pi ft\right)+j\cdot\sin\left(j2\pi ft\right)\right\}dt$$

that corresponds to $n_c(t)$ in (4.10), that is a complex, circular zero-mean, random variable:

$$\begin{aligned}w &\sim N_c\left(0,\sigma_w^2\right)\\ \sigma_w^2 &= E\left[|w|^2\right] = E\left[\left|\int_{T_o}\left(n_c(t)+jn_s(t)\right)dt\right|^2\right]\end{aligned}$$

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The evaluation of σ_w is carried out as in (4.11), by considering that the two noises are incorrelated, then

$$\sigma_w^2 = 2\sigma^2$$

where

$$\sigma^2 = E\left[\left|\int_{T_o} n_c(t)dt\right|^2\right] = \int h_{nc}(f)\cdot|H(f)|^2\,df = \frac{N_0}{2}T_o$$
$$N_0 = Fk_BT^0$$

H(f) being the Fourier Transform of the integrator, whose energy is T_o.

The evaluation of the target amplitude, ρ, as from (6.12) is then affected by random noise (6.14). The estimate is:

$$\hat{\rho} = \frac{2}{T_o}|S_r(f)| = \frac{2}{T_o}\left|T_o\frac{\rho_o}{2} + w\right| = \left|\rho_o + \frac{2}{T_o}w\right| \tag{6.15}$$

We notice that is it asymptotically unbiased, since, for large SNR $\rho_0 \gg w$

$$\hat{\rho} = \sqrt{\left(\rho_o + \frac{2}{T_o}w_r\right)^2 + \left(\frac{2}{\rho_o T_o}w_i\right)^2}$$
$$\simeq \rho_o + \frac{2}{T_o}w_r$$
$$E[\hat{\rho}] \simeq \rho_o$$

and accuracy:

$$VAR[\hat{\rho}] = \frac{4}{T_o^2}VAR[w_r] = \frac{4}{T_o^2}\frac{N_0}{2}T_o = 2\frac{N_0}{T_0}$$

Similarly to the case of detection of bits in numerical communication, we can define a Signal-To-Noise ratio

$$SNR = \frac{\rho_0^2}{VAR[\hat{\rho}]} = \frac{\rho_o^2 T_0}{2N_0} = \frac{E_R}{N_0}$$

This term is referred as "visibility index" because it defines how much signal is stronger than noise.

The estimate of amplitude is shown in (6.15) to be asymptotically unbiased, for $SNR \to \infty$.

6.1.3 Accuracy of the frequency estimation

The frequency estimate:

$$\widehat{f_T} = \max|S_r(f)| =$$
$$= \max\left|\frac{\rho}{2}\cdot T_o\left(\mathrm{sinc}\left(T_o(f-f_T)\right)\cdot\exp(j\phi) + \mathrm{sinc}\left(T_o\cdot(f+f_T)\right)\cdot\exp(-j\phi)\right) + w(f)\right|$$

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can be approximated by ignoring, in the neighbor in f_T, the second *sinc* contribution, and then computing the maximum as nulling the fist derivative. If signal is much larger than noise, we ignore the modulus operator and then:

$$\frac{\rho}{2} \cdot T_o \frac{\partial}{\partial f} \operatorname{sinc}\left(T_o(f - f_T)\right) + w' = 0$$
$$\frac{\partial}{\partial f}\left\{1 - \frac{T_o^2\pi^2}{3}(f - f_T)^2\right\} = \frac{2w'}{\rho T_0}$$

the second expression has been derived by expanding in power series the sinc function. The estimate of frequency is then:

$$f = f_T + \frac{3}{2T_o^2\pi^2}\frac{2w'}{\rho T_0}$$

That is unbiased, since the derivative of noise is zero in average. The evaluation of the standard deviation is left to the reader: we will however come back to the solution later in this book.

6.1.4 Fixed target rejection

So far, we have ignored the clutter contribution and accounted only for the moving target and noise. In the general case, of two, or m,ore targets at different velocities to be separated, the estimator developed is *no more optimal*, yet, the design of an optimal one would be near impossible as many unknowns would be introduced, according to the number of interfering sources. It is then common practice, although not optimal, to continue to adopt the matched filter [1].

The extension of (6.13) to the fixed and moving targets, leads to the frequency domain:

$$Y(f) \simeq \sqrt{2P_T} \cdot T_o \left\{\rho_P \exp\left(j\psi_P\right)\delta\left(f\right) + \frac{\rho_Q\left(j\psi_q\right)}{2}\left(\delta\left(f - f_D\right) + \delta\left(f + f_Q\right)\right)\right\}$$
$$\cdot \operatorname{sinc}\left(fT_o\right) + N(f)$$

The model with the three additive contributions are represented in Fig. 6.1.4. The ration between signal and interfering target, evaluated at $f = f_D$:

$$\frac{|S(f_D - f_0)|^2}{|S(f_0)|^2} = \frac{\rho_Q^2}{\rho_P^2}\frac{1}{\left|\operatorname{sinc}\left(f_D T_o\right)^2\right|} \simeq \frac{\rho_Q^2}{\rho_P^2}\left|\left(f_D T_o\right)^2\right|$$

depends upon the ratio of the backscattered energies and the frequency $\times$ Time product (between the Doppler shift and the observation time). The approximation holds for large separation, in that case, a tapering in time domain, shown in the figure on the right. Notice that when the Doppler resolution needed is $1/T_o$, only the rectangular window is effective. However, if we can accept a factor 3, an Hamming window will provide a 100 times better attenuation.

In particular, when the two amplitudes are identical, we define the *Doppler resolution* as the capability to distinguish between two targets. In the case of rectangular (boxcar) time-doamin window, that is, for the present estimator:

$$\Delta f_D = 2 \cdot \arg\left(sinc^2\left(\Delta f_D \cdot T_o\right) = 1\right) \Rightarrow \Delta f_D = \frac{0.88}{T}$$

[1] There exists different super-resolution methods that perform better than the Fourier Transform.

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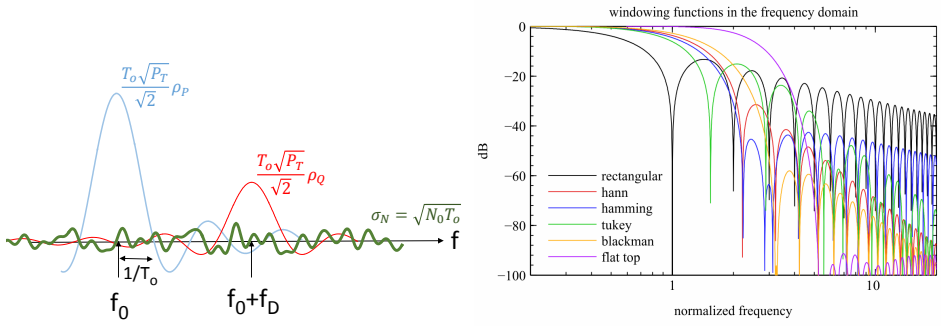


Figure 6.3: Left: representation of the spectral contribution of fixed, moving target and noise. Right: attenuation provided by different time-domain windowing

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Finally, if only the fixed target is to be removed, a notch filter at zero Doppler would be a better solution.

6.1.4.1 Example: slow moving target

As an example let us assume to detect a target with a LOS velocity as small as 1 km/h ($= 0.27 m/s$), and to select a wavelength $\lambda = 1$ cm, that corresponds to $f_0 = 30$ GHz in the microwaves, or 27 kHz with acoustic waves. We get a Doppler frequency $|f_D| \simeq 54$ Hz. We would need an integration time$>$ 20 ms, say 60 ms for 10 dB suppression of fixed target interference.

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Chapter 7

Detection

The burglar Radar example so far assumed pose an immediate problem: we have to decide if there is the burglar or not. This problem frames in the "detection" of an event, that is a binary decision, and can apply to the early warning for a Radar observing mines, or a counter sniper detection. It is the same problem of test in medicine or anti-spam in e-mail.

In our case, we take a decision basing on the estimate of the signal amplitude, say (6.15):

$$v = |y_n| = \begin{cases} |w| & \text{if } \Theta_1 \\ |\rho + w| & \text{if } \Theta_2 \end{cases} \tag{7.1}$$

where $w = w_r + jw_i$is the zero-mean complex circular Gaussian noise, and both $w_r$and w_i being normal distributed $N(0, \sigma_n^2)$.The two values of Θcorresponds to the two cases:

1. The target is absent, then we have pure noise, Θ_1
2. The target is present, Θ_2

One statistically optimal choice is to maximize of the posterior probability **MAP**:

$$P_{\Theta|v}(\Theta_1|v) \underset{\Theta_2}{\overset{\Theta_1}{\gtrless}} P_{\Theta|v}(\Theta_2|v) \tag{7.2}$$

where $P_{\Theta|v}(\Theta_1|v_k)$ is the posterior probability of the pure noise event, given the amplitude v. Eventually one can use Bayes rule to get:

$$P_\Theta(\Theta_1) f_{v|\Theta}(v|\Theta_1) \underset{\Theta_2}{\overset{\Theta_1}{\gtrless}} P_\Theta(\Theta_2) f_{v|\Theta}(v|\Theta_2)$$

where we have introduced the two likelihoods, the $f_{v|\Theta}(v|\Theta_1)$ and $f_{v|\Theta}(v|\Theta_2)$ and get the final result:

$$\frac{f_{v|\Theta}(v|\Theta_1)}{f_{v|\Theta}(v|\Theta_2)} \underset{\Theta_2}{\overset{\Theta_1}{\gtrless}} \frac{P_2}{P_1} = \eta \tag{7.3}$$

The conclusion is pretty general in binary classifiers like this and it corresponds to select the event by comparing the **likelihood ratio,** the left hand term of (7.3), with a threshold. different criteria, leads to different values of η:

- MAP, this case, η is the ratio between the prior probabilities of the target / no target events;
- Maximum Likelihood (ML), the prior probabilities are assumed 50% each and $\eta = 1$;

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- Bayesian, η is a ratio between two risks that are user-selectable;
- Constant False Alarm Ratio, the one most used in RADAR, η is selected to guarantee a level false alarm, that is a major specification of the RADAR.

In our case, described by the model (7.1), the likelihood $f_{v|\Theta}(v|\Theta_2)$ corresponds to the Rice pdf, and $f_{v|\Theta}(v|\Theta_1)$ to the particular Rice pdf with $\rho_a = 0$, also known as Rayleigh pdf:

$$f_{v|\Theta}(v|\Theta_1) = \frac{v}{\sigma_n^2} \exp\left(-\frac{v^2}{2\sigma_n^2}\right) \cdot u(v) \tag{7.4}$$

both pdf being shown in 7.1. We remind that for Rice, the mean value is $\sigma_n\sqrt{\pi/2}$and the standard deviation $\sigma_n\sqrt{(2-\pi/2)}$. This means that noise shift and spreads the pdf. The Rice distribution, for large SNR, that is $\rho \gg w$ can be approximated to a Guassian, since:

$$|\rho + w| = \sqrt{(\rho + w_r)^2 + w_i^2} \approx \rho + \frac{w_r}{\rho}$$

However, the mean value is biased:

$$E[|\rho + w|] \simeq 1 + \frac{\sigma_n^2}{2\rho}$$

and the variance is quite close to SNR:

$$VAR[|\rho + w|] \simeq \frac{\rho^2}{\sigma_n^2} - \frac{1}{4}\left(\frac{\rho^2}{\sigma_n^2}\right)^2$$

This means that, as noise increase, Rice probability gets wider.

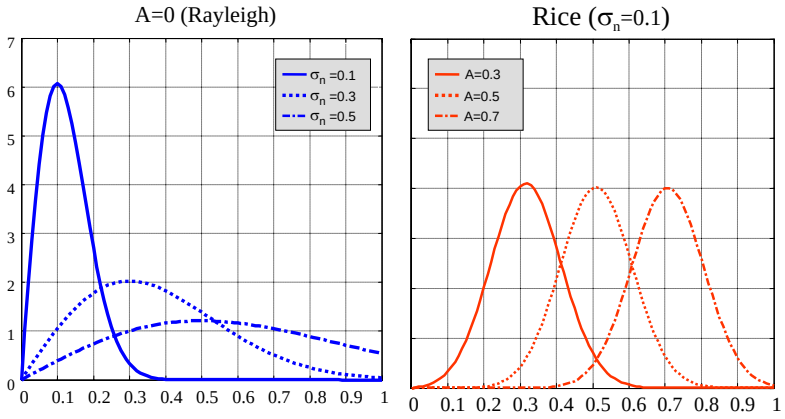


Figure 7.1: Left, Rayleigh pdf, modeling the likelihood of amplitude in absence of the target, or different noise powers. Right, Rice pdf, modeling the likelihood of amplitude in presence of the target, for different target amplitude, A.

Both pdf are jointly plotted in 7.2. Notice that, being their ratio decreasing, the optimal solution according to (7.3) is just a comparison with a threshold:

$$v \underset{\Theta_2}{\overset{\Theta_1}{\gtrless}} V_T$$

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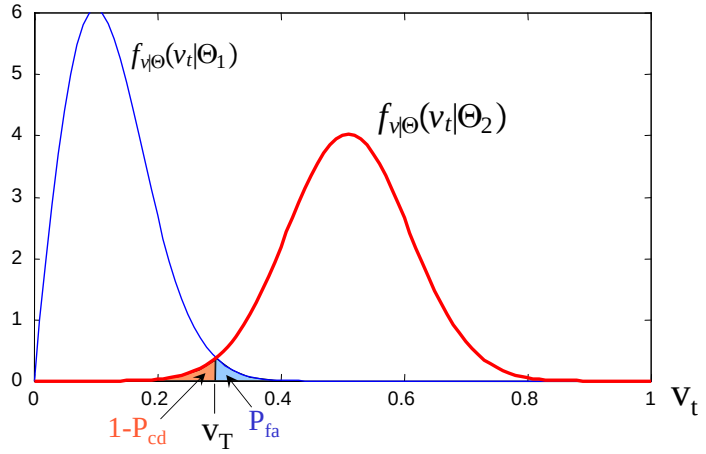


Figure 7.2: Superposition of the two likelihood on the same plot. V_T is the optimal threshold according to the ML criterion. Then, the shaded areas represents the two probabilities of error: false alarm (blue) and missed detection (orange).

In the figure, the threshold V_T has been set to for $\eta = 1$, the ML criterion. Once that the threshold is fixed, the two error probability: false alarm and missed detection, can be computed (see the shaded areas in the figure):

$$P_{fa} = \int_{v_T}^{\infty} f_{v|\Theta}\left(v|\Theta_1\right) dv = \exp\left(-\frac{v_T^2}{2\sigma_n^2}\right), \tag{7.5}$$

$$P_{md} = 1 - P_{cd} \tag{7.6}$$

where the correct detection probability is:

$$\begin{aligned} P_{cd} &= \int_{v_T}^{\infty} f_{v|\Theta}\left(v|\Theta_2\right) dv = Q_1\left(\frac{A}{\sigma_n}, \frac{v_T}{\sigma_n}\right) \\ &= \int_{v_T}^{\infty} \frac{v}{\sigma_n} \exp\left(-\frac{\left(\frac{v}{\sigma_n}\right)^2 + \left(\frac{A}{\sigma_n}\right)^2}{2}\right) I_o\left(\frac{A}{\sigma_n}\frac{v}{\sigma_n}\right) \frac{\mathrm{dv}}{\sigma_n} \\ &= \int_{\frac{v_T}{\sigma_n}}^{\infty} x \exp\left(-\frac{x^2 + \left(\frac{A}{\sigma_n}\right)^2}{2}\right) I_o\left(\frac{A}{\sigma_n}x\right) \mathrm{dx} \\ &= Q_1\left(\frac{A}{\sigma_n}, \frac{v_T}{\sigma_n}\right) \end{aligned} \tag{7.7}$$

where $Q_M(\alpha, \beta)$ is the Q-Marcum function ("marcumq" in Matlab) of order M:

$$Q_M(\alpha, \beta) = \int_{\beta}^{\infty} x \left(\frac{x}{\alpha}\right)^{M-1} \exp\left(-\frac{x^2 + \alpha^2}{2}\right) I_{M-1}\left(\alpha x\right) \mathrm{dx}, \tag{7.8}$$

I_{M-1} is the $M-1$ order modified Bessel function.

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	Decided	T1	T2		
True Value					
T1		1-Pfa	Pfa	True Negative	False Negative
T2		1-Pcd	Pcd	False Positive	Ture Positive

Table 7.1: Confusion matrix of a binary classifier

7.1 Neyman-Pearson: the CFAR detector

The classification theory allows to dimension the threshold, and the minimal power we expect from a target to be detected once that we provide requirements in form of the confusion matrix of the classifier in the table.

In RADAR, the most used criterion is the CFAR, that is satisfied by imposing in (7.5) the false alarm to be granted, P_a:

$$\begin{aligned} P_{fa} &= \exp\left(-\frac{v_T^2}{2\sigma_n^2}\right) \geq P_a, \\ \Rightarrow V_T &= \sqrt{-2\sigma_n^2 \log\left(P_a\right)} \end{aligned} \tag{7.9}$$

the value of the threshold is then to be used in (7.7) in order to compute the SNR that should exceeded to ensure a particular missed-detection probability. Then given V_T and a desired probability of correct detection, one has to invert

$$\begin{aligned} P_{cd} &= Q_1\left(\frac{\rho_{min}}{\sigma_n}, \frac{v_T}{\sigma_n}\right) \\ &= Q_1\left(\frac{\rho_{min}}{\sigma_n}, \sqrt{-2\log P_a}\right) \end{aligned} \tag{7.10}$$

to find the desired value of

$$\frac{\rho_{min}}{\sigma_n} = \sqrt{2SNR}$$

that gives the minimum Signal to Noise Ratio that guarantee both false alarm and missing detection requirements.

The inversion of Q-Marcum is provided in graphical format in 7.3. As an example, if one has to meet a false alarm of $10^{-5}$and correct decision of 99.9%, then

$$\begin{aligned} \frac{v_T}{\sigma_n} &= \sqrt{-2\log P_a} \\ Q\left(\sqrt{2SNR}, 4.8\right)\Big) &= 0.999 \end{aligned}$$

that should be inverted to get SNR, we can use the single Matlab instruction:

fminbnd(@(x)abs(marcumq(x,4.8)-0.999),0,10)

and then we get x=$\sqrt{2SNR}$ = 7.8 then SNR=14.8 dB out of the matched filter. Notice that to meet a false alarm requirement, one needs a very accurate knowledge of noise power. In our case, in the case an increase of just 1 dB in noise power will lead: according to the plots, a false alarm of 10^{-2}, one order of magnitude worse. Therefore, in the design of RADAR, not only thermal but also clutter noise should be accurately characterized.

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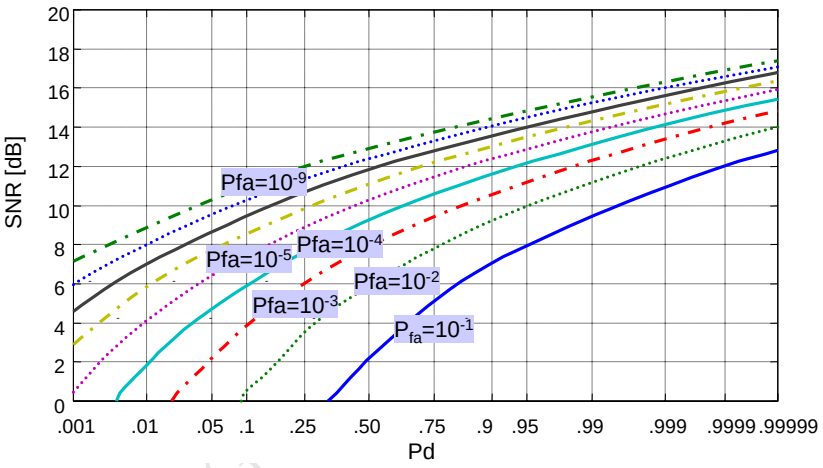


Figure 7.3: Required SNR to meet false alarm and correct detection probabilities, P_{fa} and P_{cd}.

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Chapter 8

Radar equation

The RADAR equation allows to evaluate the energy that is backscattered by the target of a given RCS and is collected by the receiving Radar. It is of primary importance for evaluating performances, say if the target is visible or not, than for Radar system **design** and dimensioning. Moreover, it is mandatory for the precise calibration, that in turn provides both the proper detection and the absolute characterization of scatterers, in the same wave that "white balance" enters in chromatic calibration of optical images. Here we will compute the energy balance buy assuming that we are in the far field of the antenna, that is at the distance we are considering, R, the antenna can be assumed as point with respect to its beamwidth: $\lambda/L \cdot R >> L$.

The general case of a bistatic Radar is represented in the geometry of Fig. 8.1. Let us now assume that the transmitted power, P_t is isotropically irradiated in the ideal medium, so that the received power at the target at distance R_t is $P_t/4\pi R_t^2$. Then we assume that the target as a *cross-section* (that is normal to the EM field) σ, that is a surface, and all the power is *isotropically re-irradiated.* The power at the receiver will be:

$$P_r = \frac{P_{tISO}}{4\pi R_t^2}\sigma\frac{A_r}{4\pi R_r^2} \tag{8.1}$$

where $A_r$is the cross-area of the receiver antenna. This simple Radar equation predicts that the received power is strongly attenuated in the path: $P_r \div R^{-4}$ (say for $R_t \sim R_r$), the "spherical divergence". The major problem for Radar is in fact this attenuation that is the square of the one we find in telecommunication systems.

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8.1 EIRP

To improve the power balance, directive antennas are used. We account for that by an adimensional factor that is the *directivity gain* of the antenna, that express the increment of power due to the concentration of power in the direction of the target. The **Effective Isotropic Radiated Power** is the power is then:

$$EIRP(\theta,\psi) = P_t G_t(\theta,\psi) \tag{8.2}$$

where the antenna directivity gain is the ratio between the EM intensity measured in a specific direction and the one that would be measured by exploiting an isotropic antenna:

$$G_d(\theta,\psi) = \frac{I(\theta,\psi)}{I_{ISO}(\theta,\psi)} = \frac{I(\theta,\psi)}{P/4\pi}$$

where P is the intensity integrated over the total surface of a unitary radius sphere:

$$P = \int_{\psi=0}^{\psi=2\pi} \int_{\theta=0}^{\theta=\pi} I(\theta,\psi)\sin\theta d\theta d\psi$$

We mostly approximate the gain to be constant within a solid angle, say $\Delta\Omega$ and zero elsewhere, so that:

$$G_d(\theta_0,\psi_0) = \frac{I_0}{I_0 \cdot \Delta\Omega/4\pi} = \frac{4\pi}{\Delta\Omega}$$

that can immediately be interpreted as the ratio between the whole solid angle, and the angle in which power is confined.

In the case of a rectangular antenna size $a\times$b :

$$\begin{aligned} G_d(\theta_0,\psi_0) &= \frac{4\pi}{\frac{\lambda}{a}\frac{\lambda}{b}} = 4\pi\frac{ab}{\lambda^2} \\ &= 4\pi\frac{A_t}{\lambda^2} \end{aligned} \tag{8.3}$$

As an example, an antenna of size 1×1 m in X band, λ =3 cm, will have a gain:

$$G_{dB} = 4\pi_{dB} + 0 + 40 - 9_{dB} = 41.45$$

The EIRP will be some 10000 larger than the effective transmitted power.

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8.2 Bistatic Radar equation

For the general case of real bistatic Radar, we need to change the ideal expression (8.1) and add the antenna gain. Furthermore we have to account for the following losses

- η_e the two-way conversion from electrical to electromagnetic field, and back;
- η_{ILL} the two-way illumination losses accounts for a non uniformly illuminated antenna, than with a wider main beam $\Delta\Omega$and a lower gain than the one predicted in (8.3);
- η_P the two-way path losses that accounts for attenuation in the path like due to hydrometers (rain, fog, clouds);

We define $\eta_T$the product of all losses. The Radar equation for the bistatic case compute the received power as:

$$P_r = \left\{\frac{P_t G_t}{4\pi R_t^2}\right\}\sigma\left\{\frac{1}{4\pi R_r^2}\right\}A_r\eta_T \tag{8.4}$$

Eventually, we take advantage of (8.3) to rewrite the RADAR equation in the following two formulations:

RADAR EQUATION as a function of the antenna gains:

$$P_r = \frac{P_t G_t G_r \lambda^2}{(4\pi)^3 R_t^2 R_r^2}\sigma\eta_T \tag{8.5}$$

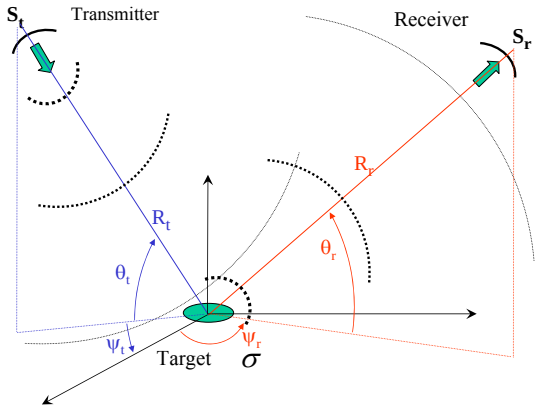


Figure 8.1: Reference geometry for the derivation of the RADAR equation.

RADAR EQUATION as a function of the antenna equivalent areas:

$$P_r = \frac{P_t A_t A_r}{4\pi\lambda^2 R_t^2 R_r^2}\sigma\eta_T \tag{8.6}$$

This second expression shows how, fixed the antenna size, the receiver power increases with the squared of the frequency, that is at short wavelength we get much more power. Notice that in general, the antenna Gain, the equivalent area and the target Radar cross section *depends on the view angle*, $(\theta_t\psi_t, \theta_r, \psi_r)$, shown in the figure.

8.3 Monostatic Radar equation

In the monostatic RADAR $R_t = R_r$ and we see that receiver power decreases with the **fourth power of the distance**

$$P_r = \frac{P_t G^2(\theta,\psi)\lambda^2}{(4\pi)^3 R^4}\sigma\eta_T \tag{8.7}$$

Instead, the power received at the target location decreases with the *second power of the distance*. This leads to the fundamental result that the target can detect the RADAR much earlier than when the RADAR detects the target.

8.4 Radar Cross Section

The definition of RCS is implicit in (8.1), that assumes σ to be the surface of a scatterer orthogonal to the EM field, who is retransmitting all the energy isotropically. In practice this should be a very rough surface, most of the scatterers are not isotropic. Therefore the RCS becomes an "equivalent" surface that could be much larger (say for a mirror placed orthogonality) or much smaller (say a target not backscattering in a monostatic RADAR) than the physical surface of the target. The two class of targets here discussed, point and distributed, corresponds to the specular and diffused reflections in optics. In a general case of a bistatic RADAR, the RCS depends upon the direction of incident wave and of the receiver, as well as the frequency and the polarization: $\sigma(f_0, pol, \theta_t, \psi_t, \theta_r, \psi_r)$. Here following we provide some examples of monostatic RCS, that is $\theta_t = \theta_r, \psi_t = \psi_r$

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8.4.1 Point targets

We define point target those scatter mechanism that makes all the backscattered energy like coming from a precise position in the 3D space (a point). This does not mean that the reflecting target has small size. A very small sphere surely acts as a point scatterer, however in that case the reflected energy is so small that is hardly relevant in the image. On the opposite, a mirror or a parabolic dish provides strong reflection id their areas are large, as coming from a single point, literally a Dirac δ, located in the "center of phase" of the target, see Fig. 5.1.

8.4.1.1 Mirror

We define as mirror a "flat" surface, as from Rayleigh criterion in section 2.4.3. If we compute the RCS in direction almost orthogonal and withing the angular aperture, in Fig. 8.2, the whole energy is confined in the solid angle:

$$\Delta\Omega = \frac{\lambda}{L_a} \cdot \frac{\lambda}{L_b} = \frac{\lambda^2}{S}$$

where L_a, L_b are the target length and width and S its surface. Than, it is equivalent to isotropic scatterer of surface :

$$\sigma = \frac{S}{\frac{4\pi}{\Delta\Omega}} = \frac{4\pi A^2}{\lambda^2}$$

In X band, a $1 \times 1 = 1m^2$scatterer would have a RCS of 14000 m^2. This means that in the specular direction, $\pm\Delta\psi/2 = \lambda/L$, the target emits an energy that is equivalent to 14000 of isotropic reflector. However, RCS is almost null outside that solid angle.

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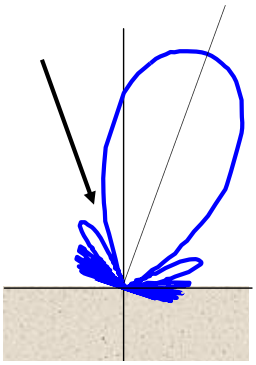


Figure 8.2: Point targets. Polar reflectivity diagram of a smooth surface whose size is comparable with the wavelength.

8.4.1.2 Dihedral

In the case of a rectangular **dihedral**, of size a x b, the area is $A = ab\sqrt{2}$,therefore:

$$\sigma = \frac{8\pi a^2 b^2}{\lambda^2}$$

The combination of the two specular reflections acts by making this σ true for very wide angular apertures, nominally the -3dB spread is 40°. This is exploited to observe trees in the forest, that, in L and P band, are very bright scatterers. We can show that the double bounce makes the phase center exact in the corner.

8.4.1.3 Trihedral and other shapes

The trihedral extends behavior of the dihedral to wide apertures in the entire 3D space. The RCS of a rectangular trihedral with length a is:

$$\sigma = \frac{4\pi a^4}{3\lambda^2}$$

this value is maintained for a cone of 40°. The RCS of a corner with $a = 1$ m, f_0 =5 GHz is $\sigma \approx 1350$ m^2. A table of RCS for common geometric objects is in Fig. 8.4.1.3.

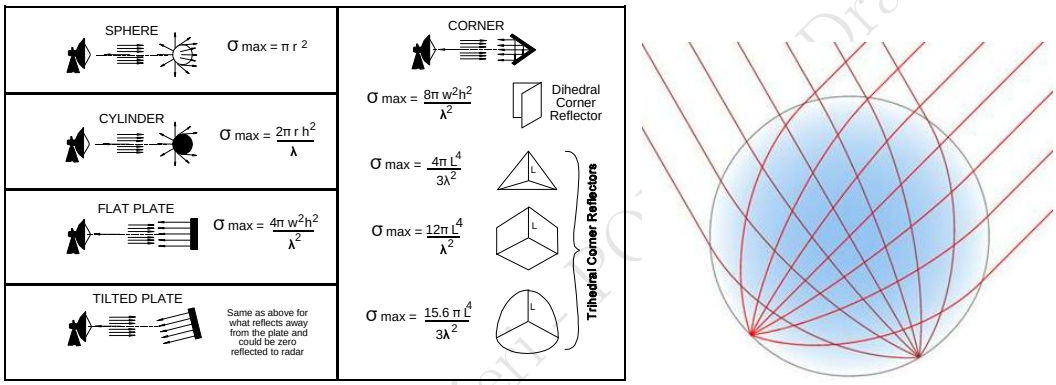


Figure 8.3: Left: RCS of different reflectors (from Granite Island Group, http://www.tscm.com/). Right, the Luneburg lens, a wide angle scatterer with a large RCS (from Wikipedia).

As a last example, we a **sphere** of radius r has isotropic RCS whose value is at the first order independent on the wavelength and the polarization, provided that $r \gg \lambda$:

$$\sigma = \pi r^2$$

If $\lambda = 2\pi r$ there is a resonance with ripples, whereas for smaller radii, the RCS reduces with λ^{-4} [1]. Notice that this is indeed a small number, when compared with other point targets. However, its symmetry makes a sphere reflection independent on the view angle.

The Luneburg lens, shown in Fig. 8.4.1.3 on the left, is filled with a dielectric whose constant changes along the radius from 2 to one (on the surface). The radius from outside the sphere has no reflection on the surface ($\epsilon_r = 1$), and enters into it. The increase in the dielectric from surface to center causes a reduction of speed and a consequence ray bending in the direction of the "denser" medium (the one with lower refraction index), as shown in the figure. The dielectric variation is dimensioned so that the focal point is on the surface, opposite to the incidence. If the focal point is entirely reflective, say a metal cap in the inner, than the sphere contribute with a very wide surface, in practice πr^2, and this happens for wide angles (depending on the cap).

[1] The same behavior is for the molecules of gas responsible of the blue sky (Rayleigh scattering).

In the general case, the computation of RCS is very complex, and many time is done by experiment in anechoic chamber. In the millimetric wavelengths, the RCS of a man is in the order of the m^2, whereas a vehicle ranges from tens to hundreds of m^2, whereas a "stealth" bomber, has an RCS comparable to a bird.

8.4.2 Distributed targets

The distributed target or fully developped speckle, has already been discussed in section 5.1. Expression (5.4) shows that the RCS of the resultant of N scatterers in the resolution cell is proportional to the energy scattered by each element and the number of elements. In the case of monostatic Radar, we define this last as **backscatter coefficient**. Its definition and dimension changes in the cases of surface and volume scattering. In the first case we assume a random distribution of targets on a surface, then the backscatter coefficient is:

$$\sigma_S^0 = \frac{\sigma}{S}$$

that is adimensional, since it represents a RCS per unit surface. In the second case, we assume that targets are distributed in a volume, like for rain–drops in a cloud - of particular interest for *meteorological RADAR*. Then backscatter is

$$\sigma_V^0 = \frac{\sigma}{V} \tag{8.8}$$

and measured in $[m^{-1}]$, as RCS per unit volume. The backscatter coefficients are derived either by EM modelling, or by measures. A comprehensive set of measures, carried out in different classes of targets, polarization and frequencies is in the book of Ulaby and Dobson [2]. An example of the backscatter in X band, in different type of scatterers and polarization is in Fig. 8.4.2.

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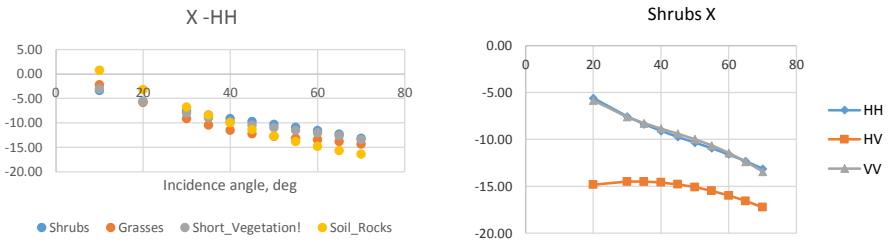


Figure 8.4: Backscatter coefficients in X band for different incidence angle.

The RADAR equations, say in the monostatic case are derived by replacing the RCS in (8.7):

$$P_{rS} = \frac{P_t G^2 \lambda^2}{(4\pi)^3 R^4} \sigma_S^0 S \tag{8.9}$$

$$P_{rV} = \frac{P_t G^2 \lambda^2}{(4\pi)^3 R^4} \sigma_S^0 V$$

For the *meteorological RADAR*, for example, the single resolution cell has volume

$$V = (R\Delta\theta)(R\Delta\phi)\left(\frac{c\tau}{2}\right)$$

where $\Delta\theta$, $\Delta\psi$ are the angular resolution and $c\tau/2$ is the range resolution:

$$\theta = \frac{\lambda}{L_\theta} \text{ e } \phi = \frac{\lambda}{L_\phi}$$

The RADAR equation becomes:

$$P_{rV} = \frac{P_t G^2 \lambda^4}{(4\pi)^3 R^2 L_\phi L_\theta} \sigma_S^0 \frac{c\tau}{2}$$

We notice the dependence upon R^{-2}, instead than R^{-4} as for conventional RADAR.

8.4.3 Polarimetric scattering

The most general case of bistatic, polarimetric scattering is shown in 8.4.3. The waveform incident to the scatterer is planar, however the polarization plane may rotate with time. The incident field can be represented with its component in the horizontal and vertical plane, where the reference system is assumed aligned with the transmitter. In order to compute the backscattered field in the direction of the receiver, we should know the behavior of the target with respect to any polarization. Again, we decompose the backscattered field in horizontal and vertical components (by keeping the same reference), and express as function of the incident, see (3.1)

$$\begin{bmatrix} E_{SH} \\ E_{SV} \end{bmatrix} = \frac{\exp\left(j\left(kr + \omega t\right)\right)}{r} \begin{bmatrix} S_{HH} & S_{HV} \\ S_{VH} & S_{VV} \end{bmatrix} \begin{bmatrix} E_{IH} \\ E_{IV} \end{bmatrix}$$

where the generic element of the square matrix (the Jones matrix), S_{PQ} expresses the complex coefficient relating the scattered component P to the incident component Q, where P and Q can be either H(orizontal) or V(vertical). The same equation can be expressed in matrix:

$$\mathbf{E}_S = \frac{\exp\left(j\left(kr + \omega t\right)\right)}{r} \cdot \mathbf{S} \cdot \mathbf{E}_I$$

The Radar Cross Section $\sigma_{PQ}$is by definition:

$$\sigma_{PQ} = 4\pi r^2 \frac{\mathbf{E}_S^* \mathbf{E}_S}{\mathbf{E}_I^* \mathbf{E}_I}$$

that leads to:

$$\sigma_{PQ} = 4\pi \left|S_{PQ}\right|^2$$

and we conclude that the **scattering matrix** (or Jones matrix) has complete information to characterize the target reflectivity. A particular case is the monostatic one, in which the Jones matrix is symmetric, then $\sigma_{HV} = \sigma_{VH}$.

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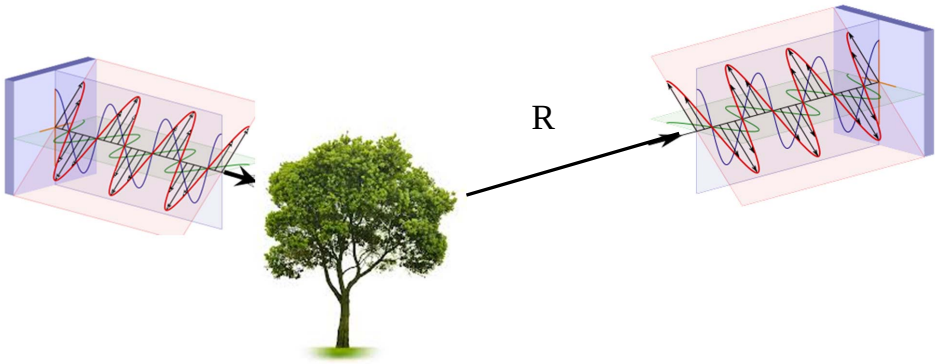


Figure 8.5: Schematic representation of the general case of a bistatic, polarimetric scattering.

Chapter 9

Pulsed Radar

The Continuous Waveform, monochromatic Radar just discussed has a very fine capability in measuring frequencies, hence discriminating targets moving a different velocities, but it is unable to find the target position. This is a consequence of the waveform used, a pure sinusoid, that is *perfectly localized in the frequency* (a Dirac δ), but *completely delocalized in time*.

On the other side, if we wish a perfect location, we would use a waveform narrow in time, and eventually, repeat periodically to track target motion. The system here is quite similar to the CW Radar shown in Fig. 6.1, however, the transmitted signal is a periodic repetition of pulses, $c(t)$, modulated:

$$s_T(t) = \left\{ \sum_m c(t - mT_P) \right\} \cdot \exp(j2\pi f_0 t) \tag{9.1}$$

We define T_p as *Pulse Repetition Interval* (PRI) and its inverse the *Pulse Repetition Frequency* (PRF), f_p. Here we assumed the complex modulation, discussed in chapter , that is a general case the comprises any real modulation. We assume now one target R_1 and noise. The signal received from the m-th transmitted pulse is:

$$s_r(t) = \rho_1 \cdot \exp(j\phi_1) \cdot c\left(t - \frac{2R_1(t)}{c}\right) \cdot \exp\left(j2\pi f_0 \left(t - \frac{2R_1(t)}{c}\right)\right) + n(t) \tag{9.2}$$

with the same assumption as for the Doppler Radar.

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9.1 TOA estimation, range ambiguity

The problem is now to estimate the Time of Arrival:

$$\tau_1 = \frac{2R_1(t)}{c}$$

and the target amplitude and phase, ρ_1, ϕ_1, from noisy data, as shown in Fig. 11.0.3. The time-varying delay make the problem rather complicated, however, we assume here that the delay of the echo do not vary so much when compared with the time extent of the single pulse, and the same also for the phase - this one being a more stringent requirement. That is: *we assume target as fixed from one tx pulse to another*,

We proceed by minimizing the energy of the error between the received signal and its noiseless model, that corresponds to the ML estimate for white noise:

$$\begin{aligned}\widehat{\tau};\hat{a};\hat{\phi} &= \arg\min_{\tau;a;\phi}\left\{\int |s_r(t)-\widehat{s}_r(t;\tau,a,\phi)|^2\,\mathrm{dt}\right\}\\ &= \arg\min_{t_0;a;\phi}\left\{\int |\widehat{s}_r(t,t_0)|^2\,dt - 2Re\int s_r(t)\widehat{s}_r^*(t;\tau,a,\phi)\mathrm{dt}\right\}\end{aligned} \tag{9.3}$$

We furthermore assume that the *echo of the return target is completely included* in the receiving window, that limits the **maximum distance** of a target to:

$$\frac{2R}{c} < (T_P - T_c) \tag{9.4}$$

$T_c$being the pulse duration. Then, we define the **unambiguous range** as

$$R_{MAX} < \frac{c}{2}(T_P - T_c) = \frac{c}{2}T_P\,(1-d_c) \tag{9.5}$$

d_c being the duty cycle. This result in the fundamental **upper limit to the PRF**:

$$f_p < \frac{c}{2R_{MAX}}\,(1-d_c) \tag{9.6}$$

In this case, the optimal set of parameters, is once again the one maximizing the correlation between data and model, from (9.2):

$$\begin{aligned}\widehat{\tau};\hat{\phi} &= \arg\max Re\left\{\int_{T_P} s_r(t)\cdot\widehat{s}_r^*(t;\tau,\phi)dt\right\}\\ &= \arg\max Re\left\{\exp(j\phi)\exp(j2\pi f_0\tau)\int_{T_P} s_r(t)\cdot\exp(-j2\pi f_0 t)\cdot c^*(t-\tau)\,dt\right\}\end{aligned}$$

The optimal estimate of the phase is the one that makes the integral a real number, the maximum being then the modulus. The optimal TOA, τ, is then the argument that maximizes the cross-correlation between the demodulated signal:

$$s_d(t) = s_r(t)\cdot\exp(-j2\pi f_0 t)$$

and the **matched filter**:

$$\begin{aligned}\widehat{t_0} &= \arg\max_{t_0}\left|\int_{T_P} s_d(t)\cdot c^*(t-t_0)\,dt\right| \\ &= \arg\max_{t_0}|s_d(t)*h_M(t)|\end{aligned} \tag{9.7}$$

$$\text{where } h_M(t) = c^*(-t)$$

Having estimated the delay, the amplitude is achieved by minimizing (9.3):

$$\hat{\rho} = \frac{|s_{rd}(t)*h_M(t)|_{t=\hat{t}0}}{E_R} \tag{9.8}$$

E_R being the energy of the transmitted pulse.

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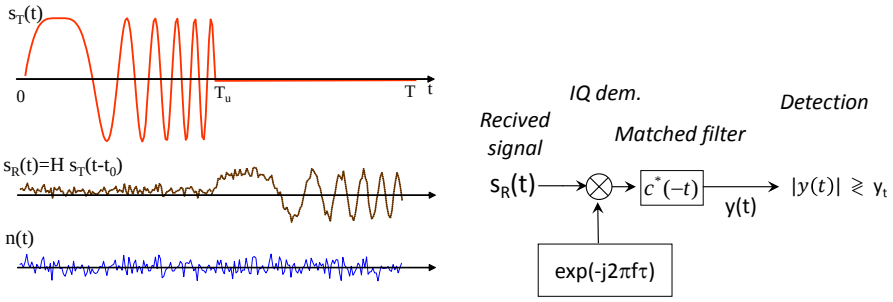


Figure 9.1: Time Of Arrival estimation: $s_T(t)$ is the transmitted signal, a pulse of length $s_R(t)$ is the received signal, a delayed version of the transmitted signal with added noise, n(t). The goal is to estimate the delay. Right: optimal, correlation based estimator.

9.1.1 Performances

If we plug the two targets model (9.2) into the estimator, we get:

$$\begin{aligned} y(t) &= \{s_r(t) \cdot \exp(-j2\pi f_0 \tau)\} * c(-t) \\ &= \rho_1 \cdot \exp(j\psi_1) \cdot r_c(t-\tau_1) + \rho_1 \cdot \exp(j\psi_1) \cdot r_c(t-\tau_1) + n_c(t) * c^*(-t) \\ \psi_n &= \phi_n - \frac{4\pi}{\lambda} R_n \end{aligned} \tag{9.9}$$

where $r_c$is the autocorrelation of the baseband pulse and n_c the complex bandpass noise. This expression can be used in many ways to evaluate performances. The visibility index, to be used for detection, is computed as for Doppler identification. We neglect the interference from the second target, and we assume only signal and noise:

$$SNR = \frac{\rho_1^2 r_c^2(0)}{E\left[|n_c(t) * c^*(-t)|^2\right]} = \frac{\rho_1^2 E_c^2}{\int h_{nc} \cdot |C(f)|^2 \, df} = \frac{\rho_1^2 E_c}{N_0} = \frac{E_r}{N_0}$$

$E_c$being the energy of the pulse. From this we can evaluate the performance of detection and of amplitude estimation as from chapter 7.

9.1.1.1 Accuracy in TOA estimation

The estimate of TOA is affected by he additive noise shifts the peak position and its amplitude as shown in 9.2. We could evaluate the accuracy of the estimate based on the matched filter, as we did for the Doppler. Instead we provide here the Cramer Rao Bound for the best unbiased estimate of TOA given the model:

$$y(t) = \rho_a r_c(t-\tau) + w(t) \tag{9.10}$$

that is:

$$\sigma_{\hat{\tau}}^2 \geq \sigma_{CRB}^2 = \frac{N_0}{8\pi^2 B_e^2 \int |C(f)|^2 \, df} = \frac{SNR^{-1}}{8\pi^2 B_e^2} \tag{9.11}$$

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where B_e is the "equivalent bandwidth" of the pulse, $c(t)$, that is defined as:

$$B_e^2 = \frac{\int (2\pi f)^2 |C(f)|^2 \, df}{\int (2\pi)^2 |C(f)|^2 \, df} \, [\mathrm{Hz}^2] \tag{9.12}$$

In the case of $c(t)$ a pulse with *constant amplitude* in the bandwidth B we get:

$$B_e^2 = \frac{1}{12} B^2$$

and thus

$$\sigma_{\hat{\tau}}^2 \geq \frac{3 \cdot SNR^{-1}}{2\pi^2 B^2} \tag{9.13}$$

Notice that the accuracy of the location **depends upon the bandwidth** of the target (the higher the bandwidth, the better the location) on the SNR, but not on the pulse duration. The duration may positively affect the energy, in the sense that **location accuracy improves with the pulse length** (provided that the bandwidth does not change), due to the better rejection of noise.

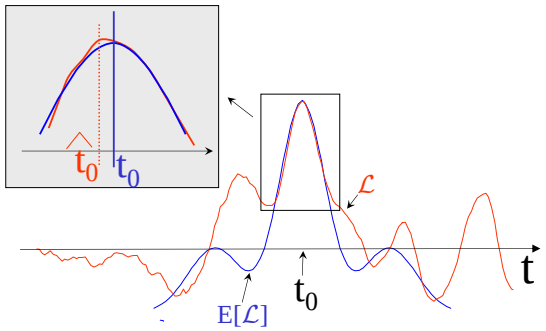


Figure 9.2: Error in delay and amplitude estimation due to noise

As an example, we assume a SNR=15 dB, to provide good detection performances, and a bandwidth of 10 MHz, like for a GPS, then:

$$\sigma_{\hat{\tau}} \geq \frac{\sqrt{3 \cdot 10^{-1.5}}}{\sqrt{2}\pi 10^6} = 2.1 \cdot 10^{-7}$$

that results in a TOA of $2.1 \cdot 10^{-7}[s] \cdot 150 \cdot 10^6[m/s] = 32$m, where half the speed of light has been used to convert time into distances in the case of the Radar.

9.1.1.2 Interferometry

It is interesting to remark here that the TOA accuracy has been derived by assuming that the phase of the target, ψ_n is (9.9) is unknown. If not, a quite better accuracy could be achieved, by looking a the phase of the matched filter output, y:

$$\hat{t}_0 = \frac{\arg\left(y\left(\hat{\tau}\right)\right)}{2\pi f}$$

Such estimate would be very precise, as a first order, the phase noise is the angle error introduced by ratio by the quadrature noise component:

$$\sigma_{\hat{\tau}} = \frac{1}{2\pi|f|}\sigma_\phi = \frac{1}{2\pi|f|}\frac{1}{\sqrt{2SNR}}$$
$$\Rightarrow \sigma_r = \frac{c}{2}\sigma_{\hat{\tau}} = \frac{\lambda}{2}\frac{1}{\sqrt{8\pi^2 SNR}}$$

with an accuracy that goes to the wavelength, in fact we really measure the number of cycles from the Radar tot he target. This feature of Radar cannot be achieved by any other devices.

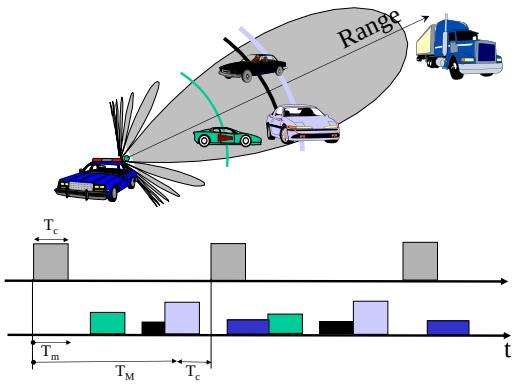


Figure 9.3: Geometry and timeline of a typical pulsed RADAR.

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9.1.2 Multi-target detection: range resolution

The implementation of the TOA estimation is shown in Fig. 9.1. In the implementation, we already assumed the **multi-target** case: that is all the values beyond a threshold. The assumption is that in any pixel (intended as resolution cell) we assume a potential target, however we are able to distinguish targets only if their distance is large compared with cross-correlation extent. We define range resolution as the minimum distances of two targets of the same amplitude to be distinguished:

$$y(t) = r_c(t - t_0) + r_c(t - t_1) \tag{9.14}$$
$$\rho_r = \frac{c}{2}|t_0 - t_1| \tag{9.15}$$

We observe that the capability of distinguishing two close targets, is due to the *3 dB width of the autocorrelation of the transmitted pulse*. As an example, if the transmitted pulse has constant amplitude in the system bandwidth:

$$|C(f)| = \text{rect}\left(\frac{f}{B}\right)$$

we get the following autocorrelation:

$$R(f) = |C(f)|^2 = |C(f)|$$
$$r(t) = E_c \cdot \text{sinc}(Bt)$$

whose 3dB lobe width gives the range resolution:

$$\rho_r = \frac{c}{2}\frac{0.88}{B} \tag{9.16}$$

Notice that the capability of distinguish two targets is not only due to the mainlobe, but also to the *sidelobes of the pulse autocorrelation*, and then the kind of window, as we discussed for Doppler resolution.

An example of the detection of three targets is shown in Fig. 9.4. The particular type of pulse, a chirp (see next section), has constant amplitude over a pretty long time extent, T_c and also constant bandwidth, B. The autocorrelation is then sinc-like with 3 dB duration 0.88/B. This means that, after matched filter, the time duration is reduced from T_c to $1/B$, a factor $TBP \gg 1$ (time-bandwidth product), and the peak scales by a factor T_c. That's why the three targets are not visible on the received data, Fig. 9.4 on the left, whereby they are quite evident on the matched filter data, in. 9.4 on the right. In Radar jargon the matched filtering is called *range compression.*

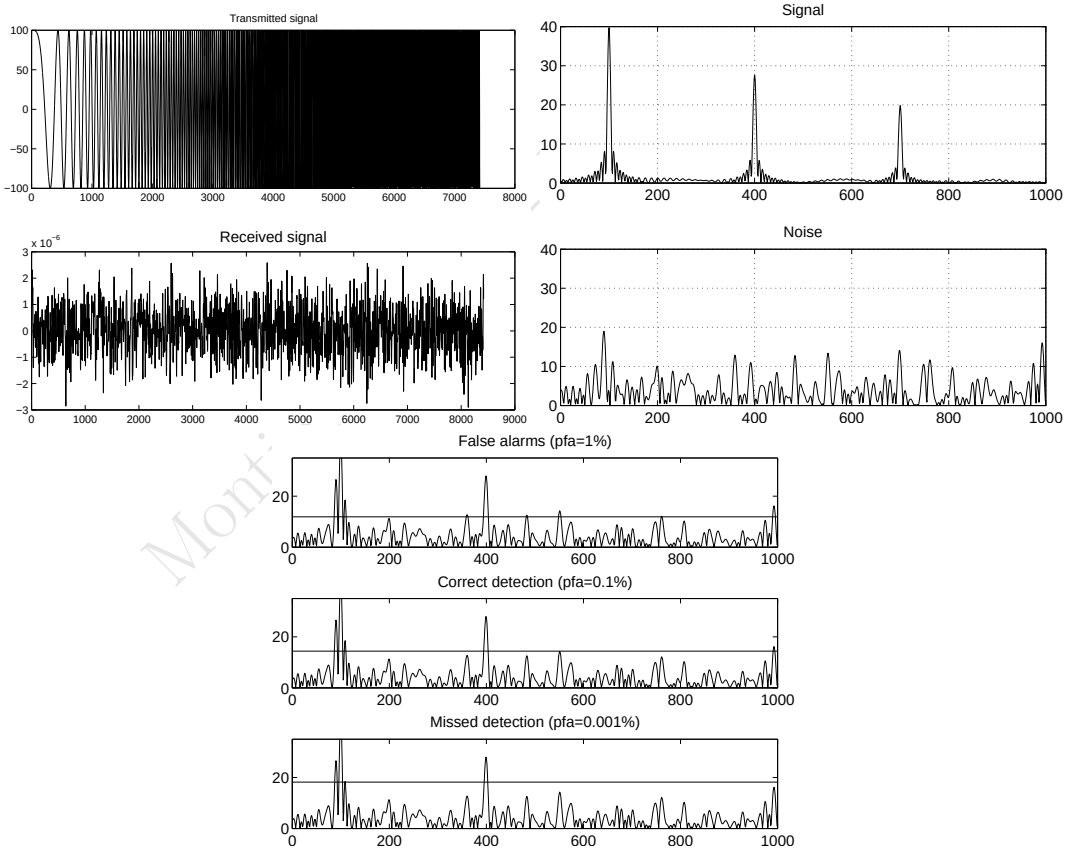


Figure 9.4: Detection of three targets with different amplitudes, measured by RCS of 400, 800 and 1600 m^2. Up-left: the pulse c(t), a chirp, and the transmitted signal, amplitude, containing three targets and noise. Notice that targets disappear. Up-right: resulting signal after matched filter, applied to the signal only and to noise only. Lower: results of detection with three different thresholds, designed for false alarm of 1%, 0.1% and 0.001%.

9.1.3 Optimal pulse design

The selection of the best waveform for the transmitted pulse is a critical issue in RADAR design. The multi-target output in (9.14) states that the theoretical best choice is a pulse whose autocorrelation is a Dirac delta, we call it "Delta Correlated":

r#1 the width of the main-lobe is narrow as to provide good *resolution* (separation between contribution of pulses),

r#2 the sidelobes should be small as to suppress the masking effect of strong echo (typical of clutter) with respect to a small one far away in time,

r#3 the spectrum should be constant in the whole bandwidth.

9.1.4 The chirp pulse

The chirp pulse has such delta-correlation properties as being derived by windowing a constant amplitude signal in both time and frequency. It derives by a Gaussian:

$$\exp(-\pi t^2) \Leftrightarrow \exp(-\pi f^2)$$

by applying a complex scale factor:

$$\exp\left(-\pi\left(\sqrt{-jk}t\right)^2\right) \Leftrightarrow \frac{\sqrt{j}}{\sqrt{k}}\exp\left(-\pi\left(\frac{f}{\sqrt{-jk}}\right)^2\right)$$

$$\exp\left(j\pi k t^2\right) \Leftrightarrow C_0 \exp\left(-j\pi\frac{f^2}{k}\right)$$

$$C_0 = \frac{\sqrt{j}}{\sqrt{k}}\exp\left(j\frac{\pi}{4}\right)$$

Thus leading to the Chirp in time and frequency:

$$c(t) = \exp\left(j\pi k t^2\right) \Longleftrightarrow C(f) = C_0 \exp\left(-j\pi\frac{f^2}{k}\right)$$

notice the curvature opposite and inverse.

We can define the instantaneous frequency of the chirp as:

$$f_i(t) = \frac{1}{2\pi}\frac{d}{dt}\angle c(t) = kt$$

that is linearly related to time by the frequency rate factor, k. It can be shown that the time domain windowed chirp:

$$c_T(t) = A\exp\left(j\pi k t^2\right)\cdot \mathrm{rect}\left(t/T\right)$$

of duration T, has constant amplitude in the bandwidth:

$$-\frac{kT}{2} \le f \le +\frac{kT}{2}$$
$$\Rightarrow B = kT$$

Therefore the autocorrelation of the chirp is, approximately, the inverse FT of a rectangle:

$$r_{cT} \simeq E\cdot \mathrm{sinc}\left(BT\right) = A^2 T\,\mathrm{sinc}\left(BT\right) \tag{9.17}$$

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E being the energy of the waveform. When a chirp is correlated with itself, it becomes compressed to a duration $\sim 1/B = 1/(KT)$, leading to the *compression factor:*

$$K = \frac{T}{\frac{1}{B}} = B \times T = kT^2$$

that corresponds to the Time-Bandwidth Product (TBP). The exact computation of the autocorrelation of the chirp:

$$\begin{aligned} r_c(t) &= \int c(t+\tau)c^*(\tau)d\tau \\ &\underset{t\geq 0}{=} A^2 \int_{-T/2}^{T/2-t} \exp\left(j\pi k(t+\tau)^2\right) \exp\left(-j\pi k\tau^2\right) d\tau \\ &= A^2 \exp\left(j\pi kt^2\right) \int_{-T/2}^{T/2-t} \exp\left(j2\pi kt\cdot\tau\right) d\tau \\ &= A^2 \frac{\exp\left(j\pi kt^2\right)}{j2\pi kt} \left[\exp(j2\pi kt\cdot\tau)\right]_{-T/2}^{T/2-t} \\ &= A^2 T\left(1-\frac{t}{T}\right) \operatorname{sinc}\left(kTt\left(1-\frac{t}{T}\right)\right) \text{ per } 0\leq t\leq T \end{aligned} \tag{9.18}$$

shows that for the sinc approximation to apply, its lobe width, 1/B should be much smaller than the support T, that is if TBP$\gg 1$, than (9.18) approximate to (9.17).

The chirp is universally exploited in RADAR, for its constant amplitude versus both time and frequency and for its capability to be compressed. This last is somewhat weird property of this linearly-frequency-modulated pulse: the longer the duration, the larger the bandwidth (!) and, therefore, the shorter the compressed pulse. A further benefit is that the sidelobes of the autocorrelation can be made much lower than those of a sinc just by applying a smooth windowing (say a raised cosinus) in either time or frequency domain. In fact, the FT of

$$\begin{aligned} c_w(t) = w(t)\cdot\exp\left(j\pi kt^2\right) &\Leftrightarrow C_w(f) = C_0 w\left(\frac{f}{k}\right)\exp\left(j\frac{\pi}{4}\right) \\ r_{cw} &= E\cdot W(kt) \end{aligned}$$

$W(\ldots)$ being the FT of $w(\ldots)$.

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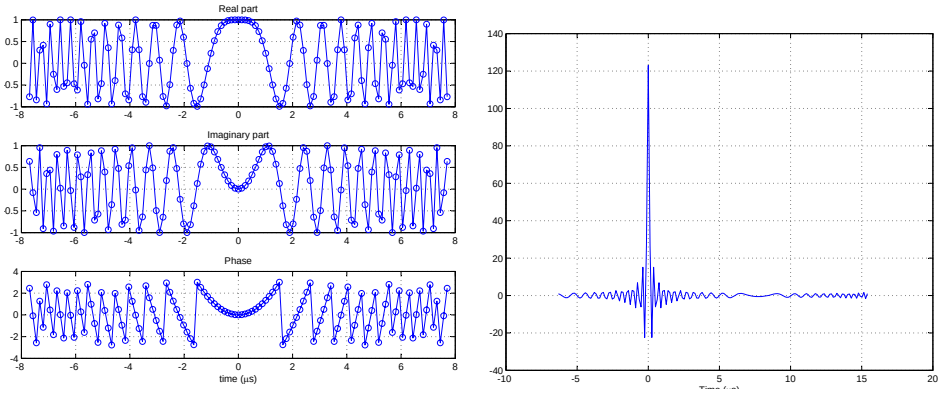


Figure 9.5: Complex chirp (left) and its autocorrelation (right).

As an example, if we need an unambiguous range of 10 km, a duty cycle of 10% and a range resolution ρ_r =1 m, then the pulse repetition interval should be:

$$T_{PRI} - T_c \geq \frac{2R_M}{c} \Rightarrow T_{PRI} = \frac{2R_M}{c\,(1-d_c)} = \frac{20}{300 \cdot 10^3 \cdot 0.9} = 74\mu s$$

that is a PRF of 13.5 kHz. The chirp duration is T_c=7.4 μs, its bandwidth is such

$$\rho_r = \frac{150}{B[MHz]} \Rightarrow B = 150MHz$$

the chirp rate is: k=150/7.4**.**10^{12}=20 $MHz/\mu s$ and the TBP is 150**.**7.4=1100 that is far larger than one.

The major drawback of such long pulses - in connection with monostatic Radar, is that a target can be observed only if the delay of their echo is > Tc (during which the transmitter is switched off), that is $d_c \cdot R_M$ = 1km.

A second drawback, that is specific of the use of a chirp is an *ambiguity in time-frequency* that prevents locating a moving target. In fact, if we assume no windowing, than a time shift τ leads to

$$\begin{aligned} c(t-\tau) &= \exp\left(j\pi k\,(t-\tau)^2\right) = \exp\left(j\pi k t^2\right)\exp\left(-j2\pi k\tau t\right)\exp\left(j2\pi\tau^2\right) \\ &= c(t)\cdot\exp\left(-j2\pi f_D t\right)\cdot\exp\left(\phi\right) \\ \text{where} f_D &= k\tau \text{ and } \phi = 2\pi\tau^2 \end{aligned}$$

that is a frequency shift by a Doppler f_D.

9.1.5 Poliphase pulses

The poliphase pulses represent a good alternative to chirp, in particular can be much shorter than a chirp, then used for short range Radar, and they are not affected by the time-Doppler ambiguity. The general expression is the M-PSK (Phase Shift Keying) waveform:

$$c(t) = \sum_{k=0}^{N-1} \exp\left(j2\pi\frac{k}{M}\right)\exp\left(j2\pi f_0 t\right)\text{rect}\left(\frac{t-T/2}{T}\right) \text{ con } k = 0\ldots M-1$$

where the phase can change in each block of T seconds. A particular case here studied is that of **bi-phase pulses**, that are actually formed by stacking pieces of sinusoids with phase either 0 or π:

$$\begin{aligned} c(t) &= \sum_{k=0}^{N-1} b_k \sin\left(2\pi f_0 t + \phi_0\right)\text{rect}\left(\frac{t-T/2}{T}\right) \\ &= \left(\sum_{k=0}^{N-1} b_k \text{rect}\left(\frac{t-T/2}{T}\right)\right) \times \sin\left(2\pi f_0 t + \phi_0\right) \end{aligned}$$

where b_k is a binary sequence with amplitude ± 1. One of the advantages of such pulses is the simplicity in its generation and correlation. The demodulated waveform is:

$$c_d(t) = \sum_{k=0}^{N-1} b_k \text{rect}\left(\frac{t-T/2}{T}\right)$$

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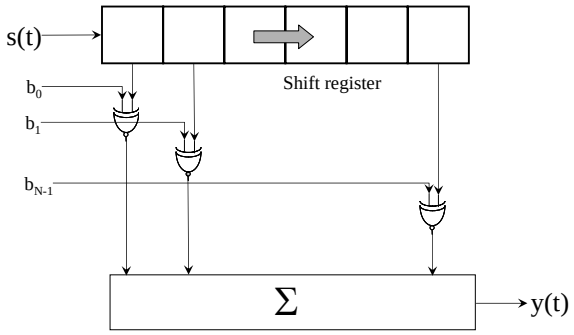


Figure 9.6: Matched filtering with a bi-phase code.

and matched filtering is implemented as in Fig. 9.6.

The autocorrelation of such pulses can be computed be expressing the demodulated pulse as a convolution between a rectangular waveform and a sequence of Dirac delta of amplitudes ± 1:

$$c(t) = c_k(t) * \text{rect}\left(\frac{t - T/2}{T}\right) \tag{9.19}$$
$$c_k(t) = \sum_{k=0}^{N-1} b_k \delta(t - kT)$$

The autocorrelation is then:

$$r_c(t) = r_{c_k}(t) * r_{rect}(t) \tag{9.20}$$

where the second is a triangular waveform (autocorrelation of rectangular) and the first is the autocorrelation of $c_k(t)$, that is a sequence of pulses with amplitudes:

$$r_{c_k}(n) = \sum_{k=0}^{N-1-|n|} b_k b_{|n|+k} \tag{9.21}$$

One than acknowledges that (9.20) is just the **linear interpolation** of (9.21), and the properties of the "compressed" pulse are derived just by looking to (9.21).

We notice that the maximum is$r_{s_k}(0) = N$, and the best pulse have sidelobes null, for the values of k that cause an even number of terms in the summation (9.21), or ± 1 for the other. Such "optimal " codes are known as "Barker", and the sequences $b_k$are shown in the table for N=2 to 13.

Examples of autocorrelation of the sequences of Barker codes are given in Fig. 9.7. The true autocorrelation is then the linear interpolation.

For lengths larger than N=13, there are not known optimal codes, one can use the *Cook & Bernfield* codes, that are suboptimal and known up to $N = 1023$ or the MLS (*Maximal Length Sequences*, ore m-sequences). These last have optimal *circular-correlation*:

$$r_{cc}(n) = \sum_m r_c(n - mN)$$

in the sense that the sidelobes have minimal amplitude:

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N	*Sequence/e*	$PSLR$ dB
2	-+; ++	-6
3	++-	-9.5
4	−+-; +++−	-12
5	+++-+	-14
7	+++−+-	-17
11	+++−−+−+-	-20.8
13	+++++−++-+-+	-22.3

Table 9.1: Barker code generator and Peak-Sidelobe Ratio of the autocorrelation.

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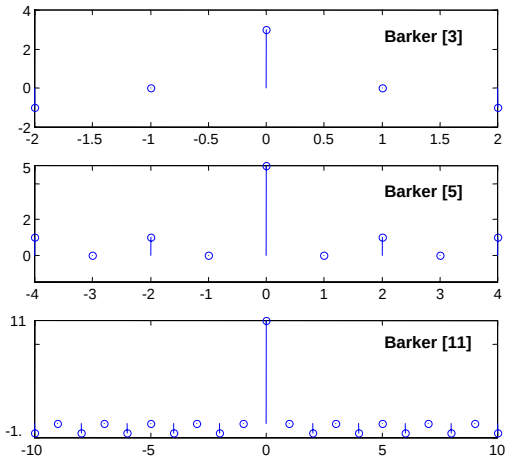


Figure 9.7: Autocorrelation of some Barker codes..

$$r_{cc}(n) = \begin{matrix} N & \text{per} \quad n = 0; \pm N; \pm 2N; \dots \\ -1 & \text{altrove} \end{matrix} \tag{9.22}$$

A subset of the MLS codes are the GOLD codes, used for GPS, that have further properties to minimize the cross-correlation between codes of the same "family". The circular correlation is not really the case of the Radar matched filter receiver, but rather the case of GNSS systems, that transmits periodic pulses with 100% duty cycles. In that case, the GOLD codes are quite useful to identify the specific transmitter between a set of possibilities.

9.2 Moving targets: Doppler Estimation

The pulsed radar is indeed capable to measures velocities, although with some limitation. Let the transmitted signal be a complex modulated pulse, g(t):

$$s_T(t) = \left\{ \sum_n g(t - nT_P) \right\} \exp\left(j2\pi f_0 t\right),$$

For the n-th pulse, we can express as:

$$\begin{aligned} s_n(t) &= \rho \exp(j\phi) \cdot g\left(t - nT_P - \frac{2R(t)}{c}\right) \exp\left(j2\pi f_0 \left(t - 2\frac{2R(t)}{c}\right)\right) \\ &\quad \cdot \exp(-j2\pi f_0 t) + n_c(t) \\ &= a \cdot g\left(t - nT_P - \frac{2R(t)}{c}\right) \exp(j2\pi f_0 t) \exp\left(j\frac{4\pi}{\lambda} R(t)\right) + n_c(t) \\ a &= \rho \cdot \exp(j\phi) \cdot \exp\left(2\pi \frac{2r_0}{\lambda}\right) \end{aligned}$$

after demodulation

$$s_d(t) = a \cdot g\left(t - nT_P - \frac{2R(t)}{c}\right) \exp\left(j\frac{4\pi}{\lambda} R(t)\right) + n_c(t) \tag{9.23}$$

In order to evaluate the result after matched filtering, we assume that the delay does not change appreciably during the time duration of the convolution, that is $\Delta R \ll \lambda$. For example in the case of linear motion, see section 6:

$$R = r_0 + v_{LOS} t$$

the resulting Doppler should make much less than cycle in the pulse duration, Tc. We assume that the attenuation due to phase variation in the matched filter time is less than 3 dB:

$$\begin{aligned} \frac{1}{Tc} \int_{Tc} \exp(j2\pi f_D t)\, dt = \operatorname{sinc}(f_D T_c) = 0.7 \Rightarrow f_D < \frac{0.6}{T_c} \\ \Rightarrow v_{LOS} < \frac{0.6}{T_c} \frac{\lambda}{2} \end{aligned}$$

As an example, for the 20 km range Radar assumed above, with 10% duty cycle, $T_c = 6.7\mu s$, and a very short wave Radar, $\lambda = 4mm$ we get a maximum line of sight velocity of 640 km/h, that is larger, for examples, of fast airplanes. Notice that the displacement of a 640 km/h

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target in the impulse of 6.7 μs is only 1 mm, the autocorrelation width being the resolution, that is 1 m.

Within this assumption after matched filtering (9.23) becomes:

$$y(t) \simeq a \cdot r_g \left(t - nT_P - \frac{2R(t)}{c} \right) \exp\left(j\frac{4\pi}{\lambda} r_0 \right) \exp\left(j2\pi f_D t \right) + n_c(t) * c(-t) \tag{9.24}$$

f we peak the autocorrelation at the maxima, for each pulse:

$$\tau = nT_P - \frac{2R(nT_p)}{c}$$

$$y\left(t = \tau\right) \simeq A \cdot \exp\left(j2\pi f_D n T_p \right) + n_c(t) * c(-t) \tag{9.25}$$

$$A = a \cdot E_g \cdot \exp\left(j\frac{4\pi}{\lambda} r_0 \right) \tag{9.26}$$

that is a discrete sinusoid at Doppler frequency, sampled at the Pulse Repetition Frequency $f_P = 1/T_P$.

9.2.1 Doppler and Range ambiguity

To avoid any aliasing, we should impose Shannon sampling:

$$B_D < f_p$$

where B_D is the total Doppler bandwidth, from the smallest to the highest frequency.

Let us assume that the target's expected motion varies in the range $-v_M < v < +v_M$. The total Doppler Bandwidth is

$$B_D = \frac{4v_{RM}}{\lambda} < f_p$$

This is a **lower bound** for the PRF, that is to be combined with the **upper one** in (9.6):

$$\begin{aligned} \frac{4v_{RM}}{\lambda} < f_p < \frac{c}{2R_M} \\ \Rightarrow \frac{4v_{RM}}{\lambda} < \frac{c}{2R_M} \\ \Rightarrow v_{RM} \cdot R_M < \frac{c\lambda}{8} \end{aligned} \tag{9.27}$$

We would be interested in observing large range intervals and fast targets, then to make the velocity × range product (9.27) as large as possible, but this is upper bounded by the product of wavelength and the speed of light. A fast propagation means in fact covering large range in short time, whereas the large wavelength is a low sensibility to Doppler, that is somewhat a drawback, but a much wide ambiguous velocity. We can observe, in this sense that Sonar are much more constrained than Radar, at same wavelength, as the velocity is thousands to millions of time faster. By this point of view, RADAR are much better.

As an example, is we assume the fastest airplane as v_{MAX}=900 km/h=250 m/s, then the maximum range that can be observed is from (9.27) 4.5 km, that is not that big for long range Radar. However, we notice that the simultaneous estimation of range and velocity is quite important when the aircraft is very close, whereas when it is far, it maybe enough to retrieve the range sampled at each pulse. Bats, for example, use estimate of velocity and range, in a sort of Kalman filter status to predict the best likely position of the mosquito in the next instant.

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Chapter 10

Doppler radar: examples and applications

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10.1 Subsidences from spaceborne SAR

The principle of Doppler estimation from pulsed can be applied also to estimate subsidences from spaceborne SAR. Such Radars make complex images by periodically revisiting each scene of the Earth. Each pixel of the image in then an observation. If such pixel represents a terrain, say in slow subsidence, moving of some mm/year, then long term observations can be combined to get the subsidence rate. We expect that the longer, the better the accuracy. But how long ?

10.1.1 Clutter and decorrelation time

So far we assumed a linear motion, so that the spectrum of a target's echo is a Dirac δ centered on the Doppler. In general we can however model the target motion as a linear term and a random fluctuation:

$$R(t) = R_0 + v_{LOS}t + d(t) \tag{10.1}$$

where $d(t)$ has some distribution, for example zero mean, normal:

$$d(t) \sim N\left(0, \sigma_d^2\right) \tag{10.2}$$

The contribution of such target to the received, demodulated signal, is:

$$s_d(t) = a \cdot \exp\left(j\left(2\pi f_D t + \frac{4\pi}{\lambda}d(t)\right)\right) = a \cdot \exp\left(j2\pi f_D t\right) \cdot \phi(t)$$
$$\phi(t) = \exp\left(j\frac{4\pi}{\lambda}d(t)\right)$$

a being the complex amplitude of the target. We have to face a process, that we is fully characterized by the statistics of ϕ(t). The power spectrum of the target contribution is

$$S_d(f) = |A|^2 \cdot \Phi\left(f - f_D\right)$$

where $\Phi\left(f - f_D\right)$ is the power spectrum of $\phi(t)$. Only in the case where displacement is constant we have a line, in all the others **the spectrum is smeared**. This implies two consequences:

1. if the target is the signal of interest, then its amplitude evaluated at frequency $f_D$is reduced, leading to a weaker detection, like and extra attenuation

2. if the target is a disturbance, the broadening of the spectrum introduces a **clutter noise** that would potentially prevent visibility of other targets.

In the first case, the output, after the matched filter and the demodulation by the nominal Doppler will be:

$$y_p = \int_{T_o} a \cdot \exp\left(j\frac{4\pi}{\lambda}d(t)\right) \mathrm{dt} \tag{10.3}$$

We can make here two cases of interest.

If d(t) is **stationary**, like due to fast *jitter of the target* (that can be intentionally, creating by a vibration, or by a transponder with random phase), then the average amplitude is:

$$E[y_p] = a \cdot \int_{T_o} E\left[\exp\left(j\frac{4\pi}{\lambda}d\right)\right] \mathrm{dt} = \mathrm{a} \cdot \mathrm{T} \cdot \gamma$$
$$\text{where}\gamma = \frac{E[y_p]}{E[y_p|_{d\equiv 0}]}$$

γ is a multiplicative term, bounded between 0 and 1, that expresses the extra attenuation due to random motion. Due to this loss, *the target cannot be detected if its amplitude is lower than threshold.*

As an example, for Normal distributed displacement, $d \sim N\left(0, \sigma_d^2\right)$:

$$\gamma = E\left[\exp\left(j\frac{4\pi}{\lambda}d\right)\right] = \exp\left(-\frac{1}{2}\sigma_\phi^2\right)$$
$$\sigma_\phi = \sigma_d \cdot \frac{4\pi}{\lambda}$$

The property of γ to be within 0 and 1 is s an example, if the target has a random fluctuation with $\sigma_d = 1$ mm, then $\gamma = 0.98$ in X band, but ˜0 in the automotive, band at 77 GHz. As for these, the lower frequencies are mostly recommended.

If d(t) is non stationary, increasing with time, like for **random walk,** the result is a limiting to the observation time, whatever the true value.

Such random walk can be due to different sources:

1. the target fluctuations, say water, leaves, or grass (vegetation),

2. the instability of the very same clock used in the modulator that can be modeled as [3]:

$$s(t) = \cos\left(2\pi f_0(t) \cdot t\right) \tag{10.4}$$

   where $f_0(t)$ is Brownian (unstable, but stable in increment),

3. the instability of the Radar track, say for Radar mounted on a rail, an airplane or a satellite,

4. the effect of propagation of light in the atmosphere, where the fluctuation of the refraction index - due to water vapor, pressure and temperature, causes the velocity of light to change, and then introduces a random delay.

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In all these cases a Brownian motion is quite a good model, assuming that at each step, the direction is changed randomly. Let us remind (5.11):

$$\gamma_s = \exp\left(-\frac{\Delta t}{\tau}\right)$$

where the time constant depends on the **intrinsic target properties**, and the wavelength, see (5.12):

$$\tau = \frac{4}{T \cdot \sigma_v^2}\left(\frac{\lambda}{4\pi}\right)^2 \tag{10.5}$$

This limit the practical observation time in (10.3) to the **coherent integration time,** as the increasing time the phase walk diverge, windowing the observation. Even an infinite observation time would be equal to a time T_e:

$$\int_{-\infty}^{\infty} a \cdot \exp\left(-\frac{|t|}{\tau}\right) \mathrm{dt} = a \cdot T_e$$

$$T_e = 2\tau$$

As an example, it has been shown in [4, 1] that short vegetation in C band has a correlation time of ˜40 days. An example of the estimation of subsidences in the are of Mt. Etna, by C-band space-borne SAR observations, is shown in Fig. 10.1.

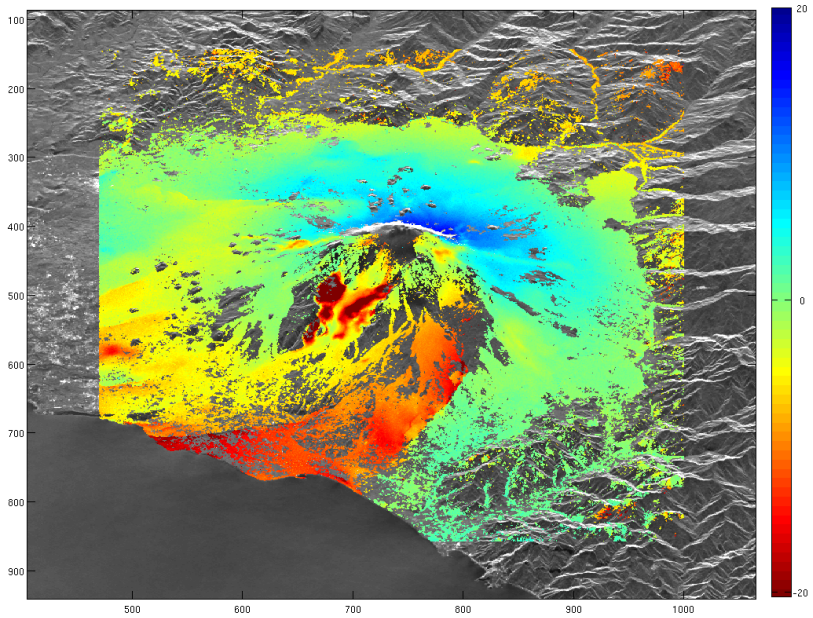


Figure 10.1: Subsidence rate (color-scale, mm/year) estimated in area of Etna by repeat-pass spaceborne SAR. Image courtesy of TREUROPA.

It is clear that the capability of estimating fractions of half wavelength, say 1/100 or so in 80 days, allows us to retrieve a subsidence rate - that is effectively a Doppler - with the accuracy of

$$\rho_v = \frac{\lambda/2 \cdot 0.01}{80/365} = \frac{53 \cdot 0.01}{80/365} = 1.2 \text{ mm/year}$$

this number makes those in the colorbar of the figure meaningful. However, addressing the right accuracy is a little more complicated and we will discuss it in the last part of this book.

10.2 Clock stability

Let us assume to transmit a single pulse, and assume a clock frequency drift:

$$s_T(t) = g(t) \exp\left(j2\pi f_{0(t)} t\right)$$

the received, baseband demodulated signal is:

$$\begin{aligned} s(t) &= A \cdot s_T\left(t - \frac{2R(t)}{c}\right) \exp\left(-j2\pi f_{0(t)} t\right) \\ &= A \cdot g(t-\tau_r) \exp\left(j\left(2\pi f_{0(t-\tau_r)}\left(t-\tau_r\right) - 2\pi f_{0(t)} t\right)\right) \end{aligned}$$

where

$$\tau_r = \frac{2R}{c}$$

In absence of drift, the signal phase would be $\phi_r = 2\pi f_0 \tau_r$, then the phase error is:

$$\phi_e = 2\pi\left(f_{0(t-\tau_r)} - f_{0(t)}\right) t - 2\pi\left(f_{0(t-\tau_r)} - f_0\right) \tau_r$$

This error is different for each target. However, in many cases of long range Radar - that are mostly affected by these drift, the range of the first target is the dominant, and there is not that change from the last. Than we assume $t \sim \tau_r$

$$\begin{aligned} \phi_e &= 4\pi \Delta f_{0(\tau_r)} \cdot \frac{R_{MIN}}{c} \\ &= 2\pi \cdot f_0 \cdot \Delta(\tau) \cdot \frac{2R_{MIN}}{c} \end{aligned} \tag{10.6}$$

$$\text{where } \Delta(\tau) = \frac{\Delta f_{0(\tau)}}{f_0}$$

The formula has been expressed as a function of the Allan deviation, $\Delta(\tau)$, that is the mean relative variation of the frequency in the time interval τ [3].

As an example, for an X band Radar, with a range of 100 km, and we assume to tolerate a phase error of 5°, the Allan deviation of the clock should be:

$$\Delta(\tau) \simeq \frac{5}{180}\pi \frac{150 \cdot 10^3}{100} \frac{1}{2\pi 10^{10}} = 2 \cdot 10^{-10}$$

in $\tau = 0.6$ ms. This is well within the present clock generator, that can even achieve an Allan deviation 5 10^{-13} in 1-10 seconds.

10.3 Vibrating targets

An example of RADAR acquisition and data analysis is shown in 10.3. A Ku-band RADAR, on the left bottom, observes a scene where a target subject to sinusoidal motion (in the blue circle, and in the bottom picture) has been placed at about 30 m from the RADAR. The scene contains also a tree with many flowers, randomly moving due to some wind, besides many reflecting objects.

The Radar transmits with a PRF of 20 Hz, while the observed range is ˜ 100 Hz. For every range bin, the amplitude of the Fourier Transform of the received data (after demodulation)

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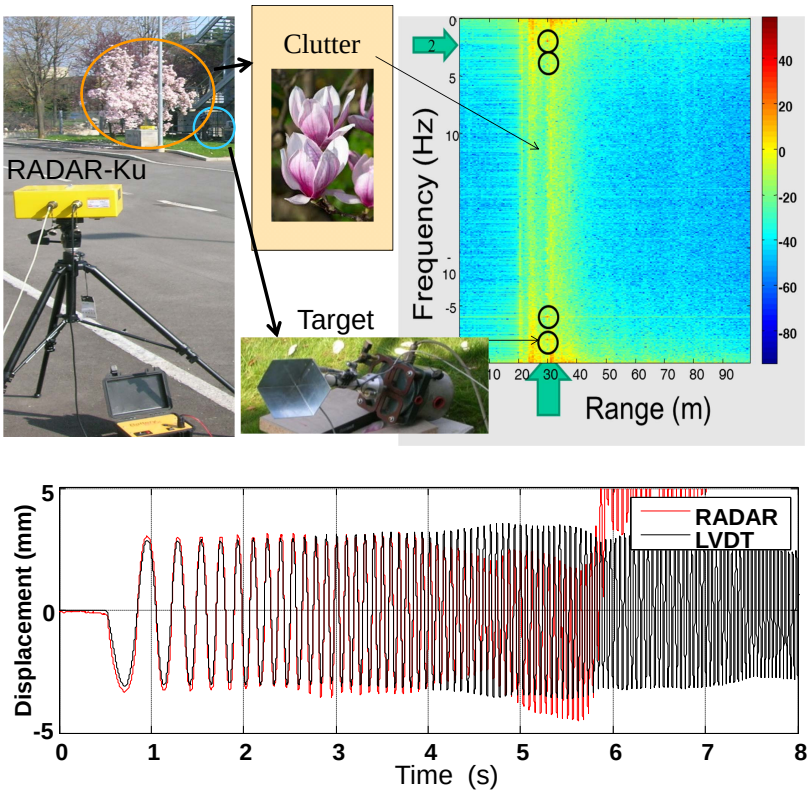


Figure 10.2: Acquisition and identification of a moving target by Ku-band RADAR. Below: comparison between the motion extracted from Radar and the one measured on the LVDT device.

is shown in the 2D image on the right. The observation time was about 100 s, that allowed for a minimal frequency of 0.01 Hz. The target is moving periodically, very close to a sinusoid with frequency $f_p = 2$ Hz, then the expected Radar return is also periodically and can be expanded in Fourier series:

$$y(t) = A \cdot \exp\left(j\frac{4\pi}{\lambda}d(t)\right) = A \cdot \sum_k c_k \cdot \exp\left(j2\pi \cdot k \cdot f_p t\right)$$

The circles in the picture evidence the first and second harmonic that is clearly distinguishable from the fixed - that contributes close to frequency 0. The movement, extracted by measuring the phase of the contribution at the proper range, is compared in the same figure with the one measured on the device, by a precise LVDT. The two measures fits within say micron, however, notice, after some time, a discrepancy due to the displacement of the whole device that the LVDT could not track.

In the figure, the **clutter due to random motion** of flowers is clearly identified by the wide spectrum in the range 25-30 m, in this case not as strong as the moving target.

A second example, a similar experiment, but carried out in absence of wind, is shown in Fig. 10.3. Here the contribution of moving target is clearly distinguishable from fixed one, disregarding the ratio of amplitudes. The capabilities of RADAR (and SONAR) to separate moving from fixed target leads to its major killer applications, like bats blind chase for insects.

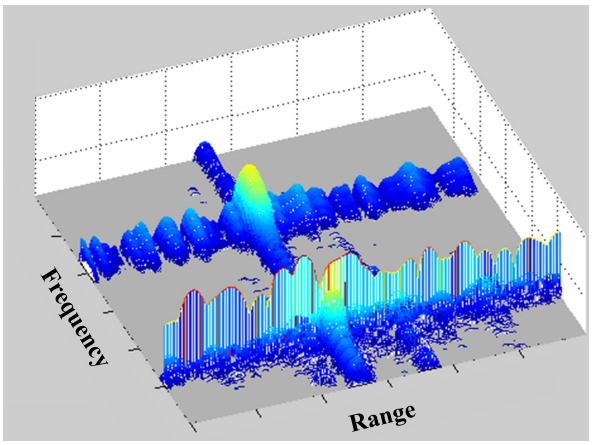


Figure 10.3: Contributions of fixed targets and a single target subject to sinusoidal vibrations in the range - frequency domain.

10.4 Bats

Bats chase fleas or mosquitoes,and other small insects. The use ultrasounds, then a Sonar, and tune the wavelength to be reflected by those small insects. The transmitted frequency - that depends upon the species - is typically 30-150 kHz, corresponding in free air ($v_s \sim 340$ m/s), to $\lambda = 2 - 10$ mm. A typical wing rate of an insect is in the range 10-100 Hz, therefore we need $T_o \gg 10$ ms to distinguish from fixed target.

The spectrogram of the transmitted pulse trains is in Fig.10.4 for different species. Notice that the pulses have some pure sinusoidal component in the initial phases, noticeable for the Rhinolophus, who chases spider in the terrain or insects in dense vegetation, and needs high sensibility. Bat's hearing can be modeled as a fixed bandwidth filter, therefore the filtering of zero Doppler (fixed targets), or negative Doppler (targets departing) is performed by reducing the transmitted frequency according with the target-sensor velocity.

As the bats gets close to the prey, he needs faster tracking, however its knowledge of the target is much more accurate, therefore it can increase the PRF. At the same time, the pulse bandwidth is increased to get better resolution. Notice that the waveform adapts its PRF with time. At the final step of the chase, when the bat is capturing its prey, there is "buzz" that corresponds to the transmissions of a train of δ pulses.

One important aspect is the stability of the oscillation. For microwaves we have seen that very low Allan deviation can be achieved by exploiting quartz. This is not so easy for optic (time coherent lasers) nor for Sonar. Let us anyhow fix the target range at 30 m, and tolerate a (big) phase error of $\pi/2$, then we derive form (10.4):

$$\Delta(\tau) \simeq \frac{\pi}{2}\frac{170}{30}\frac{1}{2\pi \cdot 3 \cdot 10^4} \sim 4 \cdot 10^{-5}$$

That value in 0.17 s (computed at the most favorable frequency), is indeed challenging: the best achievable being in the order of 0.1%. This means that the receiver is not coherent over long distances, and limits the frequency accuracy 10-50 Hz. This reduces the effect of the matched filter - and then the noise gain, to a bandwidth of 10-50 Hz, instead than the usual $1/T_0$. That is, the equivalent observation time is limited to 100 ms.

The attenuation of the acoustic waves at that frequencies is rather strong, typically 1.3 dB/m, moreover we account for forward and backpropagation, we get almost 80 dB for a

typical range of 30 m. Such huge gap between the transmitted intensity and the bat ear sensitivity leads to the potential crash of the receiver if, for example, there is even a quite small target close to the system. Bats act like long range Radar by muting the receiver for an interval correspondent to the first echo of interest. Bats get this goal by disengaging the ossicles that transmit sound in the inner ear.An example of target location by a pulsed SONAR is provided by bats when approaching the prey.

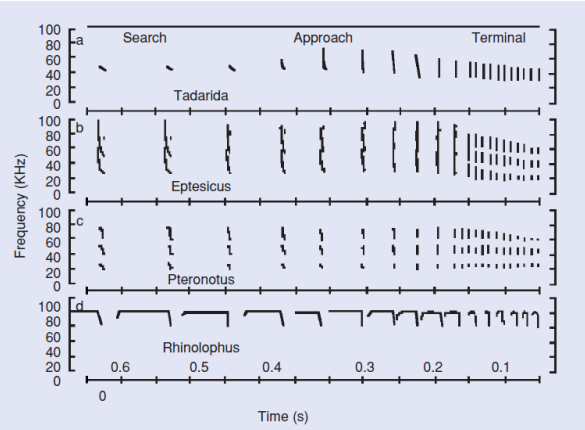


Figure 10.4: Spectrograms of sonar signals produced by four different species of bats as they advance from the search to approach and finally to the terminal phase of insect pursuit. From [5].

10.5 FMCW RADAR

The Frequency Modulated Continuous Waveform RADAR exploits a very simple, yet effective scheme to provide cheap systems with moderate range and power. The schematic block diagram of a typical FMCW RADAR is shown in Fig. 10.5. Such systems exist in both monostatic and bistatic configuration, however the simplest and most efficient, here discussed are bistatic.

The system in the figure exploit one antenna that is continuously transmitting a periodical repetition of linear-frequency-modulated pulses, i.e. chirps:

$$s_T(t) = \sum_n c(t - nT_P)\exp(j2\pi f_0 t)$$
$$c(t) = \exp(j\pi k t^2)\,\mathrm{rect}\left(\frac{t}{T_p} - \frac{1}{2}\right)$$

whose instantaneous frequency sweeps periodically as shown in the upper plot of Fig. 10.5:

$$f_i(t) = \frac{1}{2\pi}\frac{\partial}{\partial t}\arg\left(s_T(t)\right) = f_0 + k\left(t - nT_P\right)$$

The bistatic configuration allows the system to receive echoes from target that are very close to the system, in principle there is no limit to the minimum range as for pulsed RADAR. A target at range r_0 with complex reflectivity $\rho_0\exp(j\phi_0)$ (we include path loss in the am-

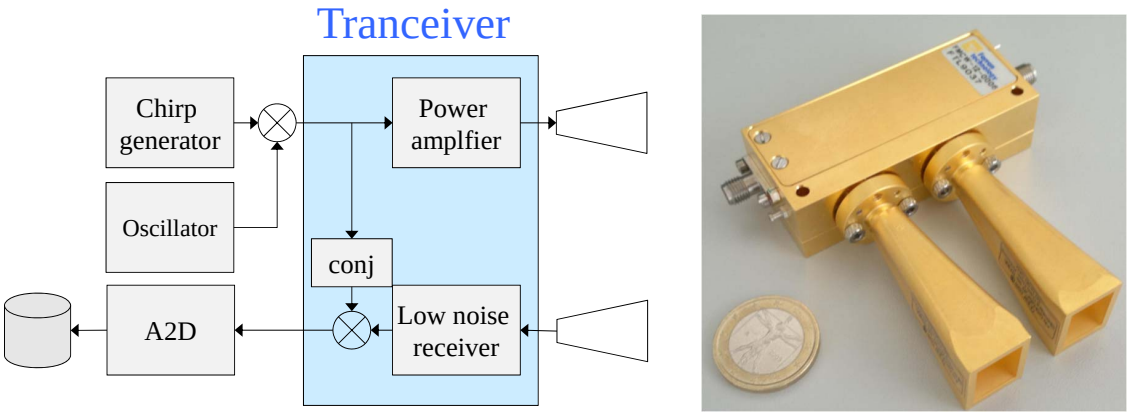


Figure 10.5: FMCW RADAR. Left: schematic block diagram. Right: example of a 77 GHz transceiver with horn antennas.

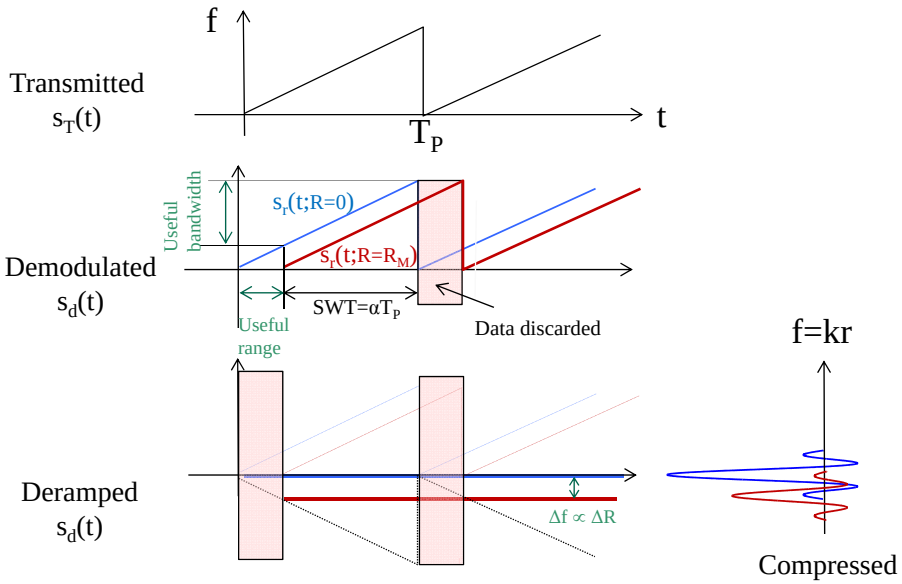


Figure 10.6: FMCW RADAR: principle in the time / frequency diagram.

plitude), will backscatter the signal:

$$s_R(t) = \rho_0 \exp(j\phi_0) \exp\left(j\pi k \left(t - t_0\right)^2\right) \exp\left(j\pi 2 f_0 \left(t - t_0\right)^2\right) \text{rect}\left(\frac{t - t_0}{T_P} - \frac{1}{2}\right)$$
$$t_0 = \frac{2r_0}{c} \tag{10.7}$$

here written only for the first echo, to be repeated periodically. Then after demodulation:

$$s_d(t) = a_0 \exp\left(j\pi k \left(t - t_0\right)^2\right) \text{rect}\left(\frac{t - t_0}{T_P} - \frac{1}{2}\right)$$
$$a_0 = \rho_0 \exp(j\psi_0)$$
$$\psi_0 = \phi_0 - \frac{4r_0}{\lambda}$$

The instantaneous frequency will be a ramp whose origin is shifted according to the target location as shown in the middle plot of Fig. 10.5.

One advantage of FMCW with respect of other CW RADAR is that there is always a significant gap in the instantaneous frequencies between the outgoing transmitted signal and the incoming received echoes, and this guarantee the separation between the two signals.

10.5.1 Deramping

The major trick in FMCW RADAR is the deramping step, that is multiplication times the conjugate phase ramp, in n, after beating times the complex conjugate of the transmitted ramp, performed by the transceiver in Fig. 10.5. The deramped signal:

$$\begin{aligned} y(t) &= s_d(t) \cdot \exp(-j\pi k t^2) \cdot \text{rect}\left(\frac{t - t_0}{T_P} - \frac{1}{2}\right) \cdot \text{rect}\left(\frac{t}{T_P}\right) \\ &= a_0 \exp\left(-j2\pi k t_0 t\right) \exp\left(j\pi k t_0^2\right) \;\; t_0 \le t \le T_P \end{aligned} \tag{10.8}$$

is converted in a pure tone, whose instantaneous frequency is proportional to the target range:

$$f_P(t) = -k t_0 \tag{10.9}$$

and whose duration changes according to the target. The maximum duration is T_P, for a target at $t_0 = 0$ and reduces with range. For the maximum range R_M the duration is:

$$T_M = T_P - \frac{2R_M}{c} = T_P\left(1 - \alpha\right) \tag{10.10}$$
$$\alpha = \frac{2R_M}{cT_P} \approx 1$$

We have introduced the definition of α that is actually the **duty cycle** of the system.

The maximum range is limited by the fact that the target will interfere with the return of the successive echo, as shown in the middle plot of Fig. 10.5. In practice, the data collected up to t_M are discarded. The sampling window acquires in the time interval:

$$T_M = \left(1 - \alpha\right) T_p \le t \le T_P$$

We observe that the "deramping" step is quite effective in reducing the bandwidth of the data from that of the demodulated data, B_d, to that of the deramped data B_y

$$B_d = \alpha k T_P \rightarrow B_y = (1 - \alpha) k T_P$$
$$\frac{B_y}{B_d} = \frac{1 - \alpha}{\alpha} \tag{10.11}$$

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This bandwidth compression is very effective as the duty cycle α approaches to 1. At the same time, after deramping, the time support of the data is increased from T_d to T_y

$$\begin{aligned} T_d = T_M = (1-\alpha)T_P \rightarrow T_y = aT_P \\ \frac{T_y}{T_d} = \frac{k(1-\alpha)}{\alpha} \end{aligned} \tag{10.12}$$

Then, by comparing (10.11) and (10.12) we see that the Time Bandwidth Product (TBP) - that is actually the number of independent samples we solve, is conserved. In other words, deramping performs a conversion from data with large bandwidth but comping from a narrow time interval, to data with very small bandwidth and large time interval. This procedure has the major advantage to allow the exploitation of very simple A2D with low sampling frequency. Such advantage makes the deramping used in many different contexts.

Notice that the central frequency is then not f_0, but rather:

$$f_c = f_0 + \frac{B_d}{2}$$

10.5.2 Range Compression

Compression of deramped data - or "range compression" - is carried out by a Fourier Transform whose role is the optimal estimate of the complex target reflectivity. Let us rephrase the deramped signal (10.8) after removing the fixed time delay of the target at far range, and its Inverse Fourier Transform:

$$\begin{aligned} y(t) &= a_0 \exp\left(-j2\pi k t_0 t\right) \exp\left(j\pi k t_0^2\right) \;\; 0 \le t \le \alpha T_P \\ Y(f) &= \left\{a_0 \exp\left(j\pi k t_0^2\right)\right\} \alpha T_P \operatorname{sinc}\left(\left(f - k t_0\right)\left(\alpha T_P\right)\right) \exp\left(j2\pi f \frac{\alpha T_p}{2}\right) \end{aligned}$$

where it is suggested to remove the last exponential by performing a (circular) shift of $\alpha T_p/2$

The first phase enclosed in brace is an unwanted range-dependent artifact, defined as "video phase", that we can compensate on the output data. Then, after removing this term, we express the output in terms of range, in place of frequencies, i.e. by making the change of variable (10.9):

$$f = kt = k\frac{2r}{c}$$

we finally get the output:

$$Y(r) = a_0 \alpha T_P \operatorname{sinc}\left(\frac{2k}{c}\alpha T_P \left(r - r_0\right)\right)$$

whose resolution is from the argument of the sinc:

$$\rho_r \simeq \frac{c}{2k\alpha T_P} = \frac{c}{2\alpha B} = \frac{c}{2B_u}$$

B being the total bandwidth and B_u the useful bandwidth shown in Fig. 10.5.

Notice that the pulse length is rather long, and the hidden assumption is that the target should not change its phase (decor-relate) in this interval.

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10.5.3 Example

Let us assume to design a FMCW rather with the following requirements.

Item	Value
Frequency	77 GHz
Max range	10 km
Max RF bandwidth	2 GHz
Range resolution	10 cm

The maximum range requires an useful pulse time as from (10.7) is 33.4 μs, the target effective bandwidth is derived from the resolution

$$\rho_r = \frac{c}{2B}$$

that is 1.5 GHz. From this, the duty cycle should be at most the ratio between the effective and the maximum bandwidth, that is 75%. If we select a larger value, we get better compression rate, then better exploitation of the energy and lower sampling frequency. However, the PRF will increase. Here we have no constraint on that, therefore we fix duty cycle at 80%, getting a pulse repetition interval as from (10.10) leading to 6 kHz PRF. The required Doppler rate k is the maximum RF bandwidth / T_p, 12 MHz/μs. Finally, the compressed pulse bandwidth is from (10.11), that amounts to 400 MHz, demanding for an A2D converter with similar bandwidth. Notice that, in absence of requirements on the PRF, we could even increase the duty cycle to 90%, leaving 3 kHz PRF and a compressed bandwidth of 200 MHz.

10.6 Monostatic versus bistatic Radar

The FMCW Radar has its force in the simplicity, and, being bistatic, in its imaging of targets since pretty small range - virtually zero. On the other hand, for far ranges, the RF bandwidth becomes huge while the large discrepancy between transmitted and received power creates interferences in the coupling between the two antennas.

The monostatic Radar does not suffer of this problems, while, on the other hand, the limiting is near range is not so relevant. The major advantage is the re-use of the same antenna, that, for this devices becomes pretty large. This is done by exploiting a *circulator*, that switches the signal path between the High Power Amplifier and the antenna, TX, and the antenna and the Low Noise Amplifier, RX, as shown in the schematic block diagram of the receiver in Fig. 10.7.

However, the difference from a Bistatic and a monostatic Radar is not merely a question of technology: a bistatic Radar, as shown in Fig. , has a quite different view, and observes different ranges and perspectives, say RCS of the target.

Mauro Chiarelli - EVALUATION Draft

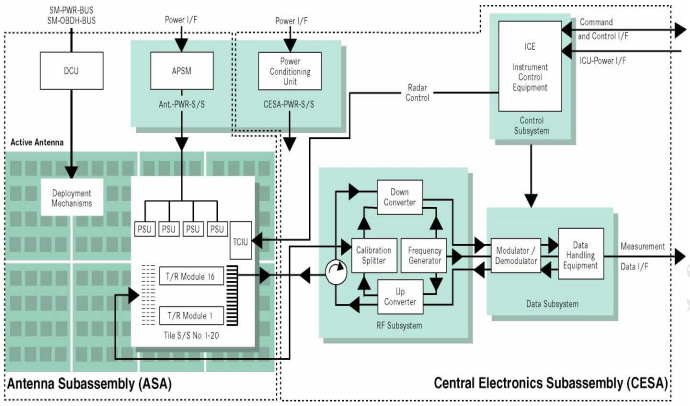


Figure 10.7: Schematic block diagram of the RADAR mounted on ESA satellite ENVISAT. © European Space Agency (ESA).

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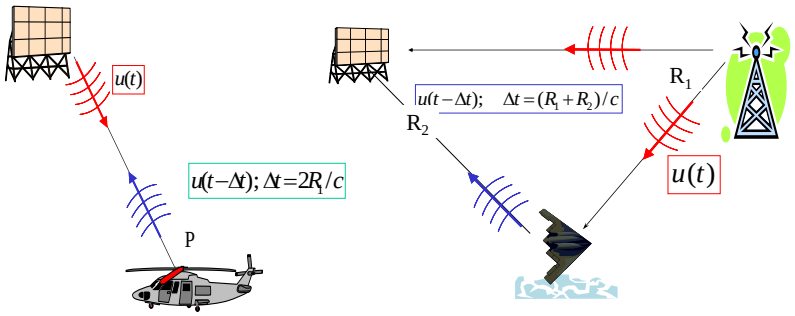


Figure 10.8: Basic principle of a monostatic (left) and a bistatic RADAR (right).

Chapter 11

The equivalent bandpass channel

Let us now define the time-limited pulse to be transmitted as *c(t)*, for example a complex chirp windowed. The natural question is how a complex waveform, like the chirp just defined can be transmitted in the real world. The point is that we do not transmit $c(t)$, but rather its modulated version:

$$\begin{aligned} s_t(t) &= \mathrm{Re}\left\{s_c(t)\right\} \\ s_c(t) &= c(t) \cdot \exp(j\omega_0 t) \end{aligned} \tag{11.1}$$

where the multiplication by $\exp(j\omega_0 t)$ is just a shift in frequency, as shown in Fig. 11.1, and thus it conserves the constant amplitude, and taking the real part correspond to the adding the Hermitian symmetric component, the negative frequencies:

$$s(t) = \frac{s_c(t) + s_c^*(t)}{2} = \mathrm{Re}\left(s_c(t)\right) \Leftrightarrow S_T(f) = \frac{S_c(f) + S_c^*(-f)}{2} \tag{11.2}$$

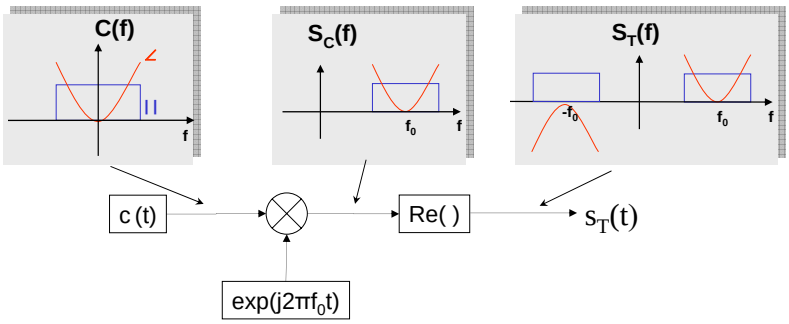


Figure 11.1: Equivalent block diagram of an IQ modulator. The input is the complex pulse c(t). The output is the pulse transferred into high frequency and combined with its Hermitian symmetric part (to make it real):

We can rearrange (11.1) as follows:

$$\begin{aligned} s_T(t) &= \mathrm{Re}\left\{\left(c_i(t) + jc_q(t)\right)\left(\cos\left(\omega_0 t\right) + j\sin\left(\omega_0 t\right)\right)\right\} \\ &= c_i(t)\cos(2\pi f_0 t) - c_q(t)\sin(2\pi f_0 t) \end{aligned} \tag{11.3}$$

where $c_i(t)$ and $c_q(t)$ are the real and imaginary component of $c(t)$:

$$c(t) = c_i(t) + jc_q(t)$$

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The implementation of (11.3) in terms of two modulations is straightforward, as Fig. 11.2 shows. This IQ modulator is widely exploited in telecommunication, in fact it can provide any form of amplitude or phase modulation. This can be shown just be writing (11.3) as follows:

$$\begin{aligned} s_T(t) &= \mathrm{Re}\left\{c(t)\cdot\exp(j2\pi f_0 t)\right\} \\ &= a(t)\cdot\cos\left(2\pi f_0 t+\phi(t)\right) \\ \text{where}\quad & a(t)=|c(t)|\,;\ \phi(t)=\angle c(t) \end{aligned} \tag{11.4}$$

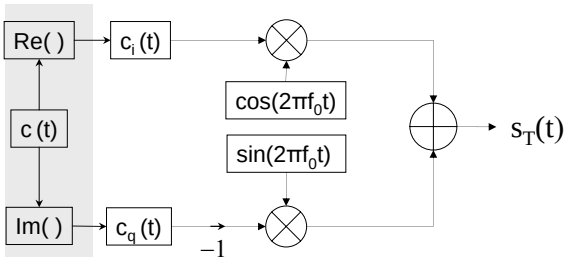


Figure 11.2: Block diagram of an IQ modulator

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11.0.1 Complex envelop

The IQ modulation in (11.3) can be generated by exploiting the Hilbert Transform (HT). Let us take the real part of the modulated chirp:

$$c_{im}(t)=c_i(t)\cos(2\pi f_0 t)$$

and let us compute the HT:

$$c_{imh}(t)=c_{im}(t)*h_H(t)$$

where $h_H(t)$ is the Hilbert kernel, that is real, and causal, yet of infinite duration, its FT being:

$$H_H(f)=j\,\mathrm{sign}(f)$$

Let then define the *complex envelop* as follows:

$$s_c(t)=c_{im}(t)-jc_{imh}(t)$$

This signal is just the same as the complex modulated chirp $s_c(t)$, as Fig. 11.3 shows. The relation is used whenever we have to convert a real signal back into its complex equivalent.

11.0.2 IQ demodulation

Let us characterize the linear system, so far assumed for RADAR propagation and reflection, by its impulse response $h(t)$, or its frequency counterpart $H(f)$. The received signal can then be expressed as follows:

$$s_R(t)=s_T(t)*h(t)+n(t) \tag{11.5}$$

$$S_R(f)=S_T(f)\cdot H(f)+N(f)=\frac{C^*(-f+f_0)+C(f-f_0)}{2}\cdot H(f)+N(f)$$

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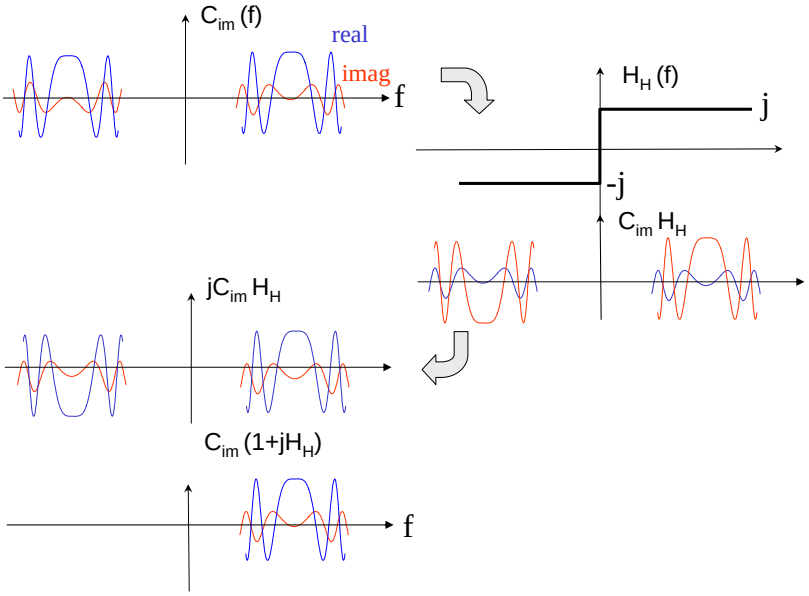


Figure 11.3: Generation of the complex envelop of signal.

$n(t)$ being the noise term. Notice that, in the simplest case of one ideal point target, $h(t)$ corresponds to a simple delay, plus an attenuation and a phase. Let us assume the IQ demodulator in Fig. 11.4. We remind that the output is a complex number achieved by combining the real part from I-channel and the imaginary from the Q-channel:

$$\begin{aligned} s_d(t) &= s_R(t) \cdot 2\cos(2\pi f_0 t) * h_{lp}(t) - j\,(s_R(t) \cdot 2\sin(2\pi f_0 t) * h_{lp}(t)) \\ &= s_R(t) \cdot 2\exp(-j2\pi f_0 t) * h_{lp}(t) \end{aligned}$$

where $h_{lp}$is the low pass filter in the system bandwidth, B. Usually we exploit two ADC after demodulation and filtering, thus $s_d(n)$ is complex sequence.

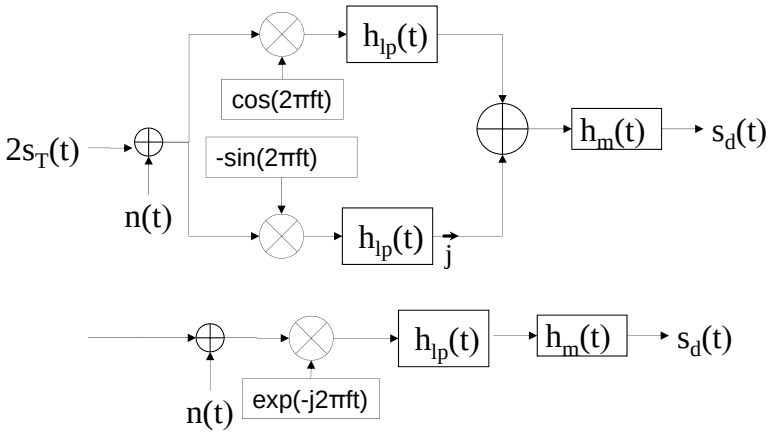


Figure 11.4: Scheme of the IQ demodulator, and its equivalent complex formulation.

Its spectrum can be expressed as follows:

$$\begin{aligned} S_d(f) &= 2 \cdot S_R(f+f_0)H_{lp}(f) \\ &= C(f)H(f+f_0)H_{lp}(f) + 2N(f+f_0)H_{lp}(f) \\ &+ C^*(-f+2f_0) \cdot H(f+f_0)H_{lp}(f) \\ S_d(f) &= C(f)H(f+f_0) + 2N(f+f_0)H_{lp}(f) \end{aligned} \tag{11.6}$$

where we have exploited the fact that $C(f)$ is itself low-pass in $-B/2 \leq f \leq B/2$, thus the term at frequency $-2f_0$ is removed by the filter, whereas the other is left unchanged, as shown in Fig. 11.4.

Expression (11.6) leads to a simple interpretation. The spectrum of the complex signal after IQ demodulation reflects a portion of the transfer function of the system in a band B, centered on f_0:

$$-B/2 \leq f \leq B/2 \Rightarrow f_0 - B/2 \leq f + f_0 \leq f_0 + B/2$$

In practice, if the transmitted pulse *C(f)* is at constant amplitude, and after the correlation with the complex conjugate:

$$\begin{aligned} y(t) &= s_d(t) \cdot c^*(-t) \\ Y(f) &= \left(C(f)H(f+f_0) + 2N(f+f_0)H_{lp}(f)\right) C^*(f) \\ &= H(f+f_0)\left|C(f)\right|^2 + 2N(f+f_0)C^*(f) \end{aligned}$$

This clarifies why $C(f)$ is desired at constant amplitude. We would get in that case:

$$Y(f) = H(f+f_0)\,\mathrm{rect}\left(\frac{f}{B}\right) + N'(f)\,\mathrm{rect}\left(\frac{f}{B}\right)$$

The first contribution shows how the RADAR output monitor a bandpass version of the channel, hence the scene reflectivity, the second contribution is just a noise term filtered in the system bandwidth. Statistically, there is no difference between N' and N.

11.0.3 The equivalent baseband channel

It is therefore possible to study the whole RADAR system by combining the IQ modulation and the IQ demodulation in the equivalent channel shown in Fig. 11.4. The upper block scheme represents the cascade of IQ modulator, channel, noise and IQ demodulator. In the middle scheme, we have moved the channel past the demodulator, that results in filtering by the demodulated channel. In the same middle diagram, we have moved the real noise n(t) past the demodulator. The effect is to make noise complex, whereas its power spectrum is just a shifted version, however PSD doubles due to the factor two in the demodulator. In the lower scheme we have moved the low pass filter before the channel, and consider that the cascade of modulator + real_part + demodulator + low pass is all pass (in the signal bandwidth):

$$\begin{aligned} y(t) &= 2Re\left\{x(t) \cdot \exp(j\omega_0 t)\right\} \cdot \exp(-j\omega_0 t) * h_{lp}(t) \\ &= \left(x(t) \cdot \exp(j\omega_0 t) + x^*(t) \cdot \exp(-j\omega_0 t)\right) \cdot \exp(-j\omega_0 t) * h_{lp}(t) \\ &= x(t) \end{aligned}$$

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The lower plot makes a RADAR system equivalent to the convolution of the baseband chirp with a frequency downn-shifted version of the real channel. Such channel is what RADAR **needs to identify, given knowledge of input and output**.

The complex noise term n_c is obtained past the demodulator:

$$\begin{aligned} n_c(t) =& \{n(t) \cdot \cos(2\pi f_0 t) - j \cdot n(t) \cdot sin(2\pi f_0 t)\} * h_{lp}(t) \\ =& n_i(t) + j n_q(t) \end{aligned}$$

Notice that both components are non stationary, before the low-pass filter, then low-passing corresponding to time averaging. The autocorrelation is:

$$\begin{aligned} r_{nI}(t, t+\tau) &= E\left[n(t) n(t+\tau) \cdot \cos(2\pi f_0 t) \cos(2\pi f_0 t + \tau)\right] \\ &= \delta(\tau) \cdot \cos^2(2\pi f_0 t) \end{aligned}$$

by assuming that n(t) is δ correlated. Then n_I is white but time varying. We can compute the averaged power

$$\begin{aligned} \overline{\sigma_c^2(t)} =& \frac{1}{T} \int_T E\left[n^2(t) \cdot \cos^2(2\pi f_0 t)\right] dt \\ =& \frac{w_n^2}{2} \\ =& \frac{N_0}{4} 2B = \frac{N_0}{2} B \end{aligned}$$

where we computed the power of the bandpass noise as the product of the power spectrum density$N_0/2$ times the total bandwidth, 2B. Therefore, being B the band of $n_c(t)$, its power spectrum density is $N_0/2$. Likewise, we can show that $n_s(t)$ has the same power spectrum density, whereas n_c, n_s are mutually incorrelated, when time averaged

$$\begin{aligned} \overline{E\left[n_c(t) n_s(t)\right]} =& \frac{1}{T} \int_T E\left[n^2(t) \cdot \cos(2\pi f_0 t) \sin(2\pi f_0 t)\right] dt \\ =& \sigma_n^2 \frac{1}{T} \int_T \frac{1}{2} \sin\left(4\pi f_0 t\right) dt = 0 \end{aligned}$$

Then, $n_c(t)$ is a complex valued noise, superposition of two incorrelated noise contribution each with power spectrum density $N_0/2$, giving a total PSD ofN_0.

The baseband output can then be expressed as follows:

$$s_d(t) = \{h(t) \cdot \exp(-j 2\pi f_0 t)\} * c(t) + n_c(t)$$

after matched filtering:

$$s_d(t) = h(t) * r_c(t) + n_c(t) * c(t)$$

that makes clear that our desired pulse is delta-correlated.

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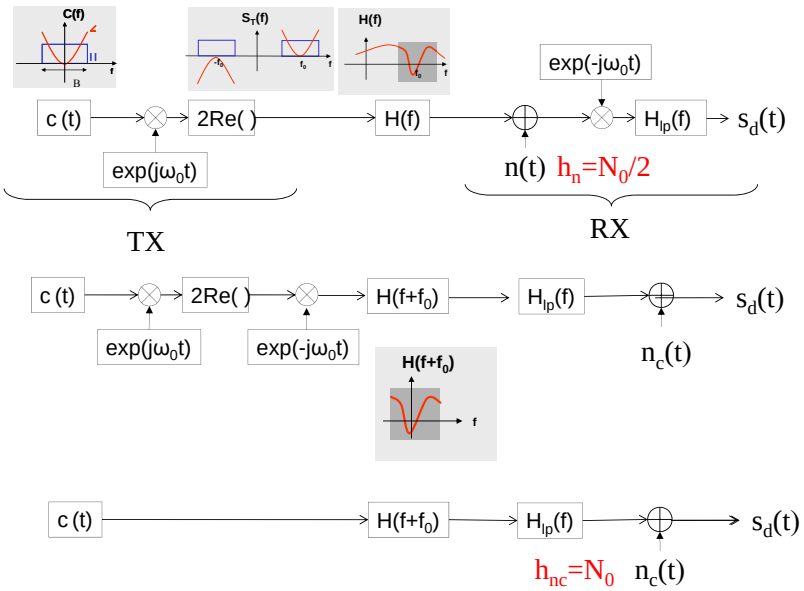


Figure 11.5: Derivation of the equivalent baseband channel for a RADAR.

Part III

Radar Imaging and SAR

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Chapter 12

Extending resolution to 2D and 3D: the geometric approach

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The monopulse Radar (or Sonar) so far studied is capable to provide location in range only, and velocity for moving targets. Imaging Radars needs to locate in the 3D. Before entering in discussion on imaging Radar, we provide some analysis on the achievement of 3D location by exploiting multiple Radar, say many ranges from different point of view, a technique well known as trilateration. The objective is both to estimate location, and its accuracy, including ambiguities, and can be carried out by pure geometric analysis. In a second section, we provide a formulation of the same problem with statistical signal processing and analysis.

12.1 TOA, DTOA and DOA

The basic principle exploited to provide 2D location is shown in Fig. 12.1. Let us assume to locate a target P, by a sensor S, (it can be a bat looking for its preys) that is moving on some track. Let us suppose that to have two measures of ranges taken at a distance B (baseline). Let us suppose B small, as we need to locate the target as fast as possible. We have now two range equations (TOA Time Of Arrival), that are more or less identical, we write the range equation as:

$$\left|\overrightarrow{R}(P)\right| = \left|\overrightarrow{S} - \overrightarrow{P}\right| = \frac{ct_0}{2}, \tag{12.1}$$

t_0 being the TOA measured. We can however make another equation by measuring differences in Time Of Arrival ($DTOA$):

$$\Delta_R = \left|\overrightarrow{P} - \overrightarrow{S_2}\right| - \left|\overrightarrow{P} - \overrightarrow{S_1}\right| \tag{12.2}$$

Notice that while the TOA equation (12.1) defines in the 3D space a sphere, with center in the Radar location and radius equal to the measured TOA, the DTOA equation (12.2) corresponds to an hyperboloid of revolution with two sheets, that has the sensor track as its major axes and the foci in the two sensor's location, S_1 and S_2. The intersections between the TOA sphere and the DTOA hyperboloid, that is a circle (we know which of the two sheets), provides the possible target's location. If the problem is 2D, as in Fig 12.1.b, there are only two possible solutions, that can be discriminated if the system is looking only on one side, *left or right*. The location in the 3D world, would then require another measure of TOA, say on an axis orthogonal to x and y in the picture.

In RADAR and SONAR, DTOA are exploited either moving the sensor, or the object. In any case, we assume that the two measures are *very close*, compared to the target. Let us then write the distance:

$$R = \sqrt{x_p^2 + (y_p - y)^2}$$
$$\left.\frac{\partial R}{\partial y}\right|_{y=0} = -\frac{y_p}{R}$$

and then:

$$\Delta_R \simeq -\frac{y_p}{R} B \simeq -B \sin\psi \tag{12.3}$$

as it could already be derived just by approximating the arc in Fig 12.1.a with a straight line. Notes that we could get the same result in vector form:

$$\Delta \left|\overrightarrow{R}(P)\right| = \left|\vec{R}_2\right| - \left|\vec{R}_1\right| \simeq \overset{\circ}{\overrightarrow{R}} \cdot \frac{\overrightarrow{R}}{\left|\overrightarrow{R}\right|}$$
$$= \overrightarrow{B} \cdot \vec{U}_R \tag{12.4}$$

$\vec{U}_R$being the range versor.

As a consequence, *the curves at constant DTOA identifies cones*, curves at *constant direction*, these cones are just the asymptotes of the hyperbolas, as shown in dotted lines in Fig 12.1.b.

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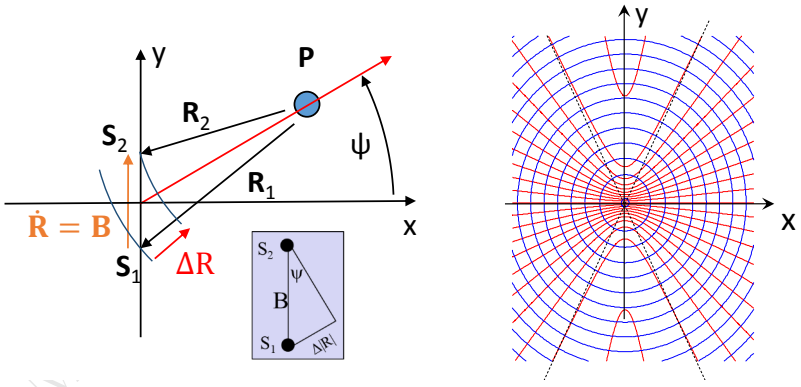


Figure 12.1: Location in the 2D/3D domain. Adding a second observation, we can measure two ranges. The difference between the two ranges corresponds to a rotational hyperboloid, that can be approximated asymptotically by a cone.

The figure Fig 12.1.b reports the curves for values of Δ_R uniformly spaced: notice that the curves are dense close to $\psi = 0$, i.e. when the target is at 90° with the sensor and quite sparse when the target is in the direction the sensor is moving. The same result can be shown by differentiating (12.3):

$$\Delta(\Delta_R) \simeq \frac{\partial(B\sin\psi)}{\partial\psi}\Delta\psi = B\cos\psi \cdot \Delta\psi \tag{12.5}$$
$$= B_P \Delta\psi$$

where B_p is the baseline projected across the target direction. The consequence is that, for improving detection, we need the two sensors looking orthogonal with respect to their

distance, like for eyes and hears. Thus, if we can approximate $\psi_0 \approx 0$ then we get from (12.3):

$$\Delta_R \simeq B \sin(\psi + \psi_0) \simeq \psi \cdot B_p \simeq \frac{B_p}{R} y \tag{12.6}$$

The DTOA provides, at most, a measure of the second coordinate, y, whereas TOA provides the first, x. However, the DTOA is much less sensitive, due to the ration B_p/R.

12.2 TOA and DTOA: a statistical signal analysis perspective

Let us approach the 3D location be exploiting three Radars, Sonars, or whatever ranging devices. The generic target to be located is identified by in the cartesian reference:

$$\mathbf{P} = \begin{bmatrix} x(t) \\ y(t) \\ z(t) \end{bmatrix}$$

the goal is to retrieve the position of the target at different times, that is its motion, and estimate the accuracy achievable. Our observables are the range - Line Of Sight distances- estimated by each Radar, as detailed in section 9.1. Let then

$$\hat{\mathbf{R}} = \begin{bmatrix} R_1 + w_1 \\ R_2 + w_2 \\ \cdots \\ R_N + w_1 \end{bmatrix} = \mathbf{R} + \mathbf{w} \tag{12.7}$$

be the observables, the true range, $\mathbf{R}$, affected by noise. We need at least N=3 to provide a proper identification in the 3D, more to improve the knowledge.

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12.2.1 Location equations

Let us model the true range as:

$$\sqrt{(x - x_k)^2 + (y - y_k)^2 + (z - z_k)^2} \approx R_k$$

that can be combined with the stochastic model for the observations, (12.7) in compact matrix formulation:

$$\mathbf{R} + \mathbf{w} = \mathbf{f}(\mathbf{P}) \tag{12.8}$$

12.2.2 Solution

In order to locate the target, we have to solve the non linear equations. This is a well known problem, that can be solved in closed form in the 2D, while there are several methods in the 3D - GNSS literature is quite rich in that.

Here we just take advantage of Newton, a general purpose approach, that iterates by linearizing (12.8) along a known position, $\mathbf{P}_n$

$$\mathbf{f}(\mathbf{P}) \approx \mathbf{f}(\mathbf{P}_n) + \left.\frac{\partial \mathbf{f}}{\partial \mathbf{P}}\right|_{\mathbf{P}=\mathbf{P}_n} \mathbf{\Delta}_P = \mathbf{f}(\mathbf{P}_n) + \mathbf{J}(\mathbf{P}_n) \cdot \mathbf{\Delta}_P \tag{12.9}$$

where $\mathbf{J}(P_0)$ is the Jacobean:

$$\mathbf{J} = \{J_{ij}\} = \left\{ \frac{\partial f_i}{\partial x_j} \right\}$$

Newton–Raphson is quite recommended when derivatives are known in closed from:

$$\frac{\partial f_k}{\partial x} = \frac{(x - x_k)}{\sqrt{(x - x_k)^2 + (y - y_k)^2 + (z - z_k)^2}}$$
$$\frac{\partial f_k}{\partial y} = \frac{(y - y_k)}{\sqrt{(x - x_k)^2 + (y - y_k)^2 + (z - z_k)^2}}$$
$$\frac{\partial f_k}{\partial z} = \frac{(z - z_k)}{\sqrt{(x - x_k)^2 + (y - y_k)^2 + (z - z_k)^2}}$$

The linear equation system (12.9) can be inverted finding a (hopefully) better solution:

$$\begin{aligned} \mathbf{f}(\mathbf{P}) &= \mathbf{0} \\ \mathbf{J}(\mathbf{P}_n) \cdot \mathbf{\Delta}_P = \mathbf{f}(\mathbf{P}_n) &\Rightarrow \mathbf{\Delta}_P = \mathbf{J}^{-1}(\mathbf{P}_n) \cdot \mathbf{f}(\mathbf{P}_n) \\ \mathbf{P}_{n+1} &= \mathbf{P}_n + \mathbf{\Delta}_P \end{aligned} \tag{12.10}$$

and iterate up to convergence. Usually if Newton converges, it is quite fast, therefore one can just limit the number of iteration. The closest the initial solution, the higher the likely of convergence. In (12.10) we assume that the Jeacobean$\mathbf{J}$ is invertible, that implies three measures, and non singular (say not two from collocated Radars). In the general case, a pseudo-inversion is suited:

$$\begin{aligned} \mathbf{J} \cdot \mathbf{\Delta}_P &= R + \mathbf{w} \\ (\mathbf{J} \cdot \mathbf{\Delta}_P - \mathbf{f}_n) &= \mathbf{w} \end{aligned}$$

The minimization of the L2 norm of the error is provided by Wiener solution [6], say imposing the in correlation between gradient and unknowns:

$$\begin{aligned} E\left[\mathbf{f}_n^T (\mathbf{J} \cdot \mathbf{\Delta}_P - \mathbf{f}_n)\right] &= 0 \\ \mathbf{\Delta}_P &= \left(\mathbf{J}^T \mathbf{R}_{fn} \mathbf{J}\right)^{-1} \mathbf{R}_{fn} = \mathbf{J}_f^{\dagger} \end{aligned}$$

where $\mathbf{J}_f^{\dagger}$ is the inversion in the norm of f.

12.2.3 Accuracy

In order to evaluate the accuracy we once again linearize, but close to the true position , $\mathbf{P}_0$, that is assuming small errors (otherwise, it is less useful to evaluate performances):

$$\mathbf{f}(\mathbf{P}) \approx \mathbf{f}(\mathbf{P}_0) + \mathbf{J}\,(P_0) \cdot \mathbf{\Delta}_P$$

and we assume that the three (or N) range measures are given with an error, w:

$$\mathbf{f}(\mathbf{P}) = \mathbf{f}(\mathbf{P}_0) + \mathbf{w}$$

Then, the error on the measures is:

$$\mathbf{w} = \mathbf{J}(P_0) \cdot \mathbf{\Delta}_P$$

computed in correspondence of the true solution.

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The relation so far found is linear and can be exploited to link the statistics of measures error and location:

$$\begin{aligned} E[\mathbf{w}] &= \mathbf{J}(P_0) \cdot E[\mathbf{\Delta}_P] \\ \mathbf{C}_w &= E[\mathbf{w}\mathbf{w}^T] = \mathbf{J}(P_0) \cdot E[\mathbf{\Delta}_P \mathbf{\Delta}_P^T] \cdot \mathbf{J}^T(P_0) \end{aligned}$$

From the first equation we see that the location error, $\mathbf{\Delta}_P$, is zero mean, provided that the range errors are so - and this is true as range error comes from thermal noise. The second equation allows for the estimation of location error covariance:

$$\mathbf{C}_{\mathbf{\Delta}_P} = E[\mathbf{\Delta}_P \mathbf{\Delta}_P^T] = \begin{bmatrix} \sigma_x^2 & & \\ & \sigma_y^2 & \\ & & \sigma_z^2 \end{bmatrix}$$

Eventually, we (pseudo-)invert the linear relation to solve for $\mathbf{C}_{\mathbf{\Delta}_P}$:

$$\mathbf{J}^T(P_0) \cdot \mathbf{C}_w \cdot \mathbf{J}(P_0) = \mathbf{J}^T(P_0)\mathbf{J}(P_0) \cdot \mathbf{C}_{\mathbf{\Delta}_P} \cdot \mathbf{J}^T(P_0)\mathbf{J}(P_0)$$

$$\left(\mathbf{J}^T(P_0)\mathbf{J}(P_0)\right)^{-1} \mathbf{J}^T(P_0) \cdot \mathbf{C}_w \cdot \mathbf{J}(P_0) \left(\mathbf{J}^T(P_0)\mathbf{J}(P_0)\right)^{-1} = \mathbf{C}_{\mathbf{\Delta}_P}$$

We assume here following the same error in all the locations, and independent

$$\mathbf{C}_w = \sigma_w^2 \mathbf{I}$$

then

$$\begin{aligned} \mathbf{C}_{\mathbf{\Delta}_P} &= \sigma_w^2 \cdot \left(\mathbf{J}^T(P_0)\mathbf{J}(P_0)\right)^{-1} \mathbf{J}^T(P_0) \cdot \mathbf{J}(P_0) \left(\mathbf{J}^T(P_0)\mathbf{J}(P_0)\right)^{-1} \\ &= \sigma_w^2 \cdot \left(\mathbf{J}^T(P_0)\mathbf{J}(P_0)\right)^{-1} \end{aligned}$$

and we look over the diagonal to find accuracies in three directions is

$$E\left[\mathbf{\Delta}_x \mathbf{\Delta}_x^T\right] = \sigma_r^2 \mathbf{C}_J = \begin{bmatrix} \sigma_x^2 & E[xy] & E[xz] \\ E[yx] & \sigma_y^2 & E[yz] \\ \cdots & \cdots & \sigma_z^2 \end{bmatrix}$$

12.2.4 Total accuracy

The accuracy has been insofar computed only along the three axis, but may be the shape of the resolution cell is highly anisotropic: it is important to compute accuracy over all the directions. For that, let:

$$\mathbf{u} = \begin{bmatrix} \cos\psi\cos\theta & \cos\psi\sin\theta & \sin\psi \end{bmatrix}^T$$

be the generic versor along any possible direction in the space.:

$$\begin{aligned} \mathbf{u}^T\mathbf{u} &= \begin{bmatrix} \cos\psi\cos\theta & \cos\psi\sin\theta & \sin\psi \end{bmatrix} \begin{bmatrix} \cos\psi\cos\theta & \cos\psi\sin\theta & \sin\psi \end{bmatrix}^T \\ &= 1 \end{aligned}$$

Then the component in that direction is

$$\mathbf{P}_u = \mathbf{P}^T\mathbf{u}$$

and the variance of the error is:

$$\begin{aligned} \boldsymbol{\sigma}_u^2 &= E\left[\left(\mathbf{\Delta}_P^T\mathbf{u}\right)^T \left(\mathbf{\Delta}_P^T\mathbf{u}\right)\right] \\ &= \mathbf{u}^T \mathbf{C}_{\mathbf{\Delta}_P} \mathbf{u} \end{aligned}$$

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12.2.5 Examples

The k-th RADAR is located with azimuth ψ_r and elevation θ_r at distance ρ_r with respect to P:

$$\mathbf{R}_k = \begin{bmatrix} \rho_k \cos\theta_k \sin\psi_k \\ \rho_k \cos\theta_k \cos\psi_k \\ \rho_k \sin\theta_k \end{bmatrix} = \begin{bmatrix} x_k \\ y_k \\ z_k \end{bmatrix}$$

Let us furthermore assume the target is in the origin, and at equal distance from the three radars R:

$$\begin{aligned} \frac{\partial f_k}{\partial x} &= \frac{(0 - x_k)}{R} = -\cos\theta_k \cos\psi_k \\ \frac{\partial f_k}{\partial y} &= \frac{(0 - y_k)}{R} = -\cos\theta_k \sin\psi_k \\ \frac{\partial f_k}{\partial z} &= \frac{(0 - z_k)}{R} = -\sin\theta_k \end{aligned}$$

. Then the Jacobean $\mathbf{J}$:

$$\mathbf{J} = -\begin{bmatrix} \cos\theta_1 \cos\psi_1 & \cos\theta_1 \sin\psi_1 & \sin\theta_1 \\ \cos\theta_2 \cos\psi_2 & \cos\theta_2 \sin\psi_2 & \sin\theta_2 \\ \cos\theta_3 \cos\psi_3 & \cos\theta_3 \sin\psi_3 & \sin\theta_3 \end{bmatrix}$$

We assume three cases of interest.

12.2.5.1 Measures along the three axis

The first with three Radar directed along the three axes:

	ψ	θ
R_1	0	0
R_2	$\pi/2$	0
R_3	$-$	$\pi/2$

Then

$$\mathbf{J} = -\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

and finally:

$$\mathbf{C}_{\mathbf{\Delta}_P} = \sigma_w^2 \cdot \left(\mathbf{J}^T(P_0)\mathbf{J}(P_0)\right)^{-1} = \begin{bmatrix} \sigma_w^2 & & \\ & \sigma_w^2 & \\ & & \sigma_w^2 \end{bmatrix}$$

The result was expected, in fact each sensors LOS is along a specific axis, and it is almost insensible to displacements in the orthogonal plane.

12.2.5.2 Closely located RADAR

As a second case, we assume that RADAR are quite close together, with a main direction ψ_0 are quite close together, with spacing $\pm\Delta\psi/2$, two of them in the plane of the target, $\theta = 0$,

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and the third looking with angle $\Delta\theta$ as in the following table.

	ψ	θ
R_1	$\psi_0 + \frac{\Delta\psi}{2}$	0
R_2	$\psi_0 - \frac{\Delta\psi}{2}$	0
R_3	ψ_0	$\Delta\theta$

Then we compute the accuracies over the three coordinates, for different horizontal and vertical apertures are shown in Fig. 12.2

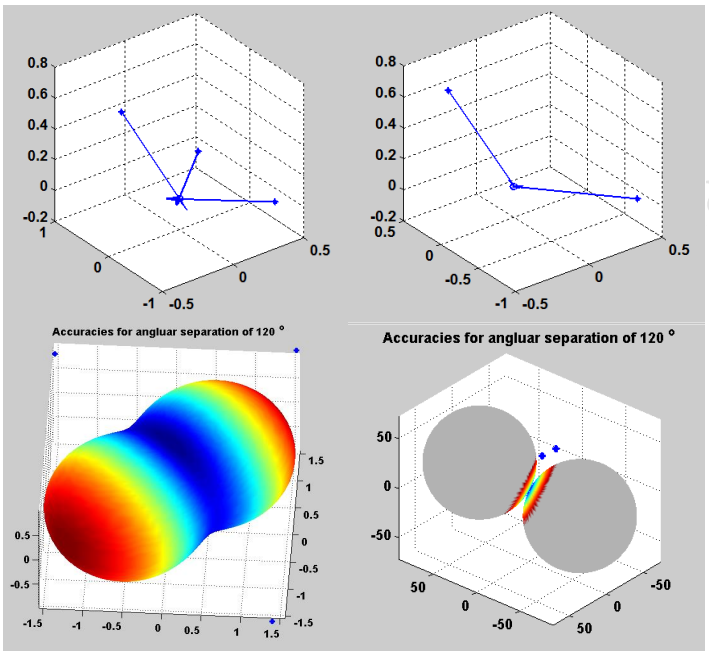


Figure 12.2: Location accuracy (lower panels), as function of the direction, measured for two configurations: three Radars with angular separation of 120°, left, and two Radars (right). The location of Radars with respect to target is shown in the upper panels.

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Chapter 13

Side Looking Aperture Radar (SLAR)

The Side Looking Aperture Radar is the simplest extension of the monopulse Radar to provide 2D images. The geometry is that already studied for a push-broom acquisition, say the radiometer, in section 4.2. The spaceborne Radar sketched in Fig. 13.1 refers to Earth Remote Sensing ERS, the first European Space Agency satellite, launched in '91, and a second one, ERS-2 launched in '95 and lasted up to 2011. ERS was in a LEO orbit as shown in the figure, sun-synchronous, that implies that all the observations were made at the same local solar time, 10.30 and 22.30 for ascending / descending orbit.

Such Radar was indeed a ■Synthetic Aperture Radar■, SAR. A SAR differs with respect to the SLAR concept by the fact that it synthesize a big antenna by combining different observations in different positions. Apart for this, that implies a much better azimuth resolution, the properties of a SAR and a SLAR images does not change for the case of fixed target.

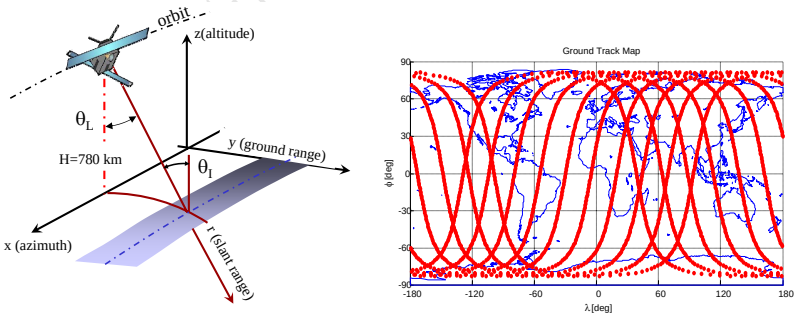


Figure 13.1: ESA Earth Remote Sensing (ERS) geometry. Right: coverage after few orbits.

13.1 Resolution, slant range and slopes distortions

The SLAR scans the space by means of an antenna that is quite long in the track (or orbit) direction, so that it can achieve the desired azimuth resolution, as from (3.8):

$$\rho_a = \Delta\psi_a \cdot R = \frac{\lambda}{L_a} \cdot R \tag{13.1}$$

Then, the complex samples acquired by each scan are stored column-wise in the raw data matrix. Each column correspond to an across-track acquisition, in the case, here assumed that the antenna is pointing broadside - that provides the best resolution as seen in the previous section. The *fast time* coordinate is then related to the sensor-target distance, or slant range, as shown in 13.2 on the right.

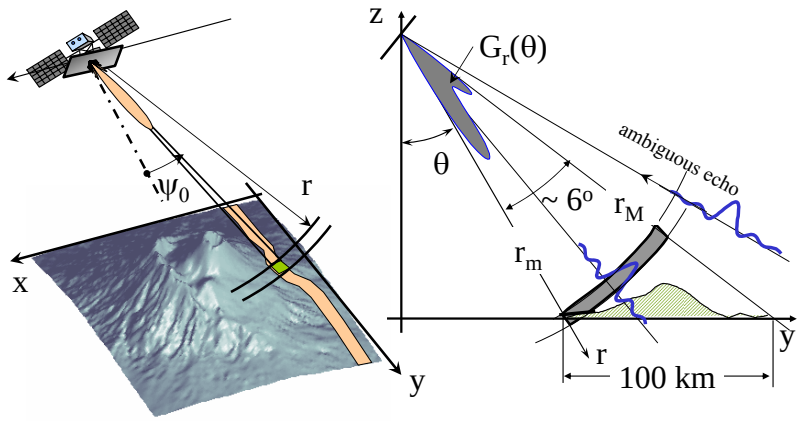


Figure 13.2: 3D geometry for a SLAR/SAR system, on the left, and 2D geometry in the along-track, height plane *(y,z)* (normal to the sensor track) on the right.

If we assume that scattering is from the surface, there is one-to-one correspondence between the slant range, and the ground location of the targets, that is *ground-range*, the y-coordinate in the image. Being the Radar a ranging device-that is the echo is placed in the data matrix according to the distance, the ground-to-slant rage depends on the local slopes, introducing a slope-dependent distortion of the image that is shown in Fig. 13.3.

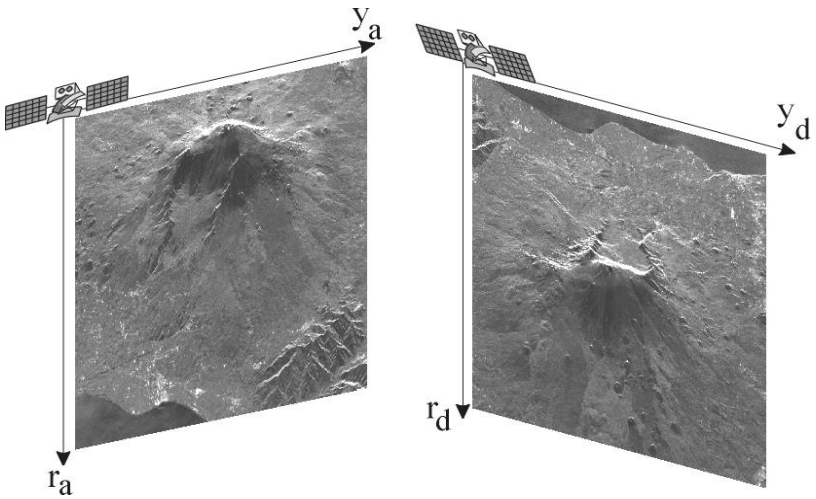


Figure 13.3: Examples of two SAR images of the same scene, Mt. Etrna, taken by two different view point. Notice the complementary, slope-dependent distortions.

The **ground range resolution** depends on the local slope, and can be related to the

slant range resolution as shown in Fig. 13.4 (left):

$$\rho_g = \frac{\rho_r}{\sin\theta_i} = \frac{\rho_r}{\sin(\theta-\alpha)} = \frac{c}{2B\sin(\theta-\alpha)} \tag{13.2}$$

θ_i being the local incidence angle and $\theta$the incidence angle with respect to a flat surface.

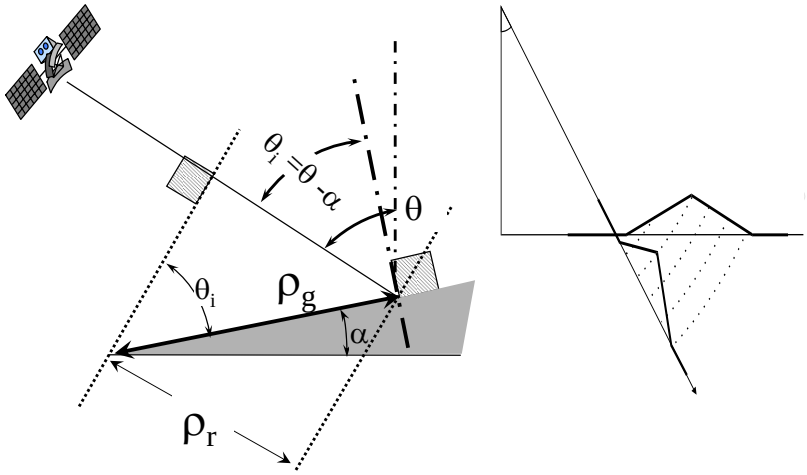


Figure 13.4: Left: SAR/SLAR geometry, exploited to find the relation between the ground range resolution cell and the slant range resolution cell. Right: perspective deformation of a SAR/SLAR image due to the uniform sampling in range typical of RADAR.

The result is a shrinking of the fore-slopes, for angles $-\pi < \alpha < -\pi+\theta$ due to the increase of ground resolution length. For the same reason, the back-slopes, correspondent to angles $-\pi+\theta < \alpha < 0$, are expanded, as shown in Fig. 13.5.

A quantitative evaluation of the ground range sampling variation with local slope is given in Fig. 13.6, that just exploits (13.2).

An example of the slope related distortions in a SAR/SLAR image is provided in Fig. 13.7.

13.2 Geocoding

The SLAR image comes naturally in a reference where each pixel is identified by the coordinates pair: slant range or fast time, along track or slow time. These two coordinates, that correspond to the the TOA and the DOA discussed in the previous section, are not sufficient to locate the target in the 3D space, and a further information is to be provided. This information, usually, comes from a Digital Elevation Model, that provides height for each pixel, or from another SAR/SLAR acquisition by a different view point, ideally in the plane orthogonal to the orbit.

The procedure that computes the 3D location for each pixel is named “geocoding”. In the case shown in Fig. 13.2, that is a spaceborne geometry, the 3D location is achieved by

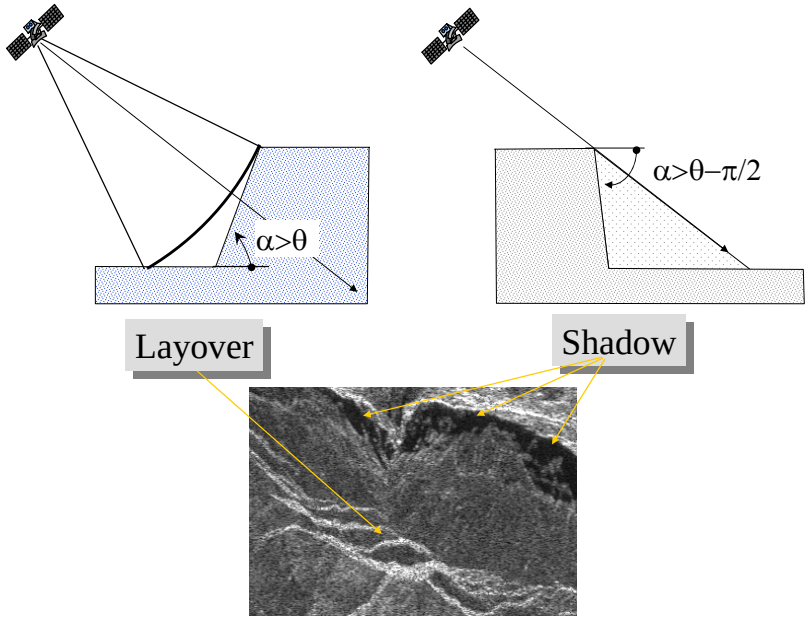


Figure 13.5: Lay-over e shadowing in SAR/SLAR images.

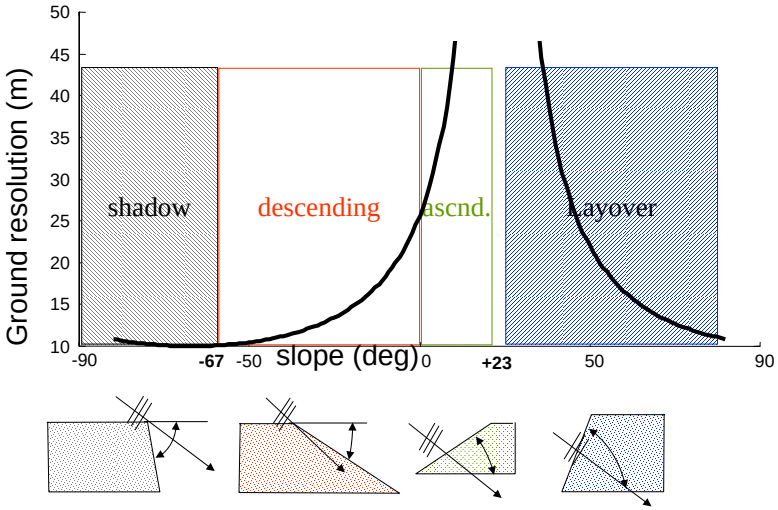


Figure 13.6: Ground resolution versus slant resolution for a 23° incidence angle RADAR.

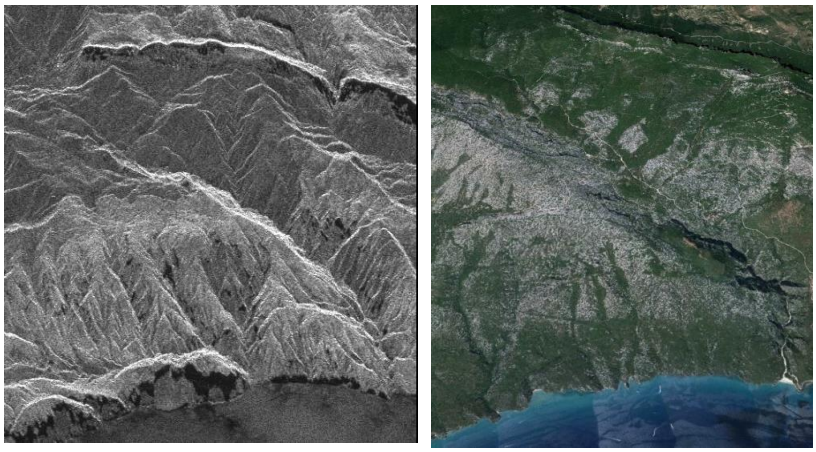


Figure 13.7: Left: ERS image of coast near Dorgali (Sardinia). Right: corresponding Google earth optical image.

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solving the system of three equations:

$$\begin{aligned} |\mathbf{R}| &= ct_0/2 \\ \dot{\mathbf{R}} \cdot \frac{\mathbf{R}}{|\mathbf{R}|} &= \psi \\ \frac{x_0^2 + y_0^2}{(R_e)^2} + \frac{z_0^2}{(R_p)^2} &= 1 \end{aligned}$$

where $\mathbf{R} = (\mathbf{S} - \mathbf{P})$ is the slant rage vector, as the difference between the satellite vector, $\mathbf{S}(\tau_0)$, and the target vector, $\mathbf{P}(x_0, y_0, z_0)$. The unknown are the the three cartesian coordinates, (x_0, y_0, z_0) of the target, the data are: the fast and slow time, (t_0, τ_0), a description of the satellite track, $\mathbf{S}(\tau_0)$, and the beam pointing angle with respect to normal, ψ, that we assume null for broadside. The three equations describe in the space respectively a sphere, a core or a plane for $\psi = 0$ and an ellipsoid modelling flat Earth, as shown in the figure. If a DEM is given, the last equation - the Earth ellipsoid, can be changed by adding the height for each target h(x,y,z).

The geocoding can be implemented in many ways. In the case of flat earth, the equation system above can be solved by Newton in a very fast way. Another option, is to provide a dense grid on the DEM surface, then for each point $P\,(x_n, y_n, z_n)$ measure the distance from the satellite (given the orbit) obtaining the corresponding coordinate pairs (t_m, τ_m). Once that the correspondence between the two sets is given, one can just pick the m-th pair closest to the actual range and time, (t, τ) and from this the 3D location, given by the corresponding n-th tern. An example of geocoding the regular grid in (t, τ) into the 3D (x, y, z) space is given in Fig. 13.2.

13.3 Radar equation for SLAR

The Radar equation for SLAR is derived from monostatic Radar equation (8.7), but assuming an homogeneous target on the ground with backscatter coefficient σ^0 and integrating over

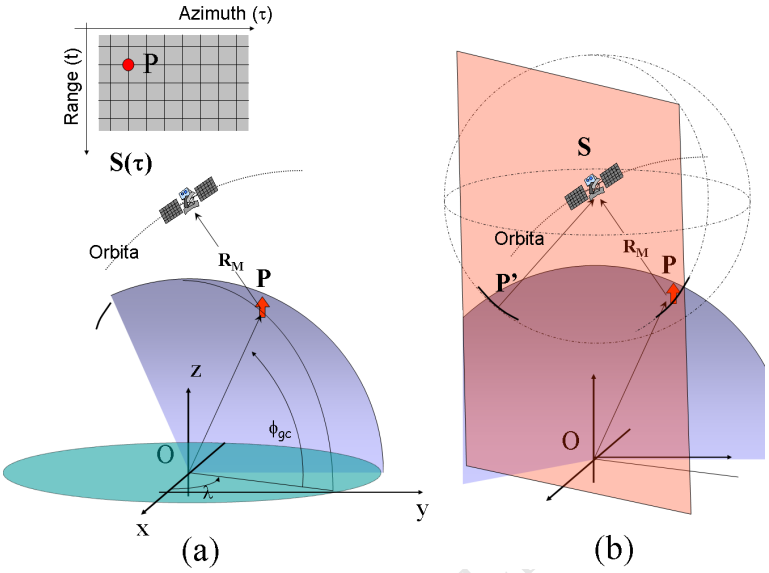


Figure 13.8: Geocoding a spaceborne SAR/SLAR image.

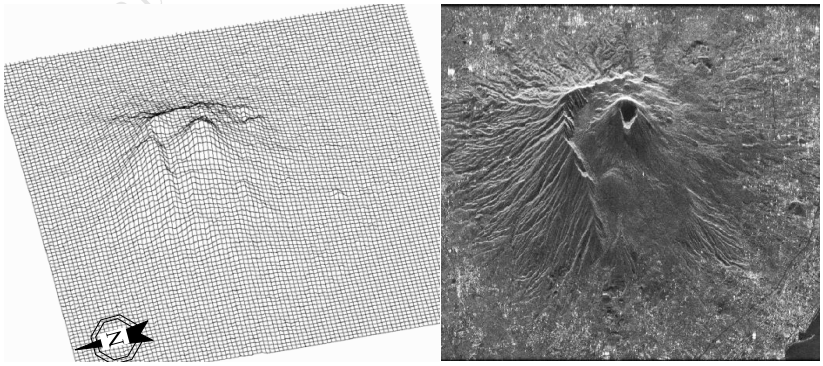


Figure 13.9: Left: SAR acquisition grid in slant, azimuth, geocoded in the 3D space. Right: SAR image of Mt. Vesuvio.

the surface illuminated by the chirp on the ground :

$$P_R = P_t \frac{G_{el}^2(r_0)\cdot\lambda^2}{(4\pi)^3 r_0^4}\eta \int \sigma^0 \frac{cT_c}{2\sin\theta} G_a^2(\psi) r(\psi) d\psi$$
$$\simeq P_t \frac{G_{el}^2(r_0)\cdot\lambda^2}{(4\pi)^3 r_0^3}\eta\sigma^0 \frac{cT_c}{2\sin\theta} G_a^2(\psi_0)\cdot\Delta\psi \tag{13.3}$$

where r_0 is the target range in the pointing direction, ψ_0, $G_{el}$the antenna gain in elevation, measured in the direction of the target, T_c the chirp duration, θ the incidence angle (we assumed flat Earth), $G_a$the normalized antenna gain in along track, and the integral expresses the RCS. Notice that we assumed the antenna pattern as separable: $G(r_0,\psi) = G_{el}(r_0)\cdot G_a(\psi)$, that applies for rectangular antennas, that are mostly used. In deriving (13.3) we have ignored the variation of range with the squint angle, $r(\psi) \sim r_0$ by assuming a small along track aperture - that is indeed the case in order to get good azimuth resolution. The noticeable result is that - for distributed targets - raw data power is inversely proportional to r^{-3}, and not r^{-4}as it would be for a point target.

Eventually we rearrange (13.3) by relating the angular aperture to the azimuth resolution, ρ_a

$$\Delta\psi = \frac{\lambda}{L_a} = \frac{\rho_a}{r_0}$$

The received power is then:

$$P_r = P_t \frac{G^2(r_0,\psi_0)\cdot\lambda^2}{(4\pi)^3 r_0^4}\eta \frac{cT_c\sigma^0\rho_a}{2\sin\theta}$$

In order to compute the SNR after matched filter, that is range compression, we observe that both power spectrum densities of distributed target reflectivity and noise are white, therefore the matched filter does not affect SNR:

$$SNR_{MF} = SNR_{raw} = \frac{P_r}{N_0 B}$$
$$= P_t \left(\frac{G^2(r_0,\psi_0)\cdot\lambda^2}{(4\pi)^3 r_0^4}\eta\right)\left(\sigma^0\rho_a\rho_{gr}\right)\left(\frac{T_c}{N_0}\right) \tag{13.4}$$

where we have introduced the range resolution as in (13.2).

Equation (13.4) can be immediately interpreted as the ratio between the power backscattered by a a distributed target over the resolution cell, and the noise integrated in the matched filter bandwidth, $1/T_c$. This equation provides the SNR on the focused image under the assumption that the system along-track sampling matches the azimuth resolution. In the case when many samples are given per single resolution cell, SNR improves by the oversampling ratio:

$$k_o = \frac{\rho_{az}}{\frac{v_s}{PRF}}$$

13.3.1 Noise Equivalent Sigma Zero

The Noise Equivalent Sigma Zero is the backscatter coefficient that provides the same power as the noise, leading to SNR=1, then:

$$P_t \left(\frac{G^2(r_0,\psi_0)\cdot\lambda^2}{(4\pi)^3 r_0^4}\eta\right)\left(\sigma^0_{NESZ}\rho_a\rho_{gr}\right)\left(\frac{T_c}{N_0}\right) = 1$$

that can be solved for σ^0_{NESZ}.

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13.4 Calibration: sigma, beta, gamma nought

The Radar equation for SAR can be used to estimate the absolute backscatter coefficient from the multi-look averaged intensity image, say an estimate of the local power. In a general case the Radar Cross Section is retrieved by Radar equation (13.3):

$$RCS = \frac{P_r}{P_t} \cdot \frac{(4\pi)^3 \, r_0^4}{G^2(r_0, \psi_0) \cdot \lambda^2 \cdot \eta} \tag{13.5}$$

Then, for general case of sloped terrain, the backscatter coefficient should be rertieved:

$$\sigma^0 = RCS \cdot \frac{2 \sin(\theta - \alpha)}{BT_c \cdot \rho_{ca} c} = \frac{\hat{RCS}}{\rho_a \rho_{rg\alpha}} \cdot \frac{1}{BT_c} \tag{13.6}$$

where $\rho_{rg\alpha}$emphasizes the true ground range. A multilooked, intensity image corrected according to (13.6) is defined as "σ^0**calibrated**". This calibration is however not trivial, since it requires knowledge of topography and assume terrain to be locally flat.

In order to avoid the problem, SLAR/SAR images are better calibrated with respect to *slant range resolution*, that is :

$$\beta^0 = RCS \cdot \frac{2}{BT_c \cdot \rho_{ca} c} = \frac{RCS}{\rho_a \rho_r} \cdot \frac{1}{BT_c}$$

This would allow a simple estimate of the backscatter:

$$\sigma^0 = \beta^0 \sin(\theta - \alpha)$$

However, for non constant slopes, other relations can be written. For example, in the case of an homogeneous volume, say for forests, the backscatter would be proportional to the surface orthogonal to line of sight. In this case we define a volumetric backscatter coefficient, γ^0as:

$$\begin{aligned} \gamma^0 &= RCS \cdot \frac{2 \tan\theta}{BT_c \cdot \rho_{ca} c} = \frac{RCS}{\rho_a \rho_r} \cdot \frac{1}{BT_c} \\ &= \beta^0 \tan(\theta) \end{aligned}$$

The rain forest is such a target, that has been found pretty stable over time, with $\gamma^0 \sim -6.4$dB, and over a wide area. For this reason it is quite used as a reference for both absolute and relative calibration, as shown in 13.4.

13.4.1 Range azimuth ambiguity

The SLAR is intended for fixed target, therefore there is no need in estimating velocity, that is Doppler. However, the observation time along track is bounded to be larger than the ambiguous slant range, as in (9.27), and the need to scan one resolution cell:

$$\begin{aligned} \frac{v_s}{\rho_a} < f_p < \frac{c}{2R_M} \\ \Rightarrow \frac{v_s}{\rho_a} < \frac{c}{2R_M} \\ \Rightarrow \frac{R_M}{\rho_a} < \frac{c}{2v_s} \end{aligned} \tag{13.7}$$

We get a constraint that is equivalent to the one found for space- or air-borne systems and radiometers in (4.5): we wish the term on the left to be infinite, but in the real world we can increase the across-track swath, the FOV, only at the price of a loss in azimuth resolution.

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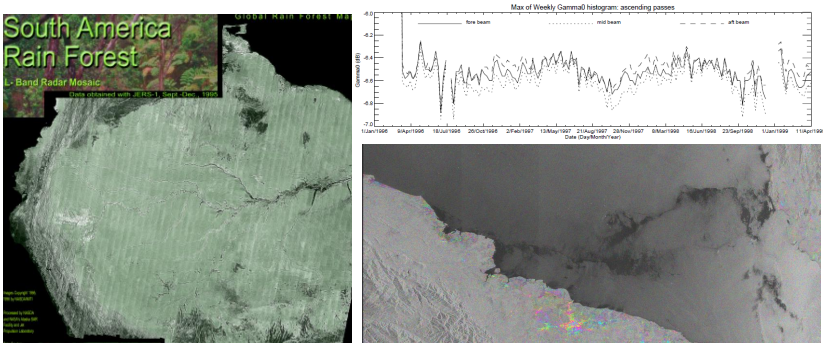


Figure 13.10: Left: mosaic of L-band SAR images of rain forest, calibrated as for , from [7]. Notice the uniform backscatter, independent on the incidence angle. The value was found close to -6.4 dB and pretty stable with time (right-upper panel). Right, lower panel: backscatter intensity, β_0, colored with interferometric fringes, of a C band image of Sumatra. Notice the variation with the local incidence angle, and the effect of currents and oil spills on the sea surface.

13.5 Example: design of a SLAR

Let us assume to design a SLAR with the specifications in Tab. 13.1.The goal is to dimension the antenna, to find the range resolution, the timing and the achievable Noise Equivalent Sigma Zero.

Data	Symbol	**Value**	**Unit**
Incidence angle (mid range)	θ_i	23	deg
Ground swath Coverage	ΔR_{gr}	1.00E+05	m
Sensor height	h_s	7.85E+05	m
Sensor velocity	v	7400	m/s
Mean reflectivity	σ^0	-6	dB
Earth Radius	R_e	6.37E+06	m
Frequency	f_0	5.33E+09	Hz
Antenna length (along track)	L_a	10	m
Duty cycle	d_c	6	%
Ground Range resolution	ρ_r	25	m
Noise Factor			
Peak power	Pt	2000	W

Table 13.1: Specifications of the SLAR system to design.

13.5.1 Antenna

Let us start designing the *antenna*, in particular we assume a rectangular reflector of size L_a (azimuth o along track) $\times$ L_b.(elevation). The aperture in azimuth is conditioned by the

azimuth resolution:

$$\rho_{az} = \Delta\psi \cdot R_t$$
$$\Delta\psi = \frac{\lambda}{L_a}$$

However, in this case L_a is given, therefore we will use the expression to compute the resolution achievable.

We can compute the mid range distance, at the nominal incidence angle, by exploiting the Carnot theorem applied to the geometry in Fig. 13.11:

$$R_s^2 = R_t^2 + R_e^2 - 2R_eR_t\cos(\pi - \theta_i)$$
$$R_s = R_e + h_s$$

that leads to $R_t = 845.km$. We get the 3dB azimuth lobe, $\Delta\psi$=5.7 mrad, and the resolution $\rho_{az} = 4.8$ km.

The elevation antenna beam should be constrained to the on ground range interval. By referring again to the geometry Fig. 13.11 we get the coverage as:

$$\Delta R_{gr} = ((\theta_{i2} - \theta_{i1}) - (\theta_{l2} - \theta_{l1}))R_e \tag{13.8}$$

We get the look angle by applying the sine theorem:

$$\frac{R_s}{\sin(\pi - \theta_i)} = \frac{R_e}{\sin\theta_l}$$

We invert the two relations, obtaining a look aperture of $\Delta\theta = 8.3$ rad. The minimal antenna length is thus

$$L_b = 1.56\frac{\lambda}{\Delta\theta}$$

where the factor 1.56 accounts for the fact that the same antenna is used for transmitting and receiving, therefore the 3dB lobe is smaller than the usual one. We get $L_b$that sum to 53 cm. In the design, we will use an antenna $L_b = 1$ with non uniform illumination to improve sidelobes.

13.5.2 Timing

Let us now compute the PRF. Its value is upper bounded by the range ambiguity:

$$T_{PRI} > \frac{2\Delta R}{c} - 2T \tag{13.9}$$
$$f_{PRF} < \frac{1}{1 + 2d_c}\frac{c}{2\Delta R}$$

and lower bounded by the sampling. Assume that we get at least one sample per resolution cell

$$f_{PRF} > \frac{v}{\rho_{az}}$$

we get 1.5 Hz. There is a large freedom in fixing the PRF. In theory we would get the highest possible value to get the highest mean power, however a reasonable margin should be added for reducing ambiguities. In order to evaluate this margin, we should have a constraint on Signal-to-Ambiguity-Ratio and estimate the antenna pattern in elevation. Such computations are too complicate for this example. We just say that we select PRF=1.7 kHz.

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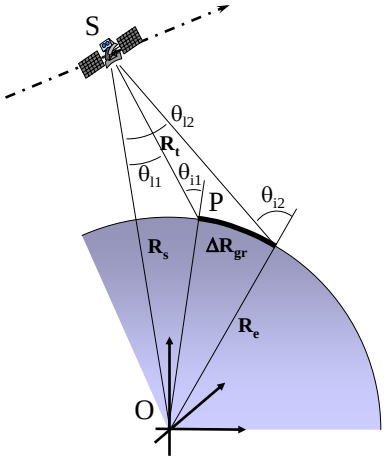


Figure 13.11: Geometry for a spaceborne SLAR.

13.5.3 Range resolution

The range resolution depends on the bandwidth and the geometry:

$$\rho_r = \frac{c}{2B \sin \theta_i}$$

We then obtain B=16 MHz.

13.5.4 Signal power at the receiver

We have now all the necessary elements for computing the transmitted power, given the NESZ. We exploit the RADAR equation:

$$P_r = \left(\frac{P_t G_T(\theta,\psi)\eta}{4\pi R^2}\right)(\sigma_o S)\left(\frac{1}{4\pi R^2}\right)\left(\frac{\lambda G_R(\theta,\psi)}{4\pi}\right)$$

where the numbers have been evaluated in Tab. 13.2. La received power is -107.7 dB_W.

13.5.5 Noise power at the receiver, NESZ

We evaluate the thermal noise power as:

$$P_n = N_0 B = FKT_0 B$$

where B is the bandwidth. The computation of the noise power is in Tab. 13.3. We get -127.3 dB_W, that is approximately 20 dB below the signal at the receiver.

13.5.6 Noise equivalent sigma zero

We can finally evaluate the noise equivalent sigma zero, that leads to SNR=0:

$$\sigma_z^0 = \sigma_{dB}^0 - SNR_{dB}$$

We get a sensitivity -25.5 dB.

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Data	**Value**	**Unit**	**dB$_W$**
Peak power	2.00E+03	W	33.0
Antenna tx losses	0.5		-3
Tx directivity	3.97E+04		46
Tx Path Loss	1.12E-13		-129.5
Losses (rain, atmosphere)	6.31E-01		-2.0
σ_0	2.5E-01		-6
Surface area	4.33E+07	m^2	76.4
Rx Path Loss	1.12E-13		-129.5
Rx directivity	3.97E+04		46
Rx losses	0.5		-3
$\lambda/4\pi$	2.52E-04	m^2	-36.0
Received Power	1.7E-11	W	-107.7

Table 13.2: Evaluation of the received power for the ENVISAT-ASAR RADAR.

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Definition	**Value**	**Dim.**	**dB**
noise factor	3		4.7
kBT$_0$		J	-204
noise bdwth	1.6E+07	Hz	72
Received Noise Power	1.85E-13	W	-127.2
SNR			19.5
Nesz			**-25.1**

Table 13.3: Evaluation of the noise power and the NESZ for the ENVISAT-ASAR sensor.

Chapter 14

Synthetic Aperture RADAR

The SLR just presented is a Radar system for imaging, that is unable to achieve a good resolution from long distances, say from airborne or spaceborne platform. A **SAR** (*Synthetic Aperture Radar*) gets this goal by synthesizing a huge antenna, as the combination of multiple along-track observations. In the conventional SAR, the sensor moves and the target is fixed, however the same principle can be applied if the sensor is fixed and the target moves, in a configuration named Inverse SAR (ISAR).

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Figure 14.1: Artistic impression of the ESA satellite SENTINEL-1.© European Space Agency (ESA).

14.1 The synthetic antenna and Doppler

A focused SAR gets fine azimuth resolution by synthesizing a wide antenna by means of many along track observations. The principle is the same of an antenna array, in 3.1, with two differences:

1) each observation is monostatic, in that at each position the system transmits and receives. This implies that we have to consider the two-way travel path, an that there is a delay between the receive of the first echo and the last in the array (scene may change),

2) at each sample, the array is synthesized by taking a sliding window on the $\pm N$ echoes centered on current position. In that way, we need to focus the array only in the broadside direction. This leads to the acquisition mode named STRIP MAP.

With reference to the geometry in Fig. 14.2, that us now reformulate the echo of the signal transmitted by the n-th antenna and reflected by the target $\mathbf{P}(r_0, \psi)$ that is identified that the range r_0 and the azimuth angle ψ. We take advantage of the *stop-and-go assumption*, for which the sensor stops, transmits the chirp, receives the echoes (may be not coming from the pulse just transmitted, but by an previous one) and then moves along track. So doing, we ignore the fact that the system is actually bistatic. We assume also an equivalent model where the target P transmits, but propagates at half speed (to account to the two-way range).

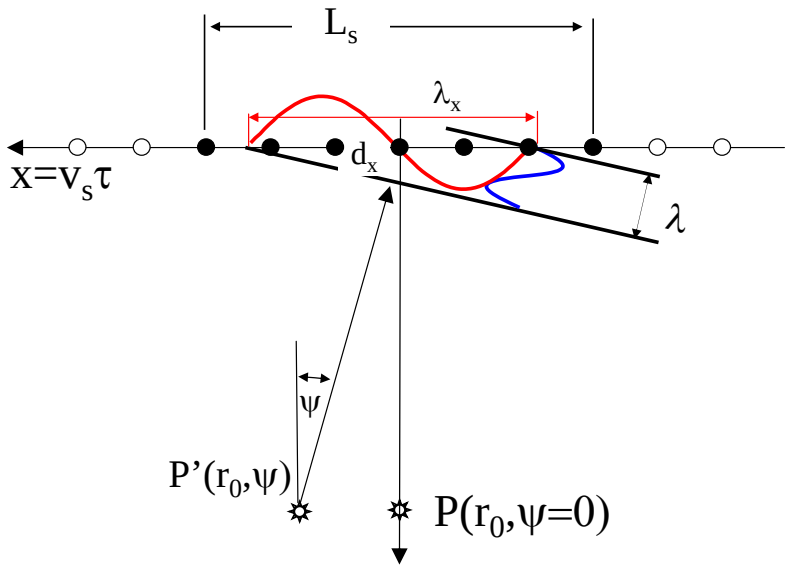


Figure 14.2: Synthetic antenna made by a set of monostatic observations.

The "re-transmitted" signal measured at the n-th sensor is, after (3.2)

$$h_P(t; n) = A(P, n) \cdot \exp\left(j\left(k \cdot 2r_n(P) - \omega t\right)\right) \tag{14.1}$$

where $A(P; n)$ is the complex number that accounts for both the RCS, the spherical divergence, the gain of the elementary antenna in n, etc. The distance between the target and the n-th antenna of the array can be approximated as in (3.3) leading to the received plane wave:

$$h_P(t; n) = A(P, n) \cdot \exp\left(j\frac{2\sin\psi_P}{\lambda} n d_x\right) \cdot \exp\left(-j\omega t\right)$$

The last exponential is removed by the demodulation, therefore, the baseband signal out of the n-th antenna is a plane wave at frequency

$$f_x = -\frac{2}{\lambda}\sin\psi_P \text{ [cyles/m]} \tag{14.2}$$

illustrated in Fig. 14.3.

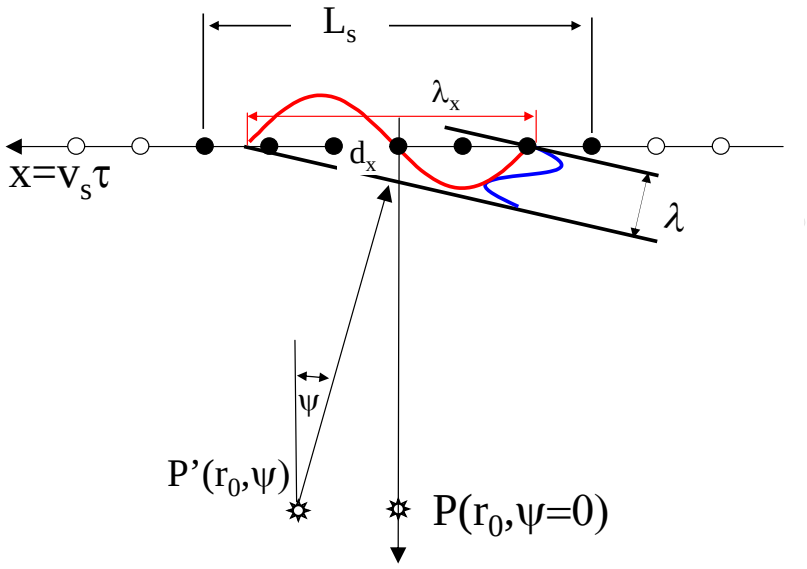


Figure 14.3: Synthetic antenna made by a set of monostatic observations.

14.1.1 Focusing

We can then apply the same analysis as for an array. Focusing the energy coming from an angle $\psi_0 is$achieved by phasing and adding contributions:

$$\begin{aligned} y(t;P) &= \sum_{n=-N}^{+N} w(n)\cdot A(P,n)\cdot\exp\left(-j2\pi f_x\cdot nd_x\right)\cdot\exp\left(-j\omega t\right) \\ &\simeq A(P,n)\cdot\exp\left(-j\omega t\right)\cdot\sum_{n=-N}^{+N} w(n)\cdot\exp\left(-j2\pi f_x\cdot nd_x\right) \\ &= s_T(P)\cdot W(\psi) \end{aligned}$$

where we have ignored the variation of the amplitude A(P,n) along the array (that we assume small, respect to the distance).

If the weights are constant in amplitude, we get a narrow beam, similar to (3.5):

$$G(\psi) = \frac{\sin\left(\pi\frac{2}{\lambda}\sin\left(\psi-\psi_P\right)N_s d_x\right)}{\sin\left(\pi\frac{2}{\lambda}\sin\left(\psi-\psi_P\right)d_x\right)} \simeq N_s\,\mathrm{sinc}\left(\frac{2\sin\left(\psi-\psi_P\right)L_s}{\lambda}\right) \tag{14.3}$$

$$L_s = N_s d_x$$

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where L_s is the length of the synthetic aperture. This is equivalent to "focusing" (approximate) with a resolution

$$\rho_{az} = \Delta\psi \cdot r_P = r_P \frac{\lambda}{2L_s} \tag{14.4}$$
$$\psi \ll \frac{\lambda}{2d_x}$$

that is can be made quite small by exploiting long synthetic apertures, Ls. Expression (14.4) compares with the resolution of a system with a real aperture antenna, yet of **double the size of the synthetic** aperture. This is a consequence of the monostatic acquisitions, that, by doubling distances, makes the system more sensitive to phase variations, like exploiting **half wavelength**.

14.1.2 Doppler and ambiguities

The fact that the synthetic antenna is formed by *moving* the sensor along track allows us to convert spatial frequencies into temporal one. The x-axis maps into the slow-time axis:

$$x = v_s \tau = v_s \cdot nT_{PRI},$$

where we accounted for sampling. Then the space frequency along the synthetic array, (14.2) is mapped into a temporal one:

$$f_a = \frac{1}{2\pi}\frac{\partial\phi}{\partial\tau} = \frac{1}{2\pi}\frac{\partial\phi}{\partial x}\frac{\partial x}{\partial\tau} = -2\frac{v_s \sin\psi_p}{\lambda} \tag{14.5}$$

where the suffix "a" reminds the along-track direction. The frequency is referred to as Doppler. The instantaneous Doppler reduces with time as the angle under which the target is observed, ψ_P, is reduced. When the target is orthogonal to the satellite, $\psi_P = 0 \Rightarrow f_a = 0$. When target is observed for an entire synthetic aperture, say $-\Delta_\psi/2 \leq \psi \leq +\Delta_\psi/2$, a total Doppler bandwidth is spanned:

$$\Delta_{fa} = 2\frac{v_s}{\lambda} 2\sin\left(\frac{\Delta_\psi}{2}\right) \approx 2\frac{v_s}{\lambda}\Delta_\psi = 2\frac{v_s}{\lambda}\frac{L_s}{r_P}$$

The inverse of the total Doppler bandwidth leads to the azimuth resolution (14.4), once scaled by the along track velocity:

$$\Delta_{fa} = \frac{v_s}{\rho_a}$$

.

The along track sampling, the pulse repetition frequency, f_p, should be larger than the total Doppler bandwidth in order to avoid alias - according to Shannon sampling theorem:

$$\Delta_{fa} = \frac{4v_s}{\lambda}\sin\left(\frac{\Delta_\psi}{2}\right) \leq f_p \tag{14.6}$$

This condition is always met if

$$f_p \geq \frac{4v_s}{\lambda} \Rightarrow f_{px} = \frac{f_p}{v_s} \geq \frac{4}{\lambda}$$

that is four samples in a wavelength. Such condition may be stringent for a SAR moving fast along track, particularly for LEO satellites. An X band spaceborne SAR would need 4

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samples in $\lambda = 3$cm, that is one sample each 75 cm, that, at a velocity of 7 km/s corresponds to roughly one sample each μs. This time interval would limit the total slant range observed to 150 m, the 2-way interval spanned by light in $1\mu s$, that is under of magnitudes less than desired.

The range and Doppler ambiguities for SAR requires that the PRF is dimensioned exactly like (13.7) such:

$$\Delta_{fa} = \frac{v_s}{\rho_a} < f_p < \frac{c}{2R_M} \Rightarrow \frac{R_M}{\rho_a} < \frac{c}{2v_s} \tag{14.7}$$

For a LEO-spaceborne case, the limit is R_M/ρ_a <21 km/m, that is a 1 meter resolution SAR would get a coverage of 21 km in slant range, that becomes 45 km on the ground for 30° incidence angle. Such limit would be less restrictive for airborne SAR, since the velocity is orders of magnitude smaller, and the maximum slant range is limited by the low altitude of the aircraft.

The constraint (14.7) roughly limits the number of resolution cells (~pixels) in a strip, being the ratio between range coverage and azimuth resolution: it corresponds to (4.5) for SAR. Notice that the resolution is referred to the single-look products, whereas several pixels should be averaged to improve the radiometric resolution. However, it is in this case recommended to improve range resolution, since in that case the limit is the bandwidth, that is however much less stringent than (14.7), particularly at high frequencies.

14.1.3 Sar antenna and resolution: a paradox

The $\lambda/4$ sampling interval may not be a problem for a ground based SAR, where the antenna moves on a rail at any velocity one wishes - provided that the image time is not critical, but it is a limit for most SARs. The limit is then relaxed by exploiting a real antenna of along track length L_a, to limit the total observed Doppler bandwidth to:

$$\Delta_{fa} = \frac{4v_s}{\lambda} \sin\left(\frac{\Delta_\psi}{2}\right) \approx \frac{2v_s}{\lambda}\Delta_\psi = \frac{2v_s}{\lambda}\frac{\lambda}{L_a} = \frac{2v_s}{L_a}$$

This value is independent on the wavelength, as an example a LEO SAR antenna of 10 m will limit the Doppler bandwidth to $\sim 1400/10 = 1400$ Hz, allowing for 700 μs pulse interval that limits maximum slant range to 700×150=105 km. This limit applies if we ignore pulse length, say very small duty cycle, and then should be reduced accordingly: a 15% duty cycle will limit the slant range to 105·0.7=70 km, that in the order of the 100 km ERS ground swath.

The introduction of a real antenna with length $L_a$limits the maximum synthetic aperture to the value:

$$L_s < r_P \Delta\psi = r_0 \frac{\lambda}{L_a}$$

as shown in Fig. 14.4.

This is turns limit the best achievable azimuth resolution to:

$$\rho_{az} \geq \frac{\lambda}{2L_s} r_0 = \frac{\lambda}{2r_0 \frac{\lambda}{L_a}} \cdot r_0 = \frac{L_a}{2} \tag{14.8}$$

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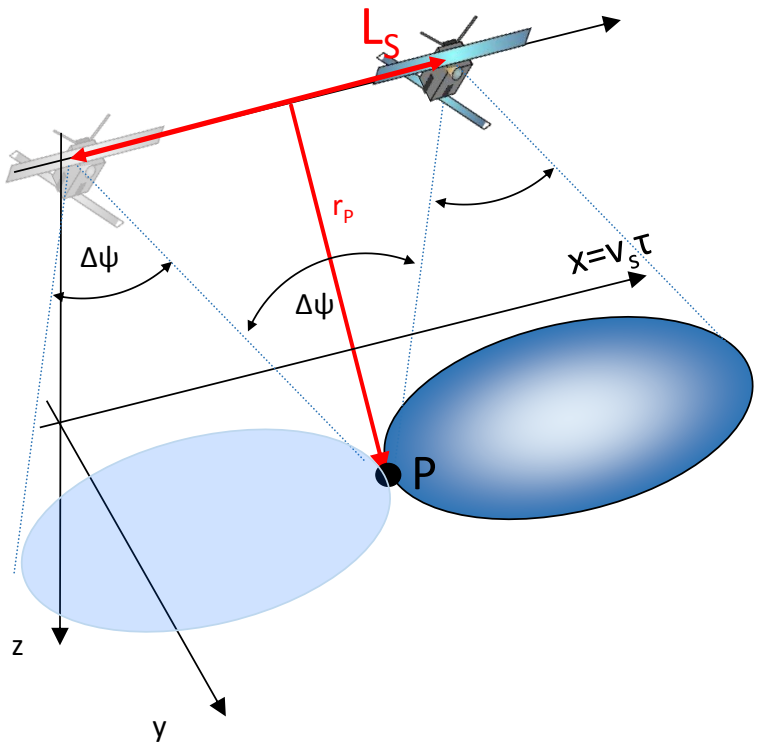


Figure 14.4: Derivation of the maximum synthetic aperture for a SAR.

This express the apparent SAR paradox, in which the azimuth resolution improves as the antenna shortens, just the opposite that we get from a usual RADAR, like a SLAR. However, there is a very simple explanation in the fact the a smaller on-board antenna correspond to a wider angle aperture and then a wider synthetic aperture, and it is this one that governs the resolution. The concept is the same of the chirp, whose correlation shortens as the chirp increases its length, and exactly for the same reason, that by increasing length we add more frequencies ending in a wider bandwidth and a better resolution.

14.2 Stripmap, Spot, ScanSAR and TOPSAR

The constraint (14.8) on the best resolution achievable can be relaxed by rotating the antenna beam along track in order to keep it pointed to a particular target, as shown in Fig. 14.5. That target is then observed for a longer synthetic aperture, and the resolution can be extended at will, in theory up to the wavelength. The drawback of SPOT-SAR mode is that only one target is imaged at full resolution: for other targets the synthetic aperture is much shorter, as shown in the figure by gray-shading, while other targets are not imaged at all. That's to say we are trading coverage (along track) by resolution, whereas the constraint (14.7) still applies[1].

The variation of the synthetic aperture with range can be exploited intentionally, to provide different trade offs between coverage and resolution. In the Hybrid configuration, [10], the antenna is rotated around a center whose ground range may change from the middle

[1]The way to relax (14.7) is to exploit distributed systems with antennas array and digital beam-forming, that implies acquiring at higher data-rates [8, 9]

of the image, that is SPOT mode, up to infinite, that is the conventional mode, also known as STRPIMAP.

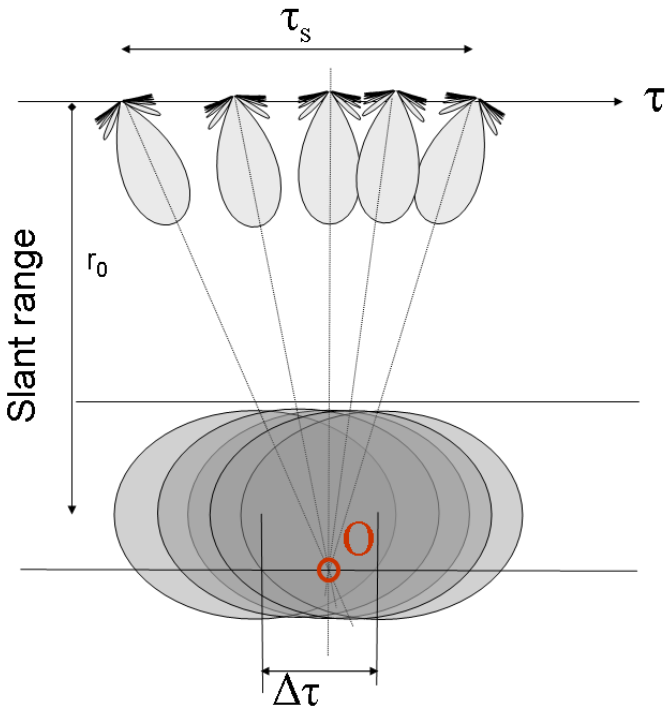


Figure 14.5: Geometry for the SPOT and HYBRID acquisition modes.

An alternative way to analyze these modes is in the along track time, Doppler domain of Fig. 14.6 and 14.7. Each target's Doppler history is approximately linear, with negative slope as from 14.5

$$f_a = -2\frac{v_s \sin\psi_p}{\lambda} \simeq -2\frac{v_s \psi_p}{\lambda} \simeq -2\frac{v_s}{\lambda R}(x - x_p) \tag{14.9}$$

Notice that f_a is null (Zero Doppler) when the satellite along track, x, is aligned with the target, x_p, shown by a point in the figure.

The antenna gain changes with the angle, and then Doppler frequency, $f_a$as in (14.9), in the general case of Hybrid acquisition we can express (14.3) as:

$$G(\psi) = g_0 \,\mathrm{sinc}\left(\frac{2\mathrm{L_s}}{\lambda}(\psi + \mathrm{k}_\psi \cdot (\tau\text{-}\tau_0))\right)$$

where $k_\psi$is the antenna rotation rate with time, and $\tau_0$the along-track focal point. In Stripmap, $k_\psi = 0$ whereas in SPOT, $k_\psi$matches the target Doppler rate, as from (14.9):

$$k_a = \frac{\partial f_a}{\partial \tau} = -\frac{2}{\lambda R}$$

Notice that it is range-dependent, than it is clear that it cannot be matched over wide strips, but a SPOT mode is intrinsically intended for small coverage.

The different acquisition modes are summarized in Fig. 14.6 and 14.7.

Finally, there are two other acquisition modes whose goal is the opposite, to get a wider coverage at the cost of reduced azimuth resolution. These modes, named ScanSAR, [11] and

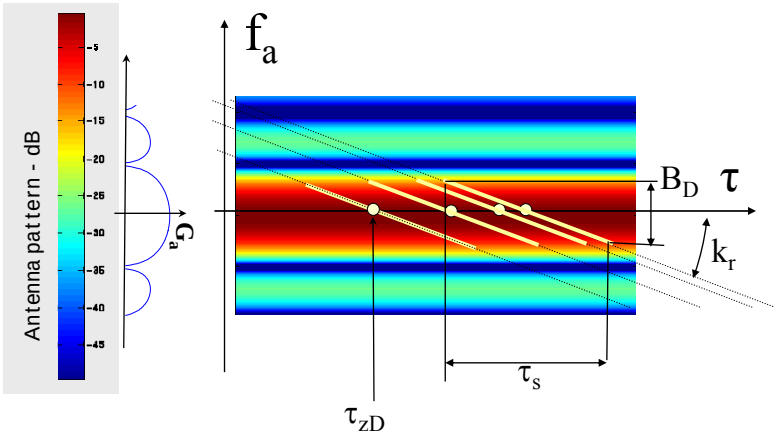


Figure 14.6: Doppler - time representation of STRIPMAP acquisition.

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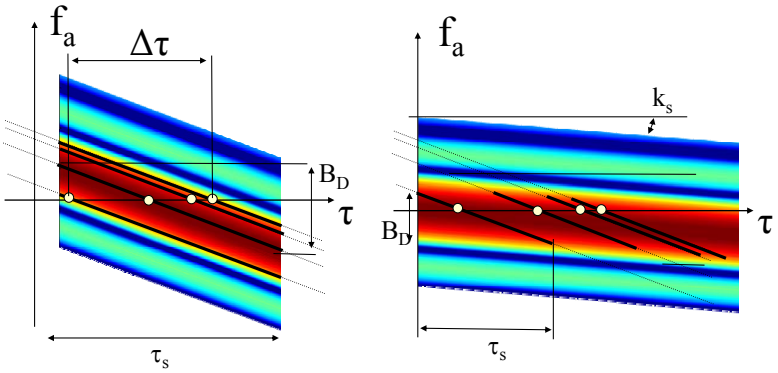


Figure 14.7: Doppler - time representation of SPOT and Hybrid acquisitions.

TOP-SAR [12], see Fig. 14.8 scans different swaths in a time domain interleaving, i.e. bu acquiring short bursts of data in each swath, but ensuring coverage of the whole scene. With reference to Fig. 14.8 on the right, this is done by ensuring that each target is imaged with the same Doppler bandwidth.

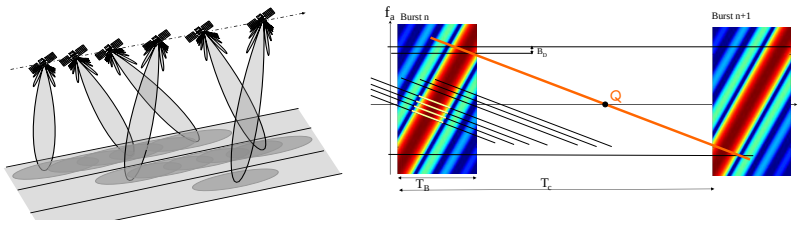


Figure 14.8: Doppler - time representation of TOPSAR acquisition.

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14.3 SAR acquisition

In this section we summarize the elements required to derive the SAR-IRF under very general assumptions. In particular we will refer in most of the case to the geometry in Fig. 14.9, that shows both a simplified system with straight orbit and flat earth, and a space-borne system with curved orbit and ellipsoidal earth.

The reference system for the **raw data** (or signal) **space** is the (*slow time, fast time*) coordinates, (τ, t), also defined as (*along track, across track*). This choice is determined by the acquisition system itself. On the other hand, the choice for the reference system in the **focused** (or model) **space** is somewhat conventional, and different choices can be made. The most widely adopted are:

1. the *zero Doppler time, zero Doppler distance*) distance (τ_0, r), also defined as (*azimuth, slant range*),

2. the Doppler, slant range, and

3. the ground georeferenced system.

In the second system, the energy of the focused target is placed along track in the location assumed by the senor when it observes that target at the center of the antenna beam. The first is a special case when we assume that the antenna beam is pointing broadside. In the third case, the energy is located at the corresponding point on the earth surface where the target is effectively.

This third choice could be the most reasonable, however there are some drawbacks that will be discussed in section 14.3.3.

A SAR focusing schemes has just the role of reversing the SAR acquisition, that is modeled by a linear system, according to the first Borne approximation. Therefore, the SAR acquisition is entirely described by the (non-stationary) Impulse Response Function (IRF). For a point-like target $\delta(\tau_0, t - 2r/c)$ we assume that the IRF is the following :

$$h_s(\tau, t; \tau_0, r) = a(\tau; \tau_0, r) \cdot p\left(t - \frac{2R}{c}\right) \cdot \exp\left(-j\frac{4\pi}{\lambda}R\right), \tag{14.10}$$

λ being the carrier wavelength and $p(t)$ the transmitted pulse, assumed range-compressed. $R(\tau; t_0, \tau_0)$ is the target-to-satellite distance, the *hodograph*:

$$R(\tau; r, \tau_0) = |S(\tau) - P(\tau_0, r)| \,. \tag{14.11}$$

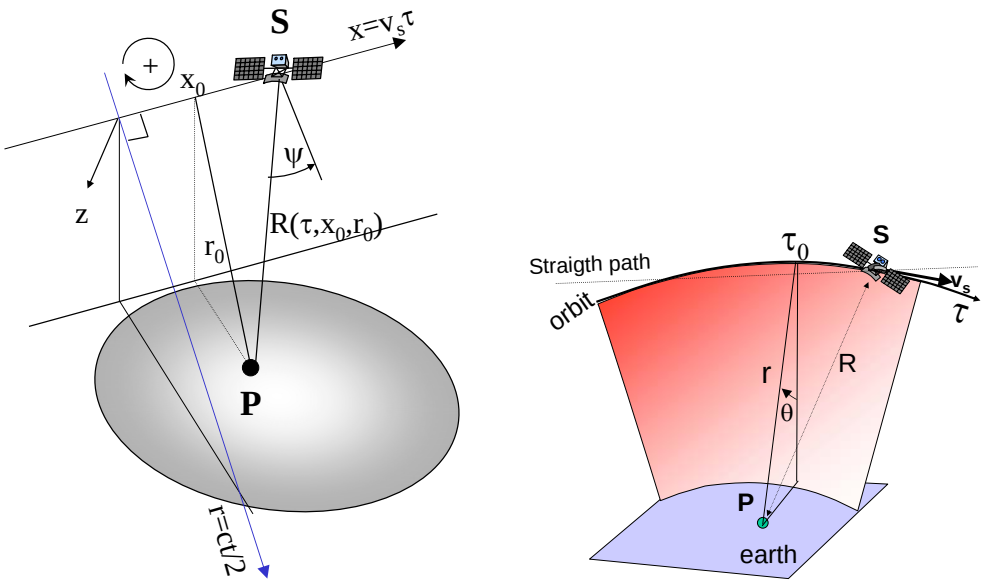


Figure 14.9: Geometry of a typical spaceborne SAR. Left: simplified system with straight orbit. Right: space-borne SAR geometry with curved orbit and ellipsoidal earth.

The notation $\mathbf{S}(\tau)$ remarks the dependence of the satellite position on the sole azimuth time, therefore we expect that the orbit is known as: $\{x(\tau), y(\tau), z(\tau)\}$. Expression (14.10) highlights the double role of the hodograph that causes a bending of the IRF, known as range migration, and a similar variation of its phase

14.3.1 Amplitude

In (14.10) we have defined as $a(\tau; \tau_0, r)$ the amplitude of the return: this can be related to the RADAR equation:

$$P_r(\tau; \mathbf{P}) = a^2(\tau; \mathbf{P}) = \frac{P_t G_t(\tau; \mathbf{P}) \cdot G_r(\tau; \mathbf{P}) \cdot \lambda^2}{(4\pi)^3 \, R^4(\tau; \mathbf{P})} \sigma(\mathbf{P}) \tag{14.12}$$

We notice that all the terms are slowly varying function of range and azimuth, and they can be ignored as far as focusing is concerned, and restored on the focused image.This is what a SAR processor usually does: amplitude is assumed constant, and it is compensated for on the focused data by a *calibration* step. An example of the effect of the antenna pattern is in the SAR image of the moon, in Fig. 14.10.

14.3.2 Space-domain and spectral-domain data support

The precise knowledge of the support of raw data in the space and wavenumber domain is fundamental for the implementation of any focusing algorithm, be it in time or in frequency. The space domain support is limited by the antenna aperture, in azimuth and elevation.

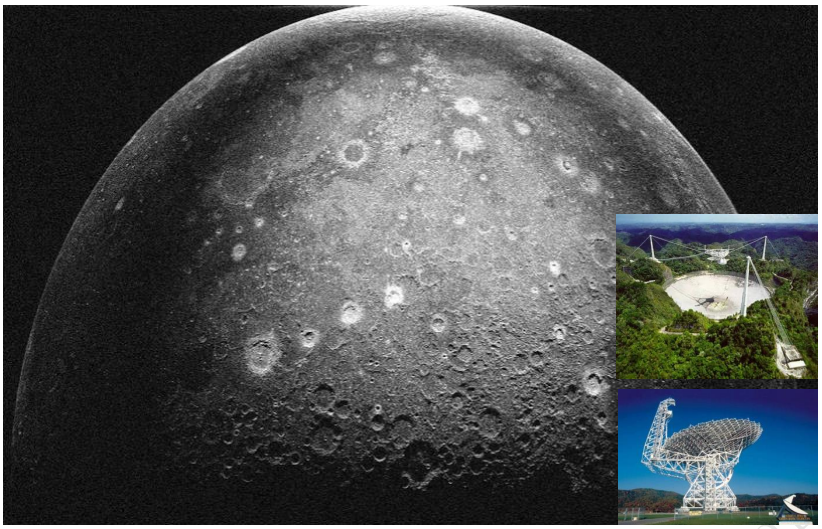


Figure 14.10: Bistatic SAR image of the Moon collected using the Arecibo Observatory and Green Bank Telescope (in the picture) acting as transmitter and receiver, in 2015. The synthetic aperture comes from the relative Earth-Moon motion. The dark ring is caused by the shape of the radar beam. The north pole of the Moon is at the bottom center of this image. Younger craters with rugged floors and rims are very bright. Image Number: WEB15031-2015 Credit: Image by Bruce Campbell, National Air and Space Museum, Smithsonian Institution.

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However, for a single, point target, *the support of the Impulse Response Function* (IRF), results from both the range migration and the spread for focused impulse response. In many cases, the range migration is dominant and this in turn depends on the antenna illumination in azimuth, as shown in Fig. 14.11. In the figure, we have assumed a generalized SAR geometry with squint angle ψ_0, while R_m and R_M are respectively the minimum and maximum slant range of the IRF portion illuminated by the antenna. The extent of the along track support is directly proportional to the antenna azimuth bandwidth, that is λ/L_a for a uniformly illuminated antenna of size L_a. We can evaluate the support in pixels, by assuming to sample data at a step equal to the nominal SAR resolution, $L_a/2$. We get the support in pixels

$$N_s > L_s \frac{2}{L_a} \simeq \frac{2\lambda r_0}{L_a^2} \tag{14.13}$$

As an example, a typical spaceborne SAR with $r_0 \sim 1000$ km and $L_a = 10$ m, gives $L_s = 25$ km in L band ($\lambda = 25$ cm), 5 km in C band C ($\lambda = 5$ cm) and 3 km in X band ($\lambda = 3$ cm). The minimum number of samples varies in the range N_s =1500-12000 depending on the band ($N_s \sim 5000$ in C band). For the GS1, the antenna illumination is not uniform, therefore we will still maintain an equivalent length L_a=10 m.

Notice that these numbers can be quite larger in airborne SARs, as the velocity is two order of magnitude smaller, allowing for higher PRF than those required to keep low range ambiguities.

The *cross-track support* is determined by the squint angle and the antenna aperture, as shown in d Fig. 14.11. If we assume that the along track support is small, compared to the closest approach, we can approximate the range migration, with respect to the closest

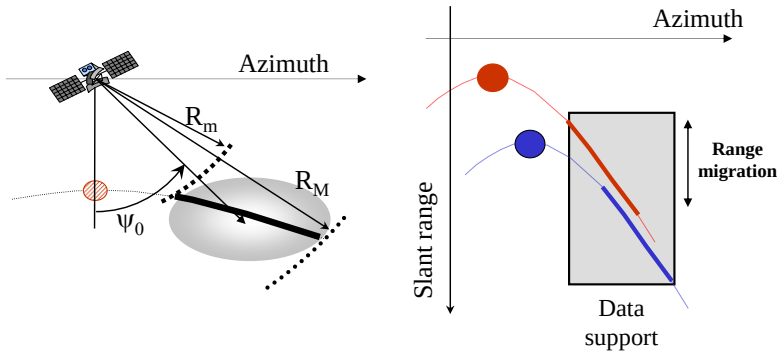


Figure 14.11: SAR acquisition geometry, on the left, and support of the raw data in the space domain, on the right.

approach as follows:

$$\Delta R(\tau) = r(\tau) - r_0 \simeq \frac{v_s^2(\tau - \tau_0)^2}{2r_0} \tag{14.14}$$

$$= \frac{r_0}{\cos\psi(\tau)} - r_0 \simeq \frac{r_0}{2}\psi^2 \tag{14.15}$$

For a zero-Doppler spaceborne SAR like GS1, we get, for $\psi = \Delta\psi/2$:

$$\Delta R_M \simeq \frac{r_0}{8}\left(\frac{\lambda}{L_a}\right)^2$$

that is typically 0.1-80 m, depending on the band ($\sim$ 4 m in C band). Notice that the cross-range IRF support is much smaller than the along track one. We can roughly evaluate the number of pixels by assuming that a ground range resolution (flat earth) equal to the azimuth:

$$N_r = \frac{\Delta R_M}{(L_a/2)\sin\theta} = \frac{r_0}{4\sin\theta}\frac{\lambda^2}{L_a^3} \tag{14.16}$$

The typical values for the range and azimuth support are in Tab. 14.1. Notice the anisotropy between range and azimuth of the IRF. The total number of samples:

$$N_r \cdot N_s \approx \left(\frac{2r_0^2}{4\sin\theta}\right)\frac{\lambda^3}{L_a^5}, \tag{14.17}$$

is proportional to the third power of the wavelength and to L_a^{-5}, that is the inverse of the fifth power of the azimuth resolution. Such extent prevent any implementation of focusing by space domain approaches, like the straight time-variant correlation, or the tomographic schemes, leading to processors based on either the full *2D wavenumber domain*, (ω-k), or the mixed *space-wavenumber domain (Range-Doppler).*

14.3.2.1 Spectral support

The support of the IRF in the 2D wavenumber domain is limited by the bandwidth of the on-board filter, and the antenna pattern. The former limits the frequency in the range $f_0 - B/2 \le f \le f_0 + B/2$, whereas the antenna limits the observation aperture to the angular

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r_0	km	1000		
L_a	m	10		
θ	^{o}	23		
Band:		*L*	**C**	*X*
f_0	MHz	1.2	**5.4**	9
λ	cm	25	**5.6**	3.3
$\Delta\psi$	^{o}	1.4	**0.32**	0.2
L_s	km	25	**5.6**	3.3
N_s	smpls	**5000**	**1132**	**666**
ΔR	m	78	**4**	1.4
N_r	smpls	**40**	**2.1**	**0.7**

Table 14.1: Determination of the space-domain support of the IRF of a typical spaceborne SAR system.

interval $\psi_0 - \Delta\psi/2 \leq \psi \leq \psi_0 + \Delta\psi/2$, where ψ_0 is the squint angle and $\Delta\psi = \lambda/L_a$ the 3 dB bandwidth. This expression assume rectilinear orbit, however, for the seek of the evaluation of the spectral support, it is accurate enough. In order to convert this angular interval into frequencies, we makes use of the relation:

$$\begin{aligned} f_\tau &= -\frac{2}{\lambda}\sin\psi = -\frac{2f}{c}\sin\psi \\ &\simeq -\frac{2f_0}{c}\psi \end{aligned}$$

We can finally derive the spectral support for the focused data, by considering that, for the plane wave:

$$w(x, r, t) = \exp\left(-j\left(k_x x + k_r r + k_z z - \omega_0 t\right)\right)$$

whose wavefront is in the plane orthogonal to (x,y), hence $k_z = 0$, the following relations hold:

$$\begin{aligned} k_x &= \frac{2\omega_0}{c}\sin\psi \\ k_r &= \frac{2\omega_0}{c}\cos\psi \end{aligned}$$

The resulting spectral support is shown in Fig. 14.12, on the left. The right part of the same figure shows the support after applying: (1) the demodulation in baseband, (2) the folding due to the sampling time step $T_{PRI} = f_{PRF}^{-1}$. Both elements should be properly accounted for when designing a processor in the wavenumber domain.

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14.3.3 The slant range - zero Doppler reference

We have observed that the SAR acquires data from a 3D geometry, but observes only two dimensions, thus it introduces a loss of information. One good choice for focusing is to register the data in a domain that matches the 2D observations.

Let us assume a straight orbit, as shown in Fig. 14.13. Let us then assume all the targets that are located in the intersection between a cone that has the track as its axis, and a plane orthogonal to the orbit, are isochronous, independently on the actual sensor position, therefore their contribute will always be added up to the same pixel, and cannot be distinguished by focusing.

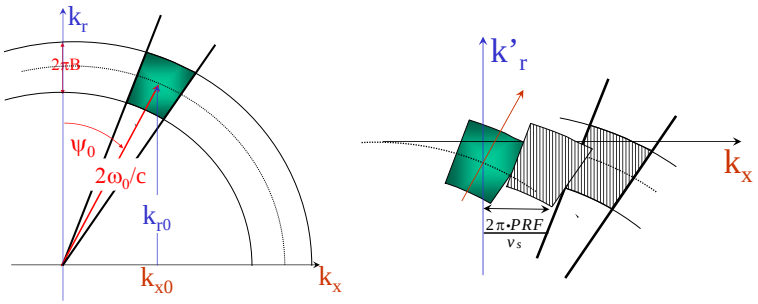


Figure 14.12: Spectral support of the SAR IRF, before and after applying the baseband demodulation and the folding due to the sampling along the orbit.

Thus, a natural reference for the focused SAR image is to put the target's contribution in a domain (range, Doppler), that is the distance from the sensor when the target is observed with an angle corresponding to the center of the antenna beam. The actual Doppler is arbitrary, as the beam center is not narrow, moreover it can change slightly when the sensor repeats the orbit. Therefore it is wise to register the image in a reference that is kept fixed, whatever the attitude of sensor. This would be a "slant range, nominal Doppler" one. However, as the majority of the system are designed to operate at zero nominal Doppler, the slant range, zero Doppler reference has become popular. Thus we makes use of the couple of coordinates τ_0, t_0, the first, τ_0, is the location of the sensor when it sees that target broadside (zero-Doppler), that is when the sensor-target velocity is zero. The second coordinate, t_0 is the sensor-target 1-way delay measured when the target is at zero Doppler.

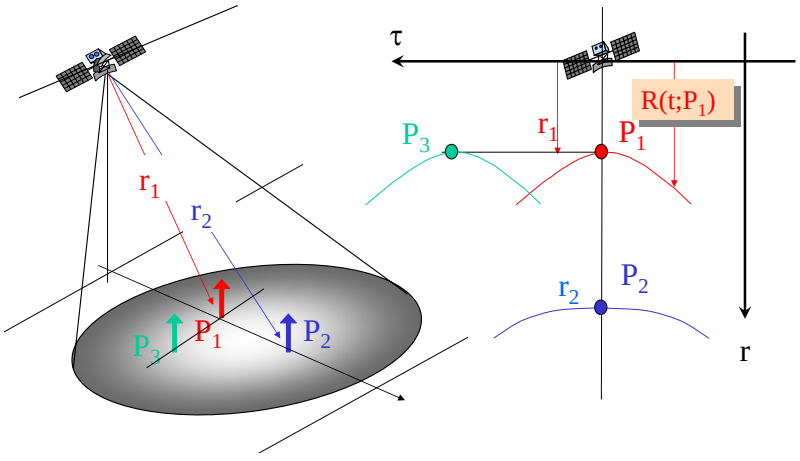


Figure 14.13: Dependency of the SAR IRF by the 3D geometry. *Left:* if a rectilinear orbit is assumed, all the target on the circle will contribute always to a same pixel, the actual pixel would depend on the sensor position along-track. *Right:* when the orbit is no-more rectilinear, the acquisition depends on the 3D location of each target, and it is no more cylindrical.

14.4 SAR focusing methods

Focusing of SAR data can be achieved as a *correlation* between the received data and a reference *matched* in phase to the IRF as this procedure can be shown to be optimal in MMSE sense. Being the reference space-variant, focusing requires a space variant convolution:

$$\begin{aligned}\widehat{a}(x,r) &= \int\int d(\tau,t)\cdot h_s^*(\tau,t;r,x)\,d\tau dt \\ &= d(t,\tau;r,x) * *h_s^*(-t,-\tau;r,x)\end{aligned} \tag{14.18}$$

$d(\tau,t)$ being the complex data; $\widehat{a}(x,r)$ the estimate of the focused data, and assuming that all targets emit the field at time $t{=}0$ (exploding reflectors, see [13]), and that $r=0$ is the location of the sensor.

This convolution is quite inefficient to be implemented directly in the time domain, due to the length of the spectral support of the IRF, that, according to table 14.1 sums to sizes from hundreds to tens of thousands samples.

Two approaches provides efficient processing in these cases:

1) focusing in the mixed Range Doppler domain, also known as **Range Doppler** focusing;

2) focusing in the full 2D wavenumber domain, also known as $\omega - k$ focusing [14], [13].

The former approach, that implements the matched filter (14.18) as a correlation in time (range), and a product in azimuth frequency (Doppler) is more efficient than the second if (1) the IRF support in range is shorter than, say 10 samples (see the fast convolution approach in [15]), and (2) the IRF can be considered *invariant* in that extent, that is the error provided by assuming stationary IRF is a small fraction of a radians.

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14.5 Noise equivalent sigma zero

SAR NESZ is quite similar to SLAR. The Signal to Noise Ratio can be expressed as follows:

$$SNR_a = P_M \cdot \left(\frac{G_{el}^2(r_0)\lambda^2\eta}{(4\pi)^3\, r_0^4}\right)\cdot\left(\sigma_o \frac{c}{2B_r\sin\theta}\frac{L_a}{2}\right)\cdot\left(\frac{T_S}{N_0}\right) \tag{14.19}$$

$$P_M = P_t T_c f_{prf} \tag{14.20}$$

The four terms can be interpreted as follows:

1. P_M is the average power, equivalent to a continuous transmission, that is the product of the peak Power times the duty cycle $d_c = T_c/T_{pri} = T_c \cdot f_{prf}$ (the chirp duration times the pulse repetition frequency),

2. the second term is the combined effect of the antenna gain, G_{el}, the spherical divergency, and total losses, η,

3. the third term is the RADAR Cross Section of the target, the product σ_0, the backscatter coefficient, the ground range resolution (θbeing the incidence angle and $B_r$the transmitted bandwidth), and the azimuth resolution (L_a being the antenna length),

4. the four term is the noise power in the time extent of the matched filter, that is the duration of the synthetic aperture:

$$T_S = \frac{\lambda r_0}{v_s L_a} \tag{14.21}$$

Notice that, in processing, a spectral windowing can superposed in azimuth, resulting in a change in both the resolution (that is then no more $L_a/2$) and the synthetic aperture time: we could account for that as an additional loss. The noise equivalent σ^0can then be obtained by imposing unitary SNR:

$$SNR = \frac{P_M G_{el}^2(r_0) \cdot \lambda^3 \cdot c}{4\,(4\pi)^3\, r_0^3 \cdot FkT_o \cdot B_r \sin\theta \cdot v_s} \cdot \sigma^0 = 1$$

whose solution is:

$$\sigma_{nesz} = \frac{4\,(4\pi)^3\, r_0^4 FKT_0 B_r \sin\theta \cdot v_s}{P_M c G_{el}^2(r_0) \cdot \lambda^3}$$

14.6 End-to-end equivalent transfer function

The SAR acquisition and focusing can be modeled as a linear system, space varying. For each potion of the target in the model space, say zero Doppler time and distance, we could write a transfer function that is the product of SAR acquisition and SAR focusing TF, as shown in Fig. 14.12. We should also account for the fact that data are down-converted to baseband, and that spectral folding introduces a further conversion in both range and azimuth. In particular, conventionally data are down-converted in range direction, by demodulating then with respect to central wave number: $k_{r0} = 2\omega_0/c$, whereas no explicit conversion is applied to azimuth spectra, but for the implicit folding introduced by sampling: $k_{xd} = 2\pi k f_{prf}/v_s$.

The spectral support of those down-converted data is almost rectangular, and centered on

$$f_r = 0$$
$$f_x = -\frac{2v}{\lambda}\sin\psi_{dc}$$

ψ_{dc} being the center of antenna beam (also defined as Doppler centroid). Notice that that center is actually the combined result of the antenna illumination and a possible Doppler dependent weighting applied in processing. Within the assumption of small bandwidth and zero Doppler, the end-to-end transfer function is modeled as:

$$h_e(x,r) \simeq h_{e0}\,\mathrm{sinc}(r/B_r) \cdot \mathrm{sinc}(x/B_x) \tag{14.22}$$

that is a 2D sinc oriented along x, and r. Such properties is the resultant of the location in TOA/DTOA, discussed in section 12.1, where we have shown that, for broadside looking the two direction of TOA (range) and DTOA (azimuth) are orthogonal. This is no more the case of a squinted SAR: a more accurate representation of the support, that accounts for non zero Doppler and large bandwidth is shown in Fig. 14.12.

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Chapter 15

SAR interferometry

A single SAR observation provide the complex RCS in a bidimensional space, therefore it is unable to locate the target in the 3D world. Moreover, the image's phase is the summation of different contributions: the target RCS and a term proportional to the optical path, hence it is not much useful. In SAR Interferometry (InSAR), two or more acquisitions are provided. The added equation makes us able to locate the target in the 3D world or to separate the target's phase from the optical path. Yet, we in interferometry we exploit the phase, and not the distance (that would be at very coarse resolution) hence we will still be subject to aliasing, and phase unwrapping techniques are required.

Two different kind of applications come from InSAR: when the two acquisitions are displaced in space (and say, at the same time), we are able to locate the target in the 3D, hence making **DEM**. On the other hand, when two acquisitions are displaced in time, we are able to measure a Line Of Sight (LOS) variation in target's displacement, hence deformations or along-track-motions.

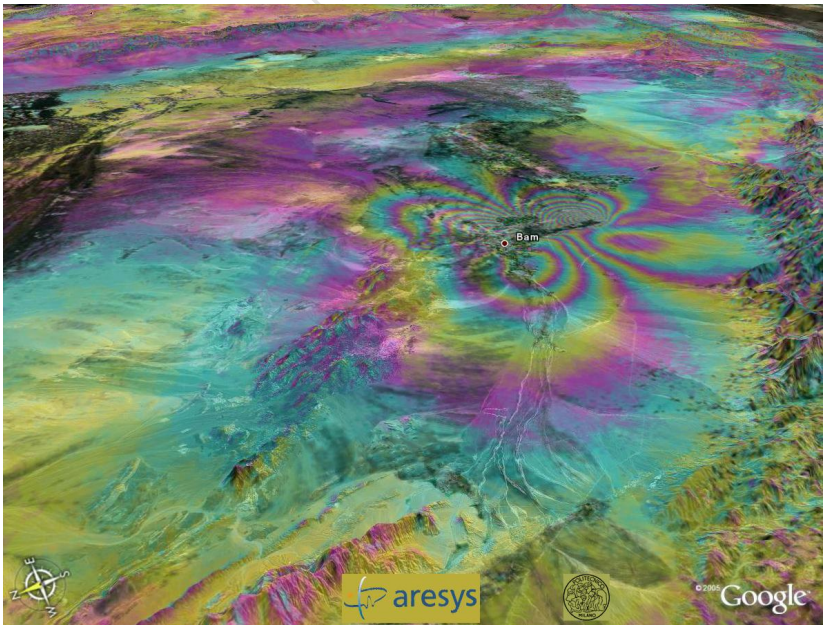


Figure 15.1: The interferometric fringes are due to deformation occurred between the two observations and caused by Bam earthquake (26/12/03). Each fringe cycle is 2.8 cm LOS displacement variation. The amplitude is from Google Earth.

15.1 The SAR Interferogram

The typical interferometric geometry is shown in Fig. 15.2. The same target is observed twice by SAR, either by two simultaneously SAR (single-pass interferometry), or by repeated observations. We usually refer to those as the Master and the slave observations (likewise, there is a master and a slave sensor).

These two observations slightly differ by the view angle: we define **baseline** the distance between the two sensors (say the antenna phase centers) when they observe the same target. The Baseline is then a vector in the 3D space, shown in 15.2 left, however, if we project the baseline in the plane containing the define the master and the target, and we take the component normal to the view (the sensor-target direction), we have the **normal baseline**, (often defined **baseline**) that plays a fundamental role in interferometry.

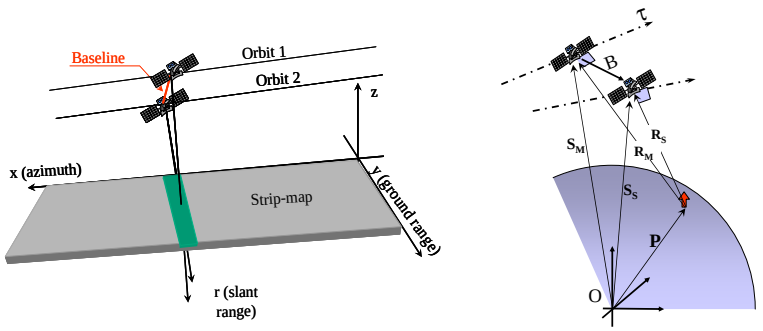


Figure 15.2: Interferometric SAR geometry referred to a spaceborne system.

15.1.1 The interferogram's phase: a single difference

Let us refer to the geometry in the plane orthogonal to the sensor track in Fig. 15.3. Let us assume that both images are focused in range and azimuth, so that the SAR system can be though as a high resolution slide-looking RADAR, and we may apply the end-to-end channel equivalence in discussed in the SAR session. In the case of two **mono static** SAR we may model both acquisitions as follows:

$$s_M(\mathbf{P}) = g(\mathbf{P}) \cdot u(t-\tau_M) e^{-j\omega_0(t-\tau_M)} e^{j\varphi(\mathbf{P})} + n_M(t) \text{ where } \tau_M = 2r_M(\mathbf{P})/c \quad (15.1)$$

$$s_S(\mathbf{P}) = g(\mathbf{P}) \cdot u(t-\tau_S) e^{-j\omega_0(t-\tau_S)} e^{j\varphi(\mathbf{P})} + n_S(t) \text{ where } \tau_S = 2r_S(\mathbf{P})/c \quad (15.2)$$

where $\mathbf{P}$ is the target, $r_M(\mathbf{P})$ and $r_S(\mathbf{P})$ are the sensor-target distances (the optical paths), τ_M and τ_S the corresponding delays, $n_M(t)$ and $n_S(t)$ the two noise terms that add in the receivers, $\varphi(\mathbf{P})$ the target's phase, and $g(\mathbf{P})$ the contribution of targets RS and the attenuation predicted by the RADAR equation. Notice that we assume that the two systems observe exactly the same target, in particular it is important that the phase center is exactly the same (that happens if the two view angles are very close).

As we know, the two range, τ_M and τ_S are well measured by the two SAR observations. However the interest is here for the DTOA, $\tau_M - \tau_S$. In particular, we assume that the two systems operate with very small fractional bandwidth, so that the accuracy that can be achieved be measuring phases is much higher than that observed on the amplitudes. Thus we assume a time reference of the master: $\tau = \tau_M$ and *we co-register the two images*, in

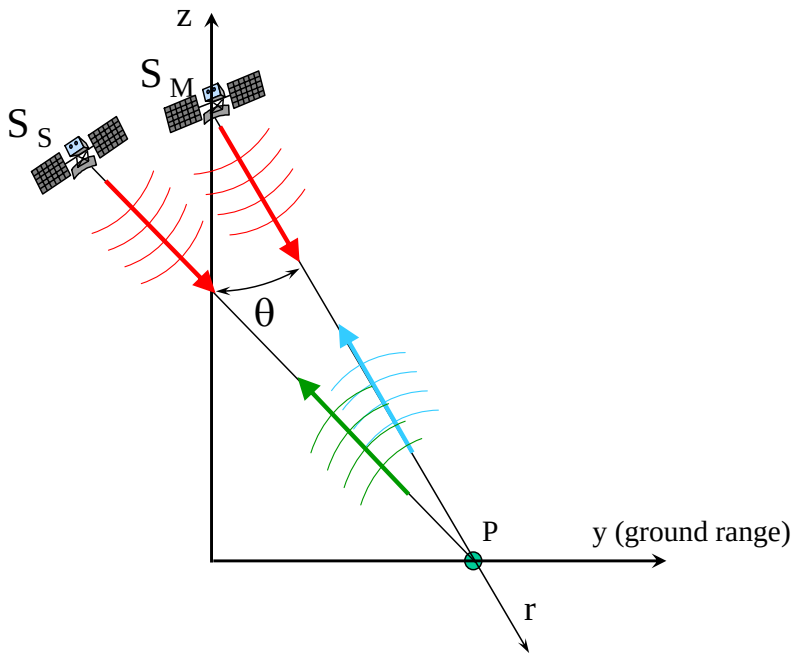


Figure 15.3: Interferometric SAR geometry in the plane orthogonal to the track.

that we locate in the slave the pixel corresponding to the same target P (that is very slightly displaced as $\tau_S \simeq \tau_M$), and we compute the interferogram as the Hermitian cross-product between the master and the slave. In the **noiseless case**:

$$i(\mathbf{P}) = s_M(\mathbf{P}) \cdot s_S^*(\mathbf{P}) \propto |u(t-\tau)|^2 e^{-j\omega_0(\tau_S - \tau_M)} = |u(t-\tau)|^2 e^{-j\omega_0 \Delta\tau(\mathbf{P})} \tag{15.3}$$

where

$$\Delta\tau = \tau_S - \tau_M = \frac{2\Delta r(\mathbf{P})}{c}$$

and the optical path difference is:

$$\Delta r(\mathbf{P}) = r_S(\mathbf{P}) - r_M(\mathbf{P})$$

The interferogram phase is thus the single-difference between the two images phase:

$$\phi_{int} = \arg\left(i(\mathbf{P})\right) = -\omega_0(\tau_S - \tau_M) = -\omega_0 \Delta\tau(\mathbf{P}) = -\frac{4\pi}{\lambda}\Delta r(\mathbf{P})$$

In a **bistatic** case, we would get:

$$\tau_S = \left(r_M(\mathbf{P}) + r_S(\mathbf{P})\right)/c$$
$$\Rightarrow \phi_{int} = -\frac{2\pi}{\lambda}\Delta r(\mathbf{P})$$

the sensitivity of phase versus the distance is halved, as we have to account only for a single-way path and not the two-way as in mono static case. In the following we adopt the mono static assumption by default.

An example of a SAR interferogram is in Fig. 15.4, achieved by simulation. Notice that the phase is given in principal value, $-\pi \leq \phi \leq \pi$, or wrapped, and its true value is very difficult to reconstruct. In the LEO spaceborne case, the average sensor-target distance is 10^6 meters, and the wavelength is typically centimetric, in case of $\lambda = 2$ cm, we have 10^8 cycles, and we understand that recovering the principal value is really impossible.

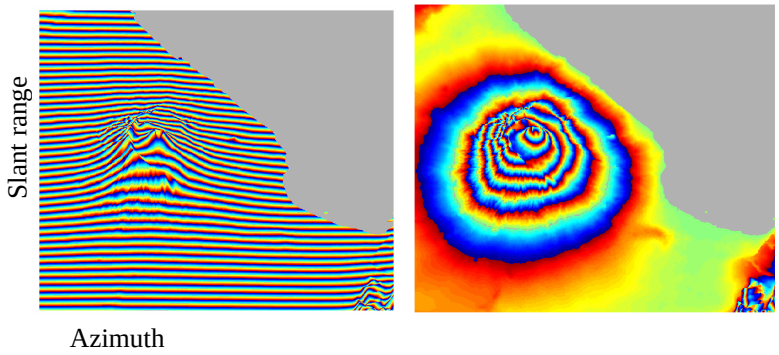


Figure 15.4: Left: simulated noiseless interferometry of Mt. Vesuvius. Normal baseline is 50 m and ambiguity elevation is 185 m per fringe cycle. Right: the same interferogram after flattening.

15.1.1.1 Double Differences

The effective use of the interferogram phase is then conditioned to finding *phase variation with respect to a reference point*, or *Ground Control Point*, that should be so close to the target in order to allow a practical unwrapping, say that we can handle some tens of cycles in the *differential phase*. We are then to a double-difference approach, where a first difference removes the unknown target's phase, and a second, an unknown number of cycles.

With reference to the geometry in Fig. 15.5, the interferometric phases of the targets P and O (this last being a reference point) are computed as from (12.6):

$$\phi(\mathbf{P}) = -\frac{4\pi B}{\lambda R} y_P;\ \phi(\mathbf{O}) = -\frac{4\pi B}{\lambda R} y_O$$

then, *differential interferometric phase* is:

$$\phi(\mathbf{P}) - \phi(\mathbf{O}) = \frac{4\pi}{\lambda} \left[(R_M(P) - R_S(P)) - (R_M(O) - R_S(O)) \right] \tag{15.4}$$

$$\simeq -\frac{4\pi B}{\lambda R} (y_P - y_O) \tag{15.5}$$

proportional to the displacement from the Ground Control Point (GCP), orthogonal to the look, $y_P - y_O$. Notice that the baseline has a sign, and if we exchange master and slave, we change the sign of the phase.

In summary, the differential interferometric phase, that we will simply define asϕ, is the result of *double differences*, the first difference is needed to remove dependency upon the intrinsic target's phase, and the second to reduce the phase excursion to few multiple of $2\pi$to be able to solve the phase ambiguity.

15.1.2 Slant terrain

Expression (15.5) allows us to compute the interferometric phase as a function of the cross-range distance between the target P and the reference point O. Let us then compute the interferometric phase contribution of a target P on a *constant sloped terrain*. Let us refer to

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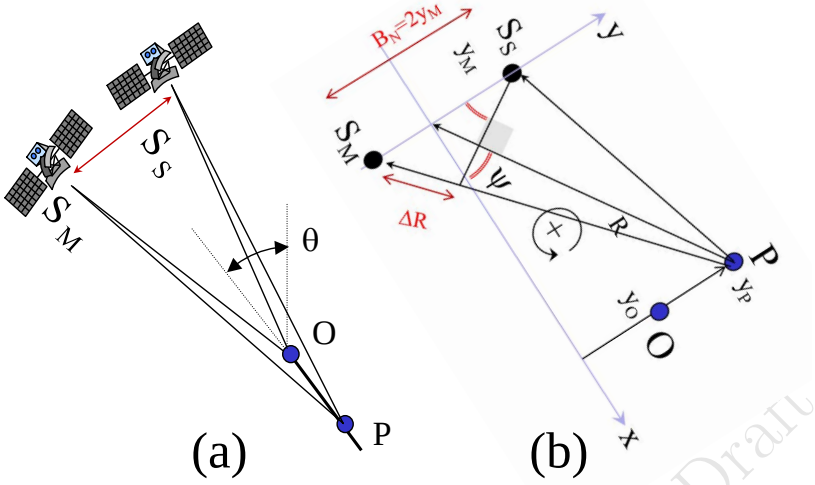


Figure 15.5: Interferometry and DTOA.

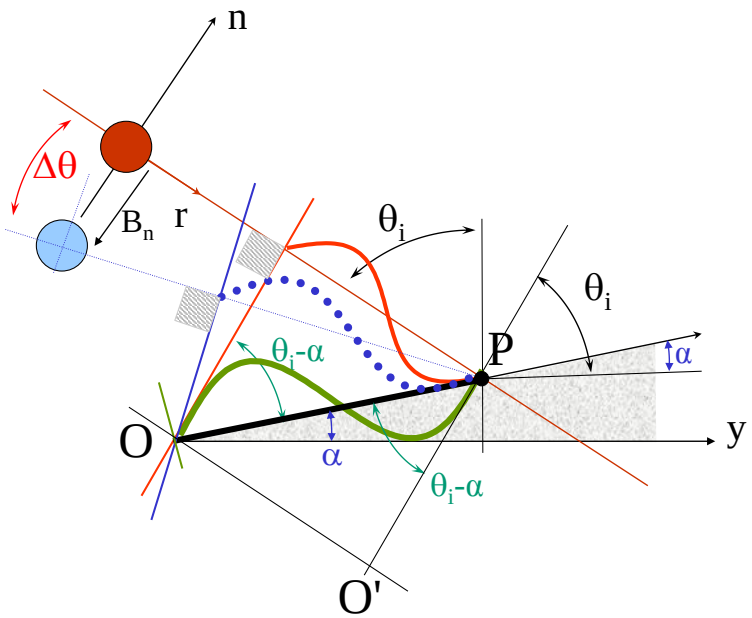


Figure 15.6: INSAR geometry, left, for computing spectral shift. Interpretation in the wavenumber domain.

the geometry in Fig. 15.1.2.a, representing a target with slope α_r. As the target moves on the slopes, its slant range increases, and also the cross-range, with respect to the reference.

We get from (15.5):

$$\phi_i(\mathbf{P}) = -\frac{4\pi}{\lambda}|\mathbf{P}-\mathbf{O}|\cos(\theta-\alpha)\cdot\Delta\theta \tag{15.6}$$

where the slant range of $\mathbf{P}$ is:

$$r(\mathbf{P}) = r_O + |\mathbf{P}-\mathbf{O}|\sin(\theta-\alpha)$$

If we express the interferometric phase as a *function of the slant range*, we get:

$$\phi_i(\mathbf{P}) = -\frac{4\pi}{\lambda}\left(r(\mathbf{P})-r_O\right)\frac{\cos(\theta-\alpha)}{\sin(\theta-\alpha)}\cdot\Delta\theta \tag{15.7}$$

that corresponds to the local frequency:

$$\begin{aligned} f &= \frac{1}{2\pi}\frac{d\phi}{dt} = \frac{1}{2\pi}\frac{d\phi}{dr}\frac{dr}{dt} = -\frac{4\pi}{2\pi\lambda}\frac{c}{2}\frac{\Delta\theta}{\tan(\theta-\alpha)} \\ &= -f_0\frac{\Delta\theta}{\tan(\theta-\alpha)} = -f_0\frac{B_n}{r_0\tan(\theta-\alpha)} \end{aligned} \tag{15.8}$$

We may conclude that a constant sloped terrain, as a first approximation (locally) contribute to the interferogram with a frequency shift that is linearly related to the baseline. The frequency shift for flat terrain is negative and its value is:

$$f_{flat} = -f_0\frac{B_n}{r_0\tan\theta}$$

and it reduces with increasing slope, up to the asymptote $-\infty$ for the layover limit: $\alpha=\theta$. However, the *upper bound for a slope* is achieved when the shift is equal to the available bandwidth:

$$\alpha \ll \theta - \arctan\left(\frac{f_0}{B}\frac{B_n}{r_0}\right)$$

For slopes in the rage that causes layover, the frequency shift is positive, and for back slopes at the limit of shadow, $\alpha=\pi/2-\theta$, the spectral shift is zero, according to Fig. 15.5.*a*.

15.1.3 Interferograms flattening

A common practice is then to compensate the interferogram for the contribution of the flat earth, so that we lead to a constant phase in equipotential gravity field (or we may compensate with respect to the ellipsoid as well). This is a variant of the double difference approach before introduced, where the reference is assumed as the "flat" surface. According to Fig. 15.4.a, the flat earth ($\alpha_r=0$) corresponds roughly to the frequency:

$$f = -f_0\frac{B_n}{r_0\tan\theta} \tag{15.9}$$

We then may approximate the "flattening" as an interferogram demodulation, shown in Fig. 15.4.b. Indeed, for a precise flattening, we have to subtract the proper contribution as computed from (15.4), where O is on flat terrain.

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15.1.4 Phases to height conversion

Phases can be converted to elevation by exploiting the geometry in Fig. 15.7, that assumes spherical earth and a flattened interferogram. An elevation q with respect to the flat earth leads to a differential phase as in (15.5):

$$\Delta\varphi_{alt} = -\frac{q}{\sin\theta} \cdot \frac{B_n}{R_0} \cdot \frac{4\pi}{\lambda} \tag{15.10}$$

The expression is approximated and can be used only for first order evaluations. In fact we have to account that the normal baseline would depend on the look angle, hence it changes with both range and the elevation itself. For the correct solution one could write the SAR geocoding and replace the earth equation by the interferometric hyperboloid. In any case, the absolute phase would be needed, or at least the phase difference with respect to a reference target.

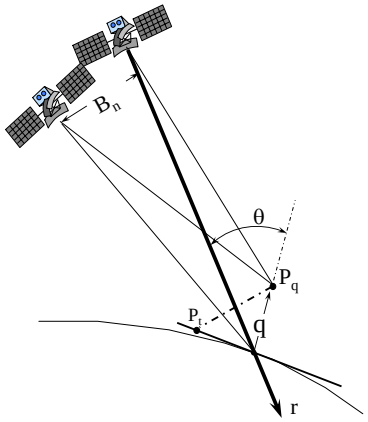


Figure 15.7: Geometry for phase to height conversion

An important element is the *ambiguity elevation*, the one that corresponds to 2π phase shift, that is the elevation difference between two iso-phase lines in the interferogram. By imposing $\Delta\varphi_{alt} = 2\pi$ in (15.10), we get:

$$q_a = -\frac{\lambda R \sin\theta}{2B_n} \tag{15.11}$$

$$|q_a|\, B_n = \frac{\lambda R \sin\theta}{2} \tag{15.12}$$

The second expression states that the product of the ambiguity elevation times the baseline is more or less constant, once that the range and the look angle are fixed. As an example, for Sentinel at 35° incidence angle, this product is $\sim 96 \times 96$, that is an elevation of ambiguity of 96 m is found for a baseline of 96 m. Fig. 15.8 shows the phase of a simulated interferogram of Mt Vesuvius, by assuming baselines Bn=50 m, 250 m. The correspondent elevation ambiguity for the first case is 96*96/50 $\simeq$ 180 m, and we count 6 -7 fringe cycles from the sea to the top of the mountain, that compares with the elevation of 1280 m. In the second case the ambiguity

elevation is 37 m thus fringes are much denser, leading to a more difficult unwrapping. On the other hand, when the baseline is small, any error on the phase, like to due to noise or propagation (atmospheric effect), is converted into a large elevation error by (15.11), hence the optimal baseline is neither too low, nor too high.

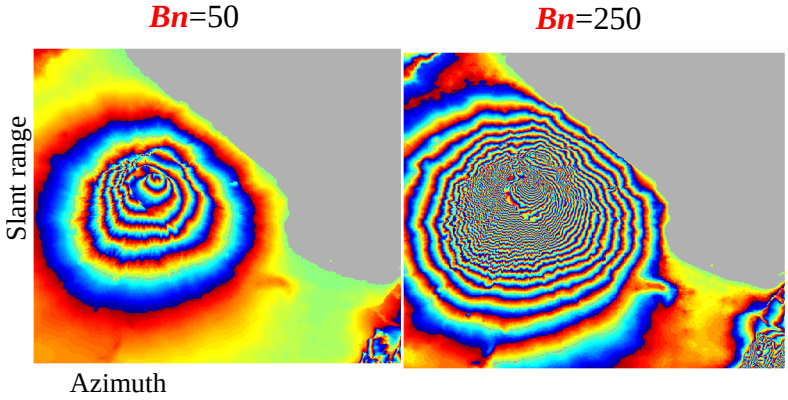


Figure 15.8: Flattened Interferogram phases achieved by simulating ER geometry and assuming Mt. Vesuvius topography. Two different baselines: 50 m on the left and 250 on the right.

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The elevation to phase conversion can then be expressed as:

$$\Delta\varphi = 2\pi \frac{q}{q_a} \tag{15.13}$$

$$q = \frac{q_a}{2\pi}\Delta\varphi \tag{15.14}$$

15.2 Phase unwrapping and DEM generation

The generation of the model of the terrain is made by exploiting interferometry to locate each target in the 3D space. The phase of the interferogram is, form (15.3) proportional to the DTOA between the target and the two sensors:

$$\arg\left(i(\mathbf{P})\right) = -\omega_0 \Delta\tau(\mathbf{P}) = -\omega_0 \frac{\Delta r}{c}(\mathbf{P})$$

If we assume ideal propagation, so that the light speed is constant and known like in free space, we can exploit this equation, that corresponds to one sheet of an hyperboloid in the 3D space (the sheet if defined by the phase sign), together with the range equation (a sphere for the mono static case or an ellipsoid for the bistatic), and the zero-Doppler location. For a broadside looking SAR, say with rectilinear orbit, the azimuth location after synthetic aperture is a plane orthogonal to the trajectory. Therefore, the target location in that plane is derived by solving the equation system:

1. master range equation, the sphere

$$\begin{aligned} |\mathbf{S}_1(\tau_1) - \mathbf{P}| &= \sqrt{(x_P - x_{S1}(\tau_1))^2 + (y_P - y_{S1}(\tau_1))^2 + (z_P - z_{S1}(\tau_1))^2} = r \\ r &\sim N\left(r_p = \frac{ct_p}{2}, \sigma_r\right) \end{aligned} \tag{15.15}$$

where is $S_1(\tau_1)$ is the location of he mater image at azimuth time τ_1, expressed then as function of the three coordinates, and x_p,y_p,z_p the range of the target in P, t_p being the travel time measured with an error may assume zero-mean, normal distributed, $\sigma_r$is the standard deviation in the target location,

2. interferometric equation, the hyperboloid

$$\begin{aligned} |\mathbf{S}_1(\tau_1) - \mathbf{P}| - |\mathbf{S}_2(\tau_2) - \mathbf{P}| &= -\frac{\Phi}{2\pi}\lambda \\ \Phi &\sim N(\Phi_p, \sigma_\phi) \end{aligned} \tag{15.16}$$

where Φ is the unwrapped phase.

The intersection of the three surfaces (the third being the plane orthogonal to the orbit) is a set of two points see Fig. 15.9, that are symmetrically placed on the right and the left of the orbit, the correct one being defined by right/left looking of the antenna. Thus we get the target location and then the DEM. The solution can be found iteratively by Newton, or by exploiting the direct inversion in [16].

Unfortunately, in (15.16), the unwrapped phase is required. Thus a phase unwrapping technique should be applied for solving the phase ambiguity. Several approaches exists in literature, that are based on the smoothness of the absolute phase field, and try to impose a consistency between the many paths that connects to points (e.g. the absolute phase field is conservative, like for gravity). Some code for 2d unwrapping is available on the web. It is important to know that the PU problem is ill conditioned, and its solution becomes almost impossible as the topography changes in a few pixels of an extent comparable with the elevation of ambiguity, or when noise standard deviation becomes close to say 1 radians.

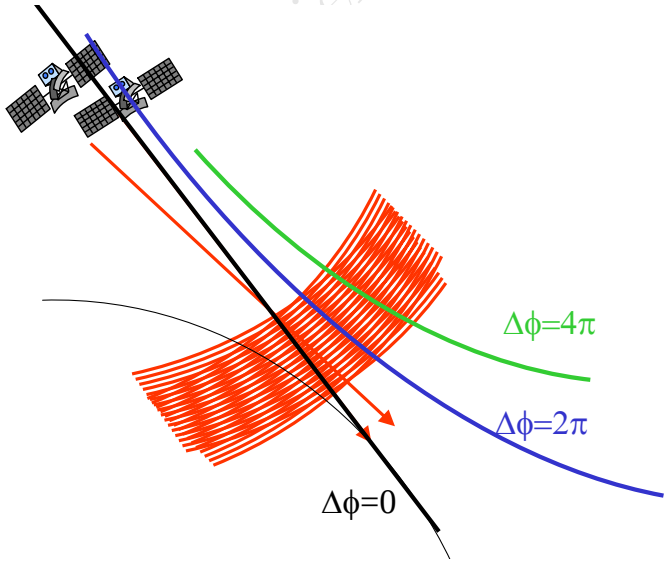


Figure 15.9: DEM generation: interferometry provides the third equation (an hyperboloid) to locate each target in the 3D world.

An example of a DEM generated from multiple SAR interferometry is shown in Fig. 15.10.

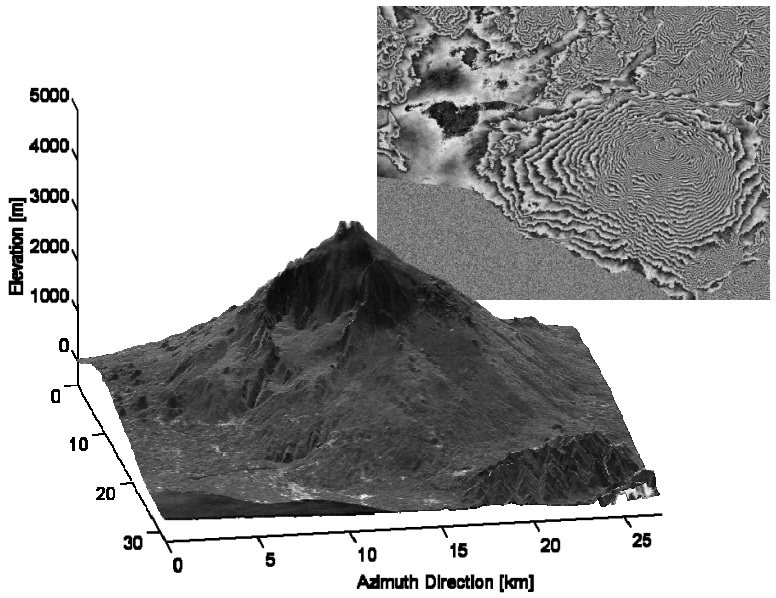


Figure 15.10: Flattened interferogram of Mt Etna and DEM derived by unwrapping several interferograms.

15.3 Differential & along-track Interferometry

DInSAR surveys are achieved by taking two observations, say by the same potion (zero Baseline), and assuming that the target Line-Of-Sight changes in the time-interval between the two. The interferometric phases can be computed by combining (15.1) and (15.2):

$$\Delta\varphi_d = \angle\left(s_M(\mathbf{P}) \cdot s_S^*(\mathbf{P})\right) = \omega_0\left(\tau_M - \tau_S\right) = \frac{2\omega_0}{c}\left(r_M - r_S\right) = \frac{4\pi}{\lambda}d$$

where *d is the Line Of Sight displacement.*

If the normal baseline is not null, we would have both the DinSAR contribution and the topographic one. In the flattened interferogram, the phase would be function of the target's height, **q**, and its LOS displacement, **d**:

$$\Delta\varphi = -\frac{4\pi}{\lambda}\frac{B_n q}{R\sin\theta} + \frac{4\pi}{\lambda}d \tag{15.17}$$

It is worthy to notice that at microwaves the effect of displacements is much remarkable than the elevation: for the C-band LEO SAR at 35^odeg, and 150 m baseline, we get:

$$\Delta\varphi = -\frac{q}{10} + 225d$$

a LOS displacement of 2.8 cm would introduce the same effect as a 2 km elevation. The major outcome of (15.17) is however that **we cannot distinguish between LOS displacements and elevation on the base of a single interferogram**, and this poses some limits in the "classical" interferometry. The problem can be avoided by making *either the temporal or the normal baselines zero*, e.g taking *two simultaneous acquisitions* for making DEMs, like for airborne systems, that exploits two across-track mounted antennas, or two (or more) *zero-baseline acquisitions to estimate LOS deformations*, like for Ground Based SAR that exploits a system mounted on a rail. However, this cannot be so easy, like for spaceborne SAR. In those cases, the multi-baseline interferometry will solve the problem.

15.3.1 Along Track Interferometry

One particular case of DInSAR in the *along track interferometry*, where two antennas are aligned in the direction of the flight. This can be achieved by a fore and aft antennas on the aircraft or by splitting the spaceborne antenna array into two half array with phase centers displaced by La/2. Whatever is stationary on the ground is invisible to ATI, giving the same contribution two the two images, but for a delay. Instead, if a *target moves across-track*, its LOS distance changes and a differential phase is observed. Let us assume that the LOS (Line of Sight) velocity is v_{LOS}, the sensor moves along track with v_s and the "baseline" is B_a (*baseline along-track*):

$$\begin{aligned}\Delta\varphi_d &= \frac{4\pi}{\lambda}d = \frac{4\pi}{\lambda}v_{LOS}\cdot\Delta\tau = \frac{4\pi}{\lambda}B_a\frac{1}{v_s}v_{LOS}\\ &= \frac{4\pi}{\lambda}B_a\frac{1}{v_s}v_{tr}\sin\theta\end{aligned} \tag{15.18}$$

where v_{tr} is the across-track velocity and θ the incidence angle.

Here again we can define a *velocity of ambiguity:*

$$\begin{aligned}2\pi &= \frac{4\pi}{\lambda}B_a\frac{1}{v_s}v_a\sin\theta\\ v_a &= \frac{\lambda}{2}\frac{v_s}{B_a\sin\theta}\\ v_{tr} &= \frac{v_a}{2\pi}\Delta\varphi\end{aligned}$$

The expression converts across-track velocities to phases. Here again, we need to dimension the velocity of ambiguity by accounting fro wrapping, but also the resolution that one wishes., that results in a constraint to the along track baseline. For example, in **X** band LEO sensor typical velocity is 7.5 km/s, if we assume $\theta = 30^0$and $B_a = 3$ m, thus half of the 6 meters antenna of Cosmo or TerraSAR-X, we get a velocity of ambiguity of 75 m/s. Such velocity would be good form traffic monitoring, but not -say- for ocean currents, for which a sensitivity of 10 cm/s is required.

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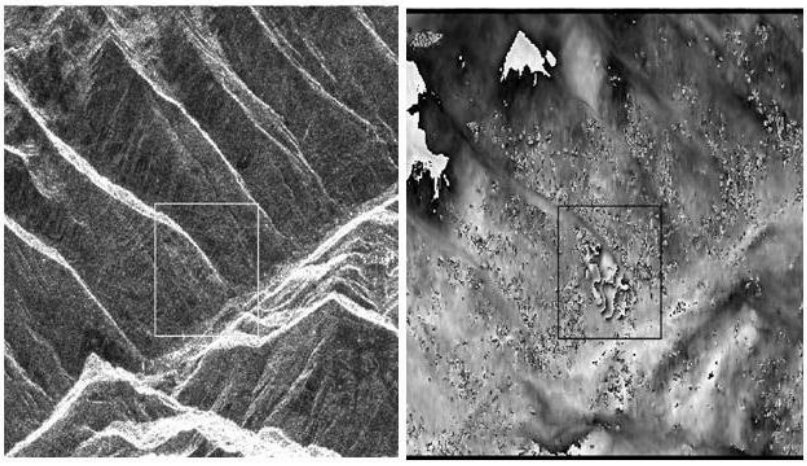


Figure 15.11: Fringes due to terrain deformation, on the right, referred to landslides of St. Etienne, in the rectangle marked in the amplitude SAR image on the left.

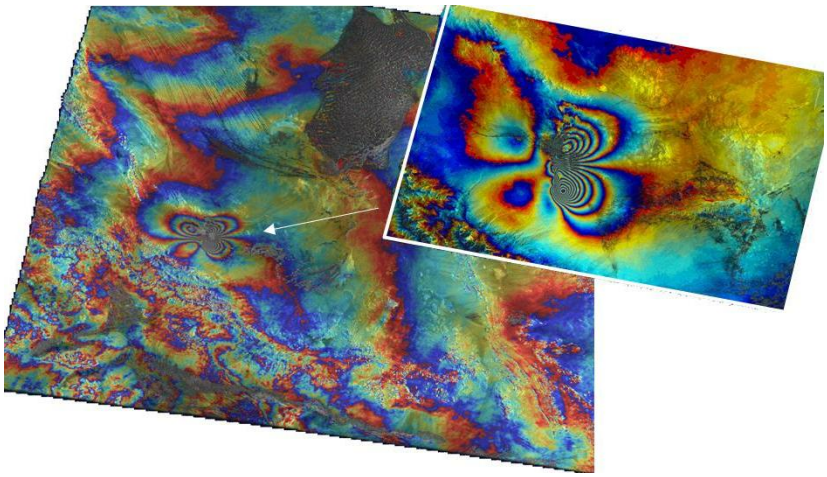


Figure 15.12: Co-seismic fringes of Bam earthquake, 26/12/03, acquired with almost zero baseline. Left: ScanSAR WSM mode (400 × 400 km) and, right, full resolution Image Mode (100 × 100 km). The change in the fringe frequency is due to the different angle of incidence.

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15.4 Noise in interferometry

So far we have assumed the ideal case, in which no sources of noise exist. We face now the real case in presence of errors, coming both by thermal and quantization noises, but also from changes in the scene observed between the acquisitions and in the acquisition geometry itself. Finally we address the change in the propagation, due to both tropospheric effects and ionospheric effects.

15.4.1 Interferometric quality: the coherence

Let us rewrite the two equations (15.115.2), referred to a single target (point or distributed) in P as follows:

$$s_M = g_p(t_M) \cdot e^{-j\phi_M(t_M)} e^{j\varphi_p(t_M)} + n_M(t_M) \tag{15.19}$$

$$s_S = g_p(t_S) \cdot e^{-j\phi_S(t_S)} e^{j\varphi_p(t_M)} + n_S(t_S) \tag{15.20}$$

where t_M and t_S are the master and slave time respectively.

Notice that even in absence of noises, n_M and n_S, the interferometric information, $\phi_M - \phi_S$ would vanish if -say- the phase of the target **changes** randomly from master to slave. In fact in absence of noise and for **independent** phases, uniform distributed in 0,π, we get from (:

$$\begin{aligned} E[i] = E\left[s_M s_S^*\right] &= g_p(t_M) g_p(t_S) \cdot e^{-j(\phi_M - \phi_S)} E\left[e^{j\varphi_{pM}} e^{-j\varphi_{pS}}\right] \\ &= g_p(t_M) g_p(t_S) \cdot e^{-j(\phi_M - \phi_S)} E\left[e^{j\varphi_{pM}}\right] E\left[e^{j\varphi_{pS}}\right] \\ &= 0 \end{aligned} \tag{15.21}$$

The best case is when the scene observed is **totally correlated,** whereas a partial decorrelation would reduce the amplitude of the signal in favor of the noise. It is quite clear that the quality in interferometry is well expressed by the absolute value of the coherence as discussed in section 5.1.2. That's to say, the interferometric quality, hence coherence, is not only affected by **noise** superposed to the images, but also to changes in the images themselves,

these changes may be occur in the time interval between the two acquisitions (**temporal changes**), or may be due to the variation of the scene when observed by different view angles (**spatial changes**). It is clear that temporal decorrelation is unavoidable in differential interferometry, whereas spatial decorrelation is the counter part for DEM generation.

The phase of the coherence is, from (5.5):

$$\begin{aligned}\angle\gamma_s(r) &= \angle E\left[s_M(P)s_S^*(P)\right] \\ &= \angle e^{-j(\phi_M-\phi_S)}E\left[e^{j\left(\varphi_{pM}-\varphi_{pS}\right)}\right] \\ &= \phi_M-\phi_S=\phi_i\end{aligned}$$

by assuming the the target's phase does not change in mean value, in the two images.

To summarize, the complex coherence phase is the information given by interferometry and the absolute value a measure of quality of that information.

15.4.2 Multi baseline interferometric model

In order to study the different sources of decorrelation, we first need a model for the generation of the SAR images exploited by interferometry.

The model shown in Fig. 15.13 is very general and applies to single baseline as well as multiple baseline interferometry.

In the figure we represent:

- the source reflectivity, that is observed at each acquisition time. In the case of a distributed target, we know $s(r,\tau)$ to be a white complex circular process, that we assume stationary.
- the phase modulation due to both geometry and deformations as in (15.17),
- the thermal, quantization and other additive noise contribution, $w(r)$,
- the function $f_n(r,\tau)$ modeling the end-to-end SAR transfer function.

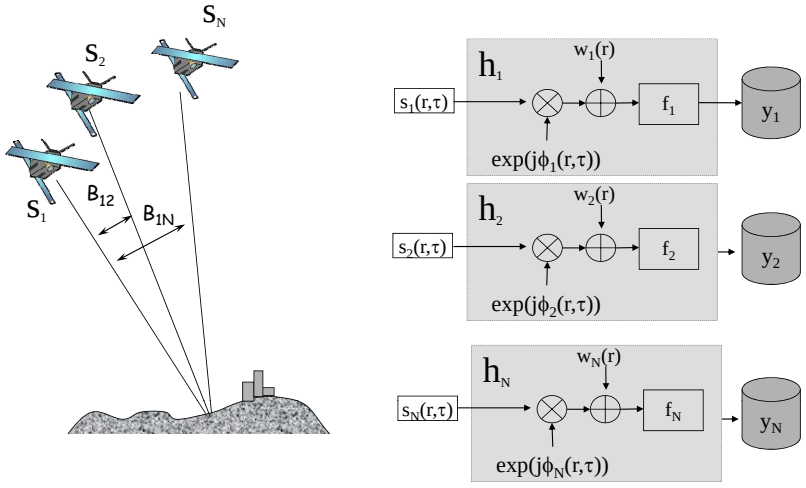


Figure 15.13: Model for multi-baseline SAR interferometry.

The objective of SAR interferometry is to estimate the phase modulation term, given the output. The source reflectivity is also an unknown, yet a nuisance as not of interest. The general formulation for the "phase" in (15.17) is:

$$\phi_o = \frac{2\pi}{c} f_0 R(S,P) \tag{15.22}$$

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This is the "optical path", the two-way distance from the Sensor S, and the target P, at the acquisition time. The term "optical" has been introduced to account for the fact that the distance is not the euclidean one, but accounts for the non ideal medium, the atmosphere, and moreover the non ideal propagation velocity. Henceforth, the distance R includes deformation, topography, and propagation effects.

If we assume that the first image is the master, and all the images are coregistered in its slant range, azimuth reference (r, τ),we may approximate each acquisition as a superposition of scatterers contribution:

$$\begin{aligned} y_n(r) &= \sum_{p=0}^{P-1} s_n(p) \exp\left(j\phi_{on}(r)\right) f(r - r_p) + w_n(r) * f_n(r) \\ &= \{s_n(r) \cdot \exp\left(j\phi_{on}(r)\right)\} * f(r) + w_n(r) * f_n(r) \end{aligned} \tag{15.23}$$

where $s_n(p)$ is the amplitude and phase of the p-th scatterer, $s_n(r)$ is the surface reflectivity. Notice that we have assumed the 1D case, along range, in this expression (and we will keep this model), for simplicity. This model can exploited to solve for topography, subsidence and atmospheric induced noise [17, 18].

15.4.3 Decorrelation sources

In the interferometric model so far discussed there can be identified three decorrelation sources, that can appear alone or mixed in the real case:

we immediately acknowledge that *thermal and quantization noises*, $w_n(r)$, acts as disturbances, however, they are not the only one. Let us cancel these noise, and compute the interferogram, thus we assume the two channel case:

- an **additive noise** term, that can be the result of thermal and quantization, as well as random clutter,
- a change in the scene observed from one time to the other, s_n or **scene decorrelation,**
- the combined effect of the phase modulation and the filtering, that's to say when targets with different optical paths contributes to the same resolution cell. We define this as volumetric decorrelation. Decorrelation due to thermal noise, scene decorrelation and changes, already discussed in section 5.1.2. We limit here to add the contribution of scatters dispersed into a volume.

15.4.3.1 Volumetric decorrelation

The volumetric decorrelation comes form the mixing of targets at different cross–range distance in the resolution cell. We have two cases:

- the first is a roughness, when targets do not lay on perfect flat surface
- the second is the case of an homogeneous scatterer distributed in a volume, like for raindrops in clouds, or snow, or crops and even forests.

Lets us study the rough case by assuming that RADAR senses a collection of N targets dispersed in the resolution cell. Each target has a random complex reflectivity, σ_p:

$$E\left[I\right] = \sum_{p=0}^{N-1} \sum_{q=0}^{N-1} g\left(r_p\right) g^*\left(r_q\right) E\left[\sigma_p \sigma_q^* \exp\left(-j\frac{4\pi}{\lambda}\left(R_M(p) - R_S(q)\right)\right)\right] \tag{15.24}$$

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where g(r) is the autocorrelation of the transmitted pulse at range r. The equation is true but for a scale factor that depends on RADAR equation that can be ignored as for evaluation of coherence.

Eventually we exploit the fact that the complex amplitudes are independent of the distances, and they are stationary (in the cell) and incorrelated:

$$E\left[\sigma_p \sigma_q^*\right] = \begin{cases} 0 & p \neq q \\ \sigma^2 & p = q \end{cases}$$

The expected value of the interferogram is:

$$\begin{aligned} E[I] &= \sum_{q=0}^{N-1} |g(p)|^2 \cdot \sigma^2 \cdot E\left[\exp\left(-j\frac{4\pi}{\lambda}\Delta R(q)\right)\right] \\ &= N \cdot |g(p)|^2 \cdot \sigma^2 \cdot E\left[\exp\left(-j2\pi\frac{q}{q_{2\pi}}\right)\right] \end{aligned}$$

This last equation has been obtained by (15.13) and by assuming that the target elevation q is a random variable, say with a known probability density, $f_q(q)$. We then express the volumetric coherence as follows:

$$\begin{aligned} \gamma &= \frac{E[I]}{\sqrt{\left(N \cdot |g(p)|^2 \cdot \sigma^2\right)^2}} \\ \gamma_V &= E\left[\exp\left(-j\phi\right)\right] \\ \phi &= k_q q \\ k_q &= \frac{2\pi}{q_{2\pi}} \end{aligned} \tag{15.25}$$

The expectation of the random variable ϕ in (15.25) can be computed by evaluating the characteristic state function, the Fourier Transform of the probability density of ϕ, for unitary argument:

$$\begin{aligned} \gamma_V &= E\left[\exp\left(-j\phi\right)\right] \\ &= \int \exp\left(-j\phi\right) f_\phi(\phi) d\phi = F_\phi\left(\omega|_{\omega=1}\right) \end{aligned}$$

As an example, for a Gaussian distribution of elevations:

$$\begin{aligned} f_q(q) = N(0, \sigma_q) &\rightarrow f_\phi(\phi) = N\left(0, k_q \sigma_q\right) \\ \gamma_V = F_\phi(1) &= \exp\left(-\frac{\sigma_\phi^2}{2}\right) = \exp\left(-2\pi^2 \frac{\sigma_q^2}{q_{2\pi}^2}\right) \end{aligned}$$

As a further example, we assume uniform slopes in $\pm\Delta_q/2$

$$\begin{aligned} f_q(q) &= \frac{1}{\Delta_q} \operatorname{rect}\left(\frac{q}{\Delta_q}\right) \rightarrow f_\phi(\phi) = \frac{1}{k_q \Delta_q} \operatorname{rect}\left(\frac{\phi}{k_q \Delta_q}\right) \\ \gamma_V &= \operatorname{sinc}\left(\frac{k_q \Delta_q}{2\pi}\right) = \operatorname{sinc}\left(\frac{\Delta_q}{q_{2\pi}}\right) \end{aligned}$$

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from which we get total decorrelation when elevations are completely dispersed in the elevation of ambiguity $q = q_{2\pi}$.

Let us now assume to have an homogeneous volumetric scatterer, then each slice of vertical thickness Δ_q attenuates the signal with exponential absorption and introduces a phase proportional to the the decorrelation would be [19]:

$$\gamma = \frac{\int_{-D}^{0} \exp\left((d_2^{-1} + jk_q)q\right)\ dq}{\int_{-D}^{0} \exp\left(d_2^{-1}q\right)\ dq} \tag{15.26}$$

$$= \frac{d_2^{-1}}{(d_2^{-1} + jk_q)} \frac{1 - \exp\left(-(d_2^{-1} + jk_q)D\right)}{1 - \exp\left(-d_2^{-1}D\right)}. \tag{15.27}$$

where D is the volume depth and d_2 is the length of the 1 Neper decay, that can be related to the is the refraction angle (in the volume), θ_v and the intrinsic 1-way extinction coefficient, σ

$$d_2 = \frac{\cos\theta_v}{2\sigma}$$

the value of $\sigma$is in Neper / m, but can be expressed in dB/m: $\sigma_{dBm} = 10 \cdot \sigma \cdot \log_{10} e$. Finally, for an indefinite volume or just say much thicker than $d_2$we get

$$\gamma_{vol} = \frac{1}{1 + j\frac{2\pi d_2}{q_{2\pi}}}$$

That introduces the same phase shift and an attenuation similar to the transfer function of a single pole system, function of the decay length, $d_2$and the pole, $s = -q_{2\pi}$ notice that coherence halves for

$$\frac{2\pi \cos\theta_v}{2\sigma} = q_{2\pi}$$

As an example for a X band system, the typical extinction of a forest is 1 dB, then $\sigma = 0.23$ and incidence angle $\theta_v = 35^0$that introduces an extinction length $d_2 = 1.77$m if the elevation of ambiguity is 30 m, and the forest $\gg 1.7$ m, the coherence is 0.93 in modulus and -20 deg in phase. Such phase accounts for an equivalent phase center that is somewhat down with respect to surface top.

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15.4.4 Phase accuracy: Cramer Rao bound

The phase accuracy can be related to coherence by exploiting the Cramer Rao bound [20]:

$$\sigma_\phi^2 \geq \sigma_{CRB}^2 = \frac{1 - \gamma^2}{2\gamma^2 R} = \frac{1 - \gamma^2}{2\gamma^2 N_L} \tag{15.28}$$

that holds for distributed targets.

The true phase estimator variance is known from literature [21], although not in closed form, and it has been represented in Fig. 15.14 as a function of both coherence and number of looks. We observe that with $N_L = 1$ look, the dispersion is rather high even at very good coherence, as speckle has a significant probability of having very small numbers (remember that phase in not defined for zero amplitude), as shown in Fig. 4.14 when M=1. However, the dispersion is strongly reduced by increasing the number of looks. When $N_L > 4$ the Cramèr-Rao (15.28) is a rather good approximation.

In the case of a point target, and SNR$\gg$1 the interferogram can be approximated by:

$$I = \left(\rho e^{j\phi_M} + n_M\right)\left(\rho e^{-j\phi_S} + n_S\right) = \rho^2 e^{j\phi_M - \phi_S} + n_M + n_S + n_M n_S \simeq \rho^2 e^{j\phi_M - \phi_S} + n_M + n_S$$

leading to a phase noise standard deviation

$$\sigma_\phi \approx \sqrt{\frac{1}{2SNR_I}} = \sqrt{\frac{\sigma_M^2 + \sigma_S^2}{\rho^2}} = \frac{1}{SNR} \tag{15.29}$$

where SNR_I and SNR are respectively the SNR on the interferogram and on a single image. Notice that the same (15.29) would be achieved by a highly correlated distributed target, i.e. by exploiting (15.28) and (14.19).

15.4.5 Elevation accuracy

The phase dispersion just estimated leads to the elevation accuracy, i.e. when interferometry is exploited for DEM generation, by exploiting the approximately linear phase-to-elevation law:

$$\sigma_h = \sigma_\varphi \frac{R\lambda \sin\theta}{4\pi B} = \frac{\lambda R \sin\theta}{2B_n} \frac{\sqrt{1-\gamma^2}}{\gamma\sqrt{2N_L}} \tag{15.30}$$

This expression holds as far as no atmospheric delay applies (see next chapter), so for simultaneous SAR imaging. We can then be able to assess the local DEM accuracy from the coherence maps as shown in Fig. 15.14.

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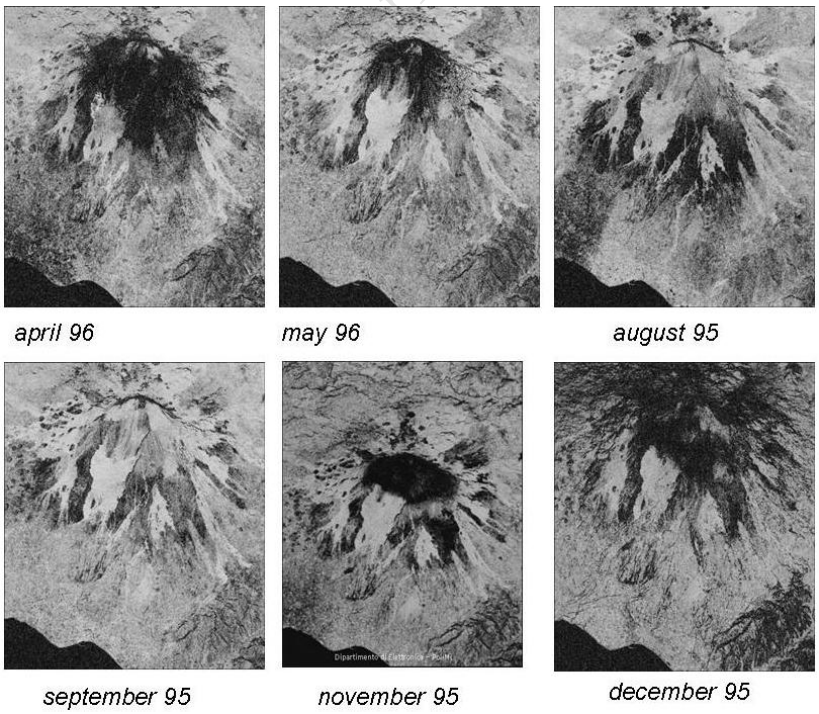


Figure 15.14: Coherence maps taken at intervals of one day in different months of the year. Area of Mt. Etna, Italy. ERS data.

15.4.6 Example: DEM generation

As an example, let us evaluate the accuracy in DEM by a C-band SAR like Sentinel-1, see tab: 15.1, The height standard deviation is derived from (15.10):

$$\sigma_q = \frac{\lambda R \sin\theta}{2B_n}\sigma_\phi \tag{15.31}$$

where σ_ϕ is computed basing on the coherence as in (15.28). The CRB holds for a sufficient number of looks, that is however necessary to provide low phase noise and then a successful unwrapping: we assume here to average N_r =2 $\times$ N_a =2 independent samples ins range, azimuth . The actual number of looks is slightly reduced by the spectral shift for sloped terrain,that limit the correlated bandwidth:

$$N_L = N_r \frac{B_r}{B_r - \Delta_B} \cdot N_a$$
$$\Delta_B = f_0 \frac{B_n}{R\tan(\theta_i - \alpha)}$$

Notice that the spectral shift Δ_B worsens the accuracy of the phase at the highest baselines (15.31), but on the other side baselines improve the sensitivity versus height. T

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15.4.7 Example: Dual Beam Interferometry

Along track interferometry provides only the estimation of the velocity in the Line Of Sight, that in turn is the projection of the ground velocity along range. In many cases the interest is in estimating both velocity components. This is achieved by making two ATI interferograms with different LOS, in order to measure two component and then reconstruct the full 2D velocity field. This principle is applied to both earth deformations (landslides, tectonic activity) in the Multiple Aperture Interferometry and to ocean currents estimation. It should be noted that interferometry is not the sole way to estimate ocean currents, as shown form the Doppler spectrum based estimate in Fig. 15.15 on the right.

The Dual Beam Interferometric system [22] shown in Fig. 15.15, left, provides a 2D estimation of ocean currents by exploiting two antennas, aft and fore, and two beams for each antenna, each of them with a different squint angle. The idea is that we have two interferograms with two different baselines (opposite).

For the fore beam, the range vector, assuming origin in at airplane height and versor are:

$$\begin{aligned}\mathbf{R} &= r_0 \cdot \begin{bmatrix} \tan\psi & \sin\theta & -\cos\theta \end{bmatrix}^T \\ \mathbf{u}_{rf} &= \frac{\mathbf{R}}{|\mathbf{R}|} = \begin{bmatrix} \sin\psi & \cos\psi\sin\theta & -\cos\psi\cos\theta \end{bmatrix}^T\end{aligned}$$

and for the aft beam ψ is opposite. We can relate v_{LOSf} (in the line of sight of the fore beam) to the two components of the velocity.

$$\begin{aligned} v_{LOSf} &= \mathbf{v}_{rf}^T \cdot \mathbf{u}_{rf} \\ &= \begin{bmatrix} \sin\psi & \cos\psi\sin\theta & -\cos\psi\cos\theta \end{bmatrix} \begin{bmatrix} v_x \\ v_y \\ 0 \end{bmatrix} \\ &= v_x \sin\psi + v_y \cos\psi \sin\theta \end{aligned}$$

General Params		
Frequency	*Hz*	5.41E+09
λ	*m*	5.55E-02
Antenna Length	*m*	10
Tx Bdwth	*Hz*	8.50E+07
Incidence angle	*deg*	30.00
Slant Range	*m*	7.88E+05
Baseline	*m*	300
Ambiguity elevation	*m*	36.4
Range		
Averaging window	*pxls*	2
Terrain slope (range)	*deg*	10
Slant range resolution	*m*	3.5
Ground range resolution	*m*	10.3
Spectral shift	*MHz*	5.66
Common band	*MHz*	79.3
Freedom Degrees	*Nf*	3.7
Azimuth		
Averaging window	*pxls*	2
Azimuth resolution	*m*	10
Coherence		
Nesz	*dB*	-26.0
backscatter	*dB*	-7.5
CNR	*dB*	18.5
gamma_CNR		*0.99*
Volume box height	*m*	10.0
gamma_vol	*gamma_v*	0.88
gamma_target_prc		0.95
Total coherence	*gamma*	0.82
DEM accuracy		
Phase noise sigma	*rad*	0.25
Phase to height	*rad/m*	36.40
Height error sigma	*m*	9.1

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Table 15.1: Evaluation of the accuracy of a DEM by C-band SAR.

and by reversing ψ symmetrically for the aft beam:

$$v_{LOSa} = -v_x \sin\psi + v_y \cos\psi \sin\theta$$

For each of the two interferograms, we relate the phase to the LOS component as from (15.18):

$$\varphi = \frac{4\pi}{\lambda}\frac{B_{ATI}}{v_s} v_{LOS} = \frac{2\pi}{v_{la}} v_{LOS}$$
$$v_{la} = \frac{\lambda v_s}{2B_{ATI}}$$

We can combine the two measures of the phases in the fore and aft interferograms, with the model by the following system with two unknowns, v_x and v_y:

$$\begin{bmatrix} \phi_f \\ \phi_a \end{bmatrix} = \frac{2\pi}{v_{la}\cos\phi} \begin{bmatrix} \sin\psi & \cos\psi\sin\theta \\ -\sin\psi & \cos\psi\sin\theta \end{bmatrix} \begin{bmatrix} v_x \\ v_y \end{bmatrix} \tag{15.32}$$
$$\mathbf{\Phi} = \mathbf{AV}$$

That can be inverted to estimate the 2D component of currents:

$$\begin{bmatrix} v_x \\ v_y \end{bmatrix} = \frac{v_{la}}{2\pi} \begin{bmatrix} \frac{1}{2\sin\psi} & \frac{1}{2\sin\psi} \\ \frac{1}{2\cos\psi\sin\theta} & -\frac{1}{2\cos\psi\sin\theta} \end{bmatrix} \begin{bmatrix} \phi_f \\ \phi_a \end{bmatrix} \tag{15.33}$$

$$v_x = \frac{v_{la}}{2\pi\sin\psi}\frac{(\phi_f + \phi_a)}{2}$$

$$v_y = \frac{v_{la}}{2\pi\cos\psi\sin\theta}\frac{(\phi_f - \phi_a)}{2}$$

As fore and aft beams have opposite baselines along x, the x-component will be the subtraction of the two phases, whereas for the y component we have same baselines, then the sum of the two terms. Notice that, for $\psi = 0$ we get back the ATI result in (15.18).

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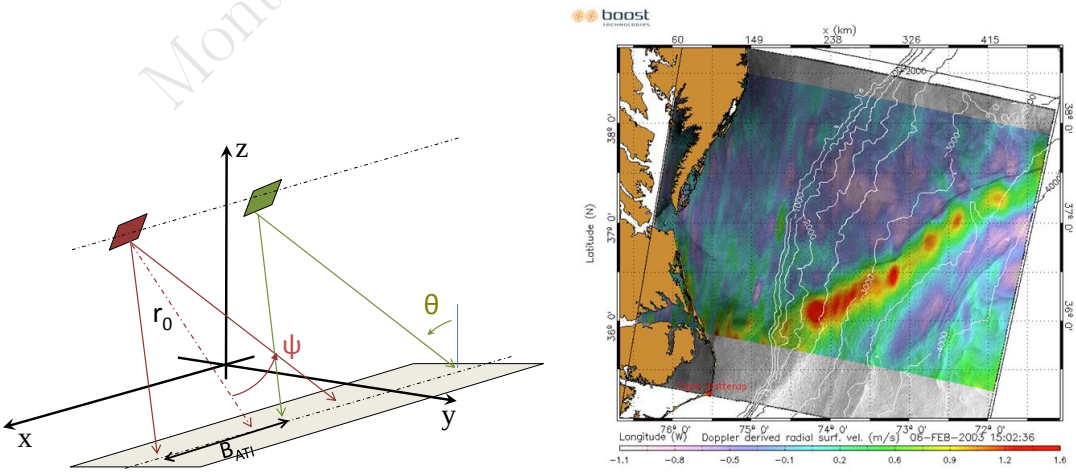


Figure 15.15: Left: Dual Beam Interferometer for estimation of 2D component of sea currents. Right: example of surface current estimated by Doppler techniques in complex SAR image (courtesy of Boost-CLS).

The accuracy for current retrieval can be derived by the expectation and the covariance matrix. The estimate is unbiased:

$$E\left[\mathbf{V}\right] = \mathbf{A}^{-1}E\left[\mathbf{\Phi}\right] = \mathbf{V}$$

as results from (15.32) and by assuming that phase is unbiased (that is if affected by zero-mean Gaussian noise, like thermal and scene decorrelation). The covariance matrix:

$$\mathbf{C}_v = E\left[\mathbf{v}\mathbf{v}^T\right] = \mathbf{A}^{-1}E\left[\boldsymbol{\phi}\boldsymbol{\phi}^T\right]\left(\mathbf{A}^{-1}\right)^T = \mathbf{A}^{-1}C_\phi\left(\mathbf{A}^{-1}\right)^T \tag{15.34}$$

$$= \begin{bmatrix} \sigma_{v_x}^2 & \sigma_{v_x}\sigma_{v_y} \\ \sigma_{v_x}\sigma_{v_y} & \sigma_{v_y}^2 \end{bmatrix} \tag{15.35}$$

is derived from the phases covariance:

$$\mathbf{C}_v = \begin{bmatrix} E\left[\phi_f\phi_f^T\right] & E\left[\phi_f\phi_a^T\right] \\ E\left[\phi_a\phi_f^T\right] & E\left[\phi_a\phi_a^T\right] \end{bmatrix} = \sigma_\phi^2\mathbf{I}$$

where σ_ϕ^2can be related to the coherence and the number of looks according to (15.28). We replace into (15.34) getting:

$$\begin{aligned}\mathbf{C}_v &= \boldsymbol{\sigma}_{\boldsymbol{\phi}}^{\mathbf{2}}\mathbf{A}^{-1}\left(\mathbf{A}^{-1}\right)^T \\ &= \sigma_\phi^2\frac{v_{la}^2}{(2\pi)^2}\begin{bmatrix} \frac{1}{2\sin^2\psi} & 0 \\ 0 & \frac{1}{2\cos^2\psi\sin^2\theta} \end{bmatrix}\end{aligned}$$

the variances of the x and y components are the two diagonal term. Then:

$$\begin{aligned}\sigma_{vx} &= \frac{v_{la}}{2\pi}\frac{1}{\sqrt{2}\sin\psi}\sigma_\phi \\ \sigma_{vy} &= \frac{v_{la}}{2\pi}\frac{1}{\sqrt{2}\cos\psi\sin\theta}\sigma_\phi\end{aligned}$$

the square root of two comes from the combination of two independent noisy measures. Notice that the along track component is much noisier than the across track, due to its lower sensitivity, unless the squint angle becomes larger than 45^0.

The variance of the interferometric phase is give by (15.28), where the coherence is computed by accounting for thermal noise and sea decorrelation (see [22] for decorrelation model):

$$\gamma = \gamma_t \cdot \exp\left(-\left(\frac{t}{\tau}\right)^2\right)$$

The exponential sea decorrelation is computed, according to Fraiser [22], where $t = B_a/v_s$is the ATI time shift, and the constant τ is approximated as:

$$\tau = 3\frac{\lambda}{u}\sqrt{erf\left(2.7\frac{\rho}{u^2}\right)}$$

ρ being the spatial resolution and u the wind speed (2-4 m/s, $<$ 10 m/s).

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15.4.8 Atmospheric Phase Screen

In SAR interferometry, the delay is the measure and the distance is obtained by converting times the propagation velocity. We have used so far the light speed in vacuum, however in the atmosphere the velocity is slightly lower, therefore

$$\begin{aligned}\phi &= 2\pi f_0 \frac{2\Delta R}{c'} = 2\pi f_0 \frac{2\Delta R}{c + e_c} \simeq 2\pi f_0 \frac{2\Delta R}{c} - 2\pi f_0 \frac{2\Delta R}{c} e_c \\ &= \phi_0 + \phi_{aps}\end{aligned}$$

The first is the phase as in free space, and the second the additional delay due to the APS, here approximated by an additive noise. The propagation delay term proportional to the LOS distance of the scatterer and by the additional one-way delay introduced by the non-idealities of the troposphere.

$$\phi_{aps} = \frac{4\pi}{\lambda} d_{atm,p} \tag{15.36}$$

The turbulence law extends for both space and time domain, then we can write:

$$E\left[\left\{d\left(t + \Delta_t, x + \Delta_x\right) - d\left(t, x\right)\right\}^2\right] = k \cdot \left(1 - \exp\left(-\rho\right)\right) \tag{15.37}$$

$$\rho = \sqrt{\left(\frac{\Delta_t}{T}\right)^2 + \left(\frac{\Delta_x}{L}\right)^2}$$

For small values of the exponent we have the power law:

$$k\left(1 - \exp\left(-\rho\right)\right) \simeq \left(1 - \left(1 - \rho\right)\right) = k\rho$$

but for large values we saturate to k= 400 mm^2, up to 1600 mm^2 in the worst cases or down to 100 mm^2 in the best, when T=10 hours. Figure 15.16 shows the statistics of APS in a week in summer as measured by Ground Based RADAR, or computed by meteorological stations [23].

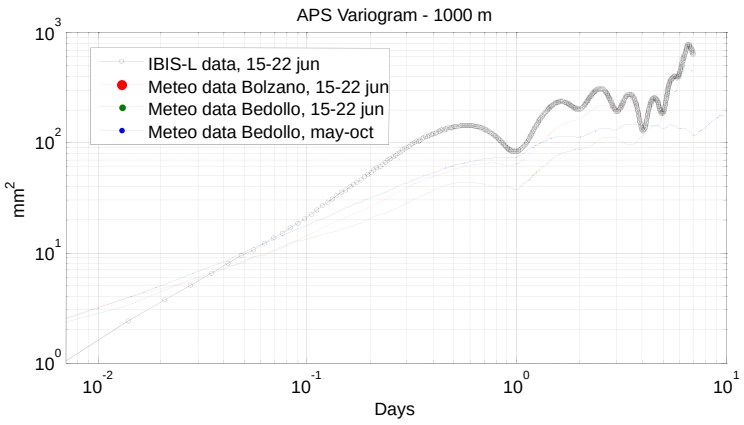


Figure 15.16: Atmospheric phase screen measured by GB RADAR or computed from meteorological station.

An example of the Atmospheric Phase Screen is in Fig. 15.17, an interferogram in Pianura Padana generated by ERS image with a very small baseline (30 m), thus an elevation of ambiguity of 270 m, The fringe pattern is clearly due to water-vapor turbulence.



Figure 15.17: Atmospheric phase screen in a repeat-pass ERS interferogram generated by a 30 m baseline, pianura padana.

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15.5 Persistent Scatterers Interferometry

The need of long temporal baselines to provide *estimates of small deformations* make SAR interferometry a challenging task fro both the presence of scene decorrelation and the Atmospheric Phase Screen. The PSI technique, introduced by Politecnico di Milano in the 90th [24] bases on the selection of those targets in the scene that are stable over long term. This selection is done by exploiting sets of repeat-pass SAR image and searching for stable amplitude and phases. As spaceborne SAR interferometry is involved, images are are observed by very different baseline, therefore the stable targets are also correlated over large celestial aperture, thus they are most point scatterers, loosely affected by volume decorrelation.

The innovation introduced by PS is not only in the selection of stable targets, but also on their use to remove the atmospheric phase screen. The PS form a dense network in space, whereas images are taken for very long term, thus it is possible to estimate and remove the APS by exploiting its statistical properties in (15.37). In practice APS is separable from slow deformations as when he second are correlated over months or years, whereas APS tends to be uncorrelated on a daily base. On the other hand, APS and deformations can also be separated in space domain, when deformations are much more local than the say one km ASP lobe.

15.5.1 Estimate and performances

In this multi-baseline case, we have N observations, and one target phase + the target location (say the height over reference DEM) unknown, Thus we could not estimate one deformation

for each PS, and we can choose either to group several PS, or to provide a parametric description for the deformation. The model for PS phase is then:

$$\boldsymbol{\phi} = \mathbf{A}\boldsymbol{\Theta} + \boldsymbol{\phi}^c + \boldsymbol{\phi}^a + \boldsymbol{\phi}^t$$

where Θ is the vector of parameters, $\mathbf{A}$ the transformation from parameters to phase, and the other terms are clutter, atmospheric and thermal noise. If we gather all the noise terms into a single one, we get:

$$\boldsymbol{\phi} = \mathbf{A}\boldsymbol{\Theta} + \mathbf{w}_{\boldsymbol{\phi}} \tag{15.38}$$

Let us assume as parameters the deformation rates and the elevation, the model in continuous time is:

$$\phi(t) = \psi_P - \frac{4\pi}{\lambda} \cdot v_P \cdot t - \frac{2\pi}{q_a} \cdot q_P + w_\phi(t)$$

We can discretize for the acquisition times $t_k$leading to (15.38), where the design matrix A , and the unknown column vector θ are:

$$\mathbf{A}\boldsymbol{\Theta} = \begin{bmatrix} 1 & k_v B_t(1) & k_q B_n(1) \\ 1 & k_v B_t(2) & k_q B_n(2) \\ \dots & \dots & \dots \\ 1 & k_v B_t(N) & k_q B_n(N) \end{bmatrix}; \Theta = \begin{bmatrix} \psi_P \\ v_P \\ q_P \end{bmatrix}$$

$$k_v = -\frac{4\pi}{\lambda};\ k_q = \frac{4\pi}{\lambda}\frac{1}{R\sin\theta}$$

where ψ is the phase proper of each target, v the deformation rate and q the target height over the reference DEM. The terms B_t and B_n are the temporal and normal baselines. If we assume white noise and same quality for all the targets, the Least Square solution is the best one, minimum variance and unbiased (asymptotically):

$$\widehat{\boldsymbol{\Theta}} = \left(\mathbf{A}^T\mathbf{A}\right)^{-1}\mathbf{A}^T\boldsymbol{\phi} = \mathbf{A}^\dagger\boldsymbol{\phi}$$

We can first show that the estimate in unbiased, for this we exploit the model (15.38):

$$E\left[\widehat{\boldsymbol{\Theta}}\right] = \mathbf{A}^\dagger E\left[\phi\right] = \mathbf{A}^\dagger\mathbf{A}\Theta + \mathbf{A}^\dagger\mathbf{E}\left[\mathbf{w}_{\boldsymbol{\phi}}\right] = \Theta$$

Thereafter, we derive the error covariance:

$$\begin{aligned} E\left[\left(\widehat{\boldsymbol{\Theta}} - \boldsymbol{\Theta}\right)\left(\widehat{\boldsymbol{\Theta}} - \boldsymbol{\Theta}\right)^T\right] &= E\left[\widehat{\boldsymbol{\Theta}}\widehat{\boldsymbol{\Theta}}^T\right] - \boldsymbol{\Theta}\boldsymbol{\Theta}^T \\ &= \mathbf{A}^\dagger E\left[\phi\phi^*\right]\mathbf{A}^\dagger - \boldsymbol{\Theta}\boldsymbol{\Theta}^T \\ &= \mathbf{A}^\dagger E\left[\left(\mathbf{A}\boldsymbol{\Theta} + \mathbf{w}\right)\left(\mathbf{A}\boldsymbol{\Theta} + \mathbf{w}\right)^*\right]\mathbf{A}^\dagger - \boldsymbol{\Theta}\boldsymbol{\Theta}^T \\ &= \mathbf{A}^\dagger\mathbf{C}_W\mathbf{A}^\dagger \end{aligned}$$

We assume that phase noise is white: $\mathbf{C}_W = \sigma_w^2 \cdot \mathbf{I}$

$$E\left[\left(\widehat{\boldsymbol{\Theta}} - \boldsymbol{\Theta}\right)\left(\widehat{\boldsymbol{\Theta}} - \boldsymbol{\Theta}\right)^T\right] = \sigma_w^2 \cdot \left(\mathbf{A}^T\mathbf{A}\right)^{-1}$$

The variances of target's phase, deformation rate and height are the term on the diagonal of $\left(\mathbf{A}^T\mathbf{A}\right)^{-1}$. Notice that the matrix becomes singular as the normal and temporal baselines are proportional: in that case it is impossible to distinguish between elevation and deformations.

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15.5.1.1 DOP

Let us define temporal baselines $B_t(i), i = 1 \dots N$ and normal baseline: $B_n(i), i = 1 \dots N$, Let us first define the terms:

$$k_v = \frac{4\pi}{1000 \cdot \lambda} \text{ [rad/mm]}; k_q = \frac{4\pi}{\lambda} \frac{1}{R \sin\theta} \text{[rad/m]}$$

The design matrix is

$$\mathbf{A} = \begin{bmatrix} 1 & k_v \frac{B_t(1)}{365.24} & k_q B_n(1) \\ 1 & \dots & \dots \\ \dots & \dots & \dots \\ 1 & k_v \frac{B_t(N)}{365.24} & k_q B_n(N) \end{bmatrix}$$

where 365.24 is the average number of days per year. We need to compute:

$$\mathbf{C} = \left(\mathbf{A}^T \mathbf{A}\right)^{-1} \tag{15.39}$$

and then derive the standard deviation of deformation rate by the third term on the diagonal:

$$\sigma_v \simeq \sqrt{c_{2,2}} \cdot \sigma_\phi \text{ [mm/year]} \tag{15.40}$$

$$\sigma_q \simeq \sqrt{c_{3,3}} \cdot \sigma_\phi \text{ [m]} \tag{15.41}$$

The scaling in mm/year and m helps in preconditioning the matrix, as the elevation, the phase and the deformations now are valued int he same range. The two terms $\sqrt{c_{3,3}}$ and $\sqrt{c_{2,2}}$ has the meaning of sensitivity of velocity and heights versus the phase noise standard deviation on the measures, and thus play the role of Dilution Of Precision in GPS.

15.5.2 Example: subsidence rate accuracy with known elevation

As a very simple case, we can assume that only the velocity is unknown:

$$\mathbf{A} = \begin{bmatrix} 1 & k_v B_t(1) \\ 1 & k_v B_t(2) \\ \dots & \dots \\ 1 & k_v B_t(N) \end{bmatrix}$$

Then

$$\mathbf{C} = \left(\mathbf{A}^T \mathbf{A}\right)^{-1} = \frac{\lambda^2 \sigma_\phi^2}{16\pi^2 \left(N \sum_n \tau_n^2 - \left(\sum_n \tau_n\right)^2\right)} \begin{bmatrix} 16\frac{\pi^2}{\lambda^2} \sum_n B_t^2(n) & -4\frac{\pi}{\lambda} \sum_n \tau_n \\ -4\frac{\pi}{\lambda} \sum_n \tau_n & N \end{bmatrix}$$

the DOP for the subsidence rate is given by the second term on the diagonal:

$$\sigma_{v_p}^2 = \frac{\lambda^2 \sigma_\phi^2}{16\pi^2} \frac{N}{N \sum_n B_t^2(n) - \left(\sum_n B_t(n)\right)^2}$$

In the case of constant revisit interval, d days

$$\begin{aligned} \sigma_{v_p}^2 &= \frac{\lambda^2 \sigma_\phi^2}{16\pi^2} \frac{N}{N \sum_{n=1}^N \left(d\left(n + n_0\right)\right)^2 - \left(\sum_{n=1}^N d(n + n_0)\right)^2} \\ &= \left(\frac{\lambda}{4\pi} \sigma_\phi\right)^2 \frac{12}{\left(N^3 - N\right) d^2} \end{aligned}$$

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that does not depends on the start time, $B_t(0)$. Notice the reduction of noise variance with term $N^3/12$ that is typical of linear regression, that exploits the leverage of the whole time interval gaining a factor T_{tot}=Nd over the standard deviation, and an additional factor $\sqrt{N}$due to the N measures.

For Sentinel-1, with $d = 12$ days sand $N = 30$ takes, that is one year of data we get:

$$\begin{aligned}\sigma_{v_p} &= \frac{\lambda}{4\pi}\sigma_\phi \frac{\sqrt{12}}{\sqrt{N^3 - N}\frac{d}{365}} \\ &\simeq 3 \cdot \sigma_\phi \text{ [mm/year]}\end{aligned} \tag{15.42}$$

Thus DOP is spaceborne SAR reduces accuracy to few mm/years, thanks to the long observation times.

Furthermore, we have to estimation the phase noise power that is the superposition of thermal, clutter and APS. We can compute the overall coherence by composing the different effects and then evaluating the phase standard deviation as from the CRB in (15.28) or its large SNR approximation (15.29).

The residual error after estimation and removal of the APS is not straightforward, as the APS is a correlated process in time and space, hence we should know the space and temporal sampling of the reference targets used to retrieve the APS. In this simplified analysis, we limit to two cases: when the distance between PS is much larger than the APS lobe and the opposite, when there are many PS in the APS lobe.

In the first case, we apply (15.40,15.42) and use for $\sigma_\phi$the APS value (15.36). For an average two-way delay of 1 cm, in C-band we have

$$\sigma_\phi \simeq 2\pi \frac{0.01}{0.056} = 1.1$$

that is by far much larger than the target long term correlation, leading to an accuracy very close to 3.3 mm/year.

On the other hand, if we assume a dense PS population, that ranges from 500 PS/km^{2}in medium resolution C band, to 10000-90000 PS/km^2 found in high resolution X band (1 x 1 m, [25]), than the APS power is reduced proportionally to the number of samples, and it is enough to have more than 100 samples to reduce the atmospheric factor to a negligible term when compared to the PS quality, leaving the figures of 0.45 to 1.8 mm/year previously found. Such performances are not achievable with GPS as the number of measures is never so dense (unless ad-hoc DGPS campaign are done).Fig. 15.18 shows a typical result achieved in exploiting a Multi Baseline stack of data by the Permanent Scatterers approach.

15.5.3 Note on phase estimation

The value of the APS phase noise found prevents the estimation of PS height and deformation rate by the ML approach, that is actually a bidimensional Fourier Transform, at least for the C-band and higher frequency. This is due to the fact the the APS error can be assimilated to a Gaussian phase noise: as the phase standard deviation increase, the wrapping causes the tail of the distribution to fold and the pdf tends to be uniform. In this extreme condition, the mean phase is completely lost, as any value is equally probable. In order to appreciate this fact, the accuracy of the phase estimate has been plotted in Figure 15.19 as a function of

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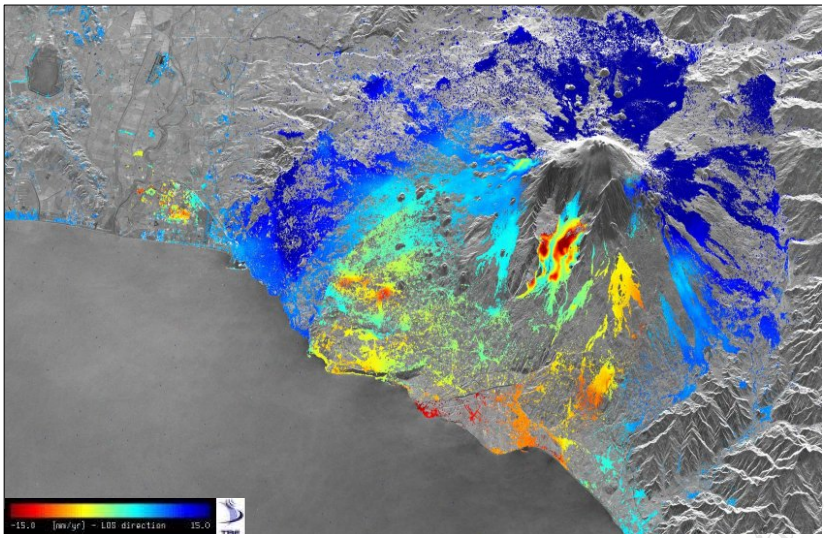


Figure 15.18: Estimate of deformation rate by PS interferometry, in mm/year: -20 red to + 20 blue. Area of Mt. Etna, 1995-2000, (www.treuropa.com).

the standard deviation of the unwrapped phase, for different values of samples (say number of PS), and compared with the bound:

$$\widehat{\sigma}_\phi = \frac{\sigma_\phi}{\sqrt{N}}$$

Notice that as the phase std approaches one radians, the estimation error diverges and approaches the limit $\pi/\sqrt{3} = 1.81$ for uniform pdf. The pdf of the wrapped phase, that is known in closed form (see [26]) is plotted in the same figure on the right.

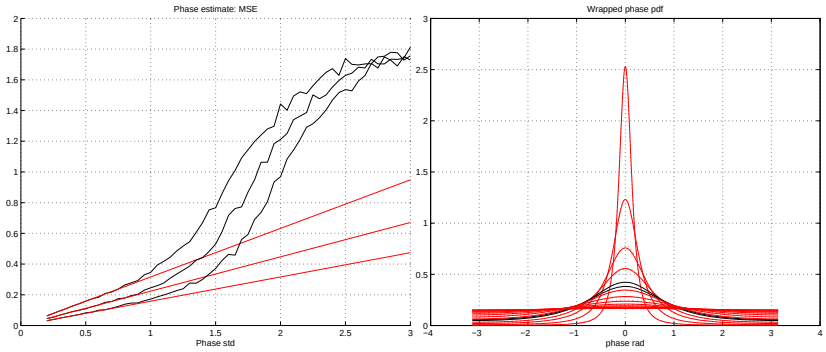


Figure 15.19: Left: standard deviation of the ML phase estimate by exploiting 10, 20 and 40 samples and for different values of the unwrapped phase std. Right: pdf of the wrapped phase for different valued of the unwrapped phase std, from 0.2 to 3 rad: the black curve refers to 1 rad, the limit value for a reliable estimation.

The average atmospheric std in C band is larger that the limit or 1 radians, and the added PS noise increase that error. This jeopardize the estimator, and prevents the success of PS technique. To avoid the problem, the approach described in [24] perform a first differentiation in space domain, that reduces the APS noise, due to is power-law low-pass spectrum. The elevation and velocity fields are then estimated, and integrated (thus reversing space differentials) to give the PS subsidence rate.

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